

Fundamentals of Enzymology

SECOND EDITION

BY

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Margaret, Jonathan, Rebekah, and Naomi, and
Evelyn, Rowena, Catherine, and Andrew.

Preface to the second edition

In preparing the second edition, we have attempted to highlight the major developments in enzymology in the seven years since the first edition appeared. Undoubtedly the most significant advances have been in the application of recombinant-DNA technology to the production of enzymes (Chapters 2 and 11), to the indirect determination of amino-acid sequences (Chapter 3) and to the analysis of enzyme structure and function using site-directed mutagenesis (Chapters 3 and 5). In many ways these advances can be said to have ushered in a 'golden age' of enzymology.

Other major advances have been made in the techniques of enzyme purification and analysis (Chapters 2 and 3), in the identification of catalytic domains within multienzyme proteins (Chapter 7), in understanding the organization of enzymes within membranes (Chapter 8), and in the application of purified enzymes in clinical analyses (Chapter 10).

We have been greatly encouraged by the response of colleagues to the first edition and hope that the new edition will continue to serve as a valuable guide for students and teachers of enzymology.

Stirling
August 1988

N.C.P.
L.S.

Preface to the first edition

Over the last few years a number of books that have dealt comprehensively with the kinetics, structures, and mechanisms of enzymes have appeared. In general, the emphasis of these books has been on understanding the behaviour of isolated enzymes and much less attention has been focused on enzymes in their natural environment in an organism. This book aims to restore the balance to some extent by discussing the properties of enzymes in systems of increasing complexity from isolated enzymes to enzymes in the cell. It should be of value to undergraduate students of biochemistry and related biological sciences. We have assumed that the reader will have an elementary knowledge of biochemistry such as might be gained from an introductory course at university.

In order to keep the length of the book within manageable limits, certain topics have been discussed only in outline. However, we have aimed to explain the principles involved in these topics in sufficient detail so that a student can pursue them if he or she wishes in the many books, research articles, and reviews referred to at the end of each chapter. We feel that it is important for undergraduates to make use of the original literature because biochemistry is such a rapidly expanding research-based subject.

In this book we have adopted the SI system of units and followed, as far as possible, the recommendations of the International Union of Biochemistry regarding the nomenclature of enzymes and substrates. Some of the recommended names may not be familiar to many biochemists, e.g. the enzyme RNA nucleotidyl transferase is more widely known as RNA polymerase. In such cases the more commonly used name is given as well as the recommended name in the Appendix at the end of the book, which also gives the reactions catalysed and Enzyme Commission numbers of all the enzymes mentioned in the book.

Stirling
January 1981

N.C.P.
L.S.

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It is particularly appropriate in a book on biological catalysts to thank the many people who helped to bring this work to fruition. We should like to thank our colleagues in the department for many useful discussions and especially to pay tribute to Dr Evelyn Stevens who improved the style of much of the book. Professors Michael Rossmann and David Blow and Drs Heiner Schirmer and Georg Schulz generously supplied original photographs of enzyme structures. Dr Hal Dixon gave us some valuable advice on nomenclature, and Dr Linda Fothergill helped with some points relating to amino-acid sequencing. Dr Neil Paterson gave some very helpful comments and advice on current medical applications of enzymology. We should also like to thank Mrs May Abrahamson for her excellent typing. In one sense, all these people acted as more than catalysts; their intervention improved the book, rather than leaving it unchanged. The errors are, of course, our responsibility.

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We would like to thank Khlayre Mullin, June Watson, and Marianne Burnett for their help in preparing the second edition, and Professor David Blow for supplying a photograph of the structure of tyrosyl-tRNA synthetase.

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A note on units

In this book we have tried, wherever possible, to use SI units. These are based on the metre-kilogram-second system of measurement. Multiples of the basic units by powers of 10 are as follows.

Prefix	Abbreviation	Prefix	Abbreviation
$10(10^1)$	deca	10^{-1}	deci
10^2	hecto	10^{-2}	centi
10^3	kilo	10^{-3}	milli
10^6	mega	10^{-6}	micro
10^9	giga	10^{-9}	nano
10^{12}	tera	10^{-12}	pico

We do not use compound prefixes, thus 10^{-9} metre (m) = 1 nm, not 1 m μ m. SI units for the various physical quantities mentioned in this book are listed below.

Quantity	SI unit
Time	<i>second (s).</i>
Length	<i>metre (m).</i> (Supplementary units retained for convenience are dm, cm) <i>kilogram (kg).</i> Note that multiples are based on 1 gram (g), i.e. mg, μ g rather than μ kg, nkg (see above compound prefix rule) <i>cubic metre (m³).</i> For convenience the litre (1 l = 1 dm ³) and millilitre (1 ml = 1 cm ³) are retained
Mass	
Volume (given in units of length cubed)	
Amount of substance	<i>mole (mol).</i> This quantity contains 1 Avogadro number of basic units (e.g. electrons, atoms, or molecules) <i>mol dm⁻³</i> used instead of molar (M) <i>mol kg⁻¹</i> used instead of molal (m)
Concentration	
Temperature	<i>kelvin (K)</i> note 0 °C = 273.15 K
Force	<i>newton (N)</i> (1 m kg s ⁻²)
Pressure	<i>pascal (Pa)</i> (1 Nm ⁻²). In this text, however, we have retained the atmosphere unit (atm)

List of abbreviations

The following standard abbreviations, which are in common use in biochemistry, are used in the text. Any others are defined where used.

A	absorbance
ACP	acyl-carrier protein
ADP	adenosine 5'-pyrophosphate (adenosine 5'-diphosphate)
AMP	adenosine 5'-phosphate (adenosine 5'-monophosphate)
ATP	adenosine 5'-triphosphate
cAMP	3':5' cyclic AMP
cDNA	complementary DNA
CoA	coenzyme A
CTP	cytidine 5'-triphosphate
dATP	2'-deoxyadenosine 5'-triphosphate
dGDP	2'-deoxyguanosine 5'-pyrophosphate (2'-deoxyguanosine 5'-diphosphate)
dGTP	2'-deoxyguanosine 5'-triphosphate
DNA	deoxyribonucleic acid
dTMP	thymidine 5'-phosphate (thymidine 5'-monophosphate or 2'-deoxyribosylthymine 5'-monophosphate)
dTTP	thymidine 5'-triphosphate (2'-deoxyribosylthymine 5'-triphosphate)
dUMP	2'-deoxyuridine 5'-phosphate (2'-deoxyuridine 5'-monophosphate)
EDTA	ethylenediaminetetra-acetate
FAD	flavin-adenine dinucleotide (oxidized form)
FADH ₂	flavin-adenine dinucleotide (reduced form)
g	gravitational field, unit of (9.81 m s ⁻²)
GDP	guanosine 5'-pyrophosphate (guanosine 5'-diphosphate)
GTP	guanosine 5'-triphosphate
h.p.l.c.	high-performance liquid chromatography
i.r.	infrared
ITP	inosine 5'-triphosphate
mRNA	messenger RNA
M _r	relative molecular mass
NAD ⁺	nicotinamide-adenine dinucleotide (oxidized form)
NADH	nicotinamide-adenine dinucleotide (reduced form)
NADP ⁺	nicotinamide-adenine dinucleotide phosphate (oxidized form)
NADPH	nicotinamide-adenine dinucleotide phosphate (reduced form)
NMN	nicotinamide mononucleotide
n.m.r.	nuclear magnetic resonance
RNA	ribonucleic acid
r.p.m.	revolutions per minute

SI unit

since the definition of standard states is

then less cumbersome, being under a pressure of 1 atm rather than 101.325 kPa.

(1 atm = 760 mm Hg = 101.325 kPa)

joule (J) (1 m² kg s⁻²).

One calorie = 4.18 J.

The gas constant, $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

(1 electron volt = 96.5 kJ mol⁻¹)

coulomb (C) (1 ampere second). Faraday

constant = 96 500 C mol⁻¹

hertz (Hz) (1 s⁻¹)

the units are kg m⁻¹ s⁻¹

(Note that 1 centipoise = 10⁻³ kg m⁻¹ s⁻¹).

electric charge

frequency

viscosity

the main implications of these units for biochemists are as follows.

Temperatures are quoted in kelvin (or absolute). Thus 25 °C is 298.15 K (in practice, this is given as 298 K).

Enthalpy, internal energy, and free-energy changes are given in J mol⁻¹. Entropies are quoted in J K⁻¹ mol⁻¹.

Molar concentrations are given in terms of mol dm⁻³ rather than M.

Enzyme activities are quoted in terms of katal (the amount of enzyme catalysing the transformation of 1 mol substrate per second). Specific activities are quoted in terms of katal kg⁻¹.

The curie (Ci) is redundant as a unit of radioactivity; 1 Ci = 3.7

× 10¹⁰ disintegrations s⁻¹. (The unit disintegrations s⁻¹ is given the name becquerel (Bq)).

For fuller discussions of SI units the following may be consulted:

quantities, units, and symbols (2nd edn). The Royal Society, London (1975).
 physicochemical quantities and units (2nd edn). M. L. McGlashan. Royal Institute of Chemistry, London (1971).

rRNA	ribosomal ribonucleic acid
tris	2-amino-2-hydroxymethyl-propane-1, 3-diol (tris(hydroxymethyl) methylamine)
tRNA	transfer RNA
UDP	uridine 5'-pyrophosphate (uridine 5'-diphosphate)
UMP	uridine 5'-phosphate (uridine 5'-monophosphate)
UTP	uridine 5'-triphosphate
u.v.	ultraviolet

Nucleotide sequences

A sequence of nucleotides can be written as, e.g., A C G C U C where each letter signifies a nucleotide.* The convention is to write the order so that the phosphodiester link runs 3' to 5' from left to right. A nucleoside 5'-phosphate is written as pX, and a 3'-phosphate as Xp. The sequence above could also be written as ApCpGpCpUpC. To indicate that there is a phosphate, pyrophosphate or triphosphate group at the 5'-end, we would write

pA C G C U C	5-phosphate
ppA C G C U C	5'-pyrophosphate
pppA C G C U C	5'-triphosphate

* The nucleotide is taken to be a ribonucleotide unit, unless the context makes it clear that it is a deoxyribonucleotide (e.g. in DNA). In a DNA sequence, the letter T refers to a thymidine phosphate (2'-deoxyribose/thymine phosphate) unit.

1

Introduction

1.1 Aims of the book

In this book we have tried to give a broad account of enzymology and have aimed to put current knowledge into perspective. Studies of enzymes have as their ultimate goal an understanding of the crucial role that these catalysts play in the metabolic processes of living organisms. Because of the complexity of such processes, it is at least necessary to gain an insight into the properties of enzymes in simpler systems, i.e. as isolated entities studied in the test tube or spectrophotometer cuvette. The chapters in the book follow a progression from the properties of isolated enzymes to the behaviour of enzymes in increasingly complex systems, leading up to the cell. We have included some discussion of the importance of enzymes in medicine and industry to emphasize that enzymology is not a purely academic subject but has increasingly wide applications.

1.2 Historical aspects

Enzymes are catalysts (i.e. they speed up the rates of reactions without themselves undergoing any permanent change). Each reaction taking place in the cell is catalysed by its own particular enzyme so that in a given cell there are a large number of enzymes. It is difficult to make a precise estimate of the number of different enzymes in each cell but it seems that a bacterial cell such as *Escherichia coli* makes about 3000 different proteins and a higher eukaryote cell about 50 000 of which the majority are enzymes. In the absence of enzymes many of these reactions would not occur even over a time period of years and life as we know it could not exist.

The word *enzyme* is derived from the Greek meaning 'in yeast' and was first used by Kühne in 1878.* At the time it was used to distinguish between what were referred to as 'organized ferments' (meaning whole microorganisms) and 'unorganized ferments' (meaning extracts or secretions from whole organisms). The term *enzyme* was thus intended to emphasize that catalytic activity was 'in yeast', i.e. a manifestation of an extract or secretion rather than of the whole organism. Although the term specified yeast, it was to be used for all 'unorganized ferments'. Of course, the catalytic activity of enzymes in microorganisms has been utilized by man for many thousands of years in processes such as fermentation and cheese-making but this was very much purely a practical use. It was only when it was shown that enzyme activity could be

* It was pointed out by Fritton¹ that the term was used as far back as the twelfth century by the Armenian philosopher, Theophrastus—although obviously not in a biochemical context.

without the need for an intact cellular structure that the study of
 could proceed along the paths already established in the study of
 In this respect Büchner's demonstration (1897) that filtrates of yeast
 could catalyse fermentation was highly significant.

Fischer in 1894 had performed some classical studies on carbohydrate
 bolizing enzymes in which he demonstrated the specificity shown by an
 ne for its *substrate* (the molecule acted on by the enzyme). On the basis of his
 riments, Fischer proposed the 'lock and key' hypothesis to describe this
 action and for many years this proved to be a fruitful way to picture the
 ing of enzyme to substrate (Fig. 1.1)

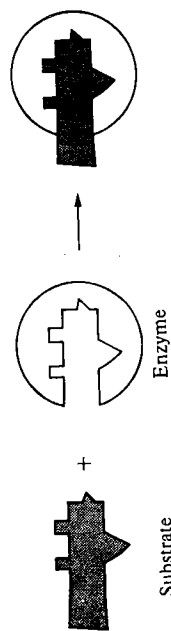


FIG. 1.1. Fischer's 'lock and key' hypothesis to explain enzyme specificity.

At this time (1890s) the chemical nature of enzymes was not clear. In fact this
 nt was only established many years later after a number of enzymes had been
 stalled and been shown to consist entirely of protein (i.e. made up of amino
 ls linked by amide bonds, see Chapter 3, Section 3.3). The first enzyme to be
 stalled (in 1926 by Sumner) was *urease*, which catalyses the hydrolysis of
 a to yield *carbon dioxide* and *ammonia*. Actually, it is now known that if
 nner had had more sensitive methods of analysis he would have found that
 preparation contained a small amount of nickel (approximately 0.1 per cent
 weight) which is essential for catalysis.² Perhaps in retrospect it was fortunate
 it he did not have such sensitive methods, since otherwise the nature of
 zymes might not have been settled for many more years.

The development of the ultracentrifuge by Svedberg (also in the 1920s) allowed
 ry high centrifugal fields capable of sedimenting macromolecules* to be
 nerated. These studies showed that proteins in solution generally consist of
 mogeneous molecules of definite M_r (in the case of enzymes the M_r values
 nge between about 10^4 and 10^7) rather than of colloidal suspensions. The
 scription of enzyme structures in precise chemical terms was then a realistic
 ssibility: this was first achieved in 1960 when the amino-acid sequence of
 onuclease³ (an enzyme catalysing the hydrolysis of ribonucleic acid) was
 duced. In 1965 the three-dimensional structure of lysozyme (an enzyme
 eaving certain bacterial cell walls) was deduced by the technique of X-ray
 ystallography (see Chapter 3, Section 3.5.1), and for the first time a mechanism
 action could be postulated in precise structural terms. We now know the

amino-acid sequence and three-dimensional structure of many enzymes and
 generalizations and comparisons between them can be undertaken.

During the late 1950s and the 1960s a number of observations were made that
 suggested that enzymes show considerable flexibility. In 1958 Koshland⁴
 proposed the 'induced fit' theory to account for the catalytic power and
 specificity shown by enzymes (see Chapter 5, Section 5.6). It also became clear
 that the catalytic activity of certain enzymes could respond to changes in
 physiological conditions. Monod and his colleagues^{5,6} proposed their 'allosteric
 model' to explain in a quantitative way how the activity of certain enzymes can
 be regulated by the binding of small molecules (*effectors*) and this provided a
 basis for understanding many features of the control of enzymes in the cell. An
 important feature of models for allosteric enzymes in general is that they
 postulate that the binding of effectors to the enzymes induces structural changes
 in the enzymes (Chapter 6, Section 6.2.2).

The first chemical synthesis of an enzyme (ribonuclease) from amino-acid
 precursors was reported in 1969.⁷ Although this represented a considerable
 achievement, it should be noted that both the chemical purity and catalytic
 activity of the preparation were rather low. The chemical synthesis can, perhaps,
 be said to represent the final proof that enzymes are no different qualitatively
 from other non-biological catalysts.

The application of recombinant DNA techniques to the study of enzymes has
 produced some remarkable new insights (see Chapter 3, Section 3.4.1 and
 Chapter 5, Sections 5.4.5 and 5.5). It has proved possible to alter catalytic activity
 and specificity in a rational manner by introducing mutations at defined
 positions using site-directed mutagenesis. This has helped in understanding the
 mechanism of enzyme action and has also opened the prospect of designing
 enzymes with specific required properties.^{3,3} For example, the specificity of
 lactate dehydrogenase has been changed to that of malate dehydrogenase by
 introducing three particular mutations at the active site (see Chapter 5, Section
 5.5.4.5).

Recently it has become evident that catalytic activity can be shown to a limited
 extent by biological molecules other than the 'classical' enzymes. In the process
 of enzyme catalysis the substrates are converted via a high-energy transition
 state to the eventual products (see Chapter 5, Fig. 5.1). A number of reports have
 shown that antibodies raised to stable analogues of the transition states of
 hydrolytic reactions (see Chapter 5, Section 5.3.4) can act as moderately effective
 catalysts for those reactions.³⁰⁻³² Fragments of RNA can also act as catalysts for
 reactions involving hydrolysis of RNA.^{8,9} The observation of these 'ribozymes'
 has important implications for theories concerning the evolution of catalytic
 function.¹⁰

For further details on the history of enzymology see references 11 and 12.

1.3 Remarkable properties of enzymes as catalysts

Enzymes display a number of remarkable properties when compared with other
 types of catalyst. The three most important are their high catalytic power, their

* The term macromolecule is generally reserved for molecules of relative molecular mass
 $M_r > 10000$

specificity, and the extent to which their catalytic activity can be regulated by a variety of naturally occurring compounds. These three properties will be illustrated below.

1.3.1 Catalytic power

Enzymes may increase the rate of a reaction by as much as 10^{14} -fold. There are not many examples where a direct comparison can be made between the rates of an enzyme-catalysed reaction and the reaction occurring under similar conditions of temperature, pH, etc. but in the absence of enzyme. This is because in the absence of an enzyme the rates may be too low to be measured easily. Where the comparison has been made, very high rate enhancements have been found, ¹³ e.g. hexokinase $> 10^{10}$, phosphorylase $> 3 \times 10^{11}$, alcohol dehydrogenase $> 2 \times 10^8$, and creatine kinase $> 10^4$. In other instances, where enzymatic and non-enzymatic catalysts are compared, the former catalyse much higher rates and in some cases do so at significantly lower temperatures (Table 1.1). The optimum conditions for enzyme catalysis are almost invariably moderate temperatures, and pHs that are not extreme. (The contrast between an enzyme-catalysed reaction and that catalysed by a non-enzymatic catalyst is well illustrated by the process of nitrogen fixation (i.e. reduction of N_2 to ammonia). Nitrogenase* catalyses this reaction at temperatures around 300 K and at neutral pH. The enzyme is a complex system comprising two dissociating protein components, one of which contains iron and the other iron and molybdenum.¹⁵ Several molecules of ATP are hydrolysed during the reduction, although the exact stoichiometry is still uncertain. By contrast, in the industrial synthesis of ammonia from nitrogen and hydrogen the conditions used are as follows:

TABLE 1.1
Examples of the catalytic power of enzymes

Substrate	Catalyst	Temperature (K)	Rate constant k (mol dm ⁻³) ⁻¹ s ⁻¹
Amide (hydrolysis)			
benzamide	H ⁺	325	2.4×10^{-6}
benzamide	OH ⁻	326	8.5×10^{-6}
benzoyl-L-tyrosinamide	α -chymotrypsin	298	14.9
Urea (hydrolysis)	H ⁺	335	7.4×10^{-7}
urease	urease	294	5.0×10^6
$2H_2O_2 \rightarrow 2H_2O + O_2$	Fe ²⁺	295	56
	catalase	295	3.5×10^7

Data taken from reference 14.

* Only certain prokaryotes can carry out nitrogen fixation and they may be either symbionts or non-symbionts. In terrestrial ecosystems, symbiotic fixation appears to exceed non-symbiotic and the most outstanding example of symbiotic fixation is that between bacteria of the genus *Rhizobium* and the roots of leguminous plants.

temperatures between 700 and 900 K, pressures between 100 and 900 atmospheres, and the presence of an iron catalyst, often promoted by traces of oxides of other metals.¹⁶ The basis of catalytic power is discussed in Chapter 5, Section 5.3.

1.3.2 Specificity

Most enzymes are highly specific both in the nature of the substrate(s) they utilize and also in the reaction they catalyse. The range of specificity varies between enzymes. There are some enzymes that have relatively low specificities (broad specificity), e.g. certain peptidases, phosphatases, and esterases, which will utilize a wide range of substrates provided they contain the required chemical bond, i.e. peptide, phosphate ester, and carboxylate ester, respectively, in these three examples. Low specificity is more commonly encountered with degradative/enzymes but is only very rarely observed with biosynthetic enzymes. The role of the former may be that of digestion, where wider specificity would be more economical. An intermediate set of enzymes show group specificity, e.g. hexokinase. This enzyme will catalyse the phosphorylation of a variety of sugars provided they are aldohexoses. However, many enzymes show absolute or near-absolute specificity, in which they will only catalyse at an appreciable rate the reaction with a single substrate (or a single pair of substrates in a bimolecular reaction), e.g. urease will only catalyse the reaction with urea, or with very similar analogues at a very much lower rate. The quantitative definition of specificity in kinetic terms is described in Chapter 4, Section 4.3.1.3.

The terms *group specificity* and *absolute specificity* can be readily appreciated in relation to low-*M_r* substrates, but when considering macromolecular substrates the position may be somewhat different in the sense that the active site of the enzyme can only interact with a part of the macromolecule. This is simply due to the relative dimensions of the active site and the macromolecule. A group of enzymes that has been studied extensively in the past decade is the restriction endonucleases. These enzymes generally recognize a sequence of four to six base pairs in DNA and then cleave the phosphodiester links in both strands but not necessarily in opposite positions. There are now known at least 400 of these enzymes differing in specificity.¹⁷ In a sense, each enzyme shows absolute specificity for the region of the substrate that is in contact with the active site, although it can act on any DNA molecule or fragment that contains the appropriate sequence.

Another distinct feature of many enzyme-catalysed reactions is their stereo-specificity; this is well illustrated in the case of NAD⁺- and NADP⁺-requiring dehydrogenases. It has been demonstrated by use of suitably labelled substrates that dehydrogenases catalyse the transfer of hydrogen from the substrate on to a particular side of the nicotinamide ring; these are designated A-side and B-side dehydrogenases (Fig. 1.2). In addition, almost all dehydrogenases act on either NAD⁺ or NADP⁺. The basis of these specificities is clear in the case of those dehydrogenases whose three-dimensional structures are known, e.g. liver alcohol dehydrogenase, lactate dehydrogenase, and glyceraldehyde-3-phosphate dehydrogenase¹⁸ and is discussed further in Chapter 5, Section 5.5.4.3.

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A note on units

In this book we have tried, wherever possible, to use SI units. These are based on the metre-kilogram-second system of measurement. Multiples of the basic units by powers of 10 are as follows.

Prefix	Abbreviation	Prefix	Abbreviation
$10(10^1)$	deca	10^{-1}	deci
10^2	hecto	10^{-2}	centi
10^3	kilo	10^{-3}	milli
10^6	mega	10^{-6}	micro
10^9	giga	10^{-9}	nano
10^{12}	tera	10^{-12}	pico

We do not use compound prefixes, thus 10^{-9} metre (m) = 1 nm, not 1 m μ m. SI units for the various physical quantities mentioned in this book are listed below.

Quantity	SI unit
Time	<i>second</i> (s).
Length	<i>metre</i> (m). (Supplementary units retained for convenience are dm, cm)
Mass	<i>kilogram</i> (kg). Note that multiples are based on 1 gram (g), i.e. mg, μ g rather than μ kg, nkg (see above compound prefix rule)
Volume (given in units of length cubed)	<i>cubic metre</i> (m^3). For convenience the litre (l = 1 dm ³) and millilitre (1 ml = 1 cm ³) are retained
Amount of substance	<i>mole</i> (mol). This quantity contains 1 Avogadro number of basic units (e.g. electrons, atoms, or molecules)
Concentration	$mol\ dm^{-3}$ used instead of molar (M) $mol\ kg^{-1}$ used instead of molal (m)
Temperature	<i>kelvin</i> (K) note $0^\circ C = 273.15\ K$
Force	<i>newton</i> (N) ($1\ m\ kg\ s^{-2}$)
Pressure	<i>pascal</i> (Pa) ($1\ N\ m^{-2}$). In this text, however, we have retained the atmosphere unit (atm)

Quantity

SI unit

since the definition of standard states is then less cumbersome, being under a pressure of 1 atm rather than 101.325 kPa.

(1 atm = 760 mm Hg = 101.325 kPa)

joule (J) ($1 \text{ m}^2 \text{ kg s}^{-2}$).

One calorie = 4.18 J.

The gas constant, $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

(1 electron volt = 96.5 kJ mol^{-1})

coulomb (C) (1 ampere second). Faraday

constant = $96\,500 \text{ C mol}^{-1}$

hertz (Hz) (1 s^{-1})

the units are $\text{kg m}^{-1} \text{ s}^{-1}$

(Note that 1 centipoise = $10^{-3} \text{ kg m}^{-1} \text{ s}^{-1}$).

Energy

Electric charge

Frequency

Viscosity

The main implications of these units for biochemists are as follows:

- Temperatures are quoted in kelvin (or absolute). Thus 25°C is 298.15 K (in practice, this is given as 298 K).
- Enthalpy, internal energy, and free-energy changes are given in J mol^{-1} .
- Entropies are quoted in $\text{J K}^{-1} \text{ mol}^{-1}$.
- Molar concentrations are given in terms of mol dm^{-3} rather than M .
- Enzyme activities are quoted in terms of katal (the amount of enzyme catalysing the transformation of 1 mol substrate per second). Specific activities are quoted in terms of katal kg^{-1} .
- The curie (Ci) is redundant as a unit of radioactivity; $1 \text{ Ci} = 3.7 \times 10^{10}$ disintegrations s^{-1} . (The unit disintegrations s^{-1} is given the name *becquerel* (Bq)).

For fuller discussions of SI units the following may be consulted:

Quantities, units, and symbols (2nd edn). The Royal Society, London (1975).
Physicochemical quantities and units (2nd edn). M. L. McGlashan. Royal Institute of Chemistry, London (1971).

The following standard abbreviations, which are in common use in biochemistry, are used in the text. Any others are defined where used.

List of abbreviations

A	absorbance
ACP	acyl-carrier protein
ADP	adenosine 5'-pyrophosphate (adenosine 5'-diphosphate)
AMP	adenosine 5'-phosphate (adenosine 5'-monophosphate)
ATP	adenosine 5'-triphosphate
cAMP	3':5' cyclic AMP
cDNA	complementary DNA
CoA	coenzyme A
CTP	cytidine 5'-triphosphate
dATP	2'-deoxyadenosine 5'-triphosphate
dGDP	2'-deoxyguanosine 5'-pyrophosphate (2'-deoxyguanosine 5'-diphosphate)
dGTP	2'-deoxyguanosine 5'-triphosphate
DNA	deoxyribonucleic acid
dTMP	thymidine 5'-phosphate (thymidine 5'-monophosphate or 2'-deoxyribosylthymine 5'-monophosphate)
dTTP	thymidine 5'-triphosphate (2'-deoxyribosylthymine 5'-triphosphate)
dUMP	2'-deoxyuridine 5'-phosphate (2'-deoxyuridine 5'-monophosphate)
EDTA	ethylenediaminetetra-acetate
FAD	flavin-adenine dinucleotide (oxidized form)
FADH ₂	flavin-adenine dinucleotide (reduced form)
^g GDP	gravitational field, unit of (9.81 m s^{-2})
GTP	guanosine 5'-pyrophosphate (guanosine 5'-diphosphate)
h.p.l.c.	high-performance liquid chromatography
i.r.	infrared
ITP	inosine 5'-triphosphate
mRNA	messenger RNA
M_r	relative molecular mass
NAD ⁺	nicotinamide-adenine dinucleotide (oxidized form)
NADH	nicotinamide-adenine dinucleotide (reduced form)
NADP ⁺	nicotinamide-adenine dinucleotide phosphate (oxidized form)
NADPH	nicotinamide-adenine dinucleotide phosphate (reduced form)
NMN	nicotinamide mononucleotide
n.m.r.	nuclear magnetic resonance
RNA	ribonucleic acid
r.p.m.	revolutions per minute



rRNA	ribosomal ribonucleic acid
tris	2-amino-2-hydroxymethyl-propane-1, 3-diol (tris(hydroxymethyl) methylamine)
tRNA	transfer RNA
UDP	uridine 5'-pyrophosphate (uridine 5'-diphosphate)
UMP	uridine 5'-phosphate (uridine 5'-monophosphate)
UTP	uridine 5'-triphosphate
u.v.	ultraviolet

Nucleotide sequences

A sequence of nucleotides can be written as, e.g., ACGCUC where each letter signifies a nucleotide.* The convention is to write the order so that the phosphodiester link runs 3' to 5' from left to right. A nucleoside 5'-phosphate is written as pX, and a 3'-phosphate as Xp. The sequence above could also be written as ApCpGpCpUpC. To indicate that there is a phosphate, pyrophosphate or triphosphate group at the 5'-end, we would write

pA	C	G	C	U	C	
ppA	C	G	C	U	C	
pppA	C	G	C	U	C	
						5'-phosphate
						5'-pyrophosphate
						5'-triphosphate

* The nucleotide is taken to be a ribonucleotide unit, unless the context makes it clear that it is a deoxyribonucleotide (e.g. in DNA). In a DNA sequence, the letter T refers to a thymidine phosphate (2'-deoxyribosylthymine phosphate) unit.

1 Introduction

1.1 Aims of the book

In this book we have tried to give a broad account of enzymology and have aimed to put current knowledge into perspective. Studies of enzymes have as their ultimate goal an understanding of the crucial role that these catalysts play in the metabolic processes of living organisms. Because of the complexity of such processes, it is at least necessary to gain an insight into the properties of enzymes in simpler systems, i.e. as isolated entities studied in the test tube or spectrophotometer cuvette. The chapters in the book follow a progression from the properties of isolated enzymes to the behaviour of enzymes in increasingly complex systems, leading up to the cell. We have included some discussion of the importance of enzymes in medicine and industry to emphasize that enzymology is not a purely academic subject but has increasingly wide applications.

1.2 Historical aspects

Enzymes are catalysts (i.e. they speed up the rates of reactions without themselves undergoing any permanent change). Each reaction taking place in the cell is catalysed by its own particular enzyme so that in a given cell there are a large number of enzymes. It is difficult to make a precise estimate of the number of different enzymes in each cell but it seems that a bacterial cell such as *Escherichia coli* makes about 3000 different proteins and a higher eukaryote cell about 50 000 of which the majority are enzymes. In the absence of enzymes many of these reactions would not occur even over a time period of years and life as we know it could not exist.

The word *enzyme* is derived from the Greek meaning 'in yeast' and was first used by Kühne in 1878.* At the time it was used to distinguish between what were referred to as 'organized ferments' (meaning whole microorganisms) and 'unorganized ferments' (meaning extracts or secretions from whole organisms). The term *enzyme* was thus intended to emphasize that catalytic activity was 'in yeast', i.e. a manifestation of an extract or secretion rather than of the whole organism. Although the term specified yeast, it was to be used for all 'unorganized ferments'. Of course, the catalytic activity of enzymes in microorganisms has been utilized by man for many thousands of years in processes such as fermentation and cheese-making but this was very much purely a practical use. It was only when it was shown that enzyme activity could be

* It was pointed out by Fruton¹ that the term was used as far back as the twelfth century by the Armenian philosopher, Theophrastus—although obviously not in a biochemical context.

expressed without the need for an intact cellular structure that the study of enzymes could proceed along the paths already established in the study of chemistry. In this respect Buchner's demonstration (1897) that filtrates of yeast extracts could catalyse fermentation was highly significant.

Emil Fischer in 1894 had performed some classical studies on carbohydrate metabolizing enzymes in which he demonstrated the specificity shown by an enzyme for its *substrate* (the molecule acted on by the enzyme). On the basis of his experiments, Fischer proposed the 'lock and key' hypothesis to describe this interaction and for many years this proved to be a fruitful way to picture the binding of enzyme to substrate (Fig. 1.1)

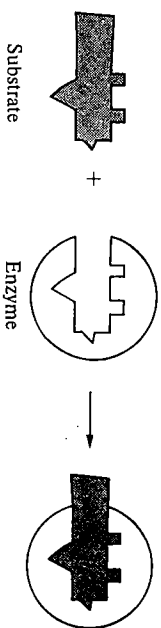


FIG. 1.1. Fischer's 'lock and key' hypothesis to explain enzyme specificity.

At this time (1890s) the chemical nature of enzymes was not clear. In fact this point was only established many years later after a number of enzymes had been crystallized and been shown to consist entirely of protein (i.e. made up of amino acids linked by amide bonds, see Chapter 3, Section 3.3). The first enzyme to be crystallized (in 1926 by Sumner) was urease, which catalyses the hydrolysis of urea to yield carbon dioxide and ammonia. Actually, it is now known that if Sumner had had more sensitive methods of analysis he would have found that the preparation contained a small amount of nickel (approximately 0.1 per cent by weight) which is essential for catalysis.² Perhaps in retrospect it was fortunate that he did not have such sensitive methods, since otherwise the nature of enzymes might not have been settled for many more years.

The development of the ultracentrifuge by Svedberg (also in the 1920s) allowed very high centrifugal fields capable of sedimenting macromolecules* to be generated. These studies showed that proteins in solution generally consist of homogeneous molecules of definite M_r (in the case of enzymes the M_r values range between about 10^4 and 10^7) rather than of colloidal suspensions. The description of enzyme structures in precise chemical terms was then a realistic possibility: this was first achieved in 1960 when the amino-acid sequence of ribonuclease³ (an enzyme catalysing the hydrolysis of ribonucleic acid) was deduced. In 1965 the three-dimensional structure of lysozyme (an enzyme cleaving certain bacterial cell walls) was deduced by the technique of X-ray crystallography (see Chapter 3, Section 3.5.1), and for the first time a mechanism of action could be postulated in precise structural terms. We now know the

* The term macromolecule is generally reserved for molecules of relative molecular mass (M_r) > 10000

amino-acid sequence and three-dimensional structure of many enzymes and generalizations and comparisons between them can be undertaken.

During the late 1950s and the 1960s a number of observations were made that suggested that enzymes show considerable flexibility. In 1958 Koshland⁴ proposed the 'induced fit' theory to account for the catalytic power and specificity shown by enzymes (see Chapter 5, Section 5.6). It also became clear that the catalytic activity of certain enzymes could respond to changes in physiological conditions. Monod and his colleagues^{5,6} proposed their 'allosteric model' to explain in a quantitative way how the activity of certain enzymes can be regulated by the binding of small molecules (*effectors*) and this provided a basis for understanding many features of the control of enzymes in the cell. An important feature of models for allosteric enzymes in general is that they postulate that the binding of effectors to the enzymes induces structural changes in the enzymes (Chapter 6, Section 6.2.2).

The first chemical synthesis of an enzyme (ribonuclease) from amino-acid precursors was reported in 1969.⁷ Although this represented a considerable achievement, it should be noted that both the chemical purity and catalytic activity of the preparation were rather low. The chemical synthesis can, perhaps, be said to represent the final proof that enzymes are no different qualitatively from other non-biological catalysts.

The application of recombinant DNA techniques to the study of enzymes has produced some remarkable new insights (see Chapter 3, Section 3.4.1 and Chapter 5, Sections 5.4.5 and 5.5). It has proved possible to alter catalytic activity and specificity in a rational manner by introducing mutations at defined positions using site-directed mutagenesis. This has helped in understanding the mechanism of enzyme action and has also opened the prospect of designing enzymes with specific required properties.³³ For example, the specificity of lactate dehydrogenase has been changed to that of malate dehydrogenase by introducing three particular mutations at the active site (see Chapter 5, Section 5.5.4.5).

Recently it has become evident that catalytic activity can be shown to a limited extent by biological molecules other than the 'classical' enzymes. In the process of enzyme catalysis the substrates are converted via a high-energy transition state to the eventual products (see Chapter 5, Fig. 5.1). A number of reports have shown that antibodies raised to stable analogues of the transition states of hydrolytic reactions (see Chapter 5, Section 5.3.4) can act as moderately effective catalysts for those reactions.³⁰⁻³² Fragments of RNA can also act as catalysts for reactions involving hydrolysis of RNA.^{8,9} The observation of these 'ribozymes' has important implications for theories concerning the evolution of catalytic function.¹⁰

For further details on the history of enzymology see references 11 and 12.

1.3 Remarkable properties of enzymes as catalysts

Enzymes display a number of remarkable properties when compared with other types of catalyst. The three most important are their high catalytic power, their

specificity, and the extent to which their catalytic activity can be regulated by a variety of naturally occurring compounds. These three properties will be illustrated below.

1.3.1 Catalytic power

Enzymes may increase the rate of a reaction by as much as 10^{14} -fold. There are not many examples where a direct comparison can be made between the rates of an enzyme-catalysed reaction and the reaction occurring under similar conditions of temperature, pH, etc. but in the absence of enzyme. This is because in the absence of an enzyme the rates may be too low to be measured easily. Where the comparison has been made, very high rate enhancements have been found, ¹³ e.g. hexokinase $> 10^{10}$, phosphorylase $> 3 \times 10^{11}$, alcohol dehydrogenase $> 2 \times 10^8$, and creatine kinase $> 10^4$. In other instances, where enzymatic and non-enzymatic catalysts are compared, the former catalyse much higher rates and in some cases do so at significantly lower temperatures (Table 1.1). The optimum conditions for enzyme catalysis are almost invariably moderate temperatures, and pHs that are not extreme. (The contrast between an enzyme-catalysed reaction and that catalysed by a non-enzymatic catalyst is well illustrated by the process of nitrogen fixation (i.e. reduction of N_2 to ammonia). Nitrogenase* catalyses this reaction at temperatures around 300 K and at neutral pH. The enzyme is a complex system comprising two dissociating protein components, one of which contains iron and the other iron and molybdenum.¹⁵ Several molecules of ATP are hydrolysed during the reduction, although the exact stoichiometry is still uncertain. By contrast, in the industrial synthesis of ammonia from nitrogen and hydrogen the conditions used are as follows:

TABLE 1.1
Examples of the catalytic power of enzymes

Substrate	Catalyst	Temperature (K)	Rate constant k (mol dm ⁻³) ⁻¹ s ⁻¹
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benzoyl-L-tyrosinamide	α -chymotrypsin	298	14.9
Urea (hydrolysis)	H ⁺	335	7.4×10^{-7}
	urease	294	5.0×10^6
$2H_2O_2 \rightarrow 2H_2O + O_2$	Fe ²⁺	295	56
	catalase	295	3.5×10^7

Data taken from reference 14.

* Only certain prokaryotes can carry out nitrogen fixation and they may be either symbionts or non-symbionts. In terrestrial ecosystems, symbiotic fixation appears to exceed non-symbiotic and the most outstanding example of symbiotic fixation is that between bacteria of the genus *Rhizobium* and the roots of leguminous plants.

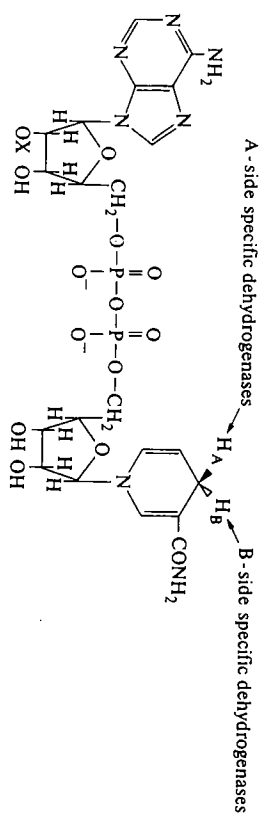
temperatures between 700 and 900 K, pressures between 100 and 900 atmospheres, and the presence of an iron catalyst, often promoted by traces of oxides of other metals.¹⁶ The basis of catalytic power is discussed in Chapter 5, Section 5.3.

1.3.2 Specificity

Most enzymes are highly specific both in the nature of the substrate(s) they utilize and also in the reaction they catalyse. The range of specificity varies between enzymes. There are some enzymes that have relatively low specificities (broad specificity), e.g. certain peptidases, phosphatases, and esterases, which will utilize a wide range of substrates provided they contain the required chemical bond, i.e. peptide, phosphate ester, and carboxylate ester, respectively, in these three examples. (Low specificity is more commonly encountered with degradative enzymes but is only very rarely observed with biosynthetic enzymes. The role of the former may be that of digestion, where wider specificity would be more economical. An intermediate set of enzymes show group specificity, e.g. hexokinase. This enzyme will catalyse the phosphorylation of a variety of sugars provided they are aldohexoses. However, many enzymes show absolute or near-absolute specificity, in which they will only catalyse at an appreciable rate the reaction with a single substrate (or a single pair of substrates in a bimolecular reaction), e.g. urease will only catalyse the reaction with urea, or with very similar analogues at a very much lower rate. The quantitative definition of specificity in kinetic terms is described in Chapter 4, Section 4.3.1.3.

The terms group specificity and absolute specificity can be readily appreciated in relation to low- M_r substrates, but when considering macromolecular substrates the position may be somewhat different in the sense that the active site of the enzyme can only interact with a part of the macromolecule. This is simply due to the relative dimensions of the active site and the macromolecule. A group of enzymes that has been studied extensively in the past decade is the restriction endonucleases. These enzymes generally recognize a sequence of four to six base pairs in DNA and then cleave the phosphodiester links in both strands but not necessarily in opposite positions. There are now known at least 400 of these enzymes differing in specificity.¹⁷ In a sense, each enzyme shows absolute specificity for the region of the substrate that is in contact with the active site, although it can act on any DNA molecule or fragment that contains the appropriate sequence.

Another distinct feature of many enzyme-catalysed reactions is their stereospecificity; this is well illustrated in the case of NAD^+ and $NADP^+$ -requiring dehydrogenases. It has been demonstrated by use of suitably labelled substrates that dehydrogenases catalyse the transfer of hydrogen from the substrate on to a particular side of the nicotinamide ring; these are designated A-side and B-side dehydrogenases (Fig. 1.2). In addition, almost all dehydrogenases act on either NAD^+ or $NADP^+$. The basis of these specificities is clear in the case of those dehydrogenases whose three-dimensional structures are known, e.g. liver alcohol dehydrogenase, lactate dehydrogenase, and glyceraldehyde-3-phosphate dehydrogenase¹⁸ and is discussed further in Chapter 5, Section 5.5.4.3.



Reduced nicotinamide adenine dinucleotide (NADH), X = H, and reduced nicotinamide adenine dinucleotide phosphate (NADPH), X = phosphate.

A-side dehydrogenases

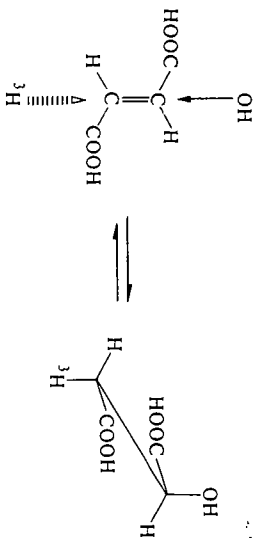
Alcohol dehydrogenase (NAD⁺)
Lactate dehydrogenase (NAD⁺)
Malate dehydrogenase (NAD⁺)
Cytochrome *b*₅ reductase (NAD⁺)
Sikimate dehydrogenase (NADP⁺)
3-Oxyacyl [acyl-carrier-protein] reductase (NADP⁺)

B-side dehydrogenases

Glycerol 3-phosphate dehydrogenase (NAD⁺)
3-Hydroxybutyrate dehydrogenase (NAD⁺)
Glucose dehydrogenase (NAD⁺)
Glyceraldehyde-3-phosphate dehydrogenase (NAD⁺)
Homoserine dehydrogenase (NADP⁺)
Glucose-6-phosphate dehydrogenase (NADP⁺)

FIG. 1.2. Stereospecificity of NAD⁺- and NADP⁺-requiring enzymes. For a more complete list see reference 18

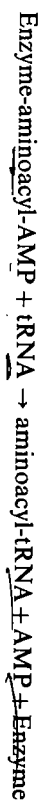
There are a number of other examples of prochirality in enzyme-catalysed reactions.²⁰ These reactions occur in such a manner that although the reactant does not have a chiral* centre the reaction occurs in a stereospecific manner. A good example is the reaction catalysed by fumarate hydratase in which fumarate is converted to malate. It can be shown by carrying out the reaction in the presence of tritiated water (³H₂O) that ³H addition occurs in a stereospecific manner. There are several other examples of this phenomenon.



A further aspect of enzyme specificity has recently been recognized and is important in protein synthesis and DNA replication. It is known that the error rate in DNA replication *in vivo* is as low as one mistake in 10⁸ – 10¹⁰ nucleotides polymerized and that the overall error rate in transcribing DNA and then translating the mRNA into protein (see Fig. 9.1) is about 1 in 10⁴ amino-acid

* A compound is said to be chiral if it cannot be superimposed on its mirror image.

residues incorporated. This specificity is higher than would be expected from the relative energies of interaction between structurally similar amino acids and the aminoacyl-tRNA synthetases and is only possible because of a proof reading or editing mechanism.²¹ The essence of this mechanism in the case of aminoacyl-tRNA synthetases will be described briefly here (a fuller account is given in Chapter 5, Section 5.5.5). The charging of tRNA catalysed by aminoacyl-tRNA synthetases occurs in two steps



The enzyme has to recognize a specific amino acid and also a specific tRNA. The latter is recognized very precisely since it is a large molecule and thus makes many contacts with the enzyme. The amino acids, which are very much smaller, are not selected so precisely. It is possible to monitor both the first and second steps in the reaction and this shows that significant errors do occur in the formation of the aminoacyl-adenylate intermediates but not in the formation of aminoacyl-tRNAs. However, any incorrect aminoacyl-adenylates formed rapidly become hydrolysed. There is evidence that the enzyme has a second site, distinct from that on which the synthesis occurs, which enables it to hydrolyse off mismatched amino acids, thus acting as an editing mechanism. A similar mechanism may also occur to ensure accurate DNA replication. The enzyme DNA nucleotidyltransferase (DNA polymerase) III also possesses exonuclease activity which acts in editing DNA replication²¹ (see also Chapter 7, Section 7.11.3).

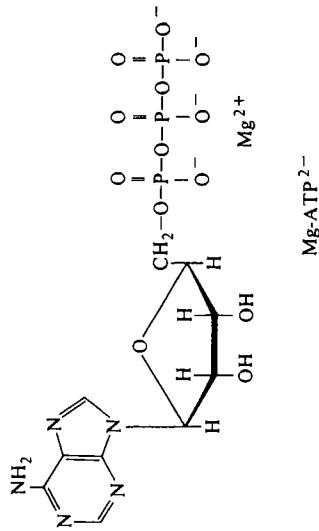
1.3.3 Regulation

The third important property of enzymes is that their catalytic activity may be regulated by small ions or other molecules. An example is the enzyme phosphatase, which catalyses the first step in the breakdown of glycogen in skeletal muscle. This is an important enzyme that enables the carbohydrate reserves to be degraded in order to generate ATP required for muscle contraction. The onset of muscle contraction is triggered by release of Ca²⁺ from the sarcoplasmic reticulum (see Chapter 8, Section 8.4.4) and this also brings about the activation of phosphatase to ensure the continued production of ATP. The mechanisms of regulation of phosphatase both by Ca²⁺ and also by hormones such as adrenalin are quite complex and are discussed in Chapter 6, Section 6.4.2.

The phenomenon of feedback inhibition is common in many biosynthetic pathways. For example, in the biosynthetic pathway leading to the synthesis of pyrimidine nucleotides, the end products UTP and CTP are able to inhibit the first enzyme in the pathway; thus they are able to limit the flow of metabolites into that pathway and so regulate their own biosynthesis. This regulation is effected through changes in the catalytic activity of early enzymes in the pathway, carbamoyl-phosphate synthase and aspartate carbamoyltransferase; this topic is discussed in Chapter 6, Sections 6.2.2.1 and 6.3.1.

1.4 Cofactors

The demonstration of enzyme crystallinity and analysis of these crystals has shown that enzymes are proteins with chain lengths generally of 100–2500 amino acids. Enzymes such as chymotrypsin (a protein-hydrolysing enzyme) or triose-phosphate isomerase (an enzyme catalysing the interconversion of two phosphorylated three-carbon compounds) are active without needing any other factor present. However, many enzymes require a non-protein component for activity; this is termed a cofactor. One group of cofactors is metal ions. Thus carboxypeptidase (an enzyme catalysing the hydrolysis of proteins from the C-terminal end) requires zinc for activity. Removal of zinc (e.g. by EDTA) leads to inactivation of the enzyme; activity can be restored by addition of zinc, or to a lesser extent, by other metals. In this case, it appears that zinc forms an integral part of the enzyme structure. The kinases (enzymes that catalyse the transfer of the γ -phosphoryl group of ATP to some acceptor molecule) have a requirement for magnesium ions, but the evidence suggests that, in these cases, the metal ion is actually bound to substrate rather than to the enzyme (i.e. the true substrate is Mg-ATP²⁻ rather than ATP). Some examples of metal ion requirements among enzymes are given in Table 1.2.



The second major class of cofactors are organic cofactors (many of which are derivatives of B vitamins). Thus most carboxylases (enzymes involved in the incorporation of CO₂) requires biotin, which is covalently linked to its enzyme. Similarly, in the case of aminotransferases (enzymes involved in interconversions of oxo-acids and amino acids), pyridoxal phosphate is bound to the enzyme as a Schiff base. Some examples of organic cofactors are listed in Table 1.3.*

* NAD(P)⁺ is often listed as a cofactor, and is generally known as a coenzyme. In fact it functions as a normal substrate for dehydrogenases and in our view is best considered as a substrate. It does, however, differ from most cell metabolites in that it becomes alternately reduced and oxidized. In this respect it does resemble redox cofactors. The same arguments apply to coenzyme A which can also be regarded as a substrate. Pantotheine, however, which is covalently attached to certain fatty-acid synthase systems (see Chapter 7, Section 7.1.1.1) is a cofactor. The problems of nomenclature of cofactors, coenzymes, and prosthetic groups is discussed further in reference 22.

TABLE 1.2
Examples of the metal ion requirements of some enzymes

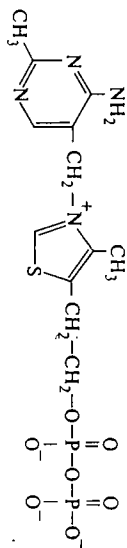
Metal	Enzyme
Na	Intestinal sucrose α -D-glucosylhydrolase
K	Pyruvate kinase (also requires Mg)
Mg	Kinases (e.g. hexokinase, pyruvate kinase), adenosinetriphosphatases (e.g. myosin adenosinetriphosphatase)
Fe	Catalase, peroxidase, nitrogenase
Zn	Alcohol dehydrogenase, carboxypeptidase
Mo	Xanthine oxidase, nitrogenase
Cu	Cytochrome c oxidase, amine oxidase

TABLE 1.3
Examples of organic cofactors

Cofactor	Linkage to apoenzyme	Type of reactions catalysed by holoenzyme
Pyridoxal phosphate	Usually Schiff base to lysine residue	Transamination, decarboxylation, racemization
Biotin	Amide bond to lysine residue	Carboxylation reactions, e.g. acetyl-CoA carboxylase, pyruvate carboxylase
Lipoic acid	Amide bond to lysine residue	Acyl transfer, e.g. pyruvate dehydrogenase and 2-oxoglutarate dehydrogenase systems

TABLE 1.3 (Contd.)

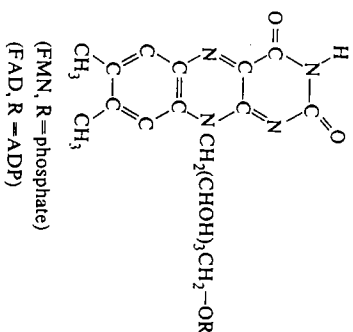
Cofactor	Linkage to apoenzyme	Type of reactions catalysed by holoenzyme
Thiamin diphosphate	Non covalent binding dissociation constant $\approx 10^{-6}$ mol dm $^{-3}$	Decarboxylation of 2-oxo-acids, e.g. pyruvate dehydrogenase and 2-oxoglutarate dehydrogenase systems



Flavin nucleotides: flavin
adenine dinucleotide (FAD) and
flavin mononucleotide
(FMN)

Non-covalent in,
e.g. amino acid oxidase;
covalent link in
succinate dehydrogenase

Redox reactions, e.g. xanthine oxidase (FAD), succinate dehydrogenase (FAD), glucose oxidase (FAD), NADPH-cytochrome reductase (FMN)



(FMN, R=phosphate)
(FAD, R=ADP)

Tightly bound cofactors (which cannot be removed by dialysis) are often termed *prosthetic* groups. An enzyme containing a cofactor or prosthetic group is termed a *holoenzyme*: when the cofactor is removed, the resulting protein is termed an *apoenzyme*. Any small molecule or other species which can reversibly bind to an enzyme is termed a *ligand* (this can include substrates, inhibitors, metal ions, etc.).

1.5 Nomenclature and classification of enzymes

1.5.1 General classification

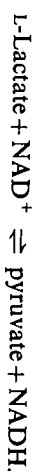
As in the development of organic chemistry, many enzymes were given 'trivial' names before any attempt was made at a system of nomenclature. In general the 'trivial' names consisted of the suffix 'ase' added to the substrate acted on (as in

the case of urease) or implied something about the reaction catalysed (as in the case of lactate dehydrogenase, which catalyses the dehydrogenation of lactate to yield pyruvate). However, some trivial names are distinctly unhelpful in this respect, e.g. old yellow enzyme, catalase, papain, trypsin, rhodanese, etc.* The present-day accepted nomenclature of enzymes is that recommended by the Enzyme Commission (set up in 1955 by the International Union of Biochemistry in consultation with the International Union of Pure and Applied Chemistry). Enzymes are named according to certain well-defined rules (for details see references 23 and 24). The six major types of enzyme-catalysed reactions are:

1. oxidation-reduction reactions, catalysed by oxidoreductases;
2. group transfer reactions, catalysed by transferases;
3. hydrolytic reactions, catalysed by hydrolases;
4. elimination reactions in which a double bond is formed, catalysed by lyases;
5. isomerization reactions, catalysed by isomerases;
6. reactions in which two molecules are joined at the expense of an energy source (usually ATP), catalysed by ligases (also known as synthetases).

The full systematic name of an enzyme not only shows the type of reaction catalysed but describes the substrate(s) acted on and any other important information. One of the enzymes from each major group will now be given to illustrate the classification scheme.

Group 1, e.g. lactate dehydrogenase. The full systematic name is L-lactate: NAD⁺ oxidoreductase, which denotes the fact that L-lactate acts as the electron donor and NAD⁺ as the electron acceptor:



Group 2, e.g. hexokinase. The full systematic name is ATP: D-hexose 6-phosphotransferase, indicating that ATP is the phosphate donor, a D-hexose the phosphate acceptor and that transfer is to the hydroxyl on the 6-carbon atom of the hexose:



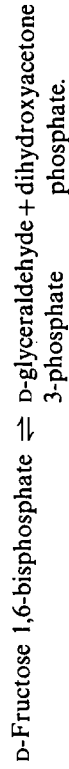
Group 3, e.g. adenosinetriphosphatase. The full systematic name is ATP phosphohydrolase, indicating that ATP, the substrate, is hydrolysed so as to split the bond which allows release of orthophosphate:



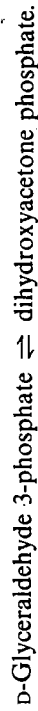
Group 4, e.g. fructose-bisphosphate aldolase. The full systematic name is D-fructose 1,6-bisphosphate D-glyceraldehyde 3-phosphate-lyase, which indicates that the substrate D-fructose 1,6-bisphosphate is cleaved in such a way as to yield

* Old yellow enzyme is now termed NADPH: acceptor oxidoreductase, catalase is H₂O₂:H₂O₂ oxidoreductase, papain and trypsin do not have systematic names because of the wide variety of substrates acted upon, and rhodanese is termed thiosulphate sulphurtransferase.

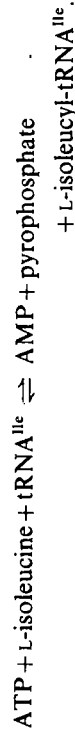
D-glyceraldehyde 3-phosphate as one of the products:



Group 5, e.g. *triosephosphate isomerase*. The full systematic name is D-glyceraldehyde 3-phosphate ketol-isomerase, indicating that the aldose D-glyceraldehyde 3-phosphate is isomerized to the ketose dihydroxyacetone phosphate:



Group 6, e.g. *isoleucyl-tRNA synthetase*. The full systematic name is L-isoleucine: tRNA^{Le} ligase (AMP-forming) and it indicates that L-isoleucine becomes bonded to a specific tRNA acceptor (tRNA^{Le}) and that in the process ATP is split to AMP and pyrophosphate*:



Many of the systematic names of enzymes are rather cumbersome and for everyday use shorter names can be used (e.g. lactate dehydrogenase, hexokinase, etc.). These shorter names are known as *recommended names*.

Enzymes are further classified by being assigned an Enzyme Commission (EC) number consisting of four parts (a, b, c, d).

The first number (a) indicates the type of reaction catalysed and can take values from 1 to 6 according to the classification of reaction types given above.

The second number (b) indicates the subclass, which usually specifies the type of substrates or the bond cleaved more precisely. In the case of oxidoreductases for example, this shows the type of chemical grouping acting as an electron donor, whereas with hydrolases it would show the type of bond that is being broken.

The third number (c) indicates the sub-subclass, allowing an even more precise definition of the reaction catalysed, in terms of the type of electron acceptor (oxidoreductases) or the type of group removed (lyases), etc.

The fourth number (d) indicates the serial number of the enzyme in its subclass.

Following these rules the EC numbers of lactate dehydrogenase, hexokinase, adenosinetriphosphatase, fructose-bisphosphate aldolase, triosephosphate isomerase, and isoleucyl-tRNA synthetase are 1.1.1.27, 2.7.1.1, 3.6.1.3, 4.1.2.13, 5.3.1.1, and 6.1.1.5, respectively. All the enzymes mentioned in this book are given their recommended names, together with their EC numbers, and the reactions catalysed are listed in the Appendix at the end of the book. (There are a few enzymes that have not yet been assigned EC numbers because the reactions they catalyse have not been precisely determined.)

* The Enzyme Nomenclature Committee of the IUB has recommended that Group 6 enzymes be termed ligases and that the use of the term synthetase be discouraged (because of the possibility of confusion between synthetase and synthase).

It should be noted that the system of nomenclature and classification of enzymes is based only on the reaction catalysed and takes no account of the origin of the enzyme (i.e. from the species or tissue it derives). Enzymes catalysing the same reaction but isolated from different species may well have substantially different amino-acid sequences and may well act via different catalytic mechanisms, but these points will not be distinguished in the classification system. For example, the adenosinetriphosphatases from the inner mitochondrial membrane and from the sarcoplasmic reticulum are not distinguished in this classification scheme, although the former is concerned with the transport of protons across the inner mitochondrial membrane and the latter with the transport of Ca²⁺ across the sarcoplasmic reticulum (for details see Chapter 8, Section 8.4.4). Both catalyse hydrolysis of ATP. Another example is the two classes of fructose-bisphosphate aldolases discussed in Chapter 5, Section 5.5.3. Therefore, for a more exact specification of an enzyme the source should also be mentioned, and if necessary the isoenzyme type. The meaning of the term isoenzyme is discussed below.

1.5.2 Isoenzymes

Within a single species there may exist several different forms of enzyme catalysing the same reaction. These could differ from one another in terms of amino-acid sequence, some covalent modification (e.g. phosphorylation of serine hydroxyl groups), or possibly in terms of three-dimensional structure (conformational changes), etc. The term *isoenzyme* should be restricted to those forms of an enzyme that arise from *genetically determined differences in amino-acid sequence* and should not be applied to those derived by modification of the same amino-acid sequence.* Thus, for example, heart muscle malate dehydrogenase (which catalyses the oxidation of malate to oxaloacetate) occurs both in the cytoplasm and in the mitochondria; these two forms of the enzyme are termed *isoenzymes* of malate dehydrogenase. Similarly, in serum or in muscle extracts, lactate dehydrogenase occurs in a variety of forms (see Chapters 5 and 10, Sections 5.5.4.1 and 10.2); these are also referred to as *isoenzymes*. It is recommended that the naming of isoenzymes should be based on the extent of migration in an applied electric field using the technique of *electrophoresis* (see Chapter 2, Section 2.6.2.2) rather than on the basis of tissue distribution (e.g. brain type, muscle type, etc.) since this distribution can vary between different species (or in a single species or tissue with the stage of development). Electrophoresis is recommended as the basis of classification because it is a widely used technique with high resolving power. Isoenzymes are numbered starting with the species having highest mobility towards the anode (see for example Fig. 10.1).

1.5.3 Multienzyme systems

Multienzymes are proteins that exhibit more than one catalytic activity and are described in some detail in Chapter 7. For the purposes of nomenclature, the

* See reference 25 for recommendations on nomenclature of isoenzymes.

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2

The purification of enzymes

2.1 Introduction

In this chapter we shall discuss the isolation of enzymes, concentrating initially on the principles underlying the more important separation methods employed in this work. We shall then illustrate how these methods are used in practice by considering eight specific examples of purification of enzymes; the examples have been chosen to give a broad coverage of the methods and problems involved in such procedures. However, it is necessary to deal first with some more general questions regarding the need for, and strategy underlying, purification procedures.

2.2 Why isolate enzymes?

It is not difficult to appreciate that if we hope to gain a detailed understanding of the behaviour of an enzyme in a complex system (be it a subcellular organelle such as the mitochondrion, a cell, or a whole organism) we must first try to understand its properties in as simple a system as possible. In many cases this simple system would consist of a solution of enzyme in a medium containing only small ions, buffer molecules, cofactors, etc. However, in some cases, e.g. enzymes that are bound to cell membranes (see Chapter 8, Section 8.4), the isolated enzyme may be inactive in the absence of phospholipid or detergent, and our simple system would need to contain these additional components.

From studies of the isolated enzyme we can learn about its specificity for substrates, kinetic parameters for the reaction (see Chapter 4), and possible means of regulation (see Chapter 6). All these features would be useful in understanding the role of the enzyme in the more complex systems (see Chapters 7 and 8). In addition, enzymes pose some extremely intriguing questions about the structure of large molecules (see Chapter 3) and mechanisms of catalysis (see Chapter 5). Detailed studies of these aspects are only possible if we have been able to purify enzymes so as to remove contaminating enzymes and other large molecules. The ready availability of isolated enzymes has been of considerable value in a number of medical and industrial applications (see Chapters 10 and 11, respectively).

2.3 Objectives and strategy in enzyme purification

2.3.1 Objectives

The aim of a purification procedure should be to isolate a given enzyme with the *maximum possible yield*, based on the percentage recovered activity compared

with the total activity in the original extract. In addition the preparation should possess the *maximum catalytic activity*, i.e. there should be no degraded or other inactivated enzyme present, and it should be of the *maximum possible purity*, i.e. it should contain no other enzymes or large molecules. In the early days of enzyme purification, crystallinity was taken to be proof of purity but there are now known to be a number of crystalline enzymes¹ that are impure, and indeed many present-day purification procedures do not involve a crystallization step (e.g. RNA nucleotidyltransferase; Section 2.8.4). Of course, crystallization is necessary if the three-dimensional structure of an enzyme is to be elucidated by X-ray crystallography (see Chapter 3, Section 3.5.1).

The catalytic activity of a preparation is determined by a suitable assay procedure (Chapter 4, Section 4.2) in which the rate of disappearance of substrate or the rate of appearance of product is determined under defined conditions of substrate concentration, temperature, pH, etc. The units of activity are usually expressed either as μmol substrate consumed, or product formed, per minute ('units' or 'international units') or as mol substrate consumed, or product formed, per second ('katal' in the SI system). There is no way of predicting the catalytic activity of a purified enzyme under a given set of conditions and so purification is carried out until the *specific activity* of the preparation (i.e. units per mg or katal per kg*) increases to reach a constant value that is not increased by further purification steps. The criteria used in assessing the purity of a preparation involve the application of analytical methods to detect contaminants (these are dealt with in more detail in Section 2.7.)

2.3.2 Strategy

The steps involved in the purification of an enzyme can be discussed in terms of the type of 'flow sheet' shown in Fig. 2.1. Within this general scheme, the procedure to be adopted for a given enzyme will involve choices of: (i) source of enzyme; (ii) methods of homogenization; and (iii) methods of separation. These choices are discussed in the following sections (2.4 to 2.6). The progress of a purification is recorded in a purification table (see Section 2.8.1.)

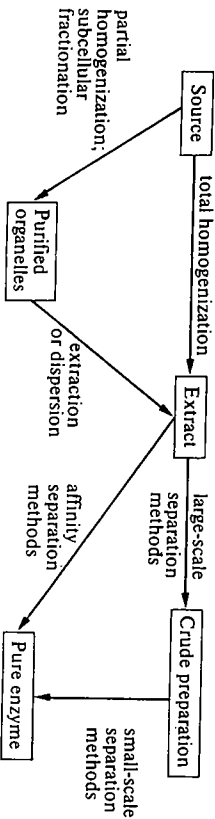


Fig. 2.1. Steps involved in the purification of enzymes.

* It should be noted that 60 units per mg corresponds to 1 katal per kg.

2.4 Choice of source

A number of factors influence the choice of starting material in an enzyme purification. Among the most important factors are the following.

2.4.1 Abundance of enzyme

In the absence of other factors it is obviously a good idea to choose a source in which the required enzyme occurs in large amounts. Thus, *lactating mammary gland* is an excellent source of enzymes such as *acetyl-CoA carboxylase* involved in fatty-acid biosynthesis,² and *kidney* is a good source of *hydrolytic enzymes* such as *alkaline phosphatase*, an enzyme that catalyses the hydrolysis of *phosphate esters* at alkaline pH.

In the case of microorganisms it is often possible to increase the level of a given enzyme by genetic or environmental manipulations.^{3,4} Thus synthesis of β -D-galactosidase can be induced in *E. coli* by growing the bacteria in a medium containing lactose as the sole source of carbon. Increasingly, recombinant-DNA techniques are being employed to construct over-producing strains; for this approach to be successful it is necessary to isolate the gene coding for the enzyme of interest and to find a suitable system in which to express that gene. Details of these techniques can be found in reference 5. An example of the usefulness of the recombinant-DNA approach involved studies⁶ on the enzyme 3-enolpyruvylshikimate-5-phosphate synthase (EPSP synthase) in *E. coli*, which is involved in the biosynthesis of aromatic amino acids. The gene for this enzyme was isolated and inserted into the multicopy plasmid⁵ PAT153. The modified plasmid was used to transform *E. coli* to give a new strain that produced approximately 100-fold more EPSP synthase than the wild-type strain; 4.8 mg pure enzyme could be isolated from 18g cells as a result of a 50-fold purification.⁶ A second example of the recombinant-DNA approach is provided by glutathione reductase (Section 2.8.7). It should be pointed out that considerable problems can be encountered in attempting to express eukaryotic genes in prokaryotes⁵ and that in some cases proteins that have been over-expressed in *E. coli* can form insoluble aggregates; recovery of active protein from these aggregates has not always been possible.⁷

2.4.2 Availability

A source with reasonable abundance of the chosen enzyme might not be readily available for various reasons (economic, geographical, etc.). Thus, *lobster muscle* is notoriously expensive, bioluminescent species are often found in far away 'exotic' locations,^{8,9} and human tissue is not readily available in many countries! Clearly, in such cases it may be necessary to make some compromise between abundance and availability.

2.4.3 Comparative studies

It may be the case that a certain enzyme has been studied in one species or within one tissue in a species; in such cases it can be highly profitable to examine the

corresponding enzyme in another species or another tissue. (In Chapter 3, Section 3.4.9.5 we shall see how comparisons of a given enzyme from different species can be used to explore aspects of enzyme evolution. A knowledge of the properties of the different isoenzymes (see Chapter 1, Section 1.5.2) of lactate dehydrogenase is of use in discussing the role of these enzymes in heart and muscle.¹⁰)

2.4.4 Subcellular location

If the enzyme catalysing a given reaction occurs at only one location within the cell* (e.g. succinate dehydrogenase and some other tricarboxylic acid-cycle enzymes occur only in the mitochondria), then the whole tissue can be homogenized or extracted in the purification procedure (Fig. 2.1), and subcellular fractionation is not essential. In some cases, however, a given enzyme may exist in a number of locations in the cell and some subcellular fractionation may well be necessary before the purification can proceed. Thus, in the case of mitochondrial adenosinetriphosphatase (Section 2.8.3) it is important to obtain a reasonably pure preparation of mitochondria from the chosen source (beef heart, a tissue rich in mitochondria) in order to avoid contamination from other adenosinetriphosphatases, for instance that involved in the transport of Na^+ and K^+ across the plasma membrane of the cell. Subcellular fractionation to prepare the constituent organelles of a cell is normally achieved by a succession of centrifugation steps with the larger organelles (nuclei, mitochondria) being sedimented first (see Chapter 8, Section 8.2.1).

(Having taken these and possibly other factors into account, the choice of source of enzyme can be made. We now proceed to look at the steps in the purification procedure outlined in Fig. 2.1. Further details of the various methods can be found in reference 11.)

2.5 Methods of homogenization

A number of methods for breaking open cells and extracting the contents are available. The choice of method usually depends on the type of tissue or organism which is used as the source of enzyme.

2.5.1 Mammalian tissue

The lack of a rigid cell wall makes homogenization of most mammalian tissues relatively easy. The tissue is often cut up into small pieces (or minced) prior to homogenization by either a rotating Teflon pestle in a glass mortar (the Potter-Elvehjem homogenizer) or a high-speed blender. Extraction is performed with isotonic† solutions in those cases where it is important to avoid the rupture

* A discussion of the subcellular locations of enzymes will be found in Chapter 8, Section 8.2.

† An isotonic solution is one which exerts the same osmotic pressure as the cell contents, containing approximately 0.3 mol dm^{-3} dissolved species (e.g. 0.3 mol dm^{-3} sucrose or $0.15 \text{ mol dm}^{-3} \text{ KCl}$).

of subcellular organelles. This is particularly important in the case of cells from tissue such as liver which possess a large number of lysosomes; these are subcellular organelles containing a number of hydrolytic enzymes such as proteases (see Chapter 8, Section 8.2.4). In other cases a medium of low ionic strength (such as $0.01 \text{ mol dm}^{-3} \text{ KCl}$) is often used for extraction. It may also be necessary in particular cases to add protease inhibitors or other substances (e.g. the reducing agent dithiothreitol) to prevent damage to the enzyme of interest during the extraction process.

(The use of the pressure homogenization technique, in which a large pressure difference is created between the interior of a cell and its surroundings, has been discussed by Avis.¹²)

2.5.2 Plant, fungal, and bacterial material

The rigid cell wall surrounding the cells in these types of material usually necessitates the use of harsher methods of extraction, e.g. grinding with abrasives such as alumina or sand, freezing and thawing, long periods of blending, or the addition of glass beads during blending. Alternatively, a French press, in which the cells are forced at high pressure through a narrow orifice, can be used. In this case, the mechanical shear forces lead to cell rupture.

However, it is often possible to disrupt the cell walls of plants, fungi, and bacteria by the use of appropriate hydrolytic enzymes. Protoplasts (cells completely or partially lacking a cell wall) may be prepared from Gram-positive bacteria, e.g. *Bacillus subtilis* by incubation with lysozyme.¹³ By a modification of this method that includes the addition of EDTA, protoplasts have also been prepared from the Gram-negative bacterium *E. coli*.¹⁴ Protoplasts have been prepared from fungi, e.g. yeast, *Neurospora* and *Aspergillus*, using enzyme extracts that generally contain a mixture of chitinase and β -glucanases.¹⁵ An alternative method may be to use a fungal mutant that lacks the normal cell wall, e.g. the slime mutant of *Neurospora*,¹⁶ and which can therefore be readily disrupted by solutions of lower osmotic pressure.

Particular problems can arise with plant tissues,¹⁷ because during homogenization the release of the contents of the vacuoles (usually acidic and containing proteases) can cause damage to enzymes. Addition of buffer can protect against such damage. In addition, plant tissues often contain certain phenolic compounds that are readily oxidized to form dark pigments that can be harmful to enzymes. The pigments can be removed by adsorption on polymers such as poly(vinylpyrrolidone) (Section 2.8.2).

2.6 Methods of separation

A successful separation of one substance from another depends on there being some property by which the substances may be distinguished. The principal properties of enzymes that can be exploited in separation methods are size, charge, solubility, and the possession of specific binding sites. Methods that

TABLE 2.1
Principal separation methods used in purification of enzymes

Property	Method	Scale
Size or mass	Centrifugation Gel filtration Dialysis: ultrafiltration	Large or small Generally small Generally small
Polarity (a) Charge	Ion-exchange chromatography Chromatofocusing Electrophoresis Isoelectric focusing Hydrophobic chromatography	Large or small Generally small Generally small Generally small Generally small
Solubility (b) hydrophobic character	Change in pH Change in ionic strength Decrease in dielectric constant	Generally large Large or small Generally large
Specific binding sites or structural features	Affinity chromatography Affinity elution Dye-ligand chromatography Immunoadsorption Covalent chromatography	Generally small Large or small Large or small Generally small Generally small

The term 'large-scale' is used to indicate that amounts of protein greater than about 100 mg can be handled at that particular step in the purification procedure.

exploit differences in these properties are listed in Table 2.1, and are described in the following sections (2.6.1 to 2.6.4).

2.6.1 Methods that depend on size or mass

2.6.1.1 Centrifugation¹⁸

Large molecules such as enzymes can be sedimented by the high centrifugal fields (up to 300 000 times gravity) generated by an ultracentrifuge. Although the rate at which any particular enzyme will sediment depends on a variety of factors including the size and shape of the molecule and the viscosity of the solution, it is found that in general the higher the M_r value the greater the rate of sedimentation. This method is not widely used in purification procedures to separate one enzyme from another because only small volumes (a few cm³) can be dealt with in ultracentrifuges that operate at high centrifugal fields. However, centrifugation is very widely used to remove precipitated or insoluble material in the course of an isolation, e.g. to remove cell debris after homogenization or to collect enzyme that has been precipitated by the addition of ammonium sulphate (see Section 2.6.3.2). When centrifugation is being used in this way, only relatively low centrifugal fields (5000 to 50 000 g) are required and consequently volumes up to several dm³ can be handled.

The use of differential centrifugation in the preparation of subcellular organelles is described in Chapter 8, Section 8.2.1.

2.6.1.2 Gel filtration^{19,20}

In gel filtration, the separation between molecules of different sizes is made on the basis of their ability to enter the pores within the beads of a beaded gel. The most widely used types of gel are Sephadex (cross-linked dextrans), Bio-Gel P (cross-linked polymers of acrylamide), Sephacryl (cross-linked polymers of dextran and acrylamide) and Sepharose (agarose, which can be cross-linked for greater rigidity). Small molecules that can enter the pores of the beads are retarded as they pass down a column containing the gel; large molecules that are unable to enter the pores pass through the column unimpeded. By varying the size of the pores (which is controlled by the degree of cross-linking in the preparation of the beaded gel), it is possible to change the range of M_r values which can be fractionated. Sephadex G-100, for example, can fractionate globular proteins in the M_r range 4000–150 000.* If the shape of the enzyme is markedly non-spherical, it will be eluted from the column in an unexpected position, i.e. not at that expected on the basis of its M_r (see Chapter 3, Section 3.2.2). In order to minimize any non-specific interactions between the enzyme and the gel, it is important that extremes of ionic strength should be avoided.²⁰ Gel filtration can be carried out on a large scale, but since large columns are rather time-consuming to run and the gels required to fill them are expensive, the method finds most application in the later (small-scale) stages of enzyme purifications.

2.6.1.3 Dialysis^{21,22} and ultrafiltration^{23,24}

A dialysis membrane such as cellophane acts as a sieve with holes large enough to permit the passage of globular proteins with M_r values up to about 20 000 but not of larger molecules. It is possible to change the pore sizes by various mechanical and chemical treatments.²⁵ Although dialysis is not generally useful for the separation of enzymes from each other, it is widely used during purification procedures to remove salts, organic solvents, or inhibitors of low M_r , from solutions of enzymes.

In ultrafiltration, small molecules and ions pass through the dialysis membrane under the influence of an applied pressure (generally N₂ gas at about 4 atm pressure). This leads to concentration of an enzyme solution, which can be useful in reducing the volume of a sample during the purification procedure.

2.6.2 Methods that depend on polarity

2.6.2.1 Ion-exchange chromatography^{26,27}

Ion-exchange chromatography depends on the electrostatic attraction between species of opposite charge. Ion exchangers usually consist of modified derivatives of some support material such as cellulose, Sephadex, etc. as shown in Fig. 2.2.

* The corresponding ranges for Sephacryl S-300 and Sepharose 6B are 10 000 to 800 000 and 10 000 to 4000 000, respectively.

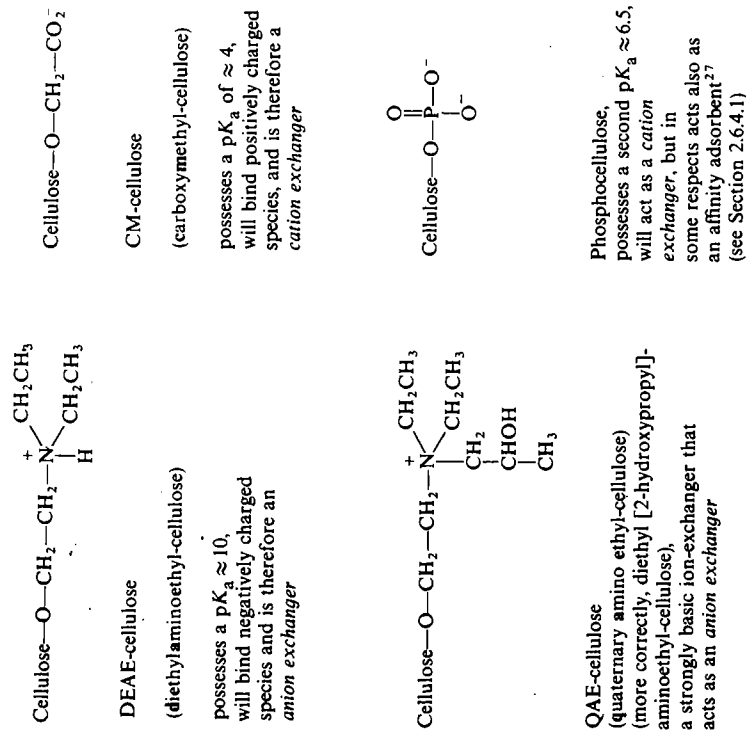


FIG. 2.2. Ion exchangers commonly used in purifications of enzymes.

During a purification procedure, the enzyme is usually applied to an ion exchanger in a solution of low ionic strength and at a pH at which the appropriate interaction will occur (i.e. the enzyme and the ion exchanger have opposite charges). Desorption of the bound species can be brought about either by changing the pH, and thus altering the charges, or, more commonly, by increasing the ionic strength of the solution so that the increased concentration of cations or anions will compete with the enzyme for the binding sites on the exchanger. Use of a gradient of increasing ionic strength permits the separation of proteins in a mixture on the basis of their ability to bind to the ion exchanger. When modified forms of Sephadex (such as DEAE-Sephadex) are used, separation is made on the basis of charge and (to a limited extent) size (see Section 2.6.1.2). It should be pointed out that DEAE-Sephadex shrinks as the ionic strength of the solution increases; this can cause problems during ion-exchange chromatography. The agarose-based ion exchangers such as DEAE-Sephareose do not suffer from this problem.

Ion-exchange chromatography can be performed on a large or a small scale. On a large scale it is often convenient to work in a batchwise manner, i.e. to

adsorb the enzyme by adding the ion exchanger to a solution and then to pour the material into a column for controlled desorption using an ionic-strength gradient. On a small scale, both adsorption and desorption are performed in a column.

The degree of purification effected by an ion-exchange step is generally up to about 10-fold (as judged by the specific activity), but examples are known where much better results have been achieved²⁸ (see also Section 2.8.6). Ion-exchange chromatography finds very wide application in present-day purification procedures. In Section 2.6.4.2 we shall describe the technique of affinity elution from ion exchangers.

In general the desorption of proteins from ion-exchangers by changes in pH has not been of great use in protein purification, partly because the buffering properties of the bound proteins distort the shape of the applied pH gradient. However, the relatively new technique of 'chromatofocusing' overcomes most of these problems.²⁹ In chromatofocusing, the pH gradient is formed by using mixtures of ampholyte-type buffers (see Section 2.6.2.3) that are of high buffering capacity. The ion-exchanger is poly(ethyleneimine) agarose, which has a wide range of titratable groups of differing pK_a values. The protein mixture is adsorbed on the ion-exchanger at the upper end of the range of pH and the pH gradient is generated on the column by addition of the acid form of the ampholyte buffer. As the pH falls, proteins are eluted from the column roughly in the order of their isoelectric points (see Section 2.6.2.3) and the technique is claimed to be of high resolving power. The costs of the materials involved make working on a large scale expensive, and thus chromatofocusing is probably most useful at the later stages of a purification procedure.

2.6.2.2 Electrophoresis³⁰

Electrophoretic separation is based on the differential movement of charged molecules under the influence of an applied potential difference. The rate of movement of a species is governed by the charge it carries and also by its size and shape. In order to minimize convection currents, the buffer (electrolyte) solution is soaked into a support (paper, cellulose powder, starch, or polyacrylamide gels). The position of a protein on the support can be determined by use of a stain such as Coomassie Blue which binds to proteins. Although the technique is normally performed on a small, analytical scale (a few milligrams or less), it is possible to use the method on a large preparative scale.³¹ After preparative electrophoresis, the separated enzymes can be eluted out of the support or run off the bottom of a column in sequence.

2.6.2.3 Isoelectric focusing^{32, 33, 34}

Isoelectric focusing is based not on the rate at which charged species move in an electric field but rather on their equilibrium position in a pH gradient. The pH gradient is established by applying a potential difference to a gel soaked with a solution containing a mixture of ampholytes that are polyamino acids with

different charge properties and hence different isoelectric* points (pI values) (Fig. 2.3). When the potential difference is applied, a negatively charged molecule will migrate towards the anode until it encounters the acid; the acid will tend to neutralize the charge so the molecule will stop. The reverse argument holds for positively charged species. By this means the ampholytes will arrange themselves in the gel so that there is a gradual increase of pH from anode to cathode.

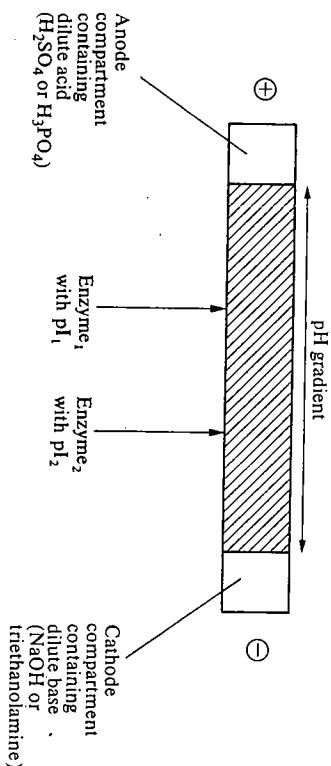


Fig. 2.3. Separation of enzymes by isoelectric focusing.

An enzyme applied to the gel will then migrate to the position in the gel where the pH equals its isoelectric point. For example if the enzyme were at a position where the pH is above its pI, then its net negative charge would ensure that it moved towards the anode until it reached its pI. It is possible to separate species that differ only slightly (as little as 0.01 pH units) in isoelectric points, and thus this method can be a very powerful tool in purification procedures. The method is generally used on an analytical scale, although it is possible to scale up for preparative work (up to gram quantities).³²

2.6.2.4³³ Hydrophobic interaction chromatography

In an aqueous environment, hydrocarbons and other non-polar molecules will tend to associate with each other rather than with water molecules (see Chapter 3, Section 3.5.5.4 for a further discussion of this 'hydrophobic' effect). The surfaces of proteins contain a high proportion of charged and polar amino-acid side-chains (see Chapter 3, Section 3.5.4.5) but usually contain at least some non-polar side-chains. It is the presence of these non-polar side-chains that accounts for the ability of many proteins to bind to non-polar molecules. Such interactions are exploited in the technique of hydrophobic-interaction chromatography.^{35, 36} In this technique, proteins are adsorbed on matrices such

* The isoelectric point of a compound is the pH at which it carries no net charge and hence does not move under an applied potential difference.

as octyl- or phenyl-Sepharose, usually at high ionic strength at which the hydrophobic interactions are stronger. Desorption of proteins can be brought about by decreasing the ionic strength and, if necessary, by addition of organic solvents or non-ionic detergents to weaken the hydrophobic interactions further. In the purification of 3-enolpyruvylshikimate-5-phosphate synthase from *E. coli* (Section 2.4.1), hydrophobic-interaction chromatography on phenyl-Sepharose was employed to give a 5-fold purification with 80 per cent recovery of activity. (The enzyme was adsorbed at 0.8 mol dm⁻³ (NH₄)₂SO₄ in a 0.1 mol dm⁻³ tris-HCl buffer at pH 7.5 and eluted using a decreasing gradient of (NH₄)₂SO₄ (0.8–0.0 mol dm⁻³) in this buffer.⁶)

2.6.3 Methods based on changes in solubility

The solubility of a compound in a given solvent depends on the balance of the forces between solute and solute and those between solute and solvent. If the former predominate the compound will be insoluble, whereas if the latter are predominant the compound will be soluble. In the course of enzyme purifications it is possible to alter the balance between these forces and hence precipitate the enzyme of interest or remove contaminating enzymes. Three of the most important ways of changing the solubility of enzymes are (i) to change the pH; (ii) to change the ionic strength; (iii) to decrease the dielectric constant. These methods can be applied on a large scale and are often used in the initial stages of a purification procedure.

2.6.3.1 Change in pH³⁷

An enzyme is least soluble at its isoelectric point, since at this pH there will be no repulsive electrostatic forces between the enzyme molecules. Adjustment of the pH to the appropriate value can therefore be used to precipitate an enzyme; this method is employed in the case of adenosinetriphosphatase (Section 2.8.3). A change in pH can also be used to precipitate unwanted enzymes and proteins (see the case of adenylate kinase, Section 2.8.1). It is important to check that the enzyme of interest is not inactivated by exposure to these changes in pH.

2.6.3.2 Change in ionic strength³⁸

Large charged molecules are generally only slightly soluble in pure (deionized) water; addition of ions promotes solubility by helping to disperse the charge carried by the large molecule. This phenomenon is known as salting in. However, if the ionic strength is increased beyond a certain point, the charged molecule will be precipitated (salting out). Although the theoretical basis for this behaviour is not too well understood, it is probable that a major factor is that at very high concentrations of salt, the concentration of water is significantly decreased, leading to a decrease in solute-solvent interactions and hence in solubility. The salt of choice in enzyme purification procedures is ammonium sulphate, which has the advantages of cheapness, high solubility in water (a saturated solution at

298 K (25°C) has a molarity of approximately 4 mol dm^{-3} and lack of harmful effects on most enzymes.* Each enzyme will usually begin to precipitate at a certain concentration of ammonium sulphate and this forms the basis of an initial fractionation procedure. For instance, if it is known from a preliminary experiment that an enzyme, X, is precipitated at around 50 per cent saturation† ammonium sulphate, then the procedure might involve addition of ammonium sulphate to 45 per cent saturation, centrifugation to remove the unwanted precipitated protein followed by addition of more ammonium sulphate to the supernatant to bring the concentration up to 55 per cent. The precipitate containing X is removed by centrifugation.

In general, the degree of purification that can be achieved by ammonium sulphate fractionation is less than 10-fold, but the method is often used in the initial stages of a purification to reduce the volume of solution to be dealt with since the precipitated enzyme can be redissolved in a small volume of buffer. In at least two cases, however, it is possible to obtain pure enzyme by ammonium sulphate fractionation (glyceraldehyde-3-phosphate dehydrogenase³⁹ and fructose-bisphosphate aldolase⁴⁰ from rabbit muscle).

Ammonium sulphate is also commonly used in crystallization of enzymes.⁴¹ The concentration of salt is raised until slight turbidity is observed; when the solution is left to stand, crystals of the enzyme should form. Some of the larger crystals (with dimensions of the order of 1 mm) may be suitable for X-ray crystallography (see Chapter 3, Section 3.5.1).

2.6.3.3 Decrease in dielectric constant^{42, 43}

Addition of a water-miscible organic solvent (such as ethanol or acetone) will decrease the dielectric constant of a solution and hence increase electrostatic forces. Although there can be rather complex effects on both solute-solute and solute-solvent forces, in general the net effect is to cause precipitation of large, charged molecules such as enzymes. This method can be used on a large scale as an initial fractionation procedure. It must be remembered, however, that addition of organic solvents can sometimes lead to the inactivation of enzymes (see Chapter 3, Section 3.7.1) and it is normally important to work at low temperatures to minimize such inactivation.

An alternative method of precipitating proteins is to add the neutral water-soluble polymer poly(ethylene glycol).^{43, 44} The polymer is thought to act by removing water from the hydration spheres of proteins.⁴⁴ Poly(ethylene glycol) preparations of average M_r 4000 or 6000 are commonly used and can give highly successful purifications (see Section 2.8.2). Difficulties can arise in removing residual traces of poly(ethylene glycol) from protein preparations but this is often unnecessary.⁴³

* It should be noted that ammonium sulphate is a weak acid. When it is added to a weakly buffered solution, ammonia or another base should be added to maintain the pH at 7 or above.
† Concentrations of ammonium sulphate are usually expressed as percentages of the concentration required to saturate the solution.

2.6.4 Methods based on the possession of specific binding sites or structural features

Enzymes normally display highly specific interactions with their substrates and other ligands (see Chapter 1, Section 1.3.2). Advantage can be taken of this specificity in the so-called affinity separation methods of enzyme purification discussed below.

2.6.4.1 Affinity chromatography⁴⁵⁻⁴⁹

In affinity chromatography, a molecule such as a substrate or competitive inhibitor (see Chapter 4, Section 4.3.2.2) that interacts specifically with the enzyme of interest is linked covalently to an inert matrix (such as agarose). When a mixture is passed down a column containing the affinity matrix, only this enzyme is retained and other enzymes and proteins are washed away. The bound enzyme can be desorbed by a pulse of substrate, which will compete for the binding sites on the enzyme, or by changing the pH or ionic strength of the solution in such a way as to weaken the binding of enzyme to the column (Fig. 2.4).

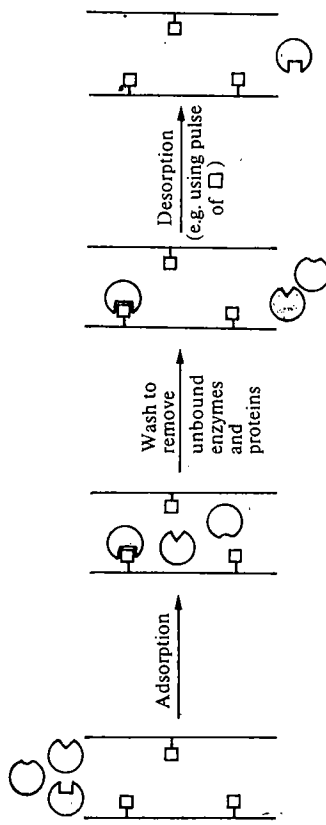


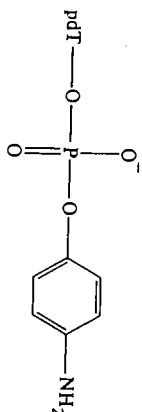
FIG. 2.4. Schematic illustration of enzyme purification using affinity chromatography.

This simple method is, in principle, capable of purifying an enzyme in a single step from a crude extract and a number of such purifications have been reported (see the example of Staphylococcal nuclease^{50, 51} outlined in Fig. 2.5).

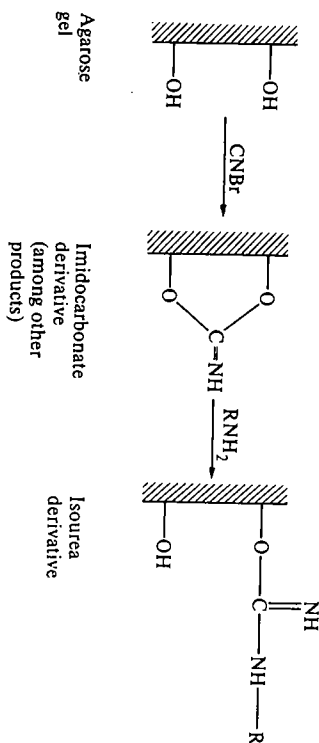
There are, however, a number of problems associated with the use of affinity chromatography in enzyme purifications; these include

1. Attaching a suitable substrate analogue or inhibitor to the matrix can be a difficult task and the reactions involved in coupling (Fig. 2.5) have not been completely characterized.
2. Linking of the ligand to the matrix may interfere with the binding to the enzyme (see Section 2.8.5) and lead to a loss of specificity in the interaction. In this connection, the need to interpose spacer arms between ligand and matrix to prevent interference by the matrix in the interaction has been recognized for some time.⁵² However, the most widely used spacer (1,6-diaminohexane)

The enzyme acts on substrates of the general type pTp-X and is competitively inhibited by pdTp. The compound



which acts as a powerful competitive inhibitor of the enzyme was attached to a cyanogen bromide-activated agarose gel via the free amino group:



The nuclease was retained by the affinity column and could be subsequently eluted at low pH. Purification could be achieved in a single step from a crude cell extract.⁵¹

Fig. 2.5. Use of affinity chromatography in the purification of Staphylococcal nuclease.

can itself participate in non-specific hydrophobic interactions (see Chapter 3, Section 3.5.5.4) with enzymes,⁵³ so the use of alternative hydrophilic spacers has been suggested.⁵⁴

3. For affinity chromatography to work successfully the strength of the interaction between matrix-bound ligand and the enzyme must be in the correct range.^{55, 56} If it is too weak, with a dissociation constant greater than about 2 mmol dm⁻³ (see Section 2.8.7), the enzyme will not be retarded by the column. However, if the interaction is too strong (dissociation constant smaller than about 10⁻⁸ mol dm⁻³), removal of the bound enzyme may only be possible under harsh conditions that lead to inactivation of the enzyme. Information obtained from inhibition studies of an enzyme in solution can be of considerable value in the successful design of affinity chromatography systems (see the example of hexokinase in Section 2.8.8).

4. Special problems are posed by enzymes that catalyse reactions involving more than one substrate. Thus it would be expected that all NAD⁺-dependent dehydrogenases would 'recognize' matrix bound NAD⁺. The differences between the various dehydrogenases occur in the binding of the second substrate, and in order to exploit these differences we normally have to

pay particular attention to the method of elution from the column. In the case of horse liver alcohol dehydrogenase, for example, specific elution of the enzyme from a column containing a matrix-bound AMP analogue could be obtained with a mixture of NAD⁺ plus pyrazole, a competitive inhibitor of the enzyme.⁵⁷

In spite of these problems, affinity chromatography has made a very significant contribution to the purification of enzymes and also of other biological molecules such as antibodies that possess specific binding sites. The technique is most likely to be successful when full use is made of information from solution studies (see Section 2.8.8).

2.6.4.2 Affinity elution⁵⁸

Affinity elution is complementary to affinity chromatography in that in the former technique the specificity of interaction is at the stage of *desorption* from the chromatographic support material, whereas in the latter the specificity occurs at the stage of *adsorption*. In affinity elution, the enzyme of interest (along with others) is adsorbed onto an ion exchanger (e.g. DEAE-cellulose, see Section 2.6.2.1), and then is eluted specifically by the appropriate substrate. The advantages over affinity chromatography are mainly of a practical nature; namely (i) the often complex task of designing and attaching the appropriate ligand to the matrix is not required, and (ii) columns of high capacity are more readily available, since ion-exchangers are much cheaper than affinity matrices. Affinity elution was used in the early 1960s in the purification of fructose-bisphosphatase from rabbit liver,⁵⁹ but the technique has been developed more extensively by others, including Scopes,^{60, 61} who has devised a method for the separation of all the enzymes involved in the glycolytic pathway (see Chapter 6, Section 6.4.1) in a single multiple-operation scheme. Thus, for instance, a mixture of glycolytic enzymes already partially fractionated by ammonium sulphate (those precipitating between 45 and 65 per cent saturation) is loaded on to a CM-cellulose column at pH 6.5; elution with phosphoenolpyruvate (0.5 mmol dm⁻³) yields pyruvate kinase, elution with D-fructose 1,6-bisphosphate (0.2 mmol dm⁻³) yields fructose-bisphosphate aldolase.^{60, 61} An analogous procedure has been reported for purification of the tricarboxylic acid-cycle enzymes from beef heart.⁶²

2.6.4.3 ⁽³⁾ Dye-ligand chromatography⁶³

In the mid 1970s it became clear that the dye Cibacron Blue F3G-A (Fig. 2.6) (which had been used to label high-M_w dextran preparations to provide a calibration for gel-filtration columns) could bind to a number of enzymes such as dehydrogenases and kinases.⁶⁴ These enzymes possess a common structural feature known as the nucleotide-binding domain (see Chapter 3, Section 3.5.3.3). Although the molecular basis for the apparent specificity has not been completely clarified,^{65, 66, 67, 68} the interaction has proved a useful tool in protein

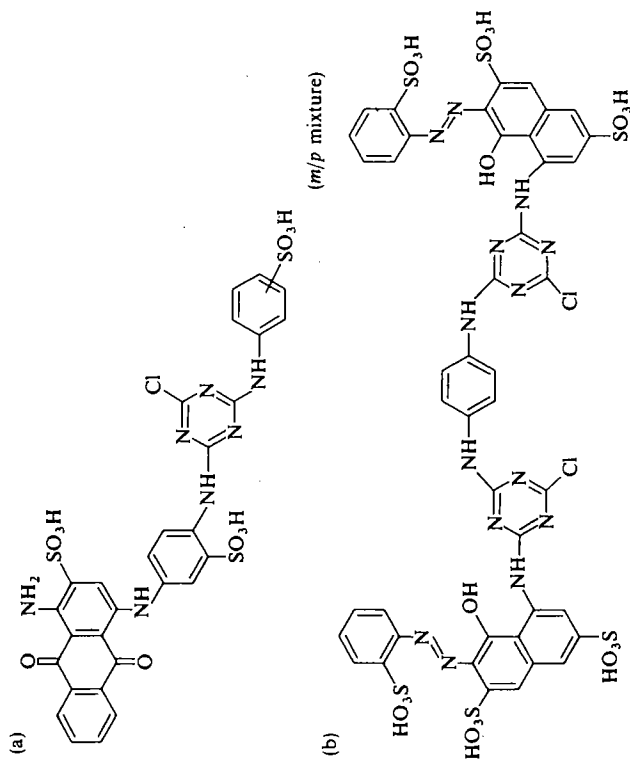


FIG. 2.6. Structures of two dyes used in dye-ligand chromatography. (a) Cibacron Blue F3G-A; (b) Procion Red HE-3B. These dyes can be linked to matrices such as agarose by reaction of the triazinyl chloride groups.

purification. The dye can conveniently be coupled to an agarose matrix via the triazinyl chloride group and the enzymes can be eluted by addition of appropriate substrate or ligand or merely by increasing the ionic strength.⁶⁴

Since the early work on Cibacron Blue, a large number of triazine dyes have been examined as potential tools for protein purification. The Procion dye series (produced by Imperial Chemical Industries) have been particularly well studied* and many of them show high degrees of selectivity for particular classes of proteins. Thus Procion Red HE-3B (Fig. 2.6) linked to Sepharose binds to NADP⁺-dependent dehydrogenases (such as glucose-6-phosphate dehydrogenase) in preference to NAD⁺-dependent dehydrogenases (such as malate dehydrogenase) and can be used for purification of the former type from crude extracts of yeast.⁶⁹ By contrast NAD⁺-dependent dehydrogenases are bound preferentially by Cibacron Blue F3G-A. Hey and Dean⁷⁰ have advocated the use of a 'tandem' strategy for protein purification in which a crude extract is first passed through a dye column that does not retain the protein of interest, and then through a second dye column that does retain the protein. The first column

* It should be pointed out that structural information about many of these dyes is not widely available, often for commercial reasons.

thus acts to remove a number of contaminating proteins that might interfere with the second step.

As well as varying the structure of the dye, the conditions of adsorption and elution can be varied.⁶³ In particular it is important to note that under certain conditions the immobilized dyes can act as effective cation exchangers. To minimize these effects, it is recommended that the operations should be carried out at an ionic strength ≥ 0.1 and at pH ≥ 7 .⁶³ An example of the use of dye-ligand chromatography in protein purification is given in Section 2.8.7.

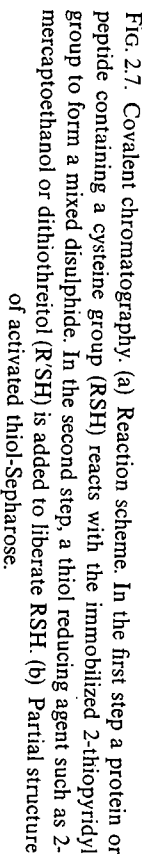
2.6.4.6 Immunoadsorption chromatography⁷¹

The high specificity shown in antibody-antigen interactions can often provide a suitable basis for purification procedures. If a small amount (ca. 0.1 mg) of a pure enzyme from one species is available, it can be used to raise antibodies in another species (usually a rabbit, goat, or mouse). A range of antibodies will be produced that vary in affinity for the enzyme because they recognize different structural features (epitopes) of the enzyme. After purification, the antibodies can be coupled to a matrix such as CNBr-activated Sepharose (see Fig. 2.5) and subsequently used to separate the enzyme antigen from a complex mixture. Desorption can be brought about by a change in pH, an increase in ionic strength or other treatments which weaken the antibody-antigen binding. (The desorption step can be the most difficult part of the entire procedure because the enzyme activity can be destroyed during the harsh conditions required to weaken the strong interactions.) The application of monoclonal antibody techniques^{72, 73} has helped to overcome a number of the problems previously encountered; thus it is often possible to choose antibodies that have an appropriate affinity for the enzyme antigen and to produce an indefinite supply of such antibodies. The monoclonal-antibody technique has been used to purify aromatic L-amino acid decarboxylase from bovine brain.⁷⁴ In this case the previously purified enzyme from bovine adrenal medulla was used as the antigen for monoclonal-antibody production. The antibody was immobilized on an activated cross-linked agarose matrix and used to provide a 25-fold purification of enzyme from extracts of bovine brain which had been already subjected to ion-exchange chromatography. Elution of enzyme from the immobilized antibody was brought about by washing with a solution containing 50 mmol dm⁻³ acetic acid and 10 per cent (v/v) ethylene glycol.

2.6.4.7 Covalent chromatography

In covalent chromatography, a covalent bond is formed between molecules in the mobile phase and the stationary-phase matrix, in contrast to the non-covalent interactions described earlier in this section. The principal applications have been in the separation and purification of cysteine-containing peptides and proteins, via the disulphide exchange reactions shown in Fig. 2.7(a).

A commonly used matrix is activated thiol-Sepharose 4B in which a 2-thiopyridyl group is attached via a glutathione spacer arm in order to minimize



Activated thiol-Sepharose can also be used to immobilize cysteine-containing proteins for affinity chromatography. This approach is described in Section 2.8.5 in connection with the purification of cytochrome *c* oxidase. A further application of covalent chromatography is mentioned in Chapter 3, Section 3.4.8, for the purification of a cysteine-containing peptide.

In Sections 2.6.1 to 2.6.4 we have discussed the principal methods employed in enzyme purification procedures. There are, however, a number of other methods, including heat denaturation,^{77, 78} fractional adsorption on calcium phosphate gels and hydroxyapatite,^{79, 80, 81} and concentration by freeze-drying (lyophilization),⁸² Further details about these methods can be found in the references given.

Having described some of the advantages and disadvantages of the various separation methods, it is necessary to consider some of the factors that influence the choice of methods and the order of their application in a particular purification procedure. At the outset it should be emphasized that there is rarely only one method or combination of methods that can be used to purify a given enzyme. The actual sequence of methods employed will depend on a variety of factors such as (i) the scale of the preparation and the yield of enzyme required; (ii) the time available for the preparation; and (iii) the equipment and expertise available in the laboratory.

In some cases, especially if there is the possibility of proteolysis during the procedure, it may be desirable to aim for as quick a purification as possible at the expense of a lower yield. If so, the use of methods based on solubility would be favoured because these are more rapid than those that involve column chromatography. Recently matrices have been developed to allow column chromatography procedures to be carried out more rapidly under high-performance liquid chromatography (h.p.l.c.) conditions. Gel filtration can be performed using Superose, a rigid cross-linked agarose-based matrix (Section 2.8.7). Other beaded hydrophilic matrices are available for ion-exchange chromatography and chromatofocusing (Section 2.6.2.1). Although these matrices can give superior resolution compared with conventional materials, it should be noted that they are expensive and require specialized apparatus. They are thus generally used in the final stages of purification procedures.

Whatever method or combination of methods is finally decided upon, the progress of the purification should be recorded in a purification table (see the example given in Section 2.8.1).

As explained in Section 2.3.1, the aim of a purification procedure is to obtain enzyme of the *maximum possible purity and maximum catalytic activity*. We shall now consider some methods to test these properties.

2.7.1 Tests for purity

Many of the methods of separation described in Section 2.6 can be used on an analytical scale to check that an enzyme preparation is homogeneous, although we should remember that an analytical test can only really be used to demonstrate the presence of impurities rather than prove their absence. Some of the more commonly employed analytical methods are listed in Table 2.2

TABLE 2.2
Some commonly employed analytical methods to check the purity of enzyme preparations

Method	Comments
Ultracentrifugation (Chapter 3, Section 3.2.1)	Not very satisfactory for detecting small (<5%) amount of impurities. Problems arise from associating-dissociating systems (Chapter 3, Section 3.2.4)
Electrophoresis (Section 2.6.2.2)	Should be run at a number of values of pH (as two enzymes might move together at a single pH)
Electrophoresis in the presence of sodium dodecylsulphate (Chapter 3, Section 3.2.3)	A good method for detecting impurities that differ in terms of subunit M_r . Very useful to detect proteolysis in preparations. Problems arise from enzymes composed of non-identical subunits since these give multiple bands (Chapter 3, Section 3.6.1.5)
Isoelectric focusing (Section 2.6.2.3)	A very sensitive method for detecting impurities. Sometimes artefacts can arise suggesting apparent heterogeneity ^{83, 84, 85}
N-terminal analysis (Chapter 3, Section 3.4.4.1)	Should indicate the presence of a single polypeptide chain. Some enzymes have a blocked N-terminus (Chapter 3, Section 3.4.7.1); others consist of more than one chain held together by disulphide bonds (e.g. chymotrypsin)

In using these methods we should be aware that the impurities may only constitute a small amount (1 per cent or so) of the total protein and may be missed altogether. It is often easier to test for the presence of contaminating enzymes by measurements of catalytic activity (e.g. testing for the presence of lactate dehydrogenase in a preparation of pyruvate kinase) than to use analytical methods to try to indicate their absence. However, if the preparation of the enzyme of interest appears to be homogeneous in a number of these analytical tests we can be reasonably confident that it is pure.

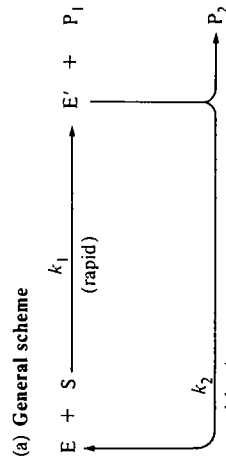
2.7.2 Tests for catalytic activity

In testing for the catalytic activity of a preparation it is important to check that the assay conditions (see Chapter 4, Section 4.2) are optimal, i.e. that any necessary activators or cofactors are present and that inhibitors are absent. It is also worthwhile to investigate the conditions under which the enzyme is stable during storage; in some cases a reducing agent such as dithiothreitol or 2-mercaptoethanol may be necessary to maintain cysteine side-chains in a reduced

state; in others storage at low temperatures, e.g. in 50 per cent (v/v) glycerol solution at 255 K (−18°C), may serve to minimize processes leading to inactivation. The possibility of degradation (e.g. caused by traces of proteases) during long-term storage should also be borne in mind (see the example in Section 2.8.6).

2.7.3 Active-site titrations

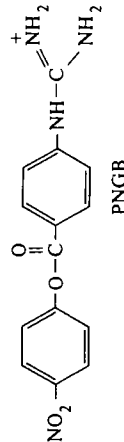
Even if an enzyme preparation is shown to be homogeneous by the various analytical methods listed in Table 2.2, there is still a possibility that some of the enzyme may be in an inactive form. In a number of cases it has proved possible to estimate the amount of active enzyme by a technique known as active-site titration.⁸⁶ In this method we observe the rapid release of a product of reaction between the enzyme and a substrate (or pseudosubstrate), when the intermediate breaks down slowly, if at all, to regenerate enzyme (Fig. 2.8). Inactive enzyme does not give rise to any product, so the concentration of product formed gives the concentration of active enzyme.



If $k_2 \rightarrow 0$, the concentration of P_1 produced gives the concentration of active enzyme directly. If k_2 is less than k_1 , but not equal to zero, it is still possible to calculate the concentration of active enzyme (see reference 91)

(b) The concentration of active enzyme in a preparation of trypsin

The reagent, *p*-nitrophenyl-*p*'-guanidinobenzoate (PNGB) reacts with



the reactive serine side chain of trypsin to form an acyl enzyme derivative (see Chapter 5, Section 5.5.1.3) and liberate *p*-nitrophenol which can be detected by its yellow colour ($\lambda_{\text{max}} = 410 \text{ nm}$). Measurement of the concentration of the *p*-nitrophenolate ion gives the concentration of the active enzyme directly. In this case, the acyl enzyme derivative does not break down to regenerate enzyme.

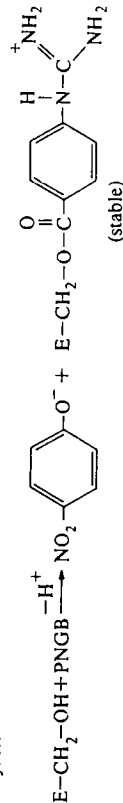


FIG. 2.8. Active-site titration of an enzyme.

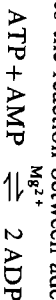
An example of the use of the method was the demonstration that commercial preparations of the protease trypsin contained only 59 and 72 per cent of active enzyme⁸⁷ (Fig. 2.8); the inactive contaminants could be removed by affinity chromatography. Active-site titration methods have been developed for a number of other enzymes, including chymotrypsin,⁸⁶ papain,⁸⁸ isoleucyl-tRNA synthetase,⁸⁹ and ornithine decarboxylase.⁹⁰

2.8 Examples of purification procedures

In this section, eight examples of enzyme purification procedures will be outlined to show how the various separation methods described in Section 2.6 are applied in actual cases. The examples illustrate the wide variety of sources, methods, and problems encountered.

2.8.1 Adenylate kinase from pig muscle⁹²

Adenylate kinase catalyses the reaction between adenine nucleotides



and is very widely distributed in nature. Pig muscle is chosen as the source because this enzyme gives crystals suitable for X-ray diffraction. The procedure is summarized in the flow sheet shown in Fig. 2.9; the progress of the purification is recorded in a purification table (Table 2.3).

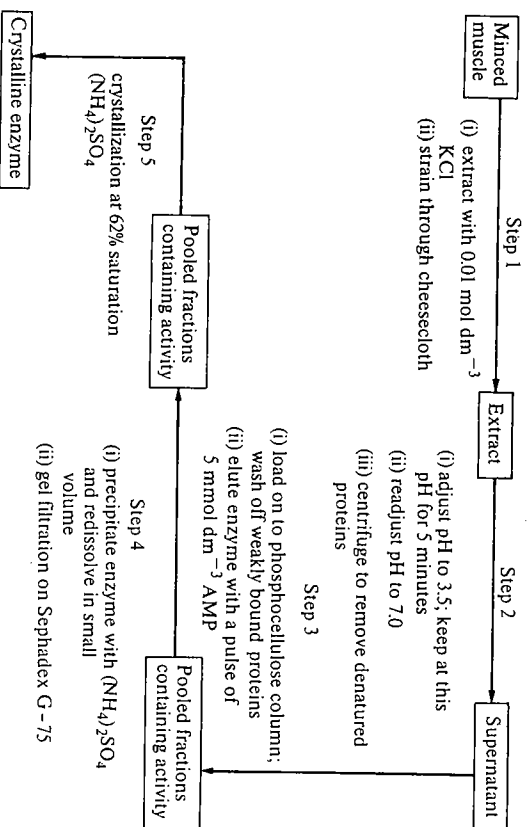


FIG. 2.9. Purification of adenylate kinase from pig muscle. All steps are performed at 273–278 K (0–5°C).

TABLE 2.3
Purification table for adenylate kinase (see Fig. 2.9) starting with 6 kg muscle

Step	Total volume (cm ³)	Total protein (mg)	Total activity (katal)	Specific activity (katal kg ⁻¹)	Yield (%)	Purification factor per step
1 (extraction)	16 600	435 000	0.0413	0.095	(100)	(1.0)
2 (pH)	15 700	112 000	0.0365	0.325	88.3	3.42
3 (phosphocellulose)	1380	1716	0.0223	13.02	54.0	40.0
4 (gel filtration)	211	462	0.0200	43.17	48.4	3.32
5 (crystallization)	—	344	0.0160	46.5	38.7	1.08

The yield is calculated from the amount of activity at each step relative to the amount in the initial extract.

The purification factor of a step is calculated on the basis of the increase in specific activity after that step (e.g. 0.325 to 13.02 in Step 3 is a 40-fold increase).

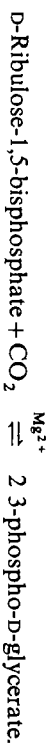
Particular points to note in this purification scheme are:

1. The cells are easily disrupted and the enzyme is extracted in solutions of low ionic strength.
2. Advantage is taken in step 2 of the exceptional stability of adenylate kinase at low pH; this enables many contaminating proteins to be removed. The stability of the enzyme is probably associated with the large amount of regular secondary structure; see Chapter 3, Section 3.5.6.3.
3. Chromatography on phosphocellulose (step 3) is made very effective by the use of affinity elution (see Section 2.6.4.2) using AMP.
4. Gel filtration is used (step 4) to remove a contaminant of *M*_r 60 000 from adenylate kinase (*M*_r 21 000). Sephadex G-75 fractionates in the *M*_r range 3000–70 000.

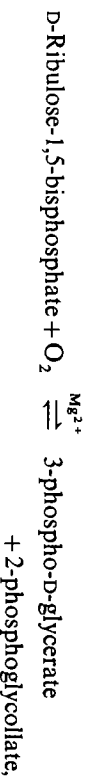
The success of the isolation procedure was judged⁹³ by the homogeneity of the product during ultracentrifugation and gel electrophoresis. Subsequent structural studies⁹² confirmed the absence of any significant contaminating proteins.

2.8.2 Ribulosebiphosphate carboxylase from spinach

Ribulosebiphosphate carboxylase catalyses the CO₂-fixation step in photosynthesis:



The enzyme also catalyses an oxygenase reaction:



and is thus often referred to as ribulosebiphosphate carboxylase/oxygenase. It has been claimed that the enzyme, which occurs at high concentrations in

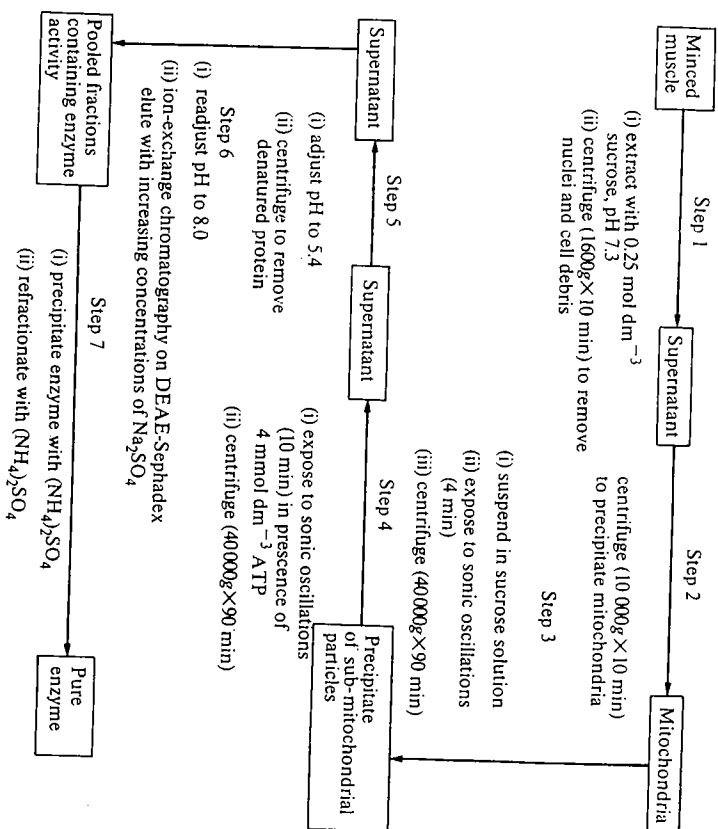


FIG. 2.12. Purification of mitochondrial adenosinetriphosphatase from beef heart.

the case of other membrane-bound enzymes alternative methods that can be used for extraction include the use of detergents or changes in ionic strength depending on the nature of the forces involved in the binding of the enzyme to the membrane (see Chapter 8, Section 8.4.1).

The purification factors in steps 4, 5, and 6 are 3.6-fold, 1.7-fold, and 3.8-fold, respectively, giving an overall purification factor from submitochondrial particles of approximately 23-fold. The final product appears homogeneous in the ultracentrifuge and on electrophoresis in the absence of denaturing agents. It should be noted that in the presence of sodium dodecylsulphate the enzyme dissociates into subunits of different size (see Chapter 3, Section 3.6.1.5) giving rise to multiple bands on sodium dodecylsulphate polyacrylamide gel electrophoresis.

2.8.4 RNA nucleotidyltransferase (RNA polymerase) from *E. coli*

RNA nucleotidyltransferase catalyses the DNA-dependent incorporation of the XMP moiety (X = nucleoside) of ATP, UTP, GTP, and CTP into high- M_r RNA

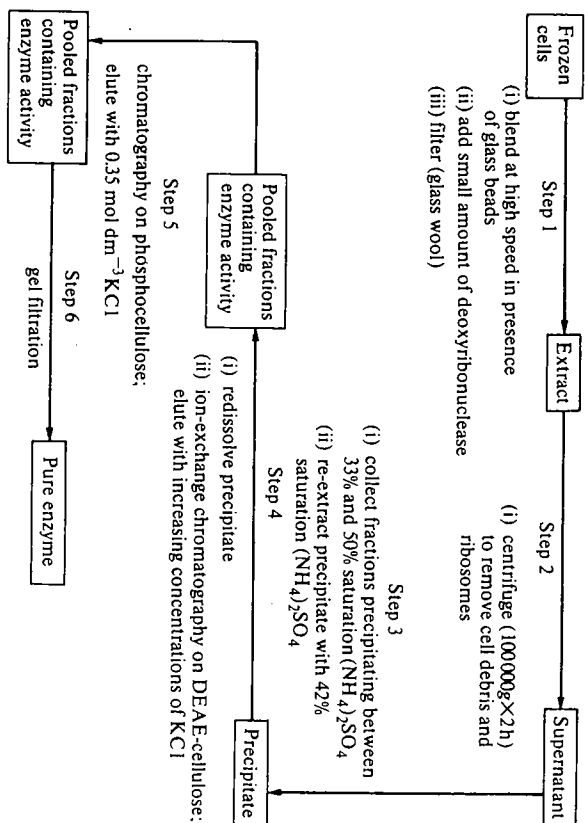


FIG. 2.13. Purification of RNA nucleotidyltransferase from *E. coli*. All steps are performed at 273–278 K (0–5 °C).

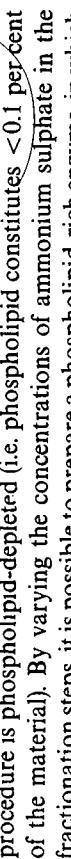
which is insoluble in acid. (For a more detailed discussion of the properties of the enzyme, see Chapter 7, Section 7.3.) The source of the enzyme in the procedure of Burgess¹⁰¹ summarized in Fig. 2.13 is *E. coli* strain K-12; the cells are grown, harvested by centrifugation at 75 per cent of maximum growth, and frozen at 253 K (–20 °C).

This procedure illustrates the harsh measures (step 1) needed to disrupt bacterial cell walls. It is possible that enzymatic methods could be used for this purpose (Section 2.5.2) but these are not easy to apply on a large scale. Because of the absence of subcellular organelles (except ribosomes) in bacteria, the extract contains significant amounts of DNA and is therefore highly viscous. Addition of deoxyribonuclease (step 1) reduces the viscosity of the medium, which is important in the later chromatographic steps. It also prevents formation of the high- M_r complex DNA-RNA-RNA nucleotidyltransferase, which would be sedimented along with ribosomes in step 2. In step 3 the oligonucleotides produced by the deoxyribonuclease treatment in step 1 are re-extracted into $(\text{NH}_4)_2\text{SO}_4$ at 42 per cent saturation. In step 6 gel filtration on a column of Bioigel A – 1.5 m (an agarose gel which fractionates in the M_r range 10 000–2 000 000) is used to remove a minor contaminant of higher M_r than the RNA nucleotidyltransferase. The overall yield of enzyme is 54 mg from 200 g of frozen cells corresponding to 56 per cent of the activity in the initial extract.

In this procedure the washes in step 1 are performed to remove non-cytochrome haemoproteins such as myoglobin. Grinding with sand (step 2) disrupts the mitochondria and the low-speed centrifugation is used to precipitate the sand, cell debris, and nuclei. Centrifugation at a higher speed (step 3) precipitates Keilin-Hartree particles, which consist of physically disintegrated mitochondrial membranes. (This preparation is named after two British bio-

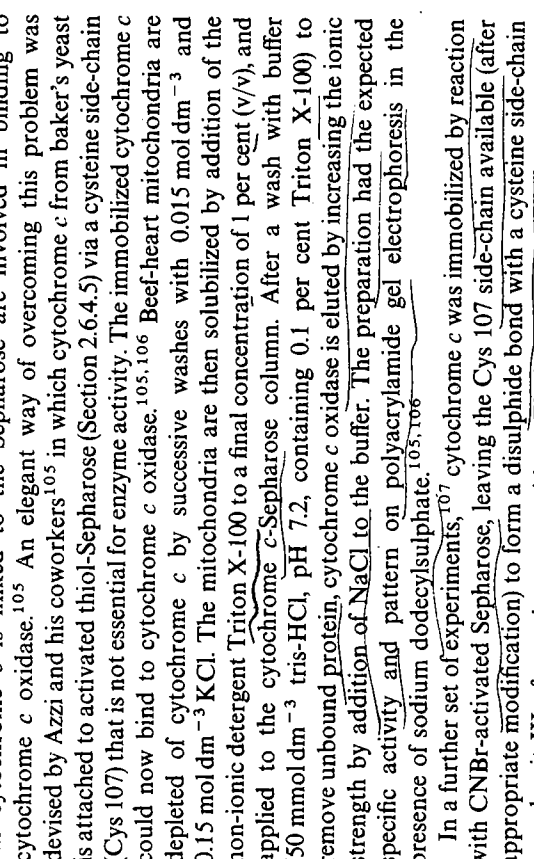
... The subsequent steps (4 and 5) in the

(cholate) and fractionation by changes in ionic strength. The combined purification factor in these two steps is 11-fold. Cytochrome *c* oxidase prepared by this



phospholipid constitutes about 20 per cent of the material. The phospholipid-depleted enzyme is inactive; activity can be restored to the level of the phospholipid-rich form by addition of suitable mixtures of phospholipids.¹⁰³

The isolation procedure for cytochrome *c* oxidase has been improved by the use of affinity chromatography. If cytochrome *c* is linked to CNBr-activated Sepharose (an agarose gel) (see Fig. 2.5), it is found that it does not in general



subunit III-depleted cytochrome *c* oxidase. This procedure opens up the possibilities of systematic studies of the role of this subunit in the proton pumping activity of the enzyme. 108

2.8.6 The *arom* multienzyme protein from *Neurospora*

This multienzyme protein* from *Neurospora crassa* contains five distinct enzyme activities, which catalyse adjacent steps in the biosynthesis of chorismic acid, the common precursor of the three aromatic amino acids tyrosine, phenylalanine, and tryptophan. Although purification of this protein had previously been reported, there was considerable doubt about the M_r of the protein and its subunit composition (see Chapter 3, Section 3.6). It was realized that these difficulties were probably due to proteolysis during the purification procedure and accordingly a new isolation procedure was developed¹⁰⁹ in which stringent precautions were taken to minimize the effects of proteases. These precautions appear to be especially necessary when purifying enzymes from fungi,¹¹⁰ since they contain large amounts of proteases. The procedure (outlined in Fig. 2.15) starts from cells that have been harvested, dried, stored at 253 K (-20°C), and powdered in a blender.

Protease inhibitors are added at each step of this procedure. In the extraction step (step 1), phenylmethanesulphonyl fluoride (PMSF) and EDTA are included; the former irreversibly inactivates serine proteases (see Chapter 5, Section 5.5.1.2), whereas the latter chelates metal ions, thereby inactivating bivalent metal-ion-dependent aminopeptidases. Most of the intracellular proteases are removed by adsorption on DEAE-cellulose under the conditions used in step 2. Deoxyribonuclease is added in step 3 to hydrolyse the DNA in the extract released from the nuclei and thus reduce the viscosity of the solution for the later chromatographic steps. In step 4 the inhibitor benzamidine is added; this is an effective reversible inhibitor of proteases with trypsin-like specificity. It is added at this step because, unlike PMSF, it is not salted out of aqueous solutions by high concentrations of ammonium sulphate.

The final yield of enzyme is about 1 mg from 20 g powdered cells, representing a yield of some 25 per cent. Purification factors for steps 2, 4, 5, and 6 are 1.3-, 5.2-, 5.9-, and 15-fold, respectively, giving an overall purification of nearly 600-fold. The purified material appeared to be homogeneous on polyacrylamide gel electrophoresis in the presence of sodium dodecylsulphate and on centrifugation in a glycerol density gradient. There was no trace of species of low- M_r (which might be formed by proteolysis), although these could be detected in samples stored at 277 K (4°C) for 19 days, suggesting that traces of proteases were still present. The material could be stored satisfactorily at 253 K (-20°C) in the presence of 50 per cent glycerol.

Peptide mapping (see Chapter 3, Section 3.4.3) suggests that the *arom* multienzyme protein consists of two identical subunits each of M_r 165 000 and each possessing five distinct enzyme activities.¹⁰⁹ This protein is thus a multienzyme polypeptide (see the discussion in Chapter 7, Section 7.11.2), in which the polypeptide chain is presumably folded into a number of globular domains each possessing an active site. The linking portions of the polypeptide chain are probably the sites of attack by proteases.

* The definition of multienzyme proteins is discussed in Chapter 7, Section 7.2.

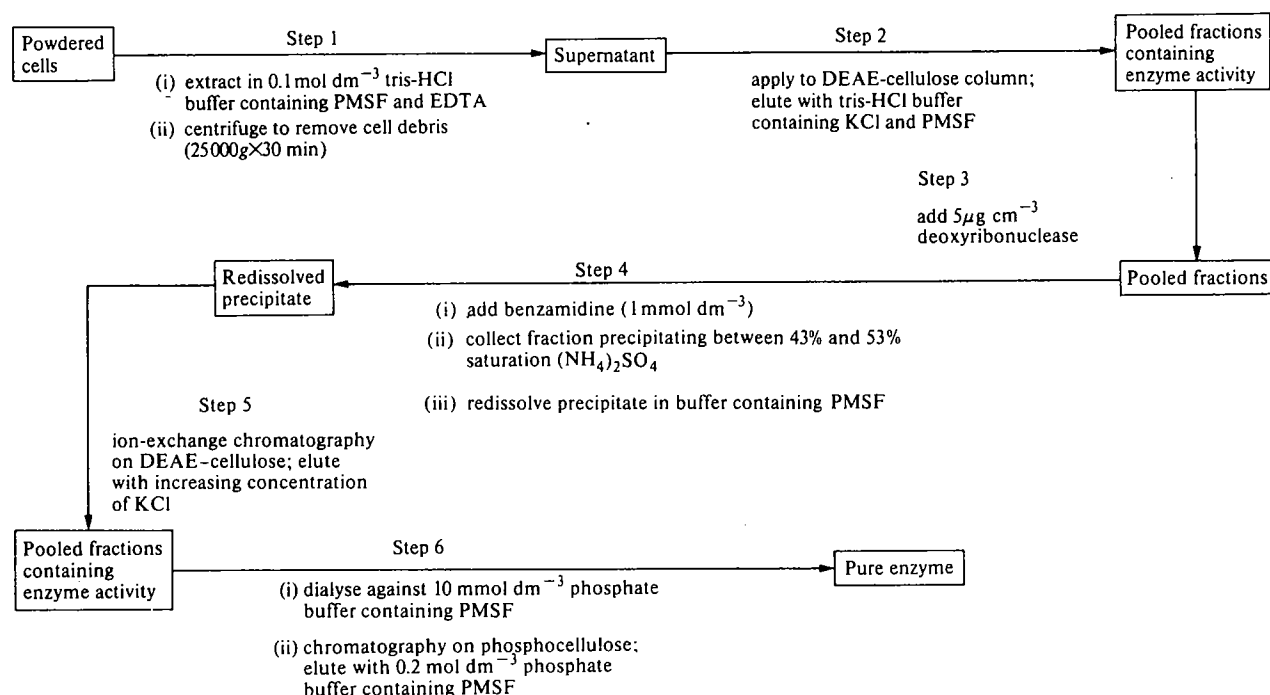


FIG. 2.15. Purification of *arom* multienzyme protein from *N. crassa*. All steps are performed at 273–278 K (0 – 5°C). PMSF is phenylmethanesulphonyl fluoride.

2.8.7 Glutathione reductase from *E. coli*

Glutathione reductase catalyses the NADPH-dependent reduction of oxidised glutathione* (GSSG):



Glutathione reductase belongs to the flavoprotein oxidoreductases, a family of enzymes (including dihydrolipoamide dehydrogenase (see Chapter 7, Section 7.7.1)) that possess a disulphide bond that is alternately oxidized and reduced as part of the catalytic mechanism. The enzyme had been previously prepared from *E. coli* using a complex two-stage affinity-chromatography procedure.¹¹¹ However, for more detailed investigations of the enzyme it was necessary to develop more a rapid procedure that could be scaled up. The preparation described by Scrutton *et al.*¹¹² made two important modifications. Firstly, by recombinant DNA techniques a strain of *E. coli* was constructed that produced 100–200 times more enzyme than the wild-type strain. (The fragment of the plasmid pGR containing the gene for the enzyme was inserted into the plasmid pKK 233-3 to form a new plasmid pKGR which was used to transform *E. coli* strain JM101). Secondly, dye-ligand chromatography (Section 2.6.4.3) was used to give a substantial degree of purification. The procedure is summarized in Fig. 2.16.

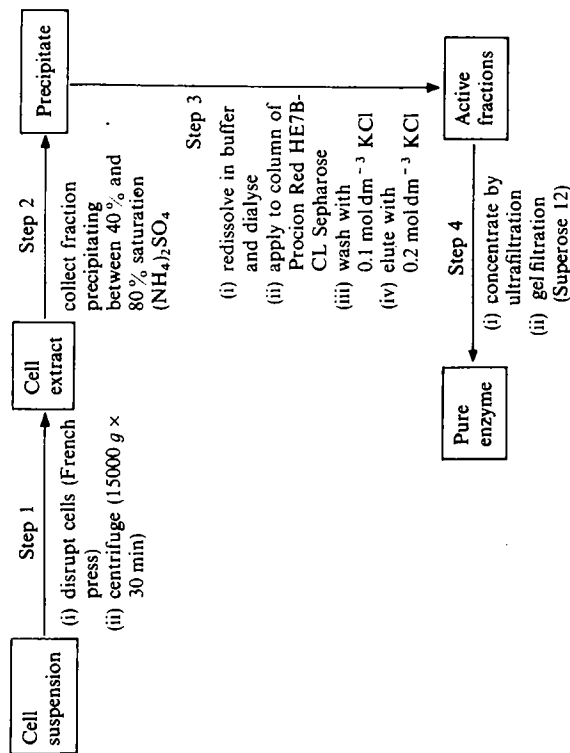


FIG. 2.16. Purification of glutathione reductase from *E. coli*. All steps are carried out at 277 K (4 °C).

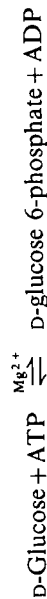
* Glutathione (GSH) is a tripeptide γ-Glu-Cys-Gly that acts as an intracellular reducing agent. In its oxidized form (GSSG), two molecules of glutathione are linked by a disulphide bond.

The procedure uses a French press (Section 2.5.2) to disrupt the cell suspension followed by centrifugation to remove cell debris.

During the addition of ammonium sulphate (step 2) the pH is maintained at 7–7.5 by addition of 2 mol dm⁻³ K₂HPO₄ as necessary. The precipitate is redissolved in a low-ionic-strength buffer (5 mmol dm⁻³ potassium phosphate, pH 7.0, containing 1 mmol dm⁻³ EDTA and 1 mmol dm⁻³ 2-mercaptoethanol) and dialysed against three changes of this buffer, before being applied to a column of Procion Red HE-7B linked to cross-linked agarose. As mentioned previously (Section 2.6.4.3), Procion dyes can show considerable specificity for different classes of enzymes and this dye had been found in preliminary work to bind glutathione reductase strongly. After washing with buffer containing 0.1 mol dm⁻³ KCl to remove weakly bound proteins, glutathione reductase is eluted by buffer containing 0.2 mol dm⁻³ KCl (step 3). In a final step the enzyme is concentrated by ultrafiltration (Section 2.6.1.3) and subjected to gel filtration on Superose 12 (which fractionates in the *M_r* range 1000–300 000) under h.p.l.c. conditions (Section 2.6.6). The overall yield is 11.6 mg enzyme from 100 cm³ of cell suspension, in turn derived from 10 dm³ of initial cell culture. The purification factors for steps 2, 3, and 4 are 1.6-, 33.6- and 1.08-fold respectively, giving an overall purification of 58.3-fold. The enzyme is homogeneous on polyacrylamide gel electrophoresis in the presence of sodium dodecylsulphate and is identical with the enzyme prepared from the wild-type strain of *E. coli*¹¹¹ by a number of criteria (specific activity, *M_r* (97 000) and N-terminal sequence analysis).

2.8.8 The hexokinase isoenzymes

There are four hexokinase isoenzymes (see Chapter 1, Section 1.5.2) which catalyse the reaction



The isoenzymes are designated hexokinases I to IV on the basis of their electrophoretic mobilities (isoenzyme I has the highest mobility towards the anode). Isoenzyme IV has a high Michaelis constant (see Chapter 4, Section 4.3.1.3) for glucose and is also known as glucokinase. In mammals the various isoenzymes possess distinct tissue distributions:

Type I	principally brain, heart, and kidney;
Type II	principally skeletal muscle and adipose tissue;
Type III	principally liver and spleen, in small amounts in other tissues;
Type IV	liver only.

Until the application of affinity chromatography techniques only isoenzyme I had been purified, e.g. from pig heart.¹¹³ Isoenzyme IV was subsequently purified¹¹⁴ from rat liver by a combination of ion-exchange chromatography and affinity chromatography using an immobilized derivative of *N*-acetylglucosamine (as shown in Fig. 2.17 with *n* = 6).

Although *N*-acetylglucosamine itself is a powerful competitive inhibitor (see Chapter 4, Section 4.3.2.2) of all four isoenzymes, it was found that isoenzymes I, II, and III were not retained by a column containing the derivative shown in Fig. 2.17 with $n=6$. A later investigation, however, showed that by varying the conditions isoenzyme II could be purified by this method.¹¹⁵ These findings prompted a detailed study of the inhibitory action of the derivatives of *N*-acetylglucosamine in solution. The results of these studies suggested explanations for the behaviour of immobilized derivatives of *N*-acetylglucosamine and helped to indicate the modifications that were necessary for successful affinity chromatography of the hexokinase isoenzymes.¹¹⁶

The values of the dissociation constants for the various derivatives of *N*-acetylglucosamine from the enzyme showed a marked dependence on the length of the alkyl spacer arm (i.e. on n in Fig. 2.17). Table 2.4 shows the results for isoenzyme I; for the other isoenzymes the values of the dissociation constants are different reflecting small differences between the structures of the active sites of the isoenzymes.

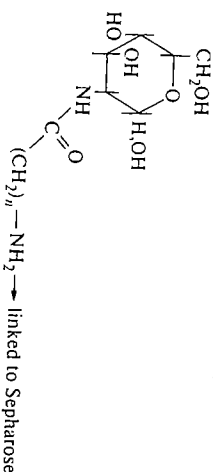


FIG. 2.17. General structure of the immobilized derivative of *N*-acetylglucosamine used in affinity chromatography of hexokinase isoenzymes.

TABLE 2.4
Dissociation constants of various *N*-acetylglucosamine derivatives from hexokinase isoenzyme I

n	Dissociation constant (mmol dm ⁻³)
3	14
4	5
6	3
8	1.3

The value of n refers to the side chain of derivatives depicted in Fig. 2.17.
The dissociation constant of the enzyme-inhibitor complex is derived from inhibition studies as described in Chapter 4, Section 4.3.2.2.

Since the concentration of immobilized ligand on the Sepharose gel is ≤ 10 mmol dm⁻³, it is possible to design an efficient chromatography system only if the dissociation constant of an enzyme-inhibitor complex is less than this value (≤ 2 mmol dm⁻³).¹¹⁶ Thus from Table 2.4 we can see that for isoenzyme I only the C_8 (i.e. $n=8$) derivative can provide the basis for an efficient chromatography system, as described in Section 2.8.8.1.

2.8.8.1 Purification of isoenzyme I from rat kidney¹¹⁶

Rat kidneys are homogenized in a triethanolamine-HCl buffer containing 0.2 mol dm⁻³ MgCl₂ and 50 mmol dm⁻³ glucose. After centrifugation the extract is partially purified by chromatography on DEAE-cellulose and then hydroxyapatite. The active fractions from these procedures are pooled, concentrated by ultrafiltration, and applied to an affinity column containing the C_8 ($n=8$) derivative at a ligand concentration of 5 mmol dm⁻³. The column is washed with triethanolamine-HCl buffer containing 70 mmol dm⁻³ KCl to remove unwanted protein; isoenzyme I is then specifically eluted by inclusion of 0.2 mol dm⁻³ glucose in the buffer. Fractions containing activity are rechromatographed on the C_8 ($n=8$) affinity column to remove a minor contaminant.

The yield of isoenzyme I is about 65 per cent with a purification factor of 21 500-fold (both expressed relative to the crude extract); the material appeared homogeneous on polyacrylamide gel electrophoresis in the presence of sodium dodecylsulphate.

By varying the length of the spacer arm (i.e. the value of n in Fig. 2.17), the concentration of immobilized ligand on the gel, and the eluting conditions it has proved possible to devise procedures to isolate all four isoenzymes in a homogeneous form.¹¹⁶ This detailed study illustrates the importance of a good understanding of the behaviour of ligands in solution for the design of affinity chromatography systems.

2.9 Conclusions from the examples of enzyme purification

The eight examples of enzyme purification discussed in Section 2.8 illustrate the range of problems faced in this type of work and some of the solutions available. It can be seen that many of the earlier procedures contain steps involving ammonium sulphate fractionation, ion-exchange chromatography, and gel filtration, whereas the more recent examples place greater emphasis on affinity methods involving the recognition of particular structural features. These affinity methods, together with developments in recombinant DNA technology,⁵ have made it feasible to purify enzymes normally present in very small amounts, which would be difficult, if not impossible to purify by the earlier methods. The choice of methods to be used in any particular case depends on a number of factors¹¹ (Section 2.6.6). If yield is the primary objective, then the number of steps in a purification should be kept to a minimum, since losses inevitably occur at each stage. (Of course the number of steps has to be sufficient to give a homogeneous

product.) If speed is the primary concern, it should be remembered that methods based on changes in solubility are in general more rapid than those involving column chromatography, e.g. ion-exchange chromatography or gel filtration, although recent developments in h.p.l.c. techniques have made such chromatographic procedures much less time-consuming (Section 2.6.6).

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3

The structure of enzymes

3.1 Introduction

In this chapter we shall consider various aspects of enzyme structure. The ultimate aim of an investigation is to establish the complete three-dimensional structure of an enzyme at atomic (or near atomic) resolution. This information provides a firm basis for understanding the properties of the enzyme, especially its catalytic activity (see Chapter 5).

The determination of this detailed structure is a daunting task for even the smallest enzyme and has been found to require several years of work. Assuming that a supply of purified enzyme is available, the principal stages of the work can be listed as:

- (1) determination of relative molecular mass, M_r * (Section 3.2);
- (2) determination of amino acid composition (Section 3.3);
- (3) determination of primary structure (Section 3.4);
- (4) determination of secondary and tertiary structure (Section 3.5);
- (5) determination of quaternary structure (Section 3.6);

The term primary structure refers to the sequence of amino acids in a polypeptide chain; it contains only one-dimensional information and tells us little about the three-dimensional structure. The terms secondary and tertiary structure refer to different aspects of the three-dimensional structure; secondary structure refers to regular elements of structure such as helix, sheet, etc. (Section 3.5) in which interactions between regions of the enzyme in close proximity in the sequence are involved. Tertiary structure refers to the folding of a chain, by which portions of the molecule well separated in the sequence are brought into close contact with each other. The term quaternary structure refers to the arrangement of the individual subunits† in an enzyme consisting of more than one subunit.

We shall discuss the various aspects of structure in the order given above, but it should be emphasized that structural investigations do not necessarily proceed in this order. It is often found that information from one type of investigation can be useful in another line of work; thus the definitive M_r value is established once

* The term *relative molecular mass* (M_r) is now used in place of *molecular weight*. M_r is a dimensionless number and is the ratio of the molecular mass of a molecule to $1/12$ the mass of one atom of ^{12}C . The latter value is known as a *dalton* (1.663×10^{-24} g). Molecular masses are often quoted in daltons or kilodaltons (abbreviated as Da or kDa, respectively).

† The term subunit refers to the smallest covalent unit. A subunit may consist of one polypeptide chain or of two or more chains linked by disulphide bonds. Thus the subunit of insulin consists of two chains; six of these subunits can aggregate to form a hexameric species.

the primary structure of the enzyme is known (Section 3.4.9.1) and the primary structure is needed to help in the elucidation of the three-dimensional structure by X-ray crystallography (Section 3.5.1). In each of the sections, the main experimental approaches involved will be indicated, together with a description of the principal results obtained and the application of these results.

3.2 The determination of M_r

Enzymes are macromolecules with M_r values ranging from about 10 000 to several million. Therefore methods for determining M_r , such as mass spectrometry or freezing point depression, which are applicable to small molecules, are not suitable for enzymes. The majority of present-day determinations of M_r values of enzymes are performed by use of one or more of the following techniques:

- * (1) ultracentrifugation;
- * (2) gel filtration;
- * (3) sodium dodecylsulphate polyacrylamide gel electrophoresis.

Of these methods, (2) and (3) are 'semi-empirical' and comparisons are made with standard molecules of known M_r . In (1), however, the M_r can be calculated using equations derived from first principles.

Other techniques less commonly employed involve measurements of osmotic pressure and light scattering. The osmotic pressure is the pressure in excess of atmospheric pressure which must be applied to prevent movement of solvent from one solution to a second solution (containing a macromolecule) when the two solutions are separated by a membrane permeable to the solvent but not to the macromolecule. More details can be found in the book by Price and Dwek.¹ (The scattering of light by a solution arises because the electrons of solute molecules can be distorted (or *polarized*) by the fluctuating electromagnetic field of the radiation, leading to scattering of the radiation in different directions. From measurements of the intensity of scattering as a function of the angle of scattering the M_r of the macromolecule can be calculated.)² If the wavelength of the radiation used is of the same order of magnitude as the dimensions of the macromolecule, we can obtain additional information on its shape.³

3.2.1 Ultracentrifugation

An ultracentrifuge is capable of generating intense centrifugal fields; in a typical machine a rotor speed of 65 000 r.p.m. corresponds to a field of the order of 300 000 times that of gravity. Under these conditions macromolecules will have a tendency to sediment, provided that the density of the macromolecule is greater than that of the solution (this generally holds for enzymes in aqueous solution). The ultracentrifuge can be used in two main ways to determine M_r values: sedimentation velocity and sedimentation equilibrium.

3.2.1 Sedimentation velocity

In this type of experiment the ultracentrifuge is operated at high speeds to generate centrifugal forces that are sufficiently intense to sediment the macromolecules. The sedimentation of an enzyme can be monitored by suitable optical means (including for instance changes in refractive index or absorption of 280 nm radiation)⁴ and from these measurements the sedimentation coefficient, s , can be calculated. This coefficient gives a measure of the sedimentation velocity in a unit gravitational field. The sedimentation coefficient cannot by itself be used to calculate the M_r of the enzyme, since the rate of sedimentation will depend on other factors such as the shape of the macromolecule. However, if we have other information, such as the value of the diffusion coefficient (D) of the macromolecule, its partial specific volume ($\bar{v}$) and the density of the solution (ρ), the M_r can be calculated⁵ from the formula:

$$M_r = \frac{RTs}{D(1 - \bar{v}\rho)}$$

where R is the gas constant, and T is the temperature.

The partial specific volume ($\bar{v}$) of a solute is defined as the volume change upon addition of 1 g of that solute to a large volume of solution, keeping all other parameters (temperature, pressure, etc.) constant. The value of $\bar{v}$ can be determined from very accurate measurements of the densities of solutions⁶ or calculated from the amino-acid composition of the enzyme⁷ by using the known values of the molar volumes of each increment of structure of the various amino-acids. The diffusion coefficient, D , can be measured in a separate experiment in which the ultracentrifuge is operated at low speeds so that no detectable sedimentation occurs.⁴

3.2.1.2 Sedimentation equilibrium

If the rotor speed is not great enough to cause complete sedimentation of the macromolecule, then after a while an equilibrium state is reached in which the tendency of the macromolecules to be sedimented is balanced by their tendency to diffuse from the region of high concentration at the bottom of the ultracentrifuge cell to the region of low concentration towards the meniscus of the solution. From measurements of the distribution of concentration, c , of the macromolecule as a function of distance, r , along the cell (i.e. distance from the axis of rotation) the M_r can be calculated provided that we also know $\bar{v}$ and ρ , as in the sedimentation velocity experiment (Section 3.2.1.1):

$$M_r = \frac{2RT}{(1 - \bar{v}\rho)\omega^2} \frac{d \ln c}{dr^2}$$

where R is the gas constant, T is the temperature, and ω is the angular velocity in rad s^{-1} . The value of $(d \ln c/dr^2)$ is obtained from the slope of the plot of $\ln c$ against r^2 .

The advantage of the sedimentation equilibrium method is that a system is being studied at equilibrium and thus there is no dependence on the shape of the solute or the viscosity of the solution. Using a short column of solution (0.15–0.30 cm) the time required to reach equilibrium is quite short (4–8 hours). Under favourable circumstances M_r can be determined to within a few per cent of the true value (i.e. the value calculated subsequently from the primary structure). For further details of these methods, a number of articles can be consulted.^{8–10} Provided that the appropriate values of $\bar{v}$ and ρ are used it is also possible to calculate the M_r of the enzyme under denaturing conditions (Section 3.7.1).

3.2.2 Gel filtration

As described in Chapter 2 (Section 2.6.1.2) certain cross-linked polymers such as Sephadex, Sephacryl and Bio-Gel are able to separate molecules according to size. Large molecules are unable to penetrate the pores of the cross-linked polymer and are eluted from a Sephadex column ahead of small molecules. For molecules of a similar shape (e.g. a series of globular proteins) there has been found to be a linear relationship between the logarithm of M_r and the distribution coefficient, K_d , over a certain range of M_r values.¹¹

The value of K_d for a given solute is defined by the relationship

$$K_d = \frac{V_e - V_0}{V_t - V_0}$$

where V_e is the elution volume of the molecule of interest, V_0 is the elution volume of a molecule completely excluded by the column (e.g. dextran blue with an M_r of 2×10^6), and V_t is the elution volume of a small molecule (e.g. glucose) which is totally included by the column. Thus $K_d = 0$ for solutes which are totally excluded and $K_d = 1$ for solutes which are totally included.

For Sephadex G-200 the 'linear' range of M_r values is from about 40 000 to about 200 000. By reference to the standard curve, the M_r of the enzyme of interest can be determined (Fig. 3.1). Other types of cross-linked polymers can be used over different ranges of M_r values.

While the gel filtration method is extremely simple to use it must be emphasized that the M_r value obtained should only be used as a guide until more definitive evidence is available. It has been known for some time^{11,12} that glycoproteins, particularly those rich in carbohydrate, show anomalous behaviour on gel filtration, and this will also be the case for proteins that are not globular in shape. Thus, the heat-stable phosphatase inhibitor protein isolated from skeletal muscle (Chapter 6, Section 6.4.2.4) appears to have an M_r of 60 000 on gel filtration through Sephadex G-100, whereas the true M_r is close to 20 000.¹³ Fortunately for our purposes it does appear that the vast majority of enzymes are globular proteins and are well behaved on gel filtration. The M_r derived by this technique seems to be accurate to within about 10 per cent.¹¹ It is also possible to measure M_r values under denaturing conditions (see Section

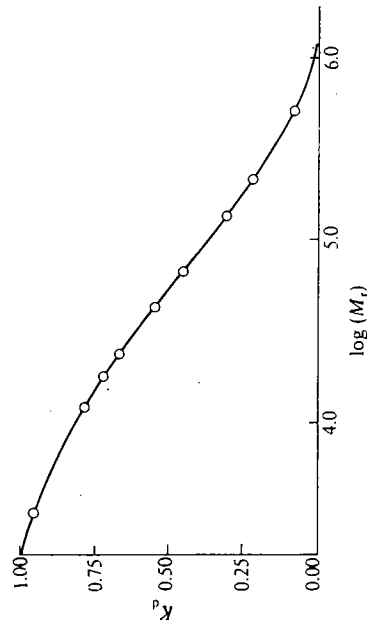


FIG. 3.1. Gel filtration of proteins on Sephadex G-200.¹¹ The proteins used (in order of increasing M_r were): glucagon, cytochrome c, myoglobin, chymotrypsinogen, ovalbumin, bovine serum albumin, lactate dehydrogenase, catalase, and β -D-galactosidase. K_d is the distribution coefficient for a solute and is defined in the text.

3.7.1) provided appropriate standard curves are constructed.^{11, 14} Gel filtration can also be performed under high-performance liquid chromatography (h.p.l.c.) conditions using silica-based column support materials. The M_r can be determined in a much shorter time (ca. 30 min) than by conventional gel filtration.¹⁵

3.2.3 Sodium dodecylsulphate polyacrylamide gel electrophoresis

The mobility of a charged molecule in an electric field is normally a function of various factors such as the size and shape of the molecule and the charge it carries and it would therefore be expected that electrophoresis would not normally give any reliable estimates of M_r .^{*} However, in the case of proteins we can overcome this difficulty by addition of the detergent sodium dodecylsulphate, SDS, which has the structure $\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^- \text{Na}^+$. (SDS is also known as sodium lauryl sulphate). A reducing agent such as 2-mercaptoethanol is also added to break disulphide bonds (see Section 3.3.2). Addition of the detergent has two principal effects.¹⁸

1. Nearly all proteins bind SDS in a more or less constant ratio, 1.4 g SDS per gram of protein. Since the negative charge carried by the SDS overwhelms any charge carried by the protein, the protein-SDS complex has a constant charge/mass ratio.
2. The three-dimensional structure of the protein is lost and the protein-SDS complex is rod-shaped with a length proportional to the M_r of the protein.

^{*} Hedrick and Smith¹⁶ have shown that the M_r of a protein can be estimated by measuring its mobility as a function of acrylamide concentration. However, this method is only reliable if the standard proteins for calibration have the same shape, degree of hydration and partial specific volume as the unknown protein.¹⁷

Since the charge and hydrodynamic properties of the protein-SDS complex are both simple functions of the M_r , it is not particularly surprising that the mobility on electrophoresis is a function of M_r alone. The fact that larger molecules have lower mobilities means that the hydrodynamic effects (i.e. sieving) predominate over the charge effects. In practice it is found that a graph of the logarithm of the M_r against mobility is linear (Fig. 3.2) and the M_r of an unknown protein can be determined by reference to the standard line. Different ranges of M_r can be examined by the use of gels of different polyacrylamide concentration or by the use of gradient gels¹⁹; thus, 10 per cent gels give good separation in the range 10 000 to 70 000 and 5 per cent gels are satisfactory in the range 25 000 to 200 000. The accuracy of the M_r obtained is estimated to be 10 per cent or better.^{18, 20}

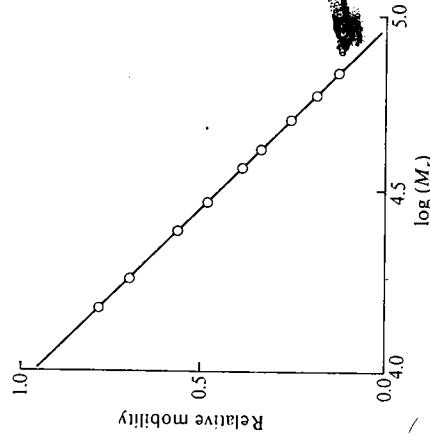


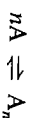
FIG. 3.2. Sodium dodecylsulphate polyacrylamide gel electrophoresis of proteins.^{18, 10} per cent polyacrylamide gels were used and mobilities are expressed relative to bromophenol blue. The proteins used (in order of increasing subunit M_r) were: lysozyme, myoglobin, trypsin, carbonate dehydratase, glyceraldehyde-phosphate dehydrogenase, fructose-bisphosphate aldolase, fumarate hydratase, catalase, and bovine serum albumin.

The comments made in Section 3.2.2 about gel filtration also apply to SDS polyacrylamide gel electrophoresis; namely that the M_r obtained should be checked by other methods. There are cases known in which anomalous behaviour is observed on electrophoresis. Thus histones, which are highly positively charged, contribute significantly to the charge of the protein-SDS complex and give rise to rather low electrophoretic mobilities.¹⁸ In addition, glycoproteins often show impaired binding of SDS and display low mobilities.¹⁸ Additional complications can arise when proteins possess highly flexible regions (see the example of the lipoyl domains in the E2 component of the pyruvate dehydrogenase complex, Chapter 7, Sections 7.7.1 and 7.7.2).

Finally, we should note that addition of SDS dissociates a protein into its constituent subunits. If the subunit contains disulphide bonds, the addition of 2-mercaptoethanol ensures that these are broken, so that the molecular weight(s) we obtain by this method will be that those of the individual polypeptide chain(s).

3.2.4 Association-dissociation systems

In the previous Sections (3.2.1, 3.2.2, and 3.2.3) we have assumed that the enzyme in question has a unique M_r , i.e. that it is monodisperse. While this is a reasonable assumption for most purified enzymes, there are systems known that are capable of undergoing association and dissociation. A good example is provided by glutamate dehydrogenase from beef liver, where the fundamental unit of M_r 336 000 (itself made up of six polypeptide chains) can form an aggregate:



where A is the hexamer of M_r 336 000 and A_n is the aggregate of M_r up to at least 2×10^6 .

The position of equilibrium in this system depends on the concentration of enzyme with the aggregate form being favoured at high concentrations.²¹ At any given concentration the M_r will contain contributions from the various species present and the measured value of this average M_r will depend on the technique employed. For instance, from osmotic pressure measurements we would determine a 'number average' M_r ($\bar{M}_n$) defined by

$$\bar{M}_n = \frac{\sum n_i m_i}{\sum n_i},$$

where n_i is the relative number of the i th species, and m_i is its relative molecular mass.

However, from sedimentation equilibrium or light-scattering measurements a 'weight average' M_r ($\bar{M}_w$) is determined.

$$\bar{M}_w = \frac{\sum n_i m_i^2}{\sum n_i m_i}.$$

Other examples of association-dissociation systems include enolase, in which a monomer-dimer equilibrium exists,²² and acetyl-CoA carboxylase, in which extensive polymerization can occur to give large filamentous aggregates.²³

It is possible to adapt some of the methods described to determine the M_r of the active unit of the enzyme when this may be in doubt, e.g. in an association-dissociation system. Cohen and Mirre^{24, 25} described the technique of *active enzyme centrifugation* in which enzyme at very low (catalytic) concentrations is sedimented through a solution of substrate. The rate of sedimentation down the ultracentrifuge cell is determined by monitoring the progress of conversion of substrate to product using absorption spectrophotometry or other convenient means of following the reaction. In addition, gel filtration can be

performed at low concentrations of enzyme if the measurement of catalytic activity is used to monitor the position of elution of enzyme from the column. Using techniques such as these it has been possible to show, for instance, that the active unit of beef-liver glutamate dehydrogenase is the hexamer of M_r 336 000.^{25, 26}

3.2.5 Uses of M_r information

The M_r of an enzyme is a fundamental piece of information because it allows us to convert the concentration of a solution from units of mass per volume (e.g. mg cm^{-3}) to units of molarity. This information can then be used in a whole variety of ways such as in considerations of composition (how many amino acids of a given type are present per molecule of enzyme?), catalytic activity (how many molecules of substrate are transformed by one molecule of enzyme per second?), and ligand binding (how many molecules of ligand are bound per molecule of enzyme?). Many of these applications will be described in the course of this and the succeeding three chapters.

Measurements of M_r made in the absence and presence of denaturing agents (see Section 3.7.1) will show whether or not the enzyme is composed of subunits and may indicate the number of such subunits (see Section 3.6.1.1). Alcohol dehydrogenase from yeast has an M_r of 140 000 on gel filtration, but on SDS-polyacrylamide gel electrophoresis the M_r is close to 35 000, indicating that the enzyme is a tetramer, composed of four subunits.

3.3 The determination of amino-acid composition

3.3.1 The amino acids

As mentioned in Chapter 1 (Section 1.2) virtually all enzymes are proteins and are thus made up of α -amino acids linked together. In some cases additional components such as metal ions, cofactors, or carbohydrate may be present. The composition of the protein part is defined in terms of the numbers of each of the 20 amino acids that occurs in each molecule of enzyme. These amino acids are shown in Table 3.1 in the form that is predominant at neutral pH, i.e. in the *zwitterion* form $^+\text{NH}_3\text{---CHR---CO}_2^-$ rather than the uncharged form $\text{NH}_2\text{---CHR---CO}_2\text{H}$.* It must be pointed out that different ionized forms of amino acids from those shown in Table 3.1 are important in certain properties, e.g. in the reaction of cysteine with an electrophile such as iodoacetamide it is the $\text{---CH}_2\text{---S}^-$ form of the cysteine side-chain that is the reactive species even though fewer than 10 per cent of the molecules are in this form at pH 7. In general, therefore, we shall, for convenience, depict amino acids as being in *their*

* Since the pK_a values of the carboxyl and amino groups attached to the α -carbon atom in the general formula are approximately 2 and 10 respectively, the 'zwitterion' form will predominate at neutral pH.

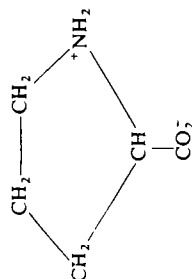
TABLE 3.1
The amino acids that are normally found in enzymes

α -Amino acids are of the general formula			
		$\begin{array}{c} \text{R} \\ \\ \text{H}_3\text{N}^+-\text{CH}-\text{CO}_2^- \end{array}$	
(except for proline). Different amino acids differ in the nature of the R group, or side-chain. One type of classification of the amino acids can be made on the basis of the chemical nature of the R group. The pK_a values of the carboxyl and amino groups attached to the α -carbon atom are approximately 2 and 10 respectively.			
Amino acid	Abbreviations	Side-chain	pK_a of side-chain
Non-polar R group			
Alanine	Ala (A)	$-\text{CH}_3$	—
Valine	Val (V)	$-\text{CH}(\text{CH}_3)_2$	—
Leucine	Leu (L)	$-\text{CH}_2-\text{CH}(\text{CH}_3)_2$	—
Isoleucine	Ile (I)	$-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$	—
Phenylalanine	Phe (F)	$-\text{CH}_2-\text{C}_6\text{H}_5$	—
Tryptophan	Trp (W)	$-\text{CH}_2-\text{C}_8\text{H}_6\text{N}$	—
Methionine	Met (M)	$-\text{CH}_2-\text{CH}_2-\text{S}-\text{CH}_3$	—
Polar, uncharged R group			
Glycine	Gly (G)	$-\text{H}$	—
Serine	Ser (S)	$-\text{CH}_2-\text{OH}$	≈ 14.0
Threonine	Thr (T)	$-\text{CH}(\text{CH}_3)-\text{OH}$	≈ 14.0
Cysteine	Cys (C)	$-\text{CH}_2-\text{SH}$	≈ 8.3
Tyrosine	Tyr (Y)	$-\text{CH}_2-\text{C}_6\text{H}_4-\text{OH}$	≈ 10.1

TABLE 3.1 (Contd.)

Amino acid	Abbreviations	Side chain	pK_a of side chain
Asparagine	Asn (N)	$-\text{CH}_2-\text{C}(=\text{O})\text{NH}_2$	
Glutamine	Gln (Q)	$-\text{CH}_2-\text{CH}_2-\text{C}(=\text{O})\text{NH}_2$	
Acidic R group			
Aspartic acid	Asp (D)	$-\text{CH}_2-\text{CO}_2^-$	≈ 3.9
Glutamic acid	Glu (E)	$-\text{CH}_2-\text{CH}_2-\text{CO}_2^-$	≈ 4.3
Basic R group			
Lysine	Lys (K)	$-(\text{CH}_2)_4-\text{NH}_3^+$	≈ 10.5
Arginine	Arg (R)	$-(\text{CH}_2)_3-\text{NH}-\text{C}(=\text{NH}_2)^+-\text{NH}_2$	≈ 12.5
Histidine	His (H)	$-\text{CH}_2-\text{C}(\text{NH})=\text{CH}-\text{N}^+=\text{CH}_2$	≈ 6.0

Proline (abbreviated Pro or P) is an imino acid with a secondary nitrogen atom. The predominant form at pH 7 is

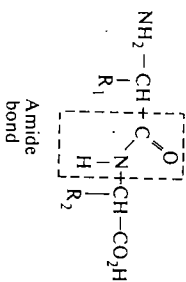


In this book we shall use the three-letter abbreviations for the amino acids. The one-letter code is frequently used in displays of amino-acid sequences to save space.

uncharged forms, and leave it to the reader to ascertain the predominant or most appropriate ionized form under a given set of circumstances.

Apart from glycine, all the amino acids have a chiral centre; indeed, isoleucine and threonine have two such centres. They are thus capable of existing in more than one enantiomeric form. It is found that in enzymes all the amino acids occur in the L-form in which the three-dimensional configuration about the α -carbon atom is as shown in Fig. 3.3(a).

D-Amino acids, which have the opposite configuration (see Fig. 3.3(b)), do occur occasionally in some small peptides; thus D-phenylalanine occurs in the antibiotic gramicidin S. The mode of insertion of D-amino acids into such



3.3.2 Linkages between amino acids

3.3.3.2 Linkages between amino acids

The bond that joins amino acid units together to form a chain is known as an *amide* or *peptide bond*. We can view the bond as arising by the elimination of water as in the reaction:

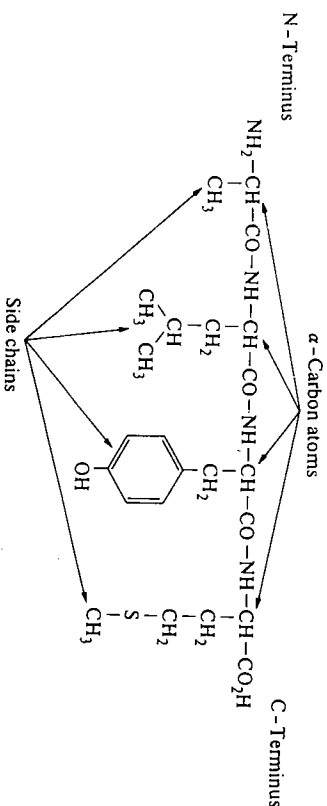


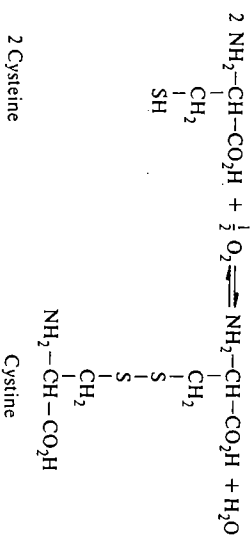
FIG. 3.4. The structure of alanylleucyltyrosylmethionine. In naming the peptide, all amino acids except the C-terminal one are given the suffix -yl.²⁸

In order to form this bond it is first necessary to activate the carboxyl group; the mechanism by which this is achieved is outlined in Chapter 9, Section 9.2.

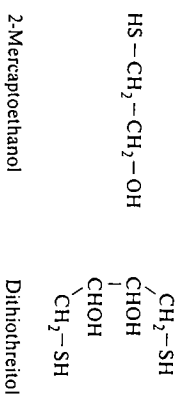
When a number of amino acids are linked together in this way we build up a peptide chain. The term polypeptide or protein, is usually applied to chains consisting of 50 or more amino-acid units. By convention, the N-terminal amino acid is written at the left-hand side and the C-terminal one at the right. The amino-acid units are numbered from the N-terminal end. Figure 3.4 shows the tetrapeptide, alanylleucyltyrosylmethionine, which would be abbreviated as Ala-Leu-Tyr-Met or ALYM in the one letter code.²⁸

Evidence for the presence of amide bonds in proteins comes from a variety of sources, e.g. u.v. and i.r. spectroscopy, the action of enzymes which are known to cleave amide bonds specifically and direct structure determinations using X-ray crystallography.

Apart from the amide bond, the only other type of covalent bond involved in linking amino acid units in enzymes is the disulphide ($-S-S-$) bond, which can be formed between two cysteine side-chains under oxidizing conditions:



The disulphide bond can be broken by the addition of reducing agents such as 2-mercaptoethanol or dithiothreitol.



3.3.3.3 Hydrolysis of amide bonds

In some enzymes, especially the small extracellular hydrolases such as lysozyme, ribonuclease, and chymotrypsin, there are no free cysteine side-chains, i.e. all are paired to form disulphide bonds. By contrast, in most intracellular enzymes such as lactate dehydrogenase the cysteine side-chains are free and there are no disulphide bonds.

Hydrolysis of the amide bonds in proteins can be effected by a variety of reagents but the most commonly used conditions are 6 mol dm⁻³ HCl at 383 K (110°C)

for 24 hours *in vacuo*. However, under these conditions some amino acids, e.g. tryptophan are partially or wholly destroyed (see Section 3.3.5). For determination of the tryptophan content of a protein, hydrolysis is carried out using *p*-toluene-sulphonic acid²⁹ or sodium hydroxide³⁰ in place of HCl. A single hydrolysis method claimed to be suitable for all amino acids (including cysteine and tryptophan) has been described.³¹ It is based on alkylation of cysteine, followed by hydrolysis by methanesulphonic acid in the presence of tryptamine, which serves as an amino-acid protectant. In view of the problems associated with the various chemical methods, attention has been paid to the possibility of carrying out complete hydrolysis of proteins using proteolytic enzymes. As yet, however, no completely satisfactory procedure has been devised.^{31, 32}

3.3.4 Analysis of amino-acid mixtures

After hydrolysis of a protein according to one of the procedures described in Section 3.3.3, the sample is dried and then analysed for the component amino acids. In the early days of protein analysis, the various amino acids were separated by complex crystallization procedures³³ and estimated gravimetrically or by microbiological assays.³⁴ These methods required many months of work and large amounts (up to several grams) of sample.

Following the development of chromatographic procedures, a great advance was made in 1958 with the development of an automated apparatus for performing amino-acid analysis,³⁵ with this apparatus it is possible to obtain the amino-acid composition of less than 1 mg of protein in a matter of a few hours. For their achievement and its application to the primary structure of ribonuclease, Stein and Moore were awarded the Nobel Prize for Chemistry in 1972. The amino-acid analyser consists essentially of an ion-exchange column, normally a sulphonated polystyrene resin. Samples to be analysed are dissolved in buffer at pH 2.2 and eluted from the column by defined volumes of citrate buffers of increasing pH, usually 3.25, 4.25, and 5.28, with the pH 5.28 buffer being of greater ionic strength than the first two. The order of elution of the amino acids (Asp, Thr, Ser, Glu, Pro, Gly, Ala, Val, Met, Ile, Leu, Tyr, Phe, Lys, His, Arg) can be seen to be broadly in line with expectations, i.e. the acidic amino acids are not retained by the negatively charged column, the neutral amino acids are retained to some extent, and the basic amino-acids are strongly retained. However, it is clear that other types of interactions between the amino acids and the column are involved, since the aromatic amino acids Tyr and Phe are eluted after the aliphatic amino acids such as Ala; this is presumably a consequence of interactions between the aromatic rings of these compounds with those of the column.³⁶

As the amino acids are eluted from the column they are detected by reaction with ninhydrin (Fig. 3.5) which gives rise to a purple colour³⁷ (monitored at 570 nm). From the colour yield the amount of the amino acid can be evaluated and hence the composition of the protein determined. A more sensitive detection method involves the reaction of the amino acids with the reagents fluoresce-

mine³⁸ or phthalaldehyde (Fig. 3.6),^{39, 40, 41} which produce highly fluorescent compounds. With these reagents amino-acid analyses can be carried out successfully on as little as 1 μ g of hydrolysed protein. Phthalaldehyde has some advantages over fluorescamine in terms of stability and sensitivity.⁴¹

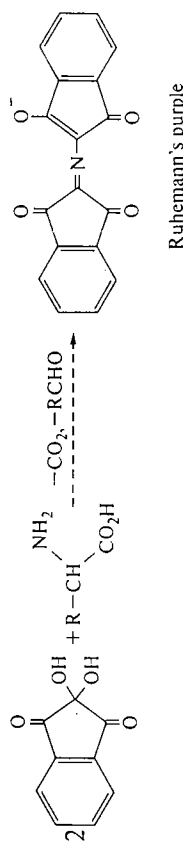


FIG. 3.5. The reaction of amino acids with ninhydrin. Proline does not give this reaction (Section 3.3.5.5).



FIG. 3.6. The reaction of amino acids with phthalaldehyde in the presence of 2-mercaptoethanol to give the highly fluorescent 1-alkylthio-2-substituted isoindole product.

H.p.l.c. has been applied successfully to the analysis of amino-acid mixtures.⁴² The mixture is reacted with reagents such as dimethylaminoazobenzene-sulphonyl chloride or phthalaldehyde and the derivatives separated by reverse-phase h.p.l.c. using an acetonitrile gradient. The detection of the derivatives is by absorbance or fluorescence, respectively. Analyses can be completed in under 40 min.

3.3.5 Problems encountered in amino-acid analysis

As mentioned in Section 3.3.3 a number of amino acids are subject to destruction during acid hydrolysis of proteins and certain precautions must be taken to obtain reliable results. A full discussion of these precautions are described in a review⁴³ that also outlines procedures for the analysis of the carbohydrate components of glycoproteins, but we shall mention some of the more commonly encountered problems in Sections 3.3.5.1 to 3.3.5.5.

3.3.5.1 Hydroxyamino acids

The hydroxyamino acids, serine, threonine, and tyrosine, are partially destroyed during acid hydrolysis (up to about 10 per cent in 24 hours). In these cases, it is usual to perform hydrolyses for different lengths of time, e.g. 24, 48, and 72 hours, and to extrapolate the results of the analyses back to zero time to obtain the true content of these amino acids.

3.3.5.2 Hindered amide bonds

Certain amide bonds in proteins, especially those between the branched side-chains of valine and isoleucine are rather resistant to hydrolysis, e.g. the Ile-Ile bond is only cleaved to the extent of about 50 per cent in 24 hours. In order to obtain reliable values of the contents of these amino acids, extended periods of hydrolysis (up to 120 hours) may be necessary.

3.3.5.3 Amide amino acids

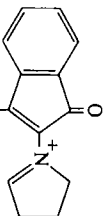
The amide amino acids asparagine and glutamine are hydrolysed to yield aspartic acid and glutamic acid, respectively. Thus the content of glutamic acid determined in the amino-acid analyser will correspond to the sum of the contents of Glu plus Gln in the original protein. The decision as to whether a particular amino acid in the sequence of the protein is in fact Glu or Gln is normally made on the basis of the electrophoretic behaviour of peptide fragments (Section 3.4.7.2) or by the chromatographic properties of the phenylthiohydantoin derivatives following Edman degradation (Section 3.4.4.3). If there is a doubt about the identity, the amino acid is designated as Glx (Z), or Asx (B) in the case of a choice between aspartic acid and asparagine.

3.3.5.4 Sulphur-containing amino acids

Special problems arise in the determination of the sulphur-containing amino acids, cysteine, cystine, or methionine, since these are especially susceptible to oxidation even when care has been taken to remove oxygen prior to hydrolysis. These amino acids can be determined reasonably reliably as oxidized derivatives if the protein is oxidized with performic acid prior to hydrolysis.⁴⁴

3.3.5.5 Proline

Proline, being an imino acid, cannot react with ninhydrin to yield the usual purple derivative (Fig. 3.5). The product formed is yellow and the amino-acid analyser is adapted to allow for monitoring at 440 nm to detect this derivative.



Product formed in reaction between proline and ninhydrin

For detection by reaction with phthalaldehyde (Section 3.3.4), proline must first be reacted with alkaline sodium hypochlorite solution.⁴¹

3.3.6 Uses of amino-acid composition data

Four particular uses of amino-acid data are described in Sections 3.3.6.1 to 3.3.6.4.

3.3.6.1 As a balance sheet

The composition of a protein can be used as a 'sum total' of amino acids to which the composition of derived peptide fragments (Section 3.4.2) must be related, since the total composition of the various fragments must equal the composition of the whole protein. This type of comparison could be used to show, for instance, that a particular fragment had not been accounted for during the purification of the fragments.

3.3.6.2 To explore similarities between enzymes

Composition data can be used to assess the degree of relatedness between enzymes, although much more detailed comparisons are possible when sequence data are available (Section 3.4.9.3). One method of assessing the relatedness has been described by Metzger⁴⁵ who defined a difference index (DI) as

$$DI = 50 \times \left[\sum_i (\text{difference in mole fractions of amino acid}_i) \right]$$

DI equals zero if there is complete identity of composition and equals 100 if there are no amino acids in common between the two enzymes. The values of DI when human glyceraldehyde-phosphate dehydrogenase is compared with the enzyme isolated from rabbit, chicken, surgeon, and *E. coli* are 2.9, 3.3, 5.4, and 7.5, respectively. These values indicate a high degree of relatedness between these enzymes; a conclusion that has been confirmed by the limited sequence data subsequently available. A somewhat similar method of comparison has been used to show that fructose-bisphosphate aldolase from yeast is related to the isoenzymes, of fructose-bisphosphate aldolase from rabbit muscle, liver, and brain.⁴⁶ A fuller discussion of the various methods of comparison has been given by Cornish-Bowden,⁴⁷ who has drawn attention to the problems encountered when comparing the compositions of proteins of differing sizes. Composition data have been used to deduce the evolutionary relationships between organisms; although the information is less complete than that obtained from comparisons of amino-acid sequences (Section 3.4.9.5), it is much more readily obtained.⁴⁷

3.3.6.3 As a guide to the properties of an enzyme

Many structural proteins have extremely simple amino-acid compositions, e.g. in the case of silk, glycine plus alanine plus serine account for 85 per cent of the total amino acids. This simplicity is correlated with the regular structure of these proteins, which is important for their function.

For enzymes, in general, no such regularities exist. Nevertheless, inspection of the amino-acid composition data can give a general idea of some of the properties of an enzyme. Thus, the protease pepsin has an overwhelming preponderance of acidic amino acids (Glu + Asp = 43) over basic amino acids (Lys + His + Arg = 4) in the enzyme isolated from the pig;⁴⁸ this can be related to

the fact that the enzyme is active at the acid pH values (1–5) found in the stomach.

Hatch evaluated the ratio of polar to non-polar amino acids in a number of proteins and found that this ratio varied between about 1 and about 3. Proteins with a low ratio, i.e. a relatively high content of non-polar amino acids appeared to have a high content of helical structure and vice versa.⁴⁹ An extreme example of an enzyme with a very high content of non-polar amino acids (ratio = 0.6) is the C₅₅-isoprenoid alcohol phosphokinase from *Staphylococcus aureus*.⁵⁰ This extremely non-polar enzyme will preferentially dissolve in the butanol layer of a butanol–water system.

3.3.6.4 To calculate the partial specific volume of an enzyme

The amino-acid composition of a protein can be used to calculate⁷ its partial specific volume, $\bar{v}$, which is used in determinations of molecular weight using the ultracentrifuge (see Section 3.2.1).

3.4 The determination of primary structure

3.4.1 The strategy underlying determination of primary structure

Up to a few years ago, determination of the primary structure of a protein or enzyme would have invariably proceeded according to the general scheme shown in Fig. 3.7.

The first protein sequence to be determined was that of insulin by Sanger in 1953;⁵¹ since then the methods have been refined and developed to a considerable extent. Sequences are now known for well over a thousand proteins and the

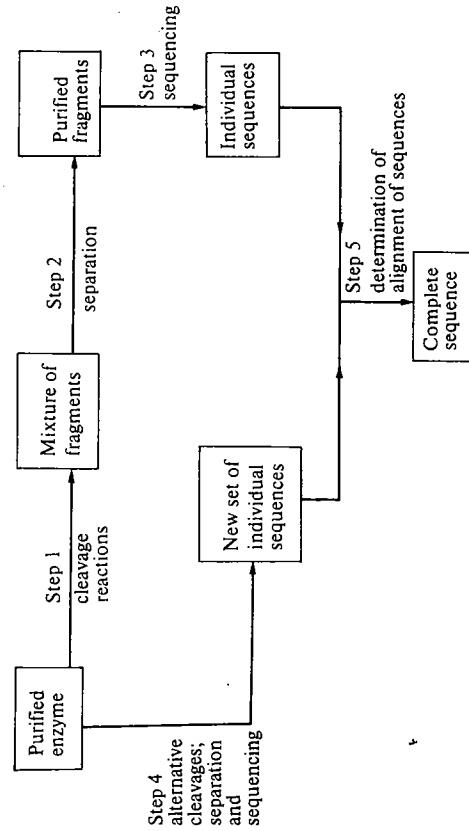


Fig. 3.7. Outline scheme for the determination of the amino acid sequence of a protein.

list is growing steadily. The largest polypeptide chains sequenced by the scheme outlined are those of phosphorylase from rabbit muscle, with 841 amino acids,⁵² and β -D-galactosidase from *E. coli*, with 1021 amino acids.⁵³ By any standards these are remarkable achievements requiring many years of dedicated effort. We shall describe the various steps referred to in Fig. 3.7 in more detail in Sections 3.4.2 to 3.4.6.

Since the late 1970s, developments in the methods of 'direct' protein sequencing have been overtaken by those in the methods of sequencing bases in deoxyribonucleic acid (DNA). If the portion of DNA that encodes a particular enzyme or protein can be isolated and sequenced, the genetic code can be used to 'convert' the sequence of nucleotide bases into the amino acid sequence of the enzyme or protein (the 'indirect' sequencing method). Details of the methods involved in isolating and sequencing the DNA can be found in a number of books and reviews.^{54–59} Isolation of suitable DNA fragments has been aided by the application of 'restriction endonucleases', enzymes that cleave DNA at a relatively small number of sites by recognizing certain sequences of bases⁶⁰, over 400 such 'restriction enzymes' of different specificity are now known.⁵⁸ If an oligonucleotide whose sequence corresponds to a portion of the protein sequence is synthesized⁶¹ and labelled with ³²P (to allow detection by autoradiography), it can be used to probe the mixture of fragments produced by the action of a restriction enzyme by the technique known as 'Southern blotting' (Fig. 3.8). In this technique the oligonucleotide probe hybridizes (forms hydrogen bonds) to the restriction fragment that contains the gene of interest. An alternative approach is to isolate the messenger RNA (mRNA) corresponding to the protein of interest. 'Reverse transcriptase' can then be used to produce the DNA that is complementary (known as cDNA) to that messenger. The latter technique has the advantage that the cDNA will correspond to the actual sequence of the protein, whereas the genomic DNA in eukaryotes may contain regions (intervening sequences or 'introns') that are excised during the formation of mature mRNA and that hence do not code for amino acids in the protein product.^{58, 59}

The DNA sequencing methods rely on the use of chemical reagents to bring about cleavage of DNA at given bases,⁶² or on the interruption of enzymatic replication of DNA by DNA polymerase using 2',3'-dideoxynucleotide triphosphates.⁶³ In both methods the products obtained are separated by electrophoresis on high-resolution gels (6–20% acrylamide in 7 mol dm⁻³ urea), which separate nucleic-acid fragments on the basis of their length. By inspection of the patterns obtained from a number of fragments obtained by causing cleavage or interruption at the different bases, it is possible to deduce the sequence of the DNA.^{62, 63} Sequences of at least 100–200 bases can be read off in this way, and it is thus not difficult to appreciate that the speed of DNA sequencing can be at least an order of magnitude greater than that of protein sequencing.

Examples of the two types of methods of protein sequencing (i.e. 'direct' and 'indirect') are given in two papers on the sequence of penicillinase from *E. coli*.^{64, 65} The 'direct' method took about four years, whereas the nucleotide

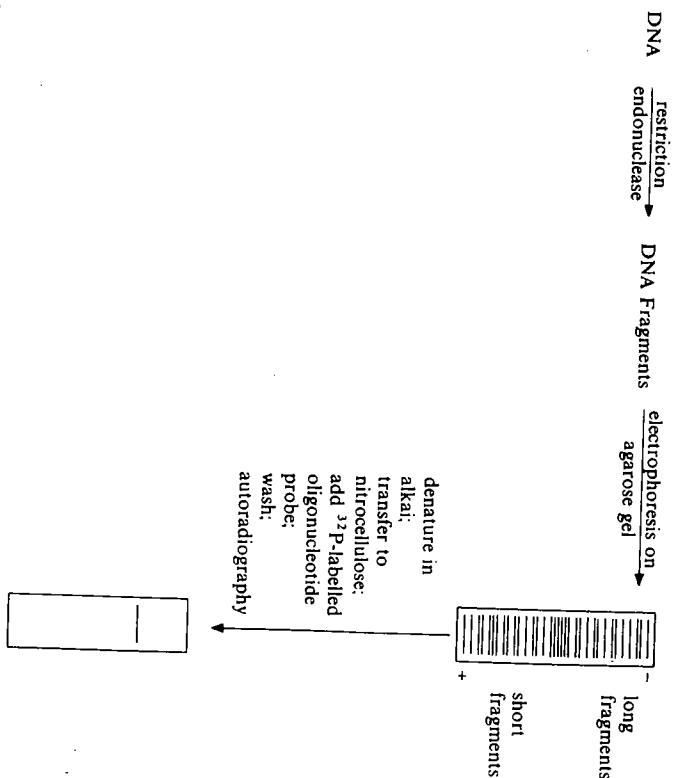


FIG. 3.8. The Southern blotting method. DNA is digested with a restriction endonuclease to give a mixture of fragments that can be separated by electrophoresis. After transfer to nitrocellulose ('blotting'), the fragment of interest can be located by adding a suitable oligonucleotide probe labelled at the 5' end with ^{32}P (using polynucleotide 5'-hydroxyl-kinase). The probe hybridizes to the fragment of interest and can be detected by autoradiography. For further details, see references 58 and 59.

sequence of the DNA was deduced in about six months and was used to resolve some of the ambiguities encountered in the 'direct' method. The two types of method have been used in a complementary way in the determination of the sequence of phosphoglycerate kinase from baker's yeast (Section 3.4.8). Despite the impressive speed of the DNA sequencing methods it is still important to have some amino-acid sequence information about a protein, in order to identify the portion of DNA that is actually transcribed and subsequently translated. Also, knowledge of the amino-acid sequence of part of the protein can be used to design a suitable oligonucleotide probe⁶¹ to locate the appropriate DNA fragment (see Fig. 3.8). In any particular case, if suitable quantities of purified protein are available, it may well be easier to sequence the protein directly. Finally, the DNA sequence cannot reveal any post-translational modifications e.g. N-terminal blocking (section 3.4.7.1), glycosylation, or formation of disulphide bonds; these can only be established by analysis of the protein itself. In Sections 3.4.2 to 3.4.6 we shall discuss the various steps involved in determination of the primary structure of an enzyme according to the general

scheme shown in Fig. 3.7. Further details can be found in the review by Allen.⁶⁶ The determination of a sequence in this way usually requires 0.1–1 μmol of enzyme, i.e. 2–20 mg of an enzyme of M_r 20 000.

3.4.2 Cleavage of the polypeptide chain (step 1 in Fig. 3.7)

The polypeptide chains of enzymes have been found to contain between about 100 and about 2500 amino-acid units. Even the smallest chains are too large to be sequenced directly from one end to the other (Section 3.4.4.4). It is therefore necessary to split the chain into a number of fragments that can each be sequenced before the sequence of the whole chain can be deduced by piecing together the sequences of the fragments. This piecing together will be much easier if we can cleave the polypeptide chain at a restricted number of points and so produce a limited number of well-defined fragments rather than cleave it randomly and so produce a collection of fragments of all sorts of lengths. Specific cleavages can be brought about by the use of certain proteases, since many of these possess high degrees of specificity; some of the more commonly used proteases are listed in Table 3.2. It is particularly important to ensure that pure preparations of proteases are used, since otherwise non-specific cleavages will

TABLE 3.2
Some proteases commonly employed in sequencing work for cleaving polypeptide chains

Protease	Nature of amino acid side-chains at cleavage site in polypeptide chain		
	amino ——— X	XY ↓ Z	————— carboxyl Z
Trypsin ⁶⁷	—	Basic side-chain (Lys, Arg)	—
Clostripain ⁶⁸ (from <i>Clostridium histolyticum</i>)	—	Arg	—
Chymotrypsin ⁶⁹	—	Aromatic (Trp, Tyr, Phe) or hydrophobic (e.g. Leu, Met) side-chain	—
Elastase ⁷⁰	—	Small hydrophobic side-chain (e.g. Ala)	—
Papain ^{71, 72}	Phe (preference)	Preference for Lys over Ala	—
Thermolysin ⁷³ (from <i>Bacillus thermoproteolyticus</i>)	—	—	Leu or Trp (preference)
Endoprotease Glu-C ⁷⁴ (from <i>Staphylococcus aureus</i> V8)	—	Glu or Glu, Asp	Bulky hydrophobic side-chain (e.g. Val, Leu, Ile, Phe)
Pepsin ⁷⁵	—	Phe (preference)	Phe, Trp, Tyr (preference)

occur; in Chapter 5, Section 5.5.1.4 we describe the use of an affinity label to ensure that trypsin preparations are free from contamination by chymotrypsin. In order to obtain complete cleavage of the chain, e.g. on the C-terminal side of every lysine and arginine in the case of trypsin, the enzyme should be treated with a denaturing agent such as guanidine hydrochloride or urea (Section 3.7.1) so that the compact three-dimensional structure of the enzyme is destroyed and the susceptible bonds are exposed to the protease.⁶⁶

The specificity of the proteolytic cleavages can sometimes be altered to advantage. For instance, lysine side-chains can be reacted with an anhydride such as maleic anhydride⁷⁶ or citraconic anhydride;⁷⁷ this prevents attack by trypsin at these points and limits cleavage to the C-terminal side of arginine side-chains, permitting the preparation of large fragments. Lysine side-chains can subsequently be regenerated by exposure to low pH (Fig. 3.9).

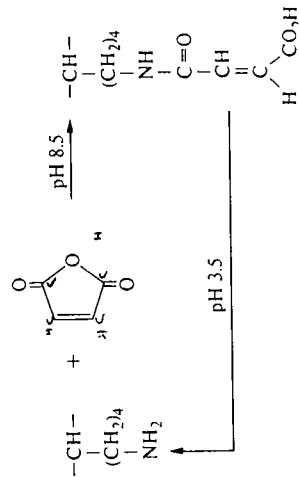


FIG. 3.9. Use of maleic anhydride to block lysine side chains reversibly. Citraconic anhydride possesses an additional methyl group attached to one of the carbon atoms in the C=C double bond.

Another possible variation involves modification of cysteine side-chains by reaction with 2-bromoethylamine to produce a side-chain analogous to that of lysine and which is in fact 'recognized' by trypsin⁷⁸ (Fig. 3.10). This modification reaction can generate additional cleavage sites and has sometimes proved useful in generating fragments from large polypeptides where successive lysine or arginine groups are well separated in the chain.⁷⁸

Polypeptide chains can also be cleaved specifically by the use of certain chemical reagents. By far the most commonly used reagent is CNBr, cyanogen bromide, which under acidic conditions will cleave a polypeptide chain on the C-terminal side of methionine side chains⁷⁹ (Fig. 3.11).

This method of cleavage is important since methionine is a relatively uncommon amino acid in proteins; hence CNBr will generally cleave a protein into a smaller number of fragments than will be produced with attack by trypsin. For instance, in the case of adenylate kinase from pig muscle, cleavage with CNBr produces six fragments, whereas attack by trypsin would yield 32

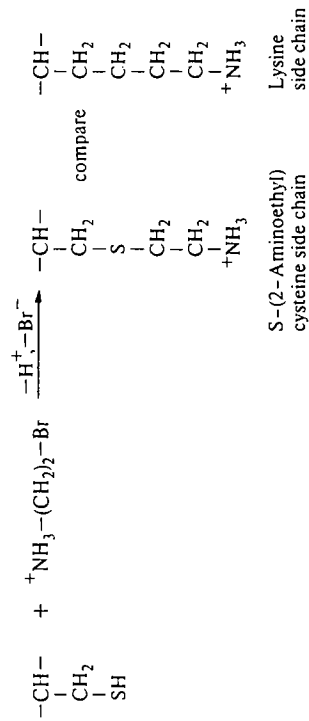


FIG. 3.10. Reaction of a cysteine side-chain with 2-bromoethylamine.

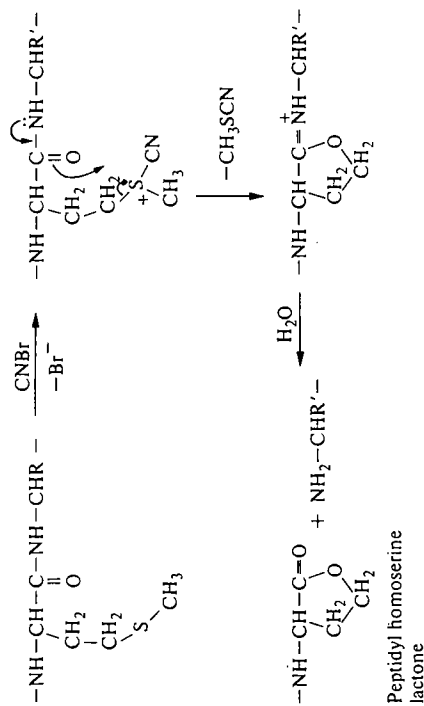


FIG. 3.11. Cleavage of a polypeptide chain on the C-terminal side of methionine by reaction with cyanogen bromide.

fragments.⁸⁰ It is obviously easier to separate and purify the components of a mixture the smaller the number of such components.

Chemical methods for bringing about selective cleavage at other types of amino-acid side-chains including cysteine (e.g. by incubation at pH 9 after cyanylation⁸¹) and tryptophan (e.g. by incubation with *o*-iodosobenzoic acid under acid conditions,⁸² see Section 3.4.8) have been described. For reviews of work in this area, the articles by Spande⁸³ and Allen⁶⁶ should be consulted.

3.4.3 Separation of the fragments (step 2 in Fig. 3.7)

The fragments produced by the various types of cleavage reactions described in Section 3.4.2 would be expected to differ in terms of their size, charge properties, chemical characteristics, etc. We could therefore utilize some of the methods described in Chapter 2 (Section 2.6) to separate and purify the various fragments present in the mixture. Methods that have been widely used include the following.



1. Separation on the basis of size by gel filtration. Sephadex G-50 would be most appropriate since it would separate species in the approximate M_r range 500 to 10 000. Peptide fragments that are eluted from a gel filtration column can be detected by measurements of absorption at 280 nm (which arises from the aromatic side chains of Tyr, Trp, and Phe) or at 215 nm (which arises from the amide bonds) or by measurements of other properties such as conductivity.⁸⁰
2. Separation on the basis of charge, which reflects the numbers and types of ionizing side chains in the various fragments. Typically the mixture of fragments is subjected to paper electrophoresis in two dimensions at different pH values, e.g. at pH 6.5 followed by pH 3.5,⁸⁴ or to ion-exchange chromatography⁸⁶ (see Chapter 2, Section 2.6.2.1).
3. Separation by a combination of paper electrophoresis, say at pH 6.5, in one dimension, followed by paper chromatography in a second dimension. A suitable solvent system for the chromatography consists of a mixture of 1-butanol, acetic acid, water, and pyridine.⁸⁵ In this procedure the fragments are separated on the basis of charge and of chemical characteristics.

In methods (2) and (3) the peptide fragments can be located by reaction with ninhydrin.⁸⁶ These methods are suitable for small-scale work with a few milligrams of the mixture of fragments and are often used after a partial separation has been achieved on a larger scale by method (1). The separation pattern of peptide fragments obtained by methods (2) or (3) is referred to as the 'fingerprint' or 'peptide map' of the protein or enzyme and is widely used to look for changes in amino-acid sequences between related proteins.

Enzymes with a polypeptide chain even of modest size can give rise to a large number of fragments each of which must be obtained in a pure state to allow the sequence to be determined. In the early 1980s the separation of the peptide fragments represented the most time-consuming part of the entire sequencing operation. However, since that time, the separation problem has been greatly aided by the application of h.p.l.c. techniques. Reverse-phase h.p.l.c., in which the stationary phase consists of silica microspheres coated with long-chain (C_8 or C_{18}) alkyl groups and the mobile phase is a gradually increasing proportion of acetonitrile or propan-2-ol in aqueous solution, is capable of rapid, high resolution of peptide mixtures. Although the equipment involved is relatively expensive, h.p.l.c. is now the method of choice for peptide separation.⁸⁷ An example of the use of h.p.l.c. is given in Section 3.4.8.

3.4.4 Sequencing of the purified fragments (step 3 in Fig. 3.7)

Determination of the amino-acid sequence of the individual fragments will be discussed under a number of headings (Sections 3.4.4.1 to 3.4.4.4).

3.4.4.1 Determination of the N-terminal amino acid

The most widely used method involves reaction of the peptide with dansyl chloride (dimethylaminonaphthylsulphonyl chloride abbreviated DNS-Cl)

under alkaline conditions, followed by acid hydrolysis to break the amide bonds (Fig. 3.12). The labelled N-terminal amino acid can be readily identified by electrophoresis or chromatography.⁸⁸ Dansyl chloride has now almost totally replaced the earlier N-terminal labelling reagent 1-fluoro-2,4-dinitrobenzene (FDNB), because dansylated amino acids are highly fluorescent and can be detected at amounts (less than 1 nmol) much lower than are required for the FDNB procedure.

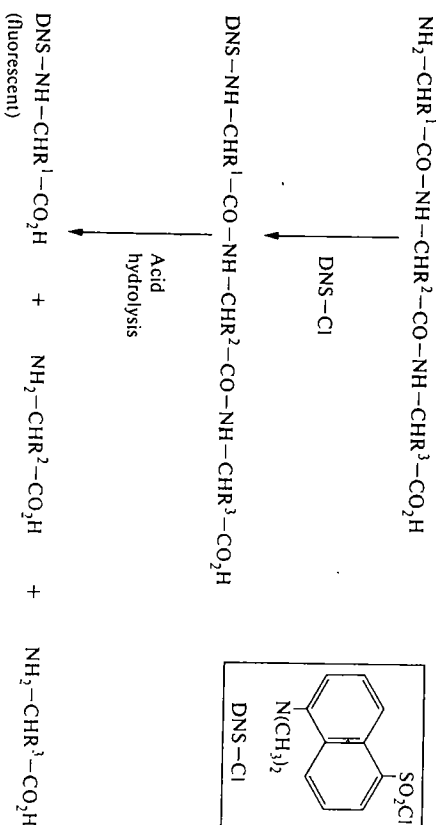


Fig. 3.12. The labelling of a tripeptide by reaction with dansyl chloride (DNS-Cl). The DNS-amino acid bond is stable to acid.

3.4.4.2 Determination of the C-terminal amino acid

Methods for determination of the C-terminal amino acid in peptides have not, in general, been as successful as those developed for the determination of the N-terminal amino acid. The most frequently used chemical method is that of hydrazinolysis, in which the peptide is reacted with hydrazine to convert all the amino acids except the C-terminal one to their hydrazides (Fig. 3.13). The unique C-terminal amino acid can be separated from the hydrazides by chromatographic procedures and identified using the amino-acid analyser (Section 3.3.4). Schroeder⁸⁹ has outlined a procedure for hydrazinolysis of peptides that has overcome some of the more notable problems previously encountered, such as low yields and difficulties in separating and identifying the amino acid liberated.

An alternative method for the determination of the C-terminal amino acid involves the use of carboxypeptidase. This enzyme removes amino acids one at a time from the C-terminal end of a polypeptide chain (see Chapter 5, Section 5.5.1.7). The amino acid released can be identified with the aid of an amino-acid analyser. The application of this method and the various difficulties associated with it, including the fact that the rate of release of the C-terminal amino acid

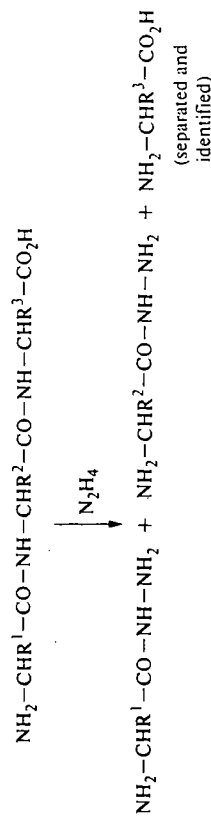


FIG. 3.13. Hydrazinolysis of a tripeptide. The C-terminal amino acid is separated and identified.

depends on the nature of its side-chain, are discussed in reviews by Ambler⁹⁰ and Ward.⁹¹

3.4.4.3 Sequential degradation of the peptide chain

The determination of the N- and C-terminal amino acids in a peptide chain provides some useful information, but in order to obtain sequence data we require a method to degrade the peptide chain in a sequential manner under carefully controlled conditions. Such a method was developed by Edman⁹² and is based on the reaction of the N-terminal amino acid with phenyl isothiocyanate (Fig. 3.14).*

The initial adduct, a phenylthiocarbamyl derivative, A, rearranges under anhydrous acidic conditions, cleaving the adjacent peptide bond and giving a heterocyclic anilinothiazolinone derivative, B. Derivative B is separated from the residual peptide by extraction with an organic solvent and converted to the more stable phenylthiohydantoin (PTH) derivative, C, before identification.⁹⁴ The Edman reaction permits the removal of amino acids one at a time from the N-terminus of a peptide chain. Although the conditions can be arranged so that the reactions involved take place with close to 100 per cent efficiency,⁹² there is a gradual accumulation of chains of differing length resulting from incomplete removal of the N-terminal amino acid at each cycle. In practice, the useful upper limit is around 30–40 cycles of degradation.

Identification of the amino acid released in each cycle of Edman degradation can be achieved directly by identification of the PTH derivative, (C in Fig. 3.14), using chromatographic procedures.⁹⁵ An alternative, indirect, method uses the Edman reaction merely to remove the N-terminal amino acid in a controlled fashion; the new exposed N-terminal amino acid is identified by reaction with dansyl chloride as described in Section 3.4.4.1. This 'dansyl-Edman' procedure⁹⁶ is illustrated in Fig. 3.15; determination of the sequence of a decapeptide, for example, can be achieved with about 20 nmoles of sample, rather less than is required for the direct method.

* Methods for sequential degradation of a polypeptide chain from the C-terminus have not generally been successful. Reaction with acetic anhydride and thiocyanate can be used for the sequential removal of up to five amino acids from the C-terminus.⁹³

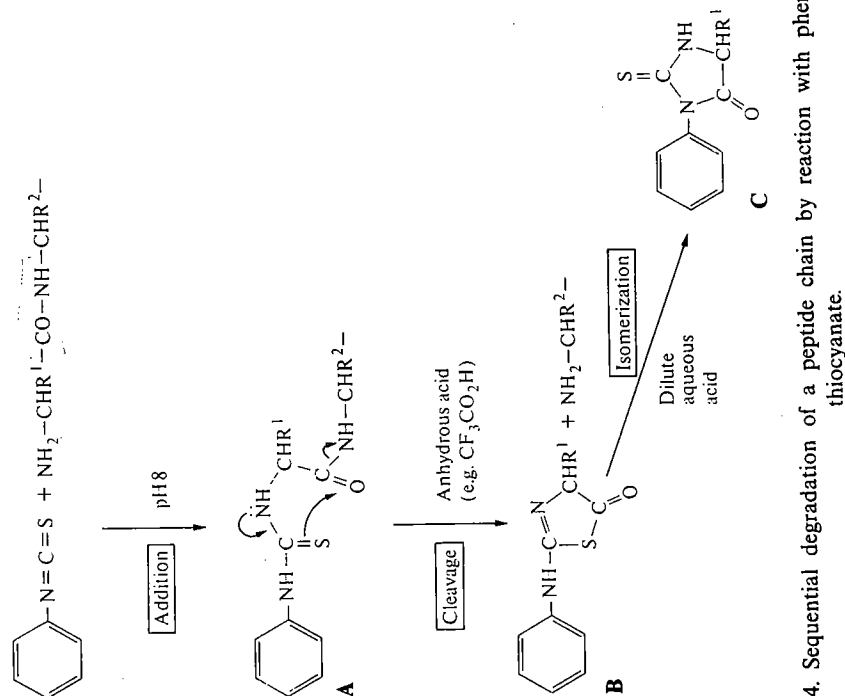


FIG. 3.14. Sequential degradation of a peptide chain by reaction with phenyl isothiocyanate.

3.4.4.4 Recent developments in methods of sequencing

Two developments have helped to speed up the determination of sequences of peptides. These are (i) automated sequencing using the Edman degradation, and (ii) mass spectrometry. We shall discuss these in turn.

(i) Automated sequencing

In 1967, Edman and Begg described the design and application of an instrument, known as a sequenator, for determining automatically the amino-acid sequences of peptides and proteins.⁹² The peptide is immobilized by being spread as a thin film in a spinning cylindrical glass cup and the necessary reagents and solvents are passed over the peptide in a programmed cycle. The PTH-amino acid formed after each cycle of the Edman degradation can be identified by thin-layer chromatography⁹² or h.p.l.c.⁹⁷ By careful attention to the conditions of the reaction, an average yield of slightly better than 98 per cent was obtained at each

The ratio of (a) to (b) is about 3 to 1. These structures were determined using only about 10 nmol (i.e. 5 µg) of sample.

3.4.5 Preparation and analysis of new fragments (step 4 in Fig. 3.7)

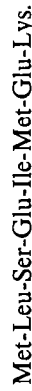
Once the peptide fragments have been purified and sequenced as described in Sections 3.4.2 to 3.4.4, the final task is to position them correctly so as to build up the overall sequence of the protein. This is usually done with the aid of a new set of fragments produced from the original protein by use of a different cleavage procedure (see Section 3.4.6). Thus, if trypsin were used to produce the first set of fragments, we might use chymotrypsin to produce the new set. These new fragments are purified and sequenced using the procedures already described (Sections 3.4.3 and 3.4.4).

3.4.6 Alignment of peptide sequences and determination of the overall sequence (step 5 in Fig. 3.7)

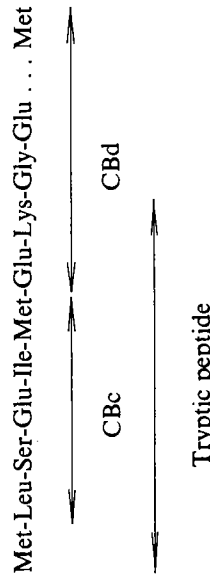
When the peptide sequences of the fragments produced by the various cleavage procedures are known, it is possible to position the peptides in the overall sequence by determining the regions of overlap of sequence. For instance, in the case of adenylate kinase from pig muscle,⁸⁰ two of the cyanogen bromide fragments CBc and CBd could be positioned with respect to each other by



the isolation of a peptide produced by the action of trypsin on the enzyme which had the sequence:



Thus the overall sequence of this portion of the polypeptide chain is



This conclusion was confirmed by the analysis of fragments produced by the action of thermolysin on the enzyme.

The following considerations may give the reader some feel for the number of overlapping amino acids that are required to position the peptides unambiguously. If the amino-acid sequence of a protein is purely random, then the

probability of occurrence of a given sequence of two amino acids, e.g. Asp-Gly, will be $1/400$ (i.e. $1/20 \times 1/20$): there are 20 different amino acids found in proteins). The probability of occurrence of a given sequence of three amino acids, e.g. Asp-Gly-Leu, will be $1/8000$. It is clear that a given sequence of three amino acids is unlikely to be repeated in a polypeptide chain of length 100 to 2500 amino acids and hence an overlap of this size should serve to position peptide fragments fairly unambiguously in the overall sequence.*

The positioning of fragments in the overall sequence of a protein is greatly helped if the sequence of a related protein is available. For instance, determination of the sequence of elastase was considerably speeded up because this enzyme shows a high degree of similarity to chymotrypsin, the sequence of which was already available.¹⁰⁶

Recently it has proved possible to position fragments in the overall sequence by making use of the electron-density maps of proteins available from X-ray crystallography (see Section 3.4.9.3). Examples of the interplay of crystallography and sequencing are to be found in the studies on phosphoglycerate mutase¹⁰⁷ and hexokinase¹⁰⁸ from yeast.

3.4.7 Some problems encountered in the determination of amino-acid sequences

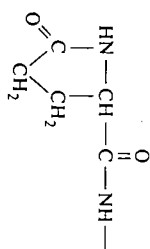
Using the procedures outlined in the previous sections (3.4.2 to 3.4.6) the sequence of a protein can usually be deduced, provided that sufficient care is taken in the purification and characterization of the various fragments. It is important to check that the composition of the enzyme deduced from the proposed sequence agrees with the composition determined by amino-acid analysis and that the sequence is in accordance with any DNA-derived sequence data available (Section 3.4.8). Any discrepancy between the various sets of data could indicate that a fragment has not been accounted for during the separation procedures. Some special problems arise in determinations of sequence; three of the most common problems are mentioned below.

3.4.7.1 Blocked N-terminal amino acid

Some proteins do not appear to possess an N-terminal amino acid when subjected to the dansyl chloride reaction (Section 3.4.4.1). This can be because the amino group is acetylated, formylated, or present as a pyroglutamyl moiety.¹⁰⁹ Examples of an acetylated N-terminal amino acid and an N-terminal pyroglutamyl group are provided by carbonate dehydratase from human erythrocytes¹⁰⁹ and peroxidase from horseradish¹¹⁰ respectively. Myristic acid

* The numerical aspects of this argument must not be taken too seriously. The 20 amino acids do not occur with equal frequency and the sequence is almost certainly not random. Thus, in adenylate kinase, which has 194 amino acids, the sequence Leu-Lys occurs three times and the sequence Met-Glu occurs twice, i.e. much above the 'random' frequency.

(C₃H₇CO₂H) is the N-terminal blocking group in the catalytic subunit of cAMP-dependent protein kinase¹¹¹ (see Chapter 6, Section 6.4.2.1).



The pyrrolutamyl group in which an N-terminal glutamic acid has formed an internal amide bond.

Allen⁶⁶ has given a general discussion of the problems of structure determination in those cases where post-translational modification has occurred.

3.4.7.2 Amide assignments

In Section 3.3.5.3 it was mentioned that the 'amide' amino acids asparagine and glutamine are converted to aspartic acid and glutamic acid, respectively, during acid hydrolysis of a protein or peptide fragment. This conversion can cause difficulties in deciding whether a particular amino acid in the sequence is asparagine or aspartic acid (or glutamine or glutamic acid). A correct assignment can usually be made on the basis of the electrophoretic mobility at pH 6.5 of a fragment containing the amino acid in question¹¹² since the side-chain of aspartic acid (glutamic acid) will carry a negative charge at this pH, whereas the side-chain of asparagine (glutamine) will carry no charge. In order to prepare the fragment of the enzyme it is necessary to perform the hydrolysis under conditions in which the amide bond in the side-chain of asparagine or glutamine will not be hydrolysed, e.g. by the use of various proteases (see Table 3.2). Using this approach it was possible to show that amino acid 102 in chymotrypsin was aspartic acid rather than asparagine.¹¹³ As we shall describe in Chapter 5 (Section 5.5.1.2), this aspartic acid side-chain plays a crucial role in the catalytic mechanism of the enzyme. Correct assignments can usually be made on the basis of Edman degradation, since the phenylthiohydantoin derivatives (Section 3.4.4.3) can readily be distinguished. Reference to the nucleotide sequence (if available) would confirm these assignments.

3.4.7.3 Location of disulphide bonds

If a protein is found to contain disulphide bonds, formed between pairs of cysteine side-chains, it is generally necessary to break these bonds either by reduction (Section 3.3.2) followed by reaction with iodoacetate or by oxidation with performic acid (Section 3.3.5.4) in order to form derivatives that are stable during the procedures involved in sequence determination. However, when these disulphide bonds are broken, the information concerning which pairs of cysteine side-chains are linked is lost, and in order to determine the location of these linkages it is necessary to isolate fragments with the disulphide bonds intact. One very useful way of detecting fragments that have these bonds intact, i.e. that contain cystine, is the 'diagonal' electrophoresis technique developed by Brown and Hartley in their studies on the structure of chymotrypsinogen.¹¹⁴ This

technique is based on the fact that oxidation of cystine (R-S-S-R) by performic acid gives rise to two molecules of cysteic acid (R-SO₃H) which, being a strong acid, will carry a negative charge at pH 6.5. Thus a peptide fragment that contains cystine will after oxidation give rise to two fragments of greater mobility towards the anode (except in the event of the two fragments having identical amino-acid composition), and will move off the 'diagonal' on electrophoresis in two dimensions at pH 6.5 (Fig. 3.17). These fragments can then readily be isolated and sequenced and the position of the disulphide bond in the overall sequence can be determined. Recently it has been shown that analysis by fast-atom bombardment mass spectrometry of proteolytic fragments prior to and after reduction of the disulphide bonds can also be used to assign the positions of the disulphide bonds.¹¹⁵

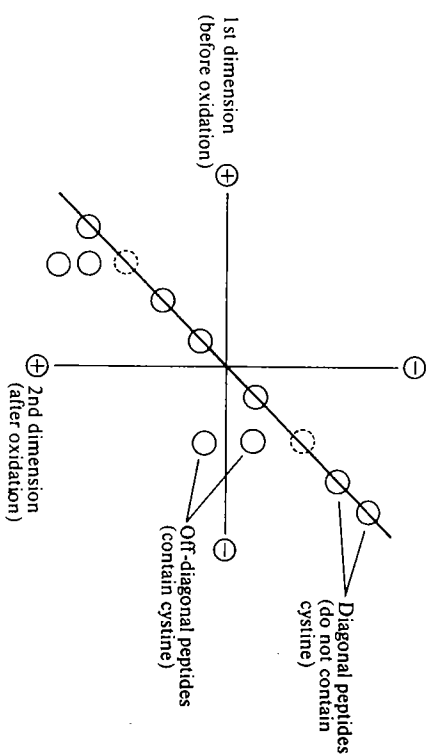


FIG. 3.17. The 'diagonal' electrophoresis method to locate cystine-containing peptides. The off-diagonal peptides would have moved to ○ if no oxidation had been performed.

3.4.8 An example of sequence determination

The various methods described in Sections 3.4.2 to 3.4.7 have been used to determine the sequences of a large number of proteins. The example we have chosen to discuss is that of phosphoglycerate kinase from baker's yeast, since this work nicely illustrates the way that 'direct' (i.e. protein) and 'indirect' (via nucleic acid) sequencing techniques can complement each other.¹¹⁶

The sequence of yeast phosphoglycerate kinase (a monomer of *M*_r 44 500) was required to permit detailed interpretation of the electron-density map determined previously by X-ray crystallography.¹¹⁷ Phosphoglycerate kinase was isolated by a procedure involving cell breakage using ammonia solution, ammonium sulphate fractionation, and affinity elution from CM-cellulose with 3-phosphoglycerate.^{118, 119} Traces of a contaminating protein were removed by gel filtration on Sephadex G-75.

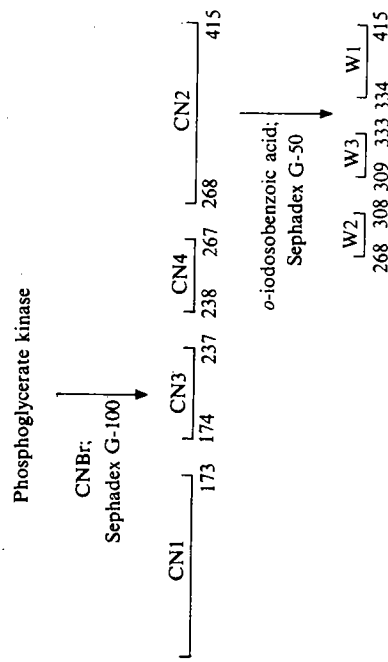


FIG. 3.18. The digestion of phosphoglycerate kinase to prepare fragments for sequencing. The numbering of fragments is described in the text.

The fragmentation steps involved in the 'direct' sequencing are illustrated in Fig. 3.18.

Amino-acid analysis revealed that the molecule possesses three methionine side-chains. Cleavage with CNBr (Section 3.4.2) yielded four fragments that could readily be separated by gel filtration on Sephadex G-100. The numbering of the fragments (CN1-4) is on the basis of their order of elution on gel filtration, CN1 being eluted first. The fractions pooled as 'CN1' were found to be heterogeneous on polyacrylamide gel electrophoresis in the presence of sodium dodecylsulphate; further purification was achieved by covalent chromatography on activated thiol-sepharose (see Chapter 2, Section 2.6.4.5), taking advantage of the fact that CN1 contained the single Cys in the molecule.

Fragments CN1, CN3, and CN4 were found by amino-acid analysis to contain homoserine, whereas CN2 did not. This placed CN2 as the C-terminal fragment. Fragments CN2, CN3, and CN4 showed free N-terminal amino groups by reaction with dansyl chloride (Section 3.4.4.1). CN1 did not show any free amino group and was therefore positioned at the N-terminal end of the molecule (see Fig. 3.18). (It had been previously established¹²⁰ by analysis of small peptides produced by proteolytic digestion that the N-terminus of yeast phosphoglycerate kinase was *N*-acetylserine.) The order of CN3 and CN4 was determined by nucleotide sequencing as described below.

The large CN2 fragment was further digested by treatment with *o*-iodobenzoic acid; under the appropriate conditions, cleavage occurs specifically at the two Trp side-chains.⁸² (Since Trp is abbreviated W in the one letter code (Table 3.1) the fragments are denoted W1-3). W1 was eluted first on gel filtration on Sephadex G-50.

The sequences of the various fragments in Fig. 3.18, i.e. CN2, CN3, CN4, W1, W2, and W3 were established by automated Edman degradation (Section 3.4.4.4) and by manual Edman sequencing of peptides derived by digestion of these

fragments with various proteolytic enzymes including trypsin, chymotrypsin, thermolysin, pepsin, and V8 protease (endoprotease Glu-C) from *Staphylococcus aureus*. The resulting mixtures of peptides were separated by various techniques, such as high-voltage electrophoresis, ion-exchange chromatography and high-performance liquid chromatography (h.p.l.c.) (Section 3.4.3). A typical h.p.l.c. profile showing the separation of tryptic peptides from fragment CN3 is shown in Fig. 3.19.

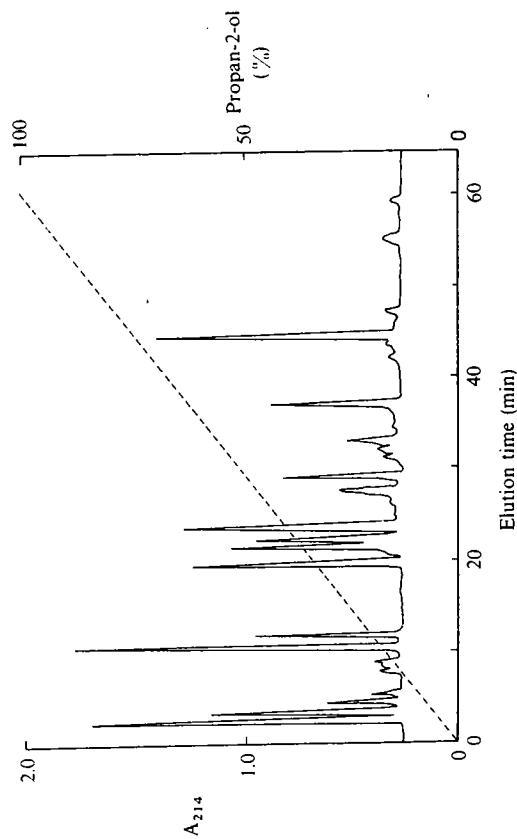


FIG. 3.19. H.p.l.c. separation of tryptic peptides from fragment CN3. Approximately 0.2 mg of the peptide mixture in 0.1 per cent trifluoroacetic acid was applied to a C₁₈ reverse-phase column and eluted with a gradient of propan-2-ol as indicated. The elution of peptides was monitored at 214 nm at which the amide bond absorbs strongly.

The order of the fragments W1, W2, and W3 in CN2 was determined by automated Edman sequencing from the N-terminus of CN2 and by the isolation of overlapping peptides, e.g. peptide S29 from the V8 protease digest of CN2 (Fig. 3.20).

Amide assignments (Gln vs. Glu; Asn vs. Asp) were made from direct identification of the phenylthiohydantoin derivatives during Edman degradation (Section 3.4.4.3).

At this point, the sequences of the fragments CN3, CN2, and CN4 and the fact that CN4 was C-terminal had been established. The remaining parts of the protein sequence were established by complementary work being undertaken on the structure of the gene coding for phosphoglycerate kinase in yeast. It had been previously established¹²¹ that this gene (the PGK gene) was located on a 2900-base (2.9-kb) fragment resulting from digestion of the vector pMA3 with the

305
310
315
-Lys-Glu-Gly-Ile-Pro-Ala-Gly-Trp-Gln-Gly-Leu-Asp-Asn-Gly-Pro-Glu-Ser-

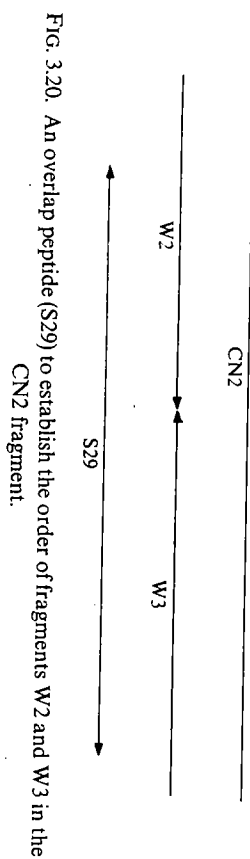


FIG. 3.20. An overlap peptide (S29) to establish the order of fragments W2 and W3 in the CN2 fragment.

restriction endonuclease Hind III*. By treatment of the pMA3 vector with the restriction endonuclease Sal I followed by an exonuclease (BAL 31) a series of overlapping gene fragments were prepared. After appropriate selection, these could be sequenced (see Fig. 3.21), to provide a complete sequence of the first 624 bases of the PGK gene (starting from ATG as the initiation codon).

By use of the genetic code, the DNA sequence could then be expressed as a protein sequence (amino acids 1 to 207).

The part of the protein sequence deduced by the DNA sequence covers the entire CN1 fragment as well as a portion of the CN3 fragment (see Fig. 3.18), thus showing the relative arrangement of these two fragments in the overall sequence.

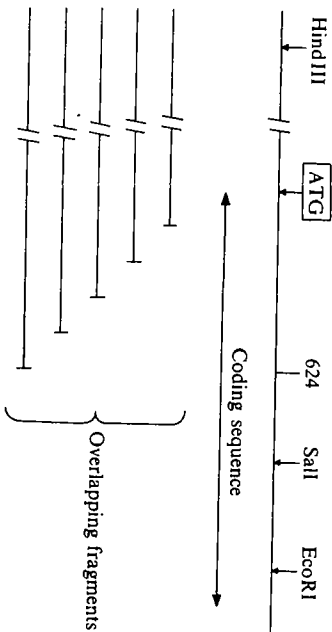


FIG. 3.21. General structure (not to scale) of the PGK gene in the pMA3 vector. The initiation codon (ATG) is shown together with the sites of cleavage by various restriction endonucleases (see reference 58 for details of these). The overlapping fragments were sequenced in the left → right direction, so that an unambiguous DNA sequence of the first 624 bases could be established.

* A complete understanding of the concepts involved in the DNA cloning and sequencing methods are not important for the purposes of this discussion. An excellent account of this area is given in reference 58.

A vector is a piece of DNA that carries genetic information and can replicate autonomously in an appropriate host. pMA3 is an artificially constructed 'shuttle vector' that can carry genetic information between different organisms, in this case *E. coli* and yeast.

In addition, nucleotide sequence data were obtained for a part of the PGK gene around the EcoRI restriction endonuclease site (Fig. 3.21); this corresponds to amino acids 281 to 415 (C-terminus). The nucleotide sequence data not only confirmed the amino-acid sequence determined directly for fragment CN2 (Fig. 3.18), but also helped to resolve some small ambiguities in the latter sequence where good overlap peptides could not readily be obtained.

The final sequence of 415 amino acids¹¹⁶ was thus obtained by the complementary approaches of protein and DNA sequencing and was consistent with all the available data.

3.4.9 Uses of sequence information

The information contained in the primary structure of an enzyme can be used in a number of ways. As we have already noted (Section 3.4.1), even a small amount of sequence information can be useful for designing an oligonucleotide probe for the gene coding for the enzyme. However, the complete sequence can give much more information; five of the more important applications are discussed in Sections 3.4.9.1 to 3.4.9.5.

3.4.9.1 To calculate the M_r of an enzyme

There are a number of experimental methods (Section 3.2) for determining the M_r of an enzyme. However, the definitive M_r is only obtained when the covalent structure, i.e. sequence of the enzyme, is established, and this calculated M_r can be used as a check on the validity of the experimental methods. For instance in the case of β -D-galactosidase from *E. coli* the M_r of the subunit had been measured as 135 000, but when the sequence became available it was clear that this value was some 15 per cent too high and that the correct value was 116 300.⁵³ The discrepancy may have arisen from the use of an incorrect value of the partial specific volume of the enzyme (Section 3.2.1.1) and from errors in the interpretation of ultracentrifuge data. A second example is afforded by the work on penicillinase from *E. coli* in which the calculated M_r was some 20 per cent higher than that measured by physical chemical methods.⁶⁴ The importance of an accurate knowledge of the M_r of an enzyme has been mentioned in Section 3.2.5.

3.4.9.2 To locate a particular amino acid in an enzyme

The primary structure of an enzyme provides a framework within which the importance of particular amino-acid side-chains can be discussed. In Chapter 5 (Section 5.4.4) we shall describe how chemical modification of an amino-acid side-chain can be used to indicate the involvement of that side-chain in the catalytic mechanism of an enzyme. For instance, the reagent diisopropyl fluorophosphate modifies one serine side-chain in chymotrypsin leading to inactivation; analysis of the peptide fragment containing this modified amino acid shows that this serine is number 195 in the sequence (see Chapter 5, Section

5.5.1.2). It is essential to have this type of information in conjunction with three-dimensional structural data (Section 3.5) to formulate detailed proposals concerning the mechanism of action of an enzyme.

3.4.9.3 To interpret data from X-ray crystallography

X-ray crystallographic studies of an enzyme (Section 3.5) yield a three-dimensional map of the electron density in the molecule. In order to interpret this pattern of electron density in terms of the arrangement of atoms it is normally necessary to know the sequence of the enzyme so that the path of the chain of α -carbon atoms and amide bonds can be traced and the positions of the side-chains located.

The X-ray structure of phosphoglycerate mutase from yeast was determined in 1974¹²² to a resolution of 0.35 nm. At this level of resolution some of the main structural features such as helices and sheets (Section 3.5.4) were clearly visible. However, it was not until sequence information on the enzyme became available that the higher-resolution electron-density maps could be interpreted in sufficient detail to allow a plausible mechanism of action to be proposed.¹⁰⁷ In this case the X-ray and sequence studies complemented each other nicely, since the electron density map helped to order the various peptide fragments in the overall sequence (Section 3.4.6).

3.4.9.4 To predict the three-dimensional structure of an enzyme

As described in Section 3.7.2 a number of experiments have shown that the primary structure contains the information necessary to specify the three-dimensional structure of an enzyme. We might therefore ask whether it is possible to use sequence data to predict the three-dimensional structure. The answer to this question is a moderately encouraging 'yes' provided that we set ourselves the more limited task of predicting which parts of the amino acid sequence might occur in regions of distinct secondary structure (helices, sheets, etc.). This type of approach is described in Section 3.7.3.

Sometimes it is possible from the sequence to guess at parts of an enzyme that may have a distinct functional role. In the case of phospholipase A_2 , it is likely that the non-polar N-terminal portion of the polypeptide chain

Ala-Leu-Trp-Gln-Phe-

is involved in the interaction of the enzyme with the non-polar portions of phospholipid substrates. This conclusion is supported by the results of chemical modification of amino-acid side-chains.¹²³ It is well recognized that certain types of N-terminal sequences are involved in directing newly synthesized proteins to particular destinations, e.g. into the lumen of the endoplasmic reticulum prior to export¹²⁴ or to the various compartments of the mitochondrion.¹²⁵ Stretches of non-polar amino acids of membrane-associated proteins have usually been assigned to membrane-spanning segments of the polypeptide chain¹²⁶ (see Chapter 8, Section 8.4). Finally, certain types of amino-acid sequence are

associated with high degrees of flexibility, e.g. the Ala/Pro-rich sequences in the lipoyl domains of the E2 component of the pyruvate dehydrogenase complex (see Chapter 7, Section 7.7.2).

3.4.9.5 To explore evolutionary relationships between enzymes

Amino-acid sequences provide a wealth of data with which to make comparisons between enzymes.¹²⁷ One of the goals of such comparisons is to learn something about the processes involved in enzyme evolution.* A change in the primary structure of an enzyme usually arises from a change in the base sequence of the DNA coding for that enzyme. We shall mention only two of the various types of genetic modification that can occur: point mutations and gene duplication. A comprehensive review of this subject has been given by Smith.¹²⁹

(i) A point mutation refers to the replacement of a single nucleotide in a triplet codon (a sequence of three bases in DNA required to specify which amino acid is incorporated into the sequence of a protein). Thus the codon AAG specifies lysine; replacement of the first base by G would give GAG which specifies glutamic acid. Comparison of the sequences of homologous enzymes, i.e. those that catalyse the same reaction, in different organisms shows that most, if not all, of the changes in sequence can be explained by point mutations.¹²⁹ With the recent developments in DNA sequencing (see Section 3.4.1), it is now also possible to examine DNA sequences in order to trace evolutionary events more directly.¹³⁰ The results confirm that the steady evolution of proteins results from the steady evolution of DNA. However, there are some additional points to be borne in mind when DNA sequences are compared that are not apparent from comparisons of protein sequences. (a) The degenerate nature of the genetic code means that changes in the third position of codons can occur at a greater rate than either the first or second positions without leading to a change in the amino acid incorporated. (b) The frequency of usage of synonymous codons (codons which specify the same amino acid) may be very different in different organisms.¹³¹ (c) The final protein sequence is specified after the intervening sequences ('introns') have been removed during formation of mature mRNA; thus a range of proteins could be produced by appropriate combinations of 'exons' (expressed sequences).¹³² We shall see in Chapter 5 (Section 5.4.5) that the technique of site-directed mutagenesis can be employed to introduce defined mutations at known sites in a protein.

Amino-acid substitutions do not appear to occur randomly along the sequence, since there is a high degree of identity around those amino acids that are known to be functionally important, e.g. those involved in the catalytic mechanism. In Chapter 5, Section 5.5.3.4 we shall see how comparisons of the sequences of fructose-bisphosphate aldolases have indicated that a histidine near

* Comparisons of three-dimensional structures of enzymes can also be used to examine these evolutionary processes (Section 3.5.3.3). It should be noted that a number of different amino-acid sequences can give rise to very similar three-dimensional arrangements, so in functional terms there may be much less restraint on amino-acid sequences.¹²⁸

the C-terminus is probably not involved in the catalytic mechanism, contrary to an earlier proposal. Presumably the pattern of amino-acid substitutions reflects a combination of two processes: *mutation* which can occur randomly along the sequence and *survival* of the mutants which is not a random process. Replacement of one amino acid by another with different characteristics (e.g. the basic side-chain of lysine replaced by the acidic side-chain of glutamic acid) would produce a non-functional enzyme if the lysine side-chain were involved in the catalytic mechanism or in the maintenance of the structure of the enzyme. The operation of natural selection would serve to ensure that organisms with such non-functional enzymes did not survive.

(ii) When enzymes catalysing similar reactions within an organism are compared it is often possible to find evidence for the occurrence of gene duplication, in which two copies of a particular gene have been produced, followed by independent evolution of these two copies by processes such as point mutation. This type of duplication process would mean that an organism would be capable of producing enzymes of differing specificity allowing a greater variety of foodstuffs to be utilized and thereby enhancing the survival prospects of the organism. The high degree of sequence homology of various proteases of differing specificity, e.g. trypsin, chymotrypsin, elastase, and thrombin, especially around those amino acids known to be involved in the catalytic mechanism (Table 3.3) suggests very strongly that these enzymes evolved from some common precursor protease by a process of gene duplication, followed by subsequent independent evolution. Other examples of such 'divergent' evolution (i.e. from a presumed common precursor) are provided by the group of thiol proteases papain, ficin and bromelain, and by the two types of alcohol dehydrogenase in mammalian liver.¹²⁹

On the other hand, comparisons between certain other enzymes suggests that a process of 'convergent' evolution may have occurred in which a similar type of

TABLE 3.3
*Amino-acid sequences around the essential serine, histidine, and aspartic acid groups in various proteases*¹³³

Enzyme	Amino-acid sequences		
	Essential serine	Essential histidine	Essential aspartic acid
Trypsin (bovine)	Gly-Asp-SER-Gly-Gly	Ala-Ala-HIS-Cys-Tyr	Asn-Asn-ASP-Ile-Met
Chymotrypsin (bovine)	Gly-Asp-SER-Gly-Gly	Ala-Ala-HIS-Cys-Gly	Asp-Asn-ASP-Ile-Thr
Elastase (porcine)	Gly-Asp-SER-Gly-Gly	Ala-Ala-HIS-Cys-Val	Gly-Tyr-ASP-Ile-Ala
Thrombin (bovine)	Gly-Asp-SER-Gly-Gly	Ala-Ala-HIS-Cys-Leu	Asp-Arg-ASP-Ile-Ala
Subtilisin BPN	Gly-Thr-SER-Met-Ala	Asn-Ser-HIS-Gly-Thr	Val-Ile-ASP-Ser-Gly

functional unit has been evolved starting from quite different precursor polypeptide chains. For example there is very little similarity in primary structure (Table 3.3) or in three-dimensional structure between the enzymes subtilisin, a protease isolated from *Bacillus subtilis*, and bovine chymotrypsin, suggesting that these two enzymes did not arise by a process of divergent evolution from a common precursor. However, both enzymes possess the 'charge relay system' of three precisely positioned side chains, Ser...His...Asp (see Chapter 5, Section 5.5.1.2) of amino acids well separated from each other in the sequence. This system is essential for the catalytic mechanism and its occurrence in the two quite different enzymes suggests that a process of 'convergent' evolution has occurred.¹²⁹ ('Convergent' evolution is sometimes referred to as 'parallel' evolution, since it is not clear that the enzymes are 'converging' towards one another with the passage of time!)

3.5 The determination of secondary and tertiary structure

Knowledge of the primary structure of an enzyme does not allow us to explain properties such as catalytic power and specificity. We must also consider how the polypeptide chain is folded up so that different parts of the chain are brought into close proximity with each other to create binding sites for substrates and unusual environments which facilitate catalysis (see Chapter 5, Section 5.3). The determination of the three-dimensional structure of an enzyme relies very heavily on X-ray crystallography since this is the only technique that has up to now been shown to be capable of giving structural information at atomic detail. We shall describe the technique in outline before considering the relationship between the structure in the crystal and that in solution. There are a number of reviews of X-ray crystallography that can be consulted for further details.^{134, 135, 136, 137, 138}

3.5.1 X-ray crystallography

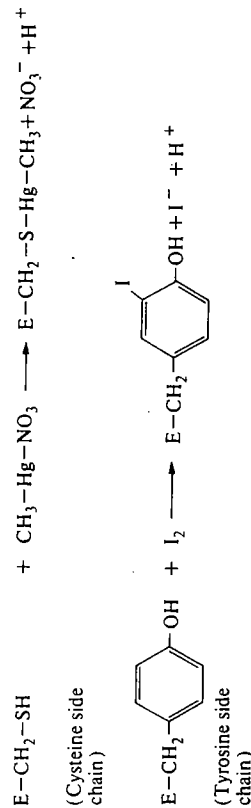
X-ray crystallography relies on the scattering of electromagnetic radiation of suitable wavelength by electrons belonging to the atoms in a molecule. In the case of a regularly arranged array of atoms, such as is present in a crystal lattice, we can have constructive or destructive interference between the scattered waves; only constructive interference will give rise to a detectable signal. X-rays (which are emitted when an electron falls to a lower orbital of an atom from a higher occupied orbital) provide suitable radiation to bring about these diffraction effects since their wavelength is comparable with the interatomic distances in a molecule.* Virtually all structures of proteins have been determined using the K_{α} radiation emitted by Cu; this is of wavelength 0.154 nm. Recently, increasing attention has been paid to the possibilities of using synchrotron radiation sources (see Chapter 5, Section 5.4.3). The intensities of these (highly specialized)

* This phenomenon is analogous to the diffraction of visible light by a diffraction grating. The spacing between the lines on the grating is of the same order as the wavelength of the incident light.

sources are much higher than those of conventional X-ray sources, thereby shortening the time required for data collection.¹³⁹

The positions and intensities of diffracted rays (in which constructive interference has occurred) are measured using either photographic film, so that a pattern of spots is obtained, or a diffractometer that can measure intensities directly. From the pattern of spots on a photographic film it is possible to deduce certain features such as the overall symmetry of the crystal and the dimensions of the repeating unit, but additional information is required in order to determine the three-dimensional structure of the enzyme, as described below.

If we know the three-dimensional structure of a molecule, it is a relatively straightforward task to calculate the positions and intensities of the diffracted rays. Proceeding in the reverse direction from the diffraction pattern to the three-dimensional structure is difficult because we need to have information on the phases of the scattered X-rays, i.e. how the waves are arriving at the detector related in terms of numbers of wavelengths from a common origin. This information on phases is lost when only the positions and intensities of the diffracted rays are recorded. However, it is possible to solve the phase problem by the method of isomorphous replacement, in which a limited number of heavy atoms are introduced at selected sites in the molecule without distorting the crystal structure.¹⁴⁰ The electron-dense heavy atom scatters the X-rays more strongly than the lighter atoms (H, C, N, O, and S) of the enzyme and will add its scattering power to the diffracted rays. By measuring the changes in intensities of the diffracted rays it is possible to extract the required information on the phases; usually, at least two heavy-atom derivatives are required for this purpose. Heavy atoms can be introduced by chemical modification of amino-acid side-chains as in the following examples:



Once the phases of the diffracted rays are known it is possible, by a process known as Fourier summation, to calculate the three-dimensional map of the electron density within the molecule.¹⁴¹ By considering only a relatively limited number of spots in the diffraction pattern, i.e. those rays diffracted through the lowest angles, a map at low resolution (e.g. 0.6 nm) can be obtained. From this low-resolution map a good idea of the overall shape of the molecule can be gained. In order to obtain a map at the degree of resolution at which amino-acid side-chains can be resolved (0.3 nm or better) it is necessary to take into account the intensities of many more spots, perhaps several thousands, arising from rays

diffracted through larger angles. Even with the advent of modern computing techniques, the analysis of diffraction data and calculation of electron-density maps is still a time-consuming process.

In order to interpret the electron-density map of an enzyme in terms of its atomic structure, it is necessary to know the amino-acid sequence of the molecule. We can then trace the path of the chain of α -carbon atoms and amide bonds and locate the various side-chains in the electron-density map. As mentioned in Section 3.4.9.3, it is not always necessary to have the complete sequence to hand, because the electron-density map can be used to order the various peptide fragments in the overall sequence. When the polypeptide chain has been fitted to the electron-density map, we can reasonably claim to have determined the three-dimensional structure of the enzyme.

When the basic requirements for X-ray crystallography are considered (Table 3.4), it is not difficult to see why the determination of the three-dimensional structure of an enzyme represents a major undertaking in terms of time and resources. Since the first high-resolution structure of a protein, myoglobin, was published in 1960¹⁴² a number of technical and theoretical developments have aided the crystallographer and the structures of well over 200 proteins have now been determined to varying degrees of resolution. From this wealth of structural information a number of interesting conclusions have been drawn (see Section 3.5.3).

The structure of an enzyme deduced by X-ray crystallography is not entirely a static one, since atomic movements such as vibration and rotation about single bonds can still occur even though the packing of molecules in the crystal will

TABLE 3.4
The basic requirements for X-ray crystallography of enzymes

Requirement	Comments
Crystals of enzyme	Crystals must be of a suitable size, i.e. of dimensions approximately 0.5 mm or larger. They can be formed by gradual changes in the ionic strength or polarity of a solution of enzyme (see Chapter 2, Section 2.6.3). The crystals must also be reasonably stable in the X-ray beam during the course of the experiment
Isomorphous heavy-atom derivatives	The derivatives are necessary to provide information on phases. They can be prepared by chemical modification of amino-acid side-chains, e.g. with mercurials, or by soaking the crystals in a solution containing heavy-metal ions. Incorporation of the heavy atom must be shown not to distort the structure, i.e. to alter only the intensities not the positions of the spots in the diffraction pattern
Computing facilities	These facilities are required to calculate the electron-density map of the molecule from information on the position, intensities, and phases of the diffracted X-rays
Primary structure	The sequence of amino acids is necessary to interpret the electron-density map in terms of the arrangement of atoms, i.e. covalent structure, of the molecule

tend to dampen these motions.¹⁴³ A good example of atomic movements in a crystal is provided by trypsinogen, the inactive precursor of the protease trypsin (see Chapter 6, Section 6.2.1.1), of which the electron-density map has no density corresponding to four major segments of the chain: N-terminus to Gly 19, Gly 142-Pro 152, Gly 184-Gly 193, Gly 216-Asn 223.¹⁴⁴ It is concluded that these segments are wagging in a flexible manner or adopting a number of conformations in the crystal leading to a 'smearing out' of electron density. In the electron-density map of trypsin these segments are well defined, presumably because the motion is now restricted. Other examples of the loss of electron density are provided by the C-terminal regions of fructose-bisphosphate aldolase and tyrosyl-tRNA synthetase (see Chapter 5, Sections 5.5.3.1 and 5.5.5.4, respectively). As described in Section 3.5.2, the flexibility of an enzyme molecule is considerably greater in solution than in the crystal.

3.5.2 The structure of an enzyme in solution

Our knowledge of the structure of an enzyme in solution is much less exact than of its structure in the crystal. Several lines of evidence, such as measurements of fluorescence quenching, hydrogen-deuterium exchange, and nuclear magnetic resonance show that proteins in solution possess fairly flexible structures.^{145, 146} It is thus more appropriate to think of an enzyme in solution as existing in a number of conformational states (related to each other by rotation about single bonds) of roughly equivalent energy. The structure in solution will represent a time average of these various conformational states of the enzyme.

There is no technique that can give information on the structures of enzymes in solution in the atomic detail provided by X-ray crystallographic studies of the solid state. However, recent advances in the application of nuclear magnetic resonance to solutions of small enzymes such as lysozyme have yielded an impressive amount of structural information.^{147, 148} Optical rotatory dispersion and circular dichroism,¹⁴⁹ which rely on the interaction of polarized light with chiral (optically active) compounds can be used to measure the amount of regular secondary structure (Section 3.5.4) in an enzyme. Chemical modification of amino-acid side-chains (see Chapter 5, Section 5.4.4) can be used to divide side-chains of a particular type into those that are 'exposed', i.e. on the surface of the enzyme, and those which are 'buried', i.e. in the interior of the enzyme.¹⁵⁰ The overall conclusion from these and other studies is that the time average structure of an enzyme in solution is on the whole very similar to its structure in the solid state (crystal).^{145, 151} and this finding provides the justification for using the crystallographic data to discuss the properties of the enzyme in solution. One observation that shows that the structures in the crystal and in solution must be reasonably similar is that certain enzymes, e.g. pancreatic ribonuclease and triosephosphate isomerase can display catalytic activity in the crystalline state. In general the rates of processes in the crystalline state are rather slower than those in solution, reflecting the slower diffusion of molecules through a crystal,

whereas the values of equilibrium properties such as dissociation constants are similar in the two phases.¹⁵¹

3.5.3 The importance of knowing the three-dimensional structure of an enzyme

The advances in our detailed understanding of the properties of enzymes over the last twenty years or so can largely be ascribed to the increase in detailed structural information provided by X-ray crystallography. These advances in understanding are discussed in the next sections (3.5.3.1 to 3.5.3.3).

3.5.3.1 To test models of macromolecular structure

The three-dimensional structure provides the experimental data with which to test theoretical models of macromolecular structure. The relative contributions of the various types of forces involved in the maintenance of the overall structure (Section 3.5.5) can be evaluated by referring to the experimentally determined structure. We can also use the experimental structure to test the success of structure prediction methods (Section 3.7.3).

3.5.3.2 To propose a mechanism of catalysis

The structure of an enzyme provides the framework within which the catalytic power of that enzyme can be understood. It is possible (see Chapter 5) to propose chemically reasonable mechanisms for a number of enzymes by combining these structural data with results from other studies such as enzyme kinetics, which tell us about the dynamic aspects of catalysis.

3.5.3.3 To explore similarities between enzymes

In Section 3.4.9.5 we described how the comparisons of amino-acid sequences have helped to explore evolutionary relationships between enzymes. Useful comparisons have also been made between the three-dimensional structures of enzymes that catalyse related reactions. Thus, Rossmann and his coworkers have demonstrated that a common nucleotide-binding structure consisting of helix and sheet elements (Section 3.5.6.4) is present in NAD⁺-requiring dehydrogenases and in kinases that bind ADP (which forms part of the NAD⁺ molecule).^{152, 153} This regularity in the three-dimensional structures would not have been deduced from comparisons of the amino-acid sequences. A more general method for making structural comparisons between enzymes has been described.^{128, 154, 155}

3.5.4 Features of structures adopted by enzymes

In this section we shall describe the principal structural elements that have been found in proteins and then proceed to outline the principles that govern the three-dimensional structures adopted by enzymes. A later section (3.5.6) will look

at the detailed structures of three particular enzymes: α -chymotrypsin, adenylate kinase, and glyceraldehyde-3-phosphate dehydrogenase.

3.5.4.1 The amide bond

The amide bond which provides the means of linking amino acids in the polypeptide chain (Section 3.3.2) is not adequately represented by the formula

$\text{O}=\text{C}-\text{NH}-$, since this fails to show that there is in fact considerable resistance to rotation about the C—N bond. Estimates of the activation energy for this rotation are of the order of 80 kJ mol^{-1} .¹⁵⁶ It is therefore more appropriate to depict the amide bond as a planar unit with extensive delocalization of the lone pair of electrons on the nitrogen atom imparting a partial double-bond character to the C—N bond (Fig. 3.22).

The form of the amide bond unit shown in Fig. 3.22 in which the two α -carbon atoms are *trans* to one another, is generally more stable than the *cis* form by some 12 kJ mol^{-1} because of the steric crowding in the latter.¹⁵⁶ However, when the amide bond involves the amino acid proline, the difference in energy between *cis* and *trans* forms is much less pronounced and the *cis* form is found occasionally in proteins.¹⁵⁶ Rotation is allowed about the single bonds linking the α -carbon atoms to the carbonyl carbon atom and to the nitrogen atom of the amide bond. We can thus describe the structure of a dipeptide unit in terms of the dihedral angles (ϕ , ψ), i.e. the angles of rotation about these bonds (Fig. 3.23)

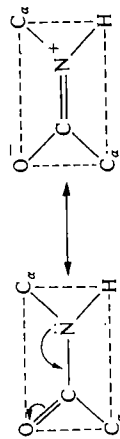


FIG. 3.22. The amide bond showing delocalization of electrons. The dotted lines represent the planar unit of two α -carbon atoms and the carbon, nitrogen, oxygen, and hydrogen atoms. The length of the amide C—N bond is 0.132 nm and the distance between the two α -carbon atoms in this *trans* configuration is approximately 0.38 nm .

Rotations about these bonds, i.e. changes in the angles ϕ and ψ , will change the distances between non-bonded atoms and thus the energy of the dipeptide unit will vary. (This is because when non-bonded atoms are brought into close proximity with each other, overlap of the electron clouds leads to repulsion and hence destabilization.) Calculations of the energy of the dipeptide unit as a function of the angles ϕ and ψ have been made,¹⁵⁷ and these show that there are certain well-defined conformations that are of lowest energy, i.e. of greatest stability. The most important of these regular secondary structures are the α -helix and β -sheet structures, which had already been proposed as the basic structural units of fibrous proteins;¹⁵⁸ thus, wool consists largely of α -helical structure and silk largely of β -sheet structure.

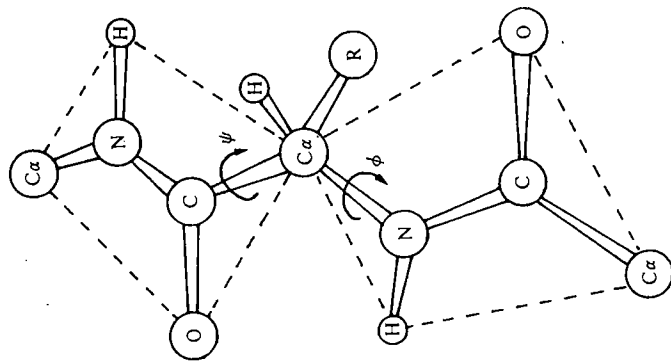


FIG. 3.23. A dipeptide unit of a polypeptide chain showing the planar amide units and the relevant angles of rotation about the bonds to the central α -carbon atom. $\phi = \psi = 180^\circ$ corresponds to fully extended polypeptide chain.

3.5.4.2 The α -helix

The α -helix is depicted in Fig. 3.24(a). It consists of a right-handed helix* (i.e. the direction of twist from the N-terminus to the C-terminus is that of a right-handed corkscrew) in which hydrogen bonds (Section 3.5.5.1) are formed between the carbonyl group of the amide bond between amino acids n and $n+1$ and the amino group of the amide bond between amino acids $n+3$ and $n+4$ (Figs. 3.24(a) and 3.24(b)). Because of the restrictions on rotation around the bonds to the α -carbon atom in proline, this amino acid cannot be incorporated into an α -helix without seriously distorting it. Proline therefore acts as a helix-breaker.

Since there are approximately 3.6 amino acids per turn of the α -helix and a total of 13 atoms in the ring closed by formation of the hydrogen bond, the α -helix is sometimes referred to as the 3.6_{13} helix. The pitch of the α -helix is 0.54 nm

* For L-amino-acids (Section 3.3.1) the right-handed helix is far more stable than the left-handed helix because of the crowding of the side-chains in the latter. Only the right-handed helix is found in proteins.

(Fig. 3.24(a)) which corresponds to a translation of 0.15 nm per amino acid. Other types of helical structure are possible, such as the 3_{10} helix, but these are less stable than the α -helix because the hydrogen bonds are somewhat distorted. The 3_{10} helix occurs only rarely in enzymes;¹⁵⁹ short lengths are found in lysozyme and in carbonate dehydratase. The ways in which α -helices can pack together in protein structures have been discussed by Richardson.¹⁶⁰

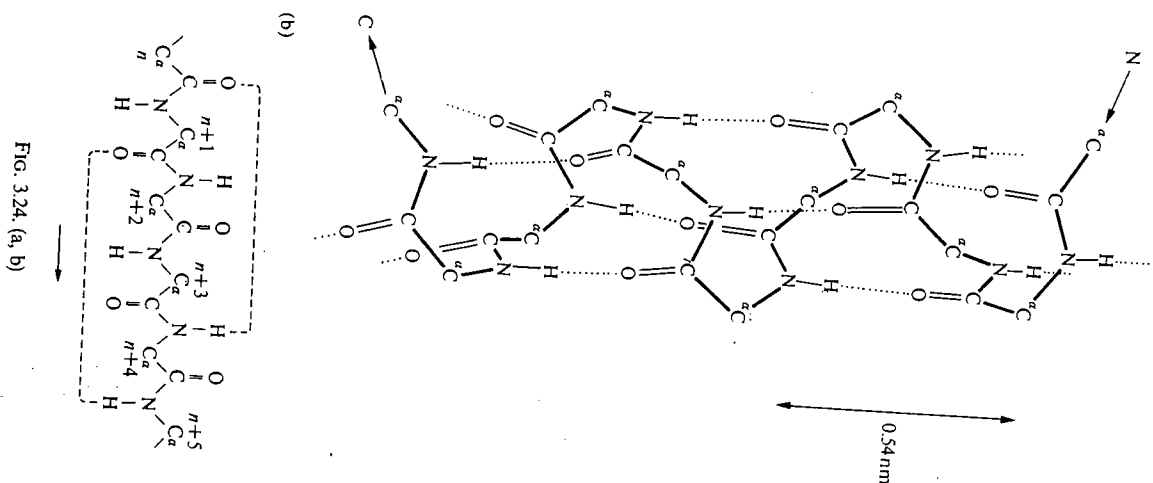


FIG. 3.24. (a, b)

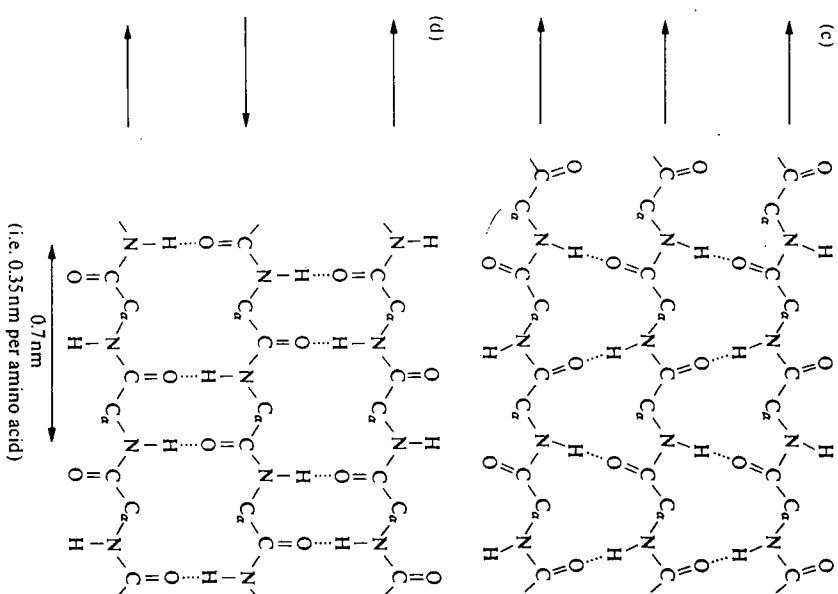


FIG. 3.24. Elements of regular structure found in proteins and enzymes, showing the arrangement of hydrogen bonds, which are depicted by dotted lines. The arrows show the direction of a polypeptide chain (N $\rightarrow$ C). (a) α -helix; the heavy lines trace the path of the polypeptide chain. (b) The hydrogen bonding arrangement in the α -helix more clearly. (c) and (d) parallel and antiparallel β -sheet structures, respectively. In three dimensions the sheets represented in (c) and (d) are in fact corrugated, with the side-chains of the amino acids projecting perpendicularly to the plane of the sheet. For the α -helix, parallel sheet and antiparallel sheet structures the values of (ϕ, ψ) (see Fig. 3.23) are approximately $(-57^\circ, -47^\circ)$, $(-119^\circ, +113^\circ)$ and $(-139^\circ, +135^\circ)$ respectively.

3.5.4.3 The β -sheet

There are two types of β -sheet structures according to whether the polypeptide chains between which hydrogen bonds are formed are running in parallel or antiparallel directions. From the patterns shown in Fig. 3.24(c) and 3.24(d) it can be seen that the hydrogen bonds in the parallel sheets are somewhat distorted; no such distortion occurs in the antiparallel sheet. Several enzymes contain

extensive networks of sheet structure: part of the arrangement found in carboxypeptidase A¹⁶¹ is illustrated in Fig. 3.25. As noted by Chothia,¹⁶² the β -sheets in globular proteins show a right-handed twist (as viewed along the polypeptide chain) rather than being flat as in silk. This twist is seen clearly in Figs. 3.31(a) and 3.32(b).

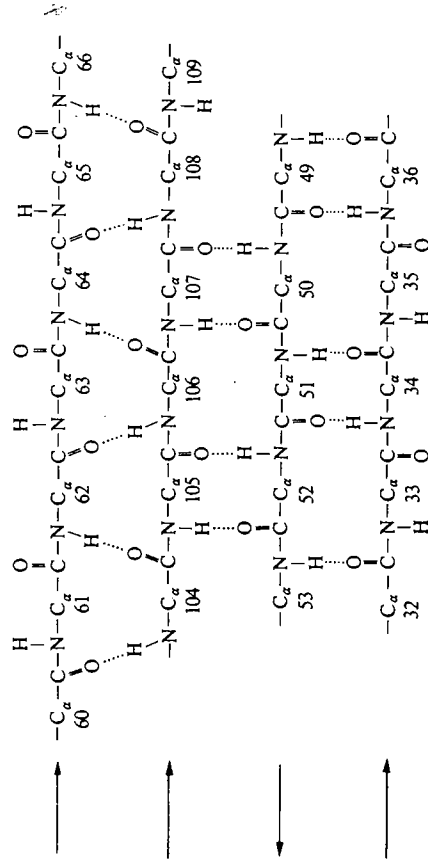


FIG. 3.25. Part of the β -sheet structure of carboxypeptidase A.¹⁶¹ The directions of the strands are shown by arrows. Hydrogen bonds are represented by dotted lines.

3.5.4.4 Other structural features

Of the various other elements of regular structure in proteins, the most abundant is the β -turn. The β -turn is a sequence of four amino acids in a protein, in which the polypeptide chain folds back on itself by nearly 180° (Fig. 3.26(a)), thus giving the protein a globular rather than a linear shape.¹⁶³ A total of 11 types of β -turn have been identified which differ in terms of the angles of rotation (ϕ , ψ in Fig. 3.23) about the various single bonds involved.¹⁶³ A β -bulge occurs where an extra amino acid is inserted in one strand of a β -sheet, disrupting the hydrogen bonding¹⁶⁴ (see Fig. 3.26(b)). A β -hairpin is a portion of polypeptide chain that folds back on itself so that the two halves constitute two adjacent strands of an antiparallel sheet¹⁶⁵ (see Fig. 3.26(c)).

The importance of these structural features in proteins has been reviewed.^{160, 166}

3.5.4.5 Principles governing the structures adopted by enzymes

Enzymes, and globular proteins in general, do not possess the simple regular structures that characterize fibrous proteins such as wool or silk. However, by examining the known three-dimensional structures, it is possible to formulate some general principles that appear to govern the structures adopted by enzymes,¹⁵⁶ these principles are summarized below.

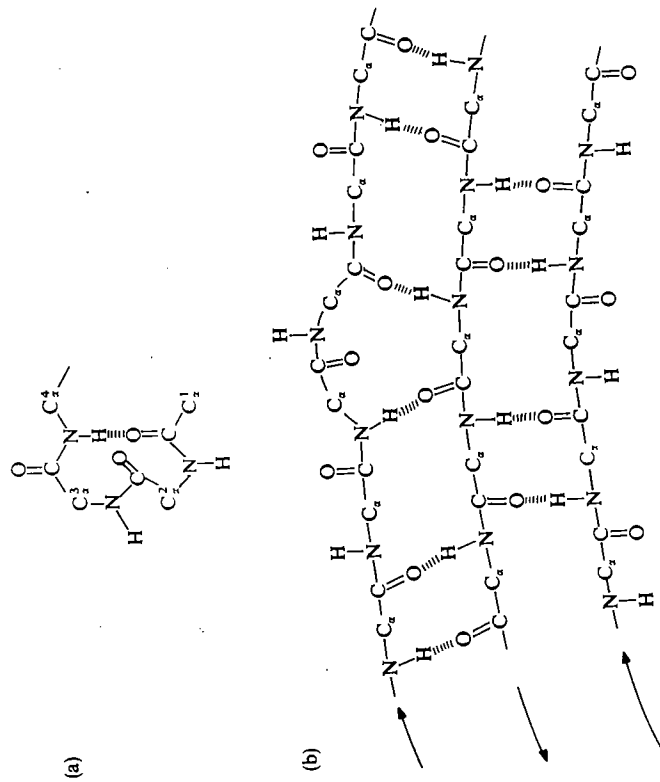


FIG. 3.26. Diagrammatic illustration of some structural features found in proteins. Dotted lines indicate hydrogen bonds. (a) A Type I β -turn;¹⁶³ (b) β -bulge;^{164, 166} (c) β -hairpin (of the Class 3 three residue type¹⁶⁶).

1. Enzymes are generally very closely packed globular structures with only a small number of internal cavities which are normally filled by water molecules.
2. The structural elements (amide bonds, helices, sheets, etc.) generally possess similar geometries to those observed in model compounds. However, some small deviations from the planarity of amide bonds have been noted in lysozyme, and α -helices can be distorted towards 3_{10} helices.

3. Non-polar amino-acid side-chains, e.g. those of Leu, Val, Phe, etc., are generally buried in the interior of the enzymes away from the solvent water. Ionized side-chains, e.g. those of Lys, Asp, etc. tend to be on the exterior of enzymes and able to interact with the solvent. Exceptions to these generalizations usually point to side-chains that have particular functions, e.g. the buried side-chain of aspartic acid 194 in chymotrypsin that is involved in a strong electrostatic interaction with the protonated α -amino group of isoleucine 16 (see Chapter 5, Section 5.5.1.2). Non-polar side-chains on the exterior of enzymes may well be involved in binding to a non-polar substrate or to a membrane.

4. Polar groups in the interior of enzymes are normally paired in hydrogen bonds. A very large percentage of the carbonyl and amino groups of amide bonds that are placed in the interior by the folding of the polypeptide chain are paired in hydrogen bond formation or form hydrogen bonds to internal water molecules and are thus rendered essentially non-polar.¹⁶⁷ While these internal hydrogen bonds may not be of great significance in terms of the energy of the folded structure, they are undoubtedly important in specifying the correct folded structure (Section 3.5.5.5).

5. Most larger enzyme molecules tend to exist in structural domains, i.e. globular units, connected by segments of polypeptide chain. Some good examples of this type of structure are found amongst the kinases and dehydrogenases,¹³⁵ which consist of two domains or 'lobes'. One of the domains is involved in binding the common substrate, i.e. NAD⁺ for dehydrogenases; the other is termed the catalytic domain and will be different in the different enzymes. It has been suggested that movements of the domains with respect to each other may be an important part of the mechanism of catalysis in these enzymes.

3.5.4.6 Classification of enzyme structures

In the previous section (3.5.4.5) we described some of the basic principles that seem to govern the types of structures adopted by enzymes. A given enzyme will possess a certain amount of helix, sheet, and other elements of regular secondary structure but this does not tell us very much about the overall structure of the enzyme. Several attempts have been made to look for patterns in the distribution of structural elements in proteins: one of the best known classifications is the four-class system introduced by Levitt and Chothia.¹⁶⁸ The four classes of protein structure are as follows.

1. All α -proteins which have only α -helix structure. Myoglobin and citrate synthase are examples of this type of protein.
2. All β -proteins which have mainly β -sheet structure. Chymotrypsin and the constant and variable regions of immunoglobulins belong to this class.
3. $\alpha + \beta$ -proteins, which have α -helix and β -sheet structural segments that do not mix but are separated along the polypeptide chain. Thermolysin, lysozyme, and ribonuclease belong to this class.

4. α/β -proteins, which have mixed or alternating segments of α -helix and β -sheet structure. The kinases and dehydrogenases with their nucleotide-binding domains are good examples of this class, which also includes other enzymes of the glycolytic pathway such as triosephosphate isomerase and phosphoglycerate mutase.

Richardson¹⁶⁰ has presented a more detailed classification of protein structures based on patterns within domains rather than of whole protein molecules.

The occurrence of ordered assemblies of structural elements found in a number of proteins suggests that there are stable types of 'supersecondary structure'. These types of structures would be of importance in directing the way in which a polypeptide chain could fold to adopt its correct three-dimensional structure (Section 3.7.2).

3.5.5 The forces involved in stabilizing the folded structures of enzymes

The forces that maintain the folded three-dimensional structure of a globular protein such as an enzyme are weak, non-covalent interactions. Experiments of the type described in Section 3.7.2 have shown that in the case of enzymes that possess disulphide bonds, these bonds are formed by oxidation of pairs of cysteine side-chains only after the three-dimensional structure of an enzyme has been acquired. It would therefore seem that disulphide bonds are important in stabilizing a folded structure that has already formed rather than directing the acquisition of this structure. The bond energy of a disulphide bond (approximately 200 kJ mol⁻¹) is much greater than that of the non-covalent interactions we shall describe in Sections 3.5.5.1 to 3.5.5.4.

The non-covalent interactions can be discussed in four categories: hydrogen bonds, electrostatic forces, Van der Waals forces, and hydrophobic forces. The division between these categories is somewhat arbitrary, since all attractive forces between atoms ultimately arise from favourable interactions between electrons and nuclei.

3.5.5.1 Hydrogen bonds

The hydrogen bond arises from the attraction between a hydrogen atom attached to an electronegative atom (O and N are the most important examples in biological systems) and another electronegative atom. Figure 3.27 shows the arrangement that is important in α -helix and β -sheet structures.

The hydrogen bond is most stable when (i) the distance between the two electronegative atoms is within closely specified limits. In the case shown in Fig. 3.27 the O...N distance is within the range 0.28–0.30 nm; and (ii) the arrangement of atoms e.g. O...H—N is linear.

A typical value for the energy of a hydrogen bond *in vacuo* is approximately 20 kJ mol⁻¹, but it is unlikely that the many internal hydrogen bonds in enzymes (Section 3.5.4.5) make any substantial contribution to the stabilization of the folded structure. This is because in the unfolded state the 'internal' polar groups

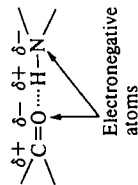


FIG. 3.27. The hydrogen bond involved in helix and sheet structures in enzymes and proteins.

could form alternative hydrogen bonds with the solvent water¹⁶⁹ and there is therefore little gain in stability when comparing the folded state with the unfolded. However, because of the strict geometrical requirements referred to above, hydrogen bonds are undoubtedly important in specifying the correct folded structure (Section 3.5.5.5).

3.5.5.2 *Electrostatic forces*

This category consists of the attractive forces that exist between charged groups, e.g. between $-\text{NH}_3^+$ (Lys) and $-\text{COO}^-$ (Asp); these interactions are also known as salt bridges.

The force between two charges is inversely proportional to the dielectric constant of the medium separating them, and the lack of detailed knowledge of the value of the dielectric constant in an enzyme makes any quantitative assessment of the contribution of electrostatic forces to the stability of the folded structure very difficult. * Although salt bridges play a crucial role in particular enzymes, e.g. in the activation of chymotrypsinogen (see Chapter 5, Section 5.5.1.2), it appears that in general the number of salt bridges is limited and hence their contribution to the stability of enzyme structures is probably rather small.

3.5.5.3 *Van der Waals forces*

Van der Waals forces are weak forces that occur when molecules or groups of atoms are in close contact with one another (less than about 0.4 nm apart in the case of atoms found in proteins†). Attractive forces result from favourable interactions between dipoles (separations of charge) that may be permanent or transient. Transient dipoles arise by the local fluctuations in electron density in

* Recently, progress has been made in determining the effective dielectric constant within proteins. The effects of changing charged amino-acid side-chains elsewhere in the protein on the pK_a of the active-site histidine side-chain (His 64) of subtilisin have been measured. Calculations suggest that the effective dielectric constant is between 30 and 60 (the values for water and benzene are 80 and 2.3, respectively). The relatively high value for the protein is thought to be due partly to the surrounding water and also to polar side-chains.^{170, 171}

† When atoms are brought too close to each other, the electron clouds repel one another. The balance of attractive and repulsive forces leads to an optimum distance, the van der Waals distance, between the atoms. This is the sum of the van der Waals radii of the atoms concerned.

atoms and can also be induced by the presence of a neighbouring dipole. A variety of interactions are possible: dipole-dipole, dipole-induced dipole, and induced dipole-induced dipole. All these forces are weak, with energies of the order of 10 kJ mol^{-1} or less, but the large number of van der Waals contacts that occur in proteins may well mean that they make a significant contribution to the stability of protein structures.

3.5.5.4 *Hydrophobic forces*

It is well known that hydrocarbons, such as methane, benzene, etc., are much more soluble in liquid hydrocarbons or most other organic solvents than in water. In fact, water has an almost unique place as an extremely poor solvent for hydrocarbon solutes.¹⁷² The probable reason for this lies in the fact that water consists of transient clusters of hydrogen bonded molecules; when a hydrocarbon is introduced into the aqueous medium there is a reorganization of the water molecules around the hydrocarbon that makes the system more ordered.¹⁷³ This increase in order, i.e. decrease in entropy, is unfavourable on thermodynamic grounds¹⁷⁴ and hence there is a tendency to force the hydrocarbon out of contact with the water and into the organic phase, so that this decrease in entropy does not occur.

In the case of proteins, these considerations dictate that the non-polar portions of the molecule, for example the side-chains of amino acids Val, Leu, Ile, Phe, Trp, should be buried in the interior of the molecule away from the solvent water. Calculations show that this burying of the non-polar side chains makes by far the major contribution to the stability of the folded structure compared with that of the unfolded structure.¹⁷⁵ For lysozyme this stabilization energy is calculated to be of the order of 1200 kJ mol^{-1} . *It should be emphasized that it is the burying of the non-polar side chains away from water that provides this stabilization, rather than interactions between the non-polar side chains themselves.*¹⁷⁶

3.5.5.5 *Some conclusions about the forces involved in the maintenance of the folded structures of enzymes*

From the discussion in the previous section (3.5.5.4) it is evident that hydrophobic forces make the greatest contribution in energy terms to the stability of the folded state compared with the unfolded state. However, hydrophobic forces do not confer any geometrical specificity on interactions, since they arise essentially from exclusion from the aqueous phase,¹⁷⁶ and thus do not serve to specify any particular folded structure. In order to specify a structure, interactions such as hydrogen bonding, that possess geometrical requirements (Section 3.5.5.1) are of great importance. The overall stability of the folded structure compared with the unfolded structure is much smaller than the values of hydrophobic stabilization energy would suggest. This difference can largely be accounted for by the fact that it is unfavourable on entropy grounds to convert a highly flexible unfolded polypeptide chain to a more ordered folded chain.¹⁷⁵

3.5.6 Examples of enzyme structures

In this section we shall describe the three-dimensional structures of the enzymes α -chymotrypsin, adenylate kinase, and glyceraldehyde-3-phosphate dehydrogenase. However, it is first necessary to mention some of the ways of depicting these structures.

3.5.6.1 Representations of enzyme structures

Considerable problems are posed by trying to depict the complex three-dimensional structures adopted by enzymes, especially when using two-dimensional representations. Undoubtedly the most informative representations are three-dimensional scale models built directly from the coordinates of the electron-density map. An example of such a model is shown in Fig. 3.28 which depicts the structure of adenylate kinase from pig muscle at a scale of 20 cm = 1 nm. This model uses wire parts that keep the bond angles and bond lengths of the planar amide bond units constant but allow rotation about the single bonds to the α -carbon atoms. Although features such as the deep cleft at the bottom right-hand side of the model are clearly visible, it is difficult to appreciate the details of the structure without having the model on hand. A similar limitation arises with 'space-filling' models in which each atom is represented by a sphere, sometimes flattened, of radius equal to its van der Waals radius. 'Space filling' models are very suitable for highlighting features of the surface of an enzyme, such as a cleft or depression, but are of little use in illustrating internal features such as the folding of the polypeptide chain.

One way of providing a more simplified representation of the structure would be to strip off the side-chains of the amino acids and merely show the folding of the polypeptide chain backbone; this approach is illustrated in Figs. 3.29(a) and 3.30 for the cases of α -chymotrypsin and adenylate kinase, respectively. Two limitations of this approach are: (i) the model does not show the nature of the side-chains in any particular region of the enzyme such as the active (catalytic) site, and (ii) it is not always easy to pick out elements of regular secondary structure. In order to meet the second limitation, it has become popular to show the folding of the polypeptide chain with particular emphasis on these elements of regular structure, helices being shown as cylinders and strands of sheet as arrows. Examples of this type of representation are shown in Figs. 3.31(a) and 3.32 for adenylate kinase and glyceraldehyde-3-phosphate dehydrogenase, respectively. Richardson has developed another approach in which α -helices are shown as spiral ribbons, the β -strands as thick arrows and the irregular structures as ropes.¹⁶⁰ Examples of this type of representation are shown in Fig. 3.31(b) and Fig. 5.23(b). These approaches have helped to focus attention on the 'super-secondary structures' (Section 3.5.4.6) found in a number of enzymes.

3.5.6.2 The structure of α -chymotrypsin from beef pancreas

α -Chymotrypsin consists of three polypeptide chains held together by five disulphide bonds: the origin of this three-chain structure is discussed in Chapter

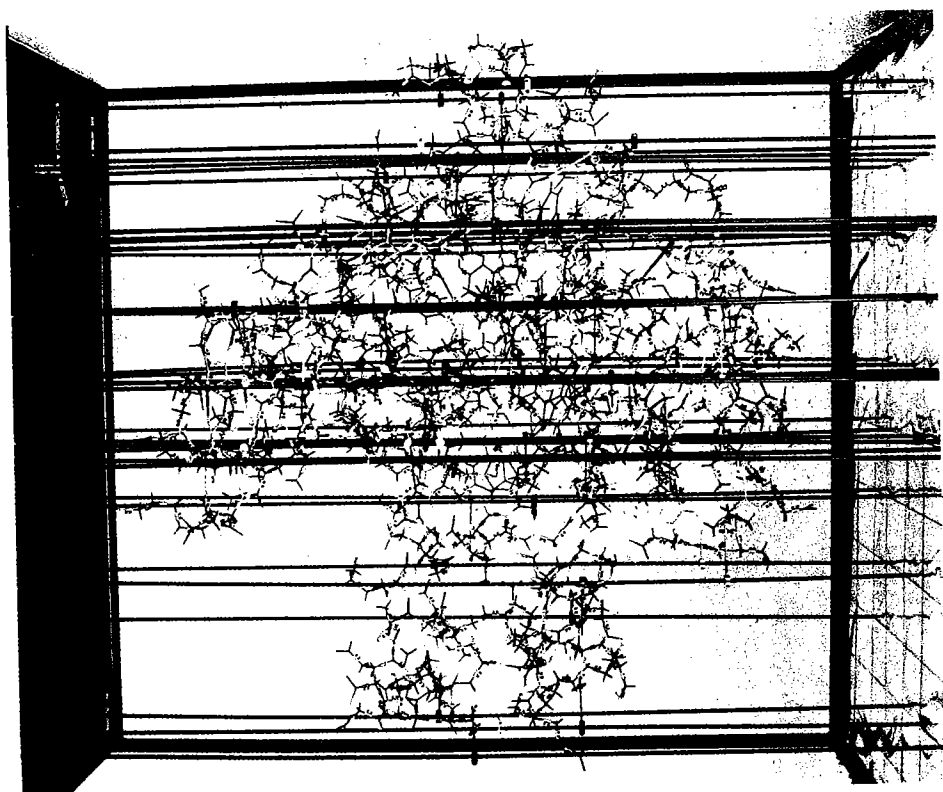


FIG. 3.28. The structure of adenylate kinase shown as a three-dimensional scale model built with wire parts (scale 20 cm = 1 nm). (Reproduced with permission from *Principles of protein structure* by G. E. Schulz and R. H. Schirmer, p. 136. Copyright Springer, New York (1979).

5, Section 5.5.1.1. The chains run between amino acids 1–13 (chain A), 16–146 (chain B), and 149–245 (chain C). The five disulphide bonds are between cysteine side-chains at positions 1 and 122, 42 and 58, 136 and 201, 168 and 182, and 191 and 220.

The structure of the enzyme is represented in Fig. 3.29(a), which shows the positions of the α -carbon atoms.¹⁷⁷ The overall shape of the molecule is that of an ellipsoid with a maximum dimension of 5.1 nm. There is a shallow depression at the active site in which there are three side-chains, Ser 195, His 57, and Asp 102 of crucial importance in the catalytic mechanism (see Chapter 5, Section 5.5.1.2).



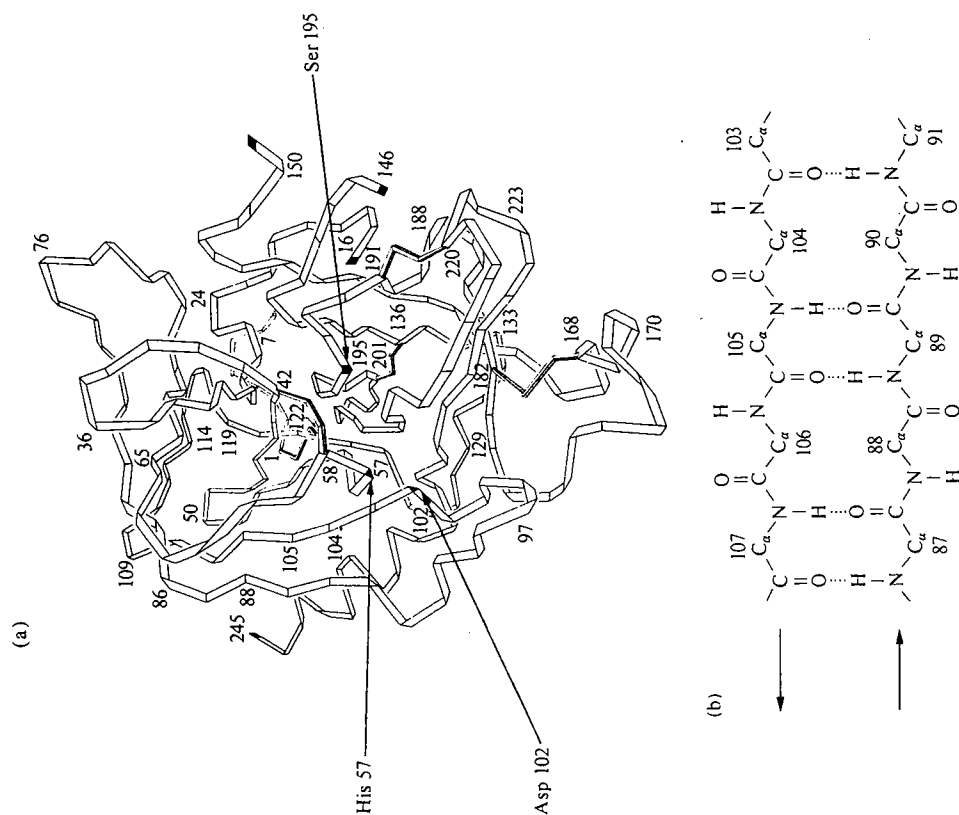


FIG. 3.29. The structure of α -chymotrypsin. (a) The positions of α -carbon atoms. (Reproduced with permission from *Enzymes*, 3rd edn (P. D. Boyer ed.) Vol. 3, p. 194. Copyright Academic Press, New York (1971).) (b) The arrangement of hydrogen bonds in part of the antiparallel sheet structure of the enzyme.

There are only two short stretches of helix between amino acids 164 and 173 and between amino acids 235 and 245. Both helices are somewhat distorted from an α -helix towards a 3_{10} helix. There is a considerable amount of antiparallel β -sheet structure in the enzyme and in fact the structure can be described as consisting of two folded units (amino acids 27-112 and 133-230) in each of which there are six strands of antiparallel sheet forming a highly distorted hydrogen-bonded 'cylinder'. While this point is difficult to see in Fig. 3.29(a), it should be relatively easy to appreciate that amino acids 87-91 and 103-107 form part of a

sheet structure with the hydrogen-bonding pattern shown in Fig. 3.29(b). A fuller discussion of the structure of enzyme has been given by Blow.¹⁷⁷

3.5.6.3 The structure of adenylate kinase from pig muscle

Adenylate kinase consists of a single polypeptide chain of 194 amino acids. Figure 3.30 shows the positions of the α -carbon atoms of the polypeptide chain.¹⁷⁸ It is immediately clear that the enzyme has a deep cleft in which the side-chains of Cys 25 and His 36 are located; these side-chains are thought to be important in the catalytic activity of the enzyme. The cleft divides the molecule into two distinct domains or 'lobes'; the domain on the right-hand side of Fig. 3.30 has three stretches of helix, whereas that on the left has five strands of sheet with interconnecting stretches of helix. The elements of regular secondary structure are more clearly represented in Fig. 3.31(a) and (b). Altogether a very high proportion (over two-thirds) of the amino acids in the molecule are in some type of regular secondary structure; this may account for the great stability of the enzyme at low pH (see Chapter 2, Section 2.8.1).

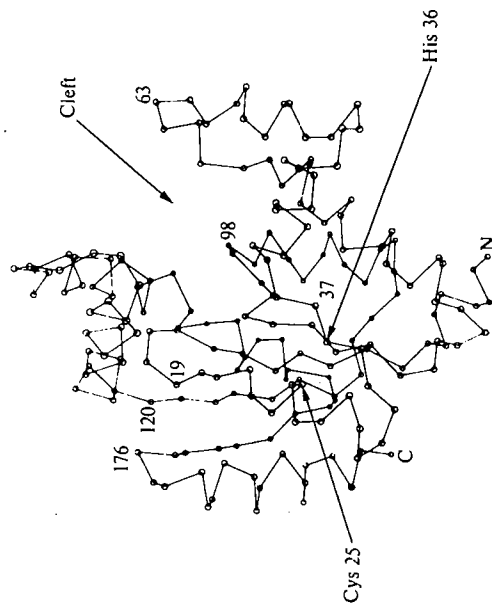


FIG. 3.30. Positions of the α -carbon atoms in adenylate kinase. The locations of Cys 25 and His 36, which are thought to be involved in the catalytic mechanism of the enzyme, are shown. (Reprinted by permission from reference 178, *Nature, Lond.* 250, 120-3. Copyright 1974, Macmillan Journals Limited.)

3.5.6.4 The structure of glyceraldehyde-3-phosphate dehydrogenase from lobster muscle

Glyceraldehyde-3-phosphate dehydrogenase is a tetramer, consisting of four subunits each of M_r 35 000. The dimensions of the tetramer are approximately

15 nm $\times$ 14 nm $\times$ 8 nm. Figure 3.32 shows the structure of a subunit in two distinct halves representing the two domains. ¹⁷⁹ It is interesting that the dividing point between the domains is close to Cys 149, the side-chain of which is involved in the catalytic activity of the enzyme.

The NAD⁺-binding domain which occupies the N-terminal part of the polypeptide chain (Fig. 3.32 (a)) is dominated by a structure consisting of strands of a twisted parallel β -sheet, the strands are labelled A-F from the N-terminus. Between these strands are stretches of helix, such as helix α_E between strands D

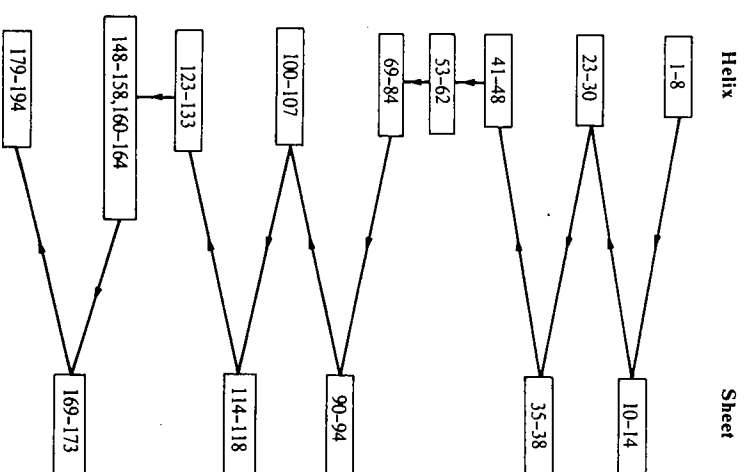
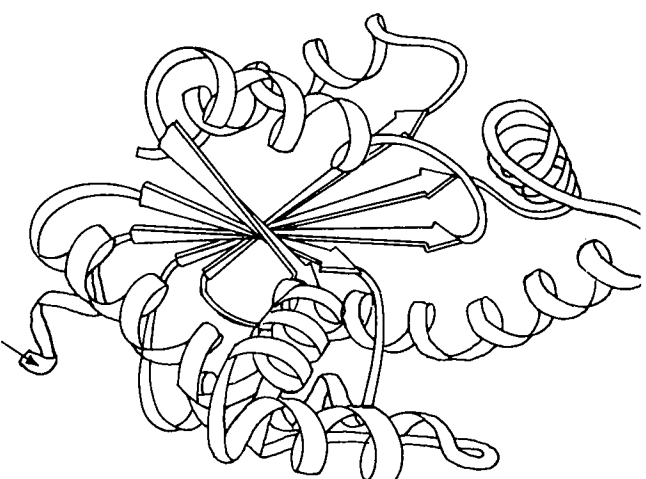
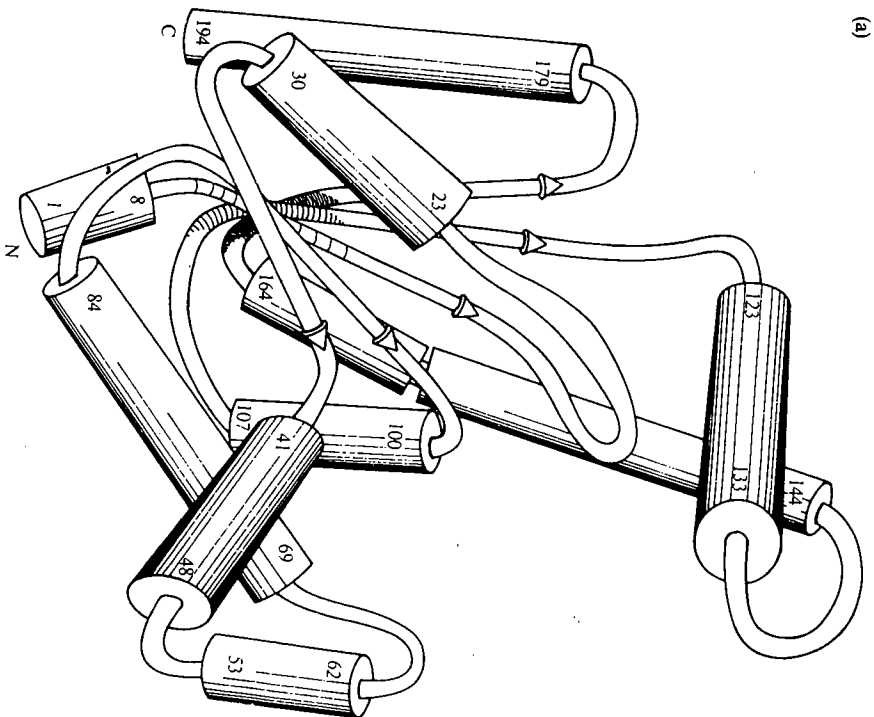


FIG. 3.31. Elements of regular structure in adenylylate kinase. (a) The path of the polypeptide chain; helices are depicted as cylinders and strands of sheet as arrows. (b) The path of the polypeptide chain; helices are depicted as ribbons and strands of sheet as arrows. (c) The amino acids involved in these elements of structure. (Part (a) reproduced with permission from *Principles of protein structure* by G. E. Schulz and R. H. Schirmer, p. 136. Copyright Springer, New York (1979). Part (b) reproduced with permission from *Advances in Protein Chemistry* 34, 167-339. Copyright, J. S. Richardson (1981).)

and E, or of extended polypeptide chain, such as between strands C and D, or of other antiparallel strands, such as those which occur between strands B and C. There is a very similar NAD^+ -binding domain in other dehydrogenases such as lactate dehydrogenase (see Chapter 5, Section 5.5.4.3) and alcohol dehydrogenase,¹⁵³ suggesting a close relationship between the common functional parts of these enzymes.

The other domain of the subunit is designated as the catalytic domain to emphasize that this domain varies between different dehydrogenases. The catalytic domain has a quite different structure (Fig. 3.32(b)) from the NAD^+ -binding domain. The dominant feature in the former is a nine-strand, mainly antiparallel, sheet structure on one side of which are three associated helices (148–166, 210–217, and 251–265) with axes roughly parallel with the axes of the sheet strands. At the C-terminus of the polypeptide chain there is a stretch of helix that packs against the central β -sheet of the NAD^+ -binding domain.

The binding site for NAD^+ is located near the interface between subunits and it may well be that some of the side-chains involved in forming the binding site of one subunit are supplied by a neighbouring subunit. This situation has also been found in other cases, e.g. triosephosphate isomerase (see Chapter 5, Section 5.5.2.2), citrate synthase¹⁸⁰ and glutamine synthetase.¹⁸¹

3.6 The determination of quaternary structure

It has been found that most large enzymes consist of a number of subunits* held together by non-covalent forces, i.e. they are *oligomeric*. In some ways the occurrence of multiple subunits in an enzyme can be considered analogous to the occurrence of folded globular domains (Section 3.5.4.5) along a single polypeptide chain. The term quaternary structure refers to the arrangement of the subunits in oligomeric enzymes (or proteins in general). In this section we shall deal with the following questions that can be asked about oligomeric enzymes. (i) How many subunits are there and of what type? (Section 3.6.1). (ii) How are the subunits arranged? (Section 3.6.2). (iii) What forces are involved in holding subunits together? (Section 3.6.3). (iv) What is the significance of multiple subunits in an enzyme? (Section 3.6.4).

3.6.1 Number and type of subunits

In this section we shall discuss the various experimental approaches to determining the number and type of subunits in an oligomeric enzyme.

3.6.1.1 Studies of M_r

An indication that an enzyme consists of multiple subunits is provided by results of studies of M_r (Section 3.2) performed in the absence and presence of denaturing agents such as guanidine hydrochloride (Section 3.7.1). In favourable

* The definition of the term subunit has been given in Section 3.1.

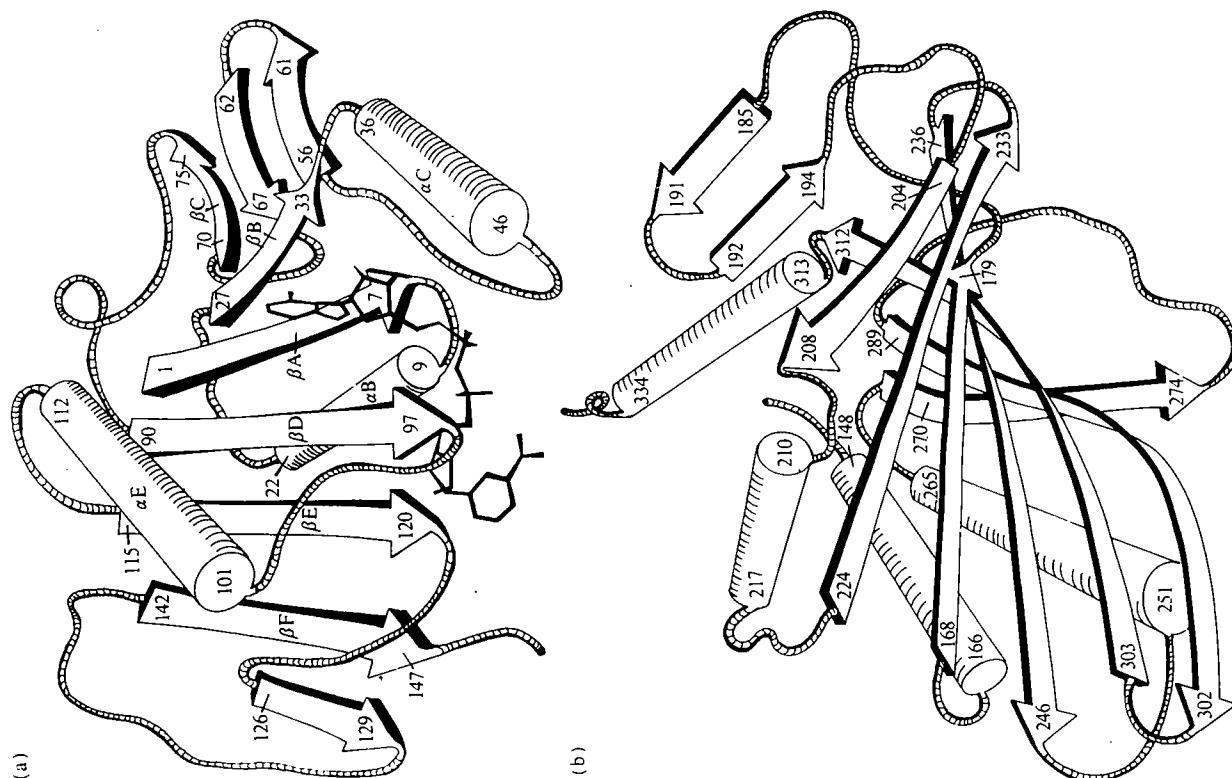


FIG. 3.32. The structure of a subunit of glyceraldehyde-3-phosphate dehydrogenase.¹⁷⁹ Helices are represented by cylinders and strands of sheet by arrows. (a) The NAD^+ -binding domain; the adenine ring binds near β_A and the nicotinamide ring binds between β_D and β_E . (b) The catalytic domain. (Part (a) is reprinted with permission from *Journal of Molecular Biology* 90, 25–49. Copyright Academic Press Inc., (London) (1974). Part (b) reprinted with permission from *Journal of Biological Chemistry* 250, 9137–62. Copyright American Society of Biological Chemists Inc. (1975).)

cases the number of subunits can be deduced directly from the M_r data. For instance, alcohol dehydrogenase from yeast has an M_r of about 145 000 as determined by ultracentrifugation. However, under denaturing conditions a value of 36 000 is obtained, suggesting that the enzyme consists of four subunits that are probably identical. In this type of work it is particularly important to make sure that traces of proteases are absent from the enzyme preparations, since such proteases may well be active under the denaturing conditions and cause extensive fragmentation of the unfolded polypeptide chains of the enzyme of interest. For instance, from the results of sodium dodecylsulphate polyacrylamide gel electrophoresis (Section 3.2.3) it was deduced that pyruvate kinase from yeast consisted of eight subunits each of M_r about 20 000.¹⁸² However, when isolated under conditions chosen to inactivate traces of proteases, the enzyme was found to consist of four subunits, each of M_r about 55 000.¹⁸³ (The unit of M_r 20 000 presumably represented the predominant fragment produced by protease action.) In Chapter 7 (Sections 7.9 and 7.11.2) we shall refer to similar difficulties in the determination of the subunit structures of tryptophan synthase and of the *arom* multienzyme polypeptide protein.

3.6.1.2 Cross-linking studies

A second method for determining the number of subunits in an oligomeric enzyme relies on the use of cross-linking agents such as dimethylsuberimide.¹⁸⁴ This compound reacts with pairs of lysine side-chains, which almost invariably occur on the surfaces of enzymes (Section 3.5.4.5) to form cross-linkages that are stable in the presence of denaturing agents (Fig. 3.33).

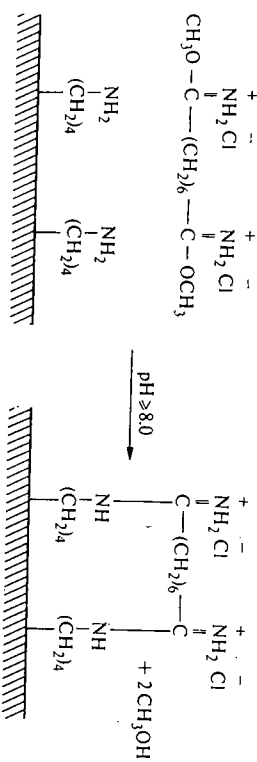


FIG. 3.33. Cross-linking of lysine side-chains by reaction with dimethylsuberimide.

If we react an enzyme that consists of, say, four subunits with this cross-linking agent, a mixture of species will be formed in which different numbers of subunits are cross-linked (Fig. 3.34). The various species can be separated by sodium dodecylsulphate polyacrylamide gel electrophoresis (Section 3.2.3) giving a total of four bands.

The number of bands will give the number of subunits directly. It is clearly

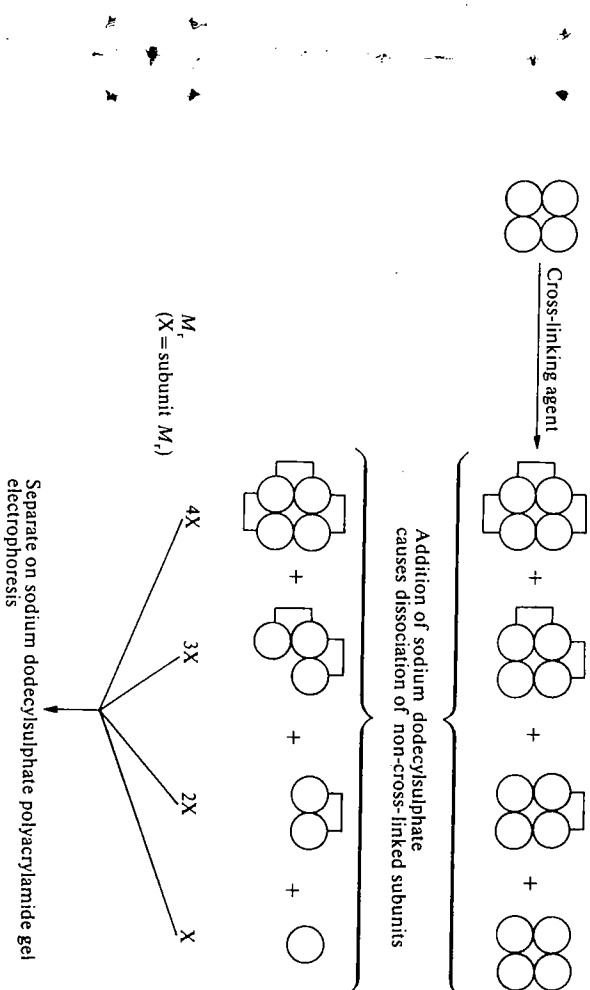


FIG. 3.34. The use of a cross-linking agent to determine the number of subunits in an oligomeric enzyme. For convenience the intrasubunit cross-linkages are not shown.

important in this work (i) to avoid complete intersubunit cross-linking, otherwise the mixture of species shown in Fig. 3.34 will not result and only one band corresponding to an M_r of $4X$ will be seen on electrophoresis; (ii) to avoid intermolecular cross-linking, otherwise large aggregates will be formed. These complications can be minimized by using low concentrations of cross-linking agent and of protein respectively.¹⁸⁵

3.6.1.3 Studies of ligand binding

The number of binding sites on an enzyme for a substrate or other ligand can often be used to indicate the number of subunits. For instance, the alcohol dehydrogenases from yeast and liver bind four and two moles of NADH per mole of enzyme, respectively, in agreement with the tetrameric (four subunit) and dimeric (two subunit) structures of the respective enzymes. Problems can arise, however, when multiple binding sites in an oligomeric enzyme do not behave independently. Thus in the case of aspartate carbamoyltransferase from *E. coli*, studies of the binding of the inhibitor CTP suggested that the enzyme contained four regulatory subunits. It was subsequently established that there are in fact six such subunits^{186, 187} and that the earlier results can be accounted for by the negative cooperativity (see Chapter 6, Section 6.2.2.2) that occurs between these binding sites.

3.6.1.4 Studies of symmetry

The type of symmetry possessed by an enzyme, deduced from X-ray crystallographic studies (Section 3.5.1), can sometimes be used to indicate the number of subunits in an enzyme or at least to exclude various proposed subunit structures. The observation of both a three-fold and a two-fold axis of symmetry in aspartate carbamoyltransferase from *E. coli*¹⁸⁷ excluded the previously proposed subunit structure of the enzyme (four catalytic plus four regulatory subunits). In conjunction with other results from studies of M_r and of the C-terminal amino acids of the subunits,¹⁸⁶ these data indicated that the enzyme consisted of six catalytic and six regulatory subunits.

3.6.1.5 Identity of subunits

If, under denaturing conditions, an enzyme gives rise to a number of species that differ in M_r , it can be concluded that the enzyme contains non-identical subunits.* In order to establish the relative numbers of each type of subunit, we can perform quantitative determinations of the N- and C-terminal amino acids of the various subunits (Sections 3.4.4.1 and 3.4.4.2). The subunit composition can then be confirmed by checking that the observed M_r of the intact enzyme equals the sum of the M_r values of the component parts, e.g. the M_r of aspartate carbamoyltransferase from *E. coli* (300 000) corresponds to the sum of the M_r of six catalytic ($6 \times 34\,000$) and six regulatory ($6 \times 17\,000$) subunits. In certain cases, however, it is not difficult to 'miss' a subunit of very different size from the other subunits: thus, in the case of phosphorylase kinase (see Chapter 6, Section 6.4.2.1) a small calcium-binding subunit was only detected a number of years after the subunit structure of the enzyme had been thought to be firmly established.

When subunits appear to be identical in terms of M_r (and/or terminal amino acids), it is desirable to check on their identity in terms of amino-acid sequence. A reasonably sound check can be made by determining the number of distinct fragments produced by the action of a specific protease such as trypsin. In each subunit the number of tryptic peptide fragments should equal the number of lysines plus the number of arginines plus one. If an enzyme consists of identical subunits, the number of tryptic peptides in the peptide map (Section 3.4.3) will equal this number ($\text{Lys} + \text{Arg} + 1$). On the other hand, if the subunits are not of identical sequence, the number of tryptic peptides will be greater than this. For instance lactate dehydrogenase from dogfish muscle had been shown to consist of four subunits each of M_r 36 000. Amino-acid composition data showed that there were 36 arginines per molecule (144 000) of enzyme, i.e. nine per subunit if the subunits were identical. Only nine distinct arginine-containing peptides could be isolated after digestion of lactate dehydrogenase by trypsin, indicating that the four subunits were identical in terms of sequence.¹⁸⁸ A similar approach

* This assumes that we are dealing with a pure preparation of enzyme and have taken steps to avoid problems arising from the action of traces of proteases.

was used to show the identity of the six subunits of glutamate dehydrogenase from beef liver.^{18,9} It is sometimes possible to underestimate the number of distinct fragments produced by the action of a protease, since there may be peptides that differ only slightly in sequence from one another and are not sufficiently resolved in the standard separation procedures (Section 3.4.3). A further complication might arise if the enzyme possesses an insoluble 'core' that is resistant to attack by the protease. The ultimate proof of subunit identity can therefore only be obtained when the complete amino-acid sequences of the subunits have been determined.

An extensive compilation has been made of the subunit stoichiometries of oligomeric enzymes.¹⁹⁰ In the majority of such enzymes, subunits are identical with one another; some of the more notable examples of enzymes that contain non-identical subunits are listed in Table 3.5. It should also be noted that almost all oligomeric enzymes that consist of identical subunits contain even numbers of such subunits. There are, however, some well-documented examples of enzymes containing odd numbers of identical subunits, such as phospho-2-keto-3-deoxygluconate aldolase from *Pseudomonas putida*¹⁹¹ and ornithine carbamoyltransferase from *Bacillus subtilis*,¹⁹² each of which are trimeric enzymes with subunit M_r values of 24 000 and 47 000, respectively.

Finally, it should be remembered that some enzymes are capable of indefinite association. The examples of glutamate dehydrogenase²¹ and acetyl-CoA carboxylase²³ have already been mentioned: a more comprehensive list of such systems has been given by Frieden.^{1,93} In such systems it is inappropriate to speak of a definite quaternary structure except perhaps under very carefully defined conditions.

3.6.2 Arrangement of subunits

The arrangement of the subunits in an oligomeric enzyme can usually be deduced from the type of symmetry possessed by the molecules (this is found by X-ray crystallography; Section 3.5.1). In general, it is found that the arrangement of the subunits is such as to maximize the number of intersubunit contacts.¹⁹⁴ For a tetrameric enzyme such as lactate dehydrogenase the preferred geometrical arrangement will be tetrahedral, rather than square or linear (Fig. 3.35). In the case of a hexameric enzyme, the preferred arrangement will be octahedral. (For a more detailed discussion of the possible geometrical arrangements of oligomeric enzymes, references 194 and 195 should be consulted.)

The detailed arrangement of the subunits and the nature of the intersubunit contacts can only be decided when the complete three-dimensional structure of the enzyme has been obtained. However, in a system that has not been studied by X-ray crystallography it is possible to learn something about the arrangement of the subunits by the use of cross-linking agents. This type of approach showed that the subunits of the Ca^{2+} - and Mg^{2+} -activated adenosinetriphosphatase of the inner membrane of *E. coli* were probably arranged as shown in Fig. 3.36. In this work, a reversible cross-linking agent, dithiobis(succinimidyl propionate)

TABLE 3.5
Some enzymes composed of non-identical subunits

Enzyme	Source	Subunit composition	Comments
Lactose synthase	Bovine mammary tissue	$\alpha\beta$	
Haemoglobin ^a	Human red blood cells	$\alpha_2\beta_2$	
Tryptophan synthase (see Chapter 7, Section 7.9.1)	<i>E. coli</i>	$\alpha_2\beta_2$	
cAMP-dependent protein kinase (see Chapter 6, Section 6.4.2.1)	Rabbit skeletal muscle	$\alpha_2\beta_2$	α and β represent catalytic and regulatory subunits, respectively
Aspartate carbamoyltransferase	<i>E. coli</i>	$\alpha_6\beta_6$	α and β represent catalytic and regulatory subunits, respectively. The molecule is assembled as $(\alpha_3)_2(\beta_2)_3$
Ribulosebiphosphate carboxylase	Spinach	$\alpha_8\beta_8$	
RNA nucleotidyltransferase (see Chapter 7, Section 7.3)	<i>E. coli</i>	$\alpha_2\beta\beta'\sigma$	
Phosphorylase kinase (see Chapter 6, Section 6.4.2.1)	Rabbit skeletal muscle	$\alpha_4\beta_4\gamma_4\delta_4$	
Adenosinetriphosphatase (see Chapter 2, Section 2.8.3)	Beef-heart mitochondria	$\alpha_3\beta_3\gamma\delta\epsilon$	

^a Haemoglobin is included (although it is not an enzyme) since a great deal is known about the inter-subunit contacts in this protein (Section 3.6.3).

FIG. 3.35. Possible arrangements of subunits in a tetrameric protein, showing the number of intersubunit contacts.

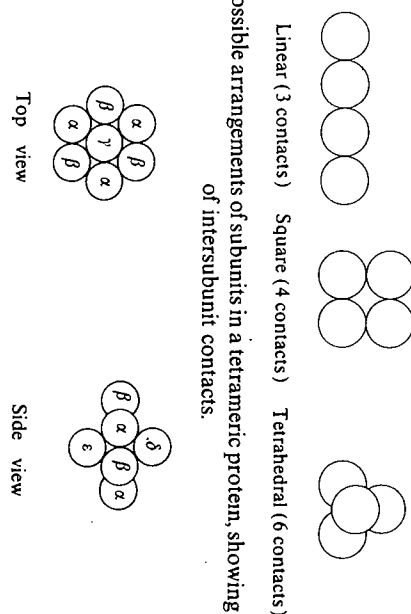


FIG. 3.36. Proposed arrangement of the Ca^{2+} - and Mg^{2+} -activated adenosinetriphosphatase from *E. coli*, as deduced by cross-linking studies.²²⁷ The subunit stoichiometry of this enzyme is $\alpha_3\beta_3\gamma\delta\epsilon$ (see Table 3.5).



was used.²²⁷ This bifunctional ester results in the cross-linking of pairs of lysine side-chains by two propionyl groups that are joined by a disulphide bond. It is thus possible to identify the components of the cross-linked species by treatment with dithiothreitol (Section 3.3.2) to break the disulphide bond, and subsequent electrophoresis in the presence of sodium dodecylsulphate. The results showed that it was possible to cross-link α and β subunits to each other or to the γ subunit. However, no α - α or β - β cross-links could be formed. Other evidence showed that the smaller δ and ϵ subunits were attached only peripherally (Fig. 3.36).

3.6.3 Forces involved in the association between subunits

Oligomeric enzymes are usually dissociated into subunits by treatment with denaturing agents (Section 3.7.1) such as guanidine hydrochloride. The forces involved in the association of the subunits are thus of the weak, non-covalent type that are involved in stabilizing the folded structure of a polypeptide chain (Section 3.5.5), i.e. hydrogen bonds, electrostatic forces, van der Waals forces, and hydrophobic forces. There is only limited information available about the nature of inter-subunit contacts because the number of oligomeric enzymes whose structures have been solved by X-ray crystallography to the required resolution is still rather small. From the studies on systems such as haemoglobin, lactate dehydrogenase, and glyceraldehyde-3-phosphate dehydrogenase, the following generalization can be made:¹⁹⁶

Number of non-polar contacts > number of hydrogen bonds
> number of electrostatic interactions.

The finding that there are large numbers of non-polar groups on the surfaces that are involved in interaction between subunits suggests that van der Waals forces

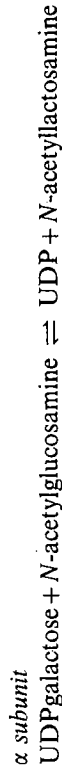
and hydrophobic forces* provide a strong tendency for subunits to associate; the importance of hydrophobic forces is also suggested by a thermodynamic analysis of protein-protein interactions.¹⁹⁷ Fisher has suggested that there is a strong correlation between the tendency for subunits to associate and the content of amino acids with non-polar side-chains.¹⁹⁹ Proteins that contain more than about 30 per cent non-polar amino acids are unable to bury the non-polar side-chains within the interior of a folded polypeptide chain. The exposed non-polar groups will then tend to interact with others on other subunits giving rise to an oligomeric system.

As mentioned previously (Section 3.5.5.5), hydrophobic forces do not confer any geometrical specificity to an interaction. Specific interactions between subunits must therefore rely on forces such as hydrogen bonding that have distinct geometrical requirements. The importance of interactions between particular side-chains in different subunits is well illustrated by the various mutant haemoglobins.¹⁹⁵ In normal haemoglobin, the side-chain of tyrosine 35 on the β -chain forms a hydrogen bond with the side-chain of aspartic acid 126 on the α -chain. In haemoglobin Philly, the tyrosine is replaced by phenylalanine, which cannot form the hydrogen bond. This loss of a hydrogen bond leads to increased dissociation of haemoglobin Philly into subunits and is associated with clinical symptoms of mild anaemia. In Chapter 5 (Section 5.5.5.5) it is shown how the replacement of phenylalanine 164, which lies near the subunit interface in tyrosyl-tRNA synthetase, by aspartic acid leads to dissociation of the subunits at pH 7.78, at which pH, the aspartic acid carries a negative charge. At low pH, the side-chains are protonated and the subunits can associate.²⁰⁰

3.6.4 The significance of multiple subunits

There seem to be at least three reasons why the possession of multisubunit enzymes is advantageous to an organism. Firstly, the presence of multiple subunits confers additional possibilities of regulating the catalytic activity of an enzyme. This is discussed in detail in Chapter 6, Section 6.2.2.1. Secondly, the assembly of different types of subunits into a large complex permits variation in catalytic properties; this is well illustrated by the examples of tryptophan synthase and RNA nucleotidyl transferase discussed in Chapter 7, Sections 7.9 and 7.3 respectively. Another example of this type of system is lactose synthase (Table 3.5), which consists of two different subunits. One of these subunits (β) is lactalbumin, which is a protein that is devoid of any catalytic activity but able to alter the specificity of the other subunit (α), a galactosyltransferase, so that it can catalyse the synthesis of lactose rather than *N*-acetyllactosamine.²⁰¹

* Hydrophobic forces are dominated by the entropy term (Section 3.5.5.4) and are thus weaker at lower temperatures. The dissociation of a number of enzymes such as adenosinetriphosphatase from mitochondria and phosphorylase b at low temperatures probably indicates that hydrophobic forces are involved in the association of subunits in these cases. Alternative explanations for such 'cold lability' have also been given.¹⁹⁸



Lactalbumin thus acts as a specifier protein that can alter the specificity of the α subunit so that when necessary (e.g. during lactation) lactose can be synthesized in preference to *N*-acetyllactosamine that is used in the biosynthesis of glycoproteins.

Many oligomeric enzymes do not display these properties; thus, lactate dehydrogenase does not show regulatory characteristics. In such cases we must look for another reason for the occurrence of multiple subunits. A third reason may well be to confer stability on the enzyme. It is difficult to gain reliable data on this point because treatments that disrupt the bonds between subunits also tend to disrupt the folded structure of the subunits themselves, so that correctly folded isolated subunits are not easy to obtain.* The observation that the surfaces of subunits contain a number of non-polar side-chains (Section 3.6.3) suggests that correctly folded subunits, were they to occur, would, in general, be rather unstable.

3.7 The unfolding and refolding of enzymes

In Sections 3.5 and 3.6 we discussed the three-dimensional structures adopted by enzymes. It is now appropriate to consider how the three-dimensional structure can be lost (Section 3.7.1) and subsequently regained (Section 3.7.2). Section 3.7.3 deals with methods of predicting the three-dimensional structure of an enzyme.

3.7.1 Unfolding of enzymes

The compact folded form of an enzyme is generally thermodynamically more stable than the modified form by only a relatively small margin of the order of 15–60 kJ mol⁻¹ 175 (Section 3.5.5.5). It is therefore not surprising that most enzymes can be relatively easily unfolded (and dissociated in the case of multisubunit enzymes) by a variety of conditions such as extremes of pH, heating, and addition of organic solvents, detergents, chaotropic agents,[†] or high concentrations of urea or guanidine hydrochloride. Loss of the three-dimensional structure of the folded enzyme is known as denaturation and agents that bring about denaturation are known as denaturing agents. Denaturation of an enzyme can be monitored by the loss of catalytic activity, by changes in

* Some progress has been made in this area using enzymes linked to insoluble supports, thereby permitting the preparation of matrix-bound subunits.²⁰² Other evidence for the existence of catalytically active subunits can be obtained from a study of the kinetics of regain of activity during refolding.²⁰³

† Chaotropic agents are ions of low charge density, e.g. trichloroacetate, that disorder water structure and facilitate the transfer of non-polar groups to the solvent water.

spectroscopic parameters such as the circular dichroism spectrum (Section 3.5.2), or by changes in other properties such as solubility.

The modes of action of denaturing agents have been extensively investigated.^{204, 205} Heating will disrupt the folded structure of an enzyme by increasing the vibrational and rotational motions of atoms. A change in pH will affect the state of ionization of amino-acid side-chains that may well be involved in the maintenance of the folded structure. Addition of organic solvents or detergents would be expected to unfold an enzyme by interacting with the non-polar amino-acid side-chains previously buried in the interior of the folded structure. Careful measurements have shown that both urea and guanidine hydrochloride also act by increasing the solubility of the non-polar amino-acid side-chains^{206, 207} while maintaining the hydrogen-bonding capacity of the aqueous solvent.²⁰⁸ Guanidine hydrochloride has some advantages over urea as a denaturing agent, since solutions of the latter slowly decompose to yield cyanate.²⁰⁹

In most cases that have been investigated, the transition between folded and unfolded states seems to be relatively sharp, i.e. there is a small range of concentration of denaturing agent over which the enzyme changes more or less completely from one form to the other. Recent studies on the unfolding of myoglobin by guanidine hydrochloride²⁰⁹ show that the number of binding sites for the denaturing agent increases as the protein unfolds; this provides a clue to the sharpness of the transition between folded and unfolded states. A two-state model, in which only the fully folded and fully unfolded states are considered, has generally provided a satisfactory description of the thermodynamic and some of the kinetic aspects of denaturation.²¹⁰

3.7.2 Refolding of enzymes

Several years ago, Anfinsen performed an important experiment that showed that a denatured enzyme could regain its correct folded structure when the denaturing agent was removed.²¹¹ This observation led to the conclusion that the primary structure of an enzyme contained the information required to specify its three-dimensional structure. (This conclusion is hardly surprising since the enzyme is synthesized *in vivo* as a polypeptide chain that folds up to assume its three-dimensional structure.) In outline, Anfinsen's experiment (Fig. 3.37) involved treating pancreatic ribonuclease with 2-mercaptoethanol to break the four disulphide bonds (Section 3.3.2) and then adding 8 mol dm⁻³ urea to produce inactive, unfolded, reduced enzyme. When the urea and reducing agent were removed by dialysis and the enzyme was oxidized to re-form the disulphide bonds, it was found that almost 100 per cent of the original enzyme activity was regained. Anfinsen argued that the complete regain of activity showed that the refolded enzyme possessed the correct disulphide bonds and therefore that the polypeptide chain had folded in such a way as to bring the correct pairs of cysteine side-chains into juxtaposition. There are 105 possible ways (8!/2⁴ × 4!) in which four disulphide bonds can be formed from eight cysteine side-chains,²¹² so

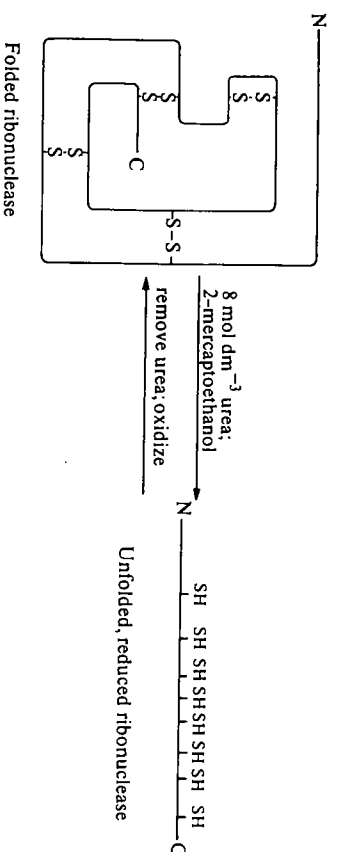


FIG. 3.37. The reduction and re-oxidation of pancreatic ribonuclease.

formation of the disulphide bonds on a random basis would lead to the regain of only about 1 per cent enzyme activity.

It has now become clear that the refolding of unfolded reduced, ribonuclease is a very complex process. Initially a number of distinct intermediates are observed in which different numbers of incorrect disulphide bonds are formed; these intermediates slowly rearrange to give the final, active product containing the correct disulphide bonds. For an account of this work, articles by Creighton^{213, 214} should be consulted. Interestingly, there is an enzyme, protein disulphide-isomerase, that catalyses the conversion of those species containing incorrect disulphide bonds to that containing the correct bonds. It seems likely that this enzyme ensures that *in vivo* only a protein containing the correct disulphide bonds will be formed.^{215, 216}

An important problem concerns the rate at which denatured proteins refold. It has been calculated²¹⁷ that it would take something like 10⁷⁷ years for a polypeptide chain of 100 amino acids to 'sample' all the possible conformations; this time should be compared with the age of the earth, i.e. 4.6 × 10⁹ years. Since proteins refold fairly quickly, on a time scale of seconds or minutes, it is clear that refolding does not involve a random search of all possible conformations until the most stable one is found. The current view²¹⁸ is that folding involves initially the formation of small local elements of structure that then act as nucleation centres to direct further folding of the polypeptide chain until the final folded state is reached. The elements of regular secondary structure such as α -helix or β -sheet (Section 3.5.4) are strong candidates for such nucleation centres. Attention has also been directed to the possibility that *cis* $\rightleftharpoons$ *trans* isomerization of prolines in the polypeptide chain (Section 3.5.4.1) may be a rate-determining step in the folding process,²¹⁹ and an enzyme (prolyl isomerase) catalysing this isomerization has been isolated.²²⁰ Studies on the refolding of denatured proteins can give valuable insights into the way the polypeptide chain acquires its three-dimensional structure during biosynthesis. However, it is important to realize that the two processes cannot always be compared, because it appears that the growing polypeptide chain generally acquires recognizable folded

structure while still attached to the ribosome. The complexities of the situation *in vivo* have been discussed.^{221, 222}

3.7.3 Structure-prediction methods

The experiment outlined in Fig. 3.37 shows that the amino-acid sequence of a protein specifies its three-dimensional structure. For some time, biochemists have been investigating the possibility of predicting the three-dimensional structure from the sequence data. An *a priori* approach, in which the most stable conformation of a polypeptide chain is calculated by taking into account interactions between non-bonded atoms, is extremely laborious and useful results have been obtained only for short peptide chains. A greater degree of success has been obtained with semi-empirical approaches in which known three-dimensional protein structures are analysed to determine the probability that a certain amino acid, or sequence of amino acids, occurs in a particular element of regular structure on a more than random basis.¹ With this information it is possible to draw up a set of rules to predict structural features in other proteins.^{2,12} The Chou-Fasman rules²²³ exemplify this approach. Amino acids are rated according to the likelihood that they occur in α -helices, β -sheets, or β -turns; for instance glutamic acid shows a strong preference for α -helices, a very low preference for β -sheets, and a middle to low preference for β -turns; proline and glycine have very low preferences for α -helices. The empirical rule for α -helix formation is that a cluster of four 'helix-forming' amino acids out of a sequence of six will nucleate a helix; the helical segment will extend in both directions until terminated by a sequence of four 'helix-breaking' amino acids. Analogous rules for the formation of β -sheet and β -turn structures have also been given.²²³

The Chou-Fasman and other prediction methods predict the elements of regular secondary structure (α -helix and β -sheet) with a success rate of about 60 per cent.²²⁴ While this achievement is noteworthy, it is clear that we are still some considerable way from being able to predict the structure of a protein.* It has been suggested that combinations of separate prediction methods can give more reliable results.²²⁶ The fact that the success rates of the prediction methods are only relatively modest implies that while short-range forces, extending over a few amino acids in the sequence, are important in specifying the final folded structure of a protein, the contribution of long-range forces between amino acids more remote in the sequence cannot be neglected.

3.8 Concluding remarks

The last two decades have witnessed an explosive growth in the amount of information concerning the structures of enzymes. This has been particularly

true for the three-dimensional structures revealed by X-ray crystallography. We are now in a position to discuss, for instance, the mechanism of enzyme catalysis in much greater detail than was previously possible. Chapter 5 illustrates the substantial progress that has been made in this direction.

Probably the aspect of protein structure over which the greatest question mark remains is that of the refolding of enzymes (Section 3.7.2) and the relationship between the amino-acid sequence and the three-dimensional structure. The X-ray crystallographic studies on kinases and dehydrogenases have shown that functionally related enzymes contain common elements of three-dimensional structure, although this regularity is not readily apparent when comparisons are made between the amino-acid sequences of these enzymes. It is reasonable to expect that, as more data are accumulated, the underlying basis of such structural regularities and of the evolutionary relationships between functionally related enzymes will become clearer.

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* An interesting example of the use of the structure-prediction methods occurred in the case of adenylate kinase. Several workers in the structure-prediction field were given the amino-acid sequence of the enzyme *before* the results of X-ray crystallography became available. A comparison of the actual structure with the various predicted structures was then made.²²⁵

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4 An introduction to enzyme kinetics

4.1 Outline of the chapter

The subject of enzyme kinetics often generates considerable trepidation in biochemistry students. Many treatments of the topic make two assumptions: firstly, that the readers can confidently handle complex algebraic equations, and, secondly, that a detailed knowledge of kinetics is an indispensable part of a biochemist's training. In this chapter we do not make these assumptions but instead introduce some of the main ideas in enzyme kinetics that are useful in understanding the mechanism of action and control of isolated enzymes (see Chapters 5 and 6) and the role of enzymes in the cell (see Chapter 8). The emphasis will be on the information that can be gained from a study of enzyme kinetics.

The first part of the chapter (Section 4.2) describes some practical aspects of enzyme kinetics, answering the question 'How do we obtain kinetic data?' The next part (Section 4.3) deals with the analysis of the results, discussing the theoretical background, equations involved, and information obtained. In these sections we shall be dealing with 'steady-state kinetics', where an enzyme is present at very small molar concentrations (usually much less than 1 per cent) compared with the molar concentration(s) of substrate(s) acted upon. Under these conditions, the equations involved are *comparatively* straightforward and the data are usually collected over a time scale of minutes, so that 'conventional' methods of mixing and observation can be employed. In Section 4.4 we refer in outline to experiments in which enzyme and substrate(s) are present at comparable concentrations. Under these conditions it is usually necessary to use special techniques to ensure rapid mixing and observation. This type of experiment has given detailed insights into the various steps in an overall reaction, particularly those occurring in the early, 'pre-steady-state' period (see Fig. 4.2). In fact, these studies are often termed 'pre-steady-state kinetics'. As described in Chapter 8 (Section 8.5.4) there are many examples known where, in the cell, the concentration of an enzyme is comparable with that of its substrate and in such cases the study of pre-steady-state kinetics is particularly relevant.

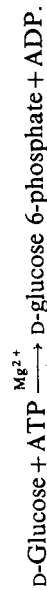
We end this section by stating three of the reasons why it is important for the biochemistry student to have some knowledge of enzyme kinetics. Firstly, kinetics, *in conjunction with other techniques*, provides valuable information on the mechanism of action of an enzyme (see Chapter 5). Secondly, it can give an insight into the role of an enzyme under the conditions that exist in the cell and the response of an enzyme to changes in the concentrations of metabolites (see

Chapter 8). Thirdly, it can help to show how the activity of an enzyme can be controlled, which may provide a valuable pointer to mechanisms of regulation under physiological conditions (see Chapters 6 and 8).

4.2 How do we obtain kinetic data?

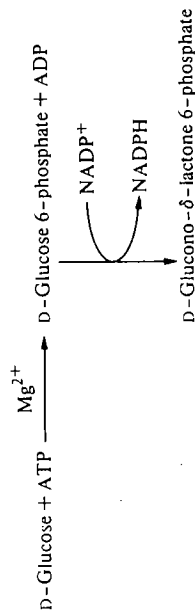
The aim of an experiment is to measure the rate of formation of product (or disappearance of substrate) under specified conditions. It is then possible to vary in turn certain parameters such as the concentration of substrate(s), pH, temperature, or concentration of modifying ligands, and collect data to analyse in terms of theoretical models.

The rate of a particular enzyme-catalysed reaction can often be measured in a number of ways, but there is normally one method that is more convenient than the others. This point can be illustrated by reference to the reaction catalysed by hexokinase:



The rate of this reaction could be monitored by removing samples from the reaction mixture at known times after addition of enzyme, stopping the reaction quickly (e.g. by addition of acid to inactivate the enzyme) and measuring the amount of product formed. Ion-exchange chromatography (see Chapter 2, Section 2.6.2.1) would be a useful technique for separating the products from the substrates in this case. Clearly such a 'stop and sample' (or *discontinuous*) assay procedure involves possible sampling errors and considerable work in separation and estimation of the products.

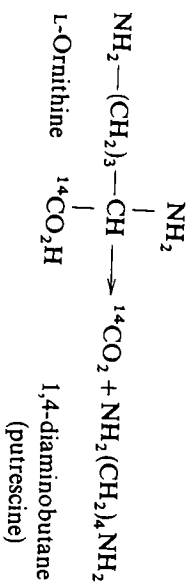
A more convenient method would involve the *continuous* measurement of some property that changes during the course of the reaction. In the case of the reaction above, there is no convenient change in absorbance, for example, but we could bring about such a change if we *couple* the production of D-glucose 6-phosphate to the reduction of NADP⁺ to NADPH using glucose-6-phosphate dehydrogenase:



NADP⁺ does not absorb at 340 nm, whereas NADPH does, so it is possible to monitor the production of NADPH (and hence of D-glucose 6-phosphate) continuously. If we use a *coupled assay procedure* such as this we must add sufficient coupling enzyme and substrate(s) so that the D-glucose 6-phosphate

formed in the first step is 'immediately' converted to D-glucose- δ -lactone 6-phosphate, i.e. so that the coupling reaction is not rate limiting. Several detailed analyses of the kinetics of coupled assay systems have indicated the conditions under which the true rate of the reaction of interest can be measured.^{1, 2, 3, 4, 5} It is, of course, obvious that the coupling enzyme should be highly purified and certainly free from detectable quantities of the enzyme we are trying to assay.

In a number of cases it is not possible to monitor the reaction continuously (either directly on the reaction of interest or by using a coupled assay) so a 'stop and sample' method must be employed. For instance, in the reaction catalysed by ornithine decarboxylase,



if L-[1- ^{14}C]ornithine is used, the CO_2 liberated will be radioactive. This CO_2 can be trapped by a suitable base (e.g. ethanolaniline dissolved in 2-methoxyethanol) and the radioactivity estimated by scintillation counting. Determination of the amount of CO_2 formed after various times of reaction allows the rate of the reaction to be calculated. Assay procedures involving radioactively labelled substrates are very sensitive and are of particular value when low concentrations of substrates are being used or when the amount of enzyme activity is low.

When the method of assay has been decided upon, it is important to take a number of precautions to obtain reliable data.^{6, 7}

1. The substrates, buffers, etc. should be of as high a purity as possible, since contaminants may affect the activity of enzymes. For example, commercial preparations of NAD^+ sometimes contain inhibitors of dehydrogenases^{8, 9} and it has been found that certain preparations of ATP contain trace amounts of vanadate (VO_4^{3-}) ions, which act as a powerful inhibitor of adenosinetriphosphatase (Na^+ , K^+ -activated).¹⁰
2. It must be ascertained that the enzyme preparation does not contain any compound (or other enzyme) that interferes with the assays. The possibility of non-enzyme-catalysed conversion of substrate to product should be tested by performing appropriate control experiments (e.g. using heat-inactivated enzyme).
3. The enzyme should be stable (i.e. not lose any significant amount of catalytic activity) during the time taken for assay. Breakdown of substrate (other than by the enzyme) should not occur.
4. Since the activity of an enzyme can be markedly affected by changes in pH, temperature, etc., it is important to ensure that these parameters are stabilized by use of buffers, thermostatted baths, etc.

5. It should be checked that, once the steady state has been achieved (Section 4.3.1.1), the measured rate of reaction is constant over the period of interest and is proportional to the amount of enzyme added.
6. The *initial* rate of reaction should be measured to avoid possible complications arising from product inhibition, occurrence of the reverse reaction and depletion of substrate.^{11, 12}

4.3 How do we analyse kinetic data?

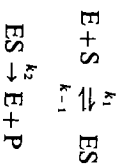
As in many other branches of experimental science, kinetic data are analysed in terms of theoretical models in order to test the correctness of the models and to deduce the values of constants in the equations derived from the models. (In this section we shall describe for a number of situations the theoretical background to steady-state kinetics, then show how to treat data in terms of the resulting equations, and finally describe the significance of the results obtained.)

4.3.1 One-substrate reactions

It is easiest to deal initially with reactions in which only one substrate is acted on by an enzyme. This would include reactions catalysed by enzymes in the following groups (see Chapter 1, Section 1.5.1): hydrolases (if H_2O is considered to be in a large excess), isomerases, and most lyases.

4.3.1.1 Theoretical background

We shall assume that catalysis occurs via rapid and reversible formation of a complex between enzyme, E , and substrate, S . (The part of the enzyme to which the substrate binds is known as the *active site* of the enzyme.) This complex then breaks down in a slow step to give the product, P , and regenerate enzyme (Scheme 1). In practice this scheme is likely to be an oversimplification (see Section 4.3.1.4).



Scheme 1. The conversion of substrate to product in an enzyme catalysed reaction. k_1 , k_{-1} , k_2 represent the rate constants for the individual steps.

(The equation describing the kinetics of this scheme can be derived by making one of two types of assumption as follows.)

(i) The equilibrium assumption

Here we assume that the $\text{E} + \text{S} \rightleftharpoons \text{ES}$ equilibrium is only slightly disturbed by the breakdown of ES to give product. This is clearly a better assumption the lower the value of k_2 relative to k_{-1} .

The equilibrium constant, K , is defined by

$$K = \frac{[E][S]}{[ES]}, \quad (4.1)$$

where $[E]$ and $[S]$ are the concentrations of free enzyme and free substrate, respectively. However, since the total concentration of substrate is much greater than the total concentration of enzyme (see Section 4.1), essentially all the substrate is free and we can set $[S]_{\text{free}}$ equal to $[S]_{\text{total}}$.

At any concentration of S we can evaluate the fraction, F , of enzyme present as ES as follows:

$$F = \frac{[ES]}{[E] + [ES]}.$$

Now, from (4.1)

$$[ES] = \frac{[E][S]}{K},$$

$$\begin{aligned} \therefore F &= \frac{[E][S]}{K} \bigg/ \left([E] + \frac{[E][S]}{K} \right) \\ &= \frac{[S]}{K + [S]}. \end{aligned} \quad (4.2)$$

If the total concentration of enzyme is $[E]_0$, then $[ES] = F[E]_0$; thus, from (4.2)

$$[ES] = \frac{[E]_0[S]}{K + [S]}.$$

The rate of product formation, v , is given by

$$\begin{aligned} v &= k_2[ES] \\ \therefore v &= \frac{k_2[E]_0[S]}{K + [S]}. \end{aligned} \quad (4.3)$$

Sometimes the equation (4.3) is written in the form

$$v = \frac{k_{\text{cat}}[E]_0[S]}{K + S}$$

where k_{cat} is a first-order rate constant equal to k_2 (k_{cat} is sometimes known as the turnover number of the enzyme). This formulation will be referred to later (see Section 4.3.1.3).

An equation of the type shown in (4.3) means that v will tend towards a maximum (or limiting) value as $[S]$ increases (Fig. 4.1).

The maximum rate will be observed when all the enzyme is in the form of the ES complex. Let the maximum (limiting) rate $= V_{\text{max}}$; this will be equal to $k_2[E]_0$; then eqn (4.3) can be rewritten as:

$$v = \frac{V_{\text{max}}[S]}{K + [S]}. \quad (4.4)$$

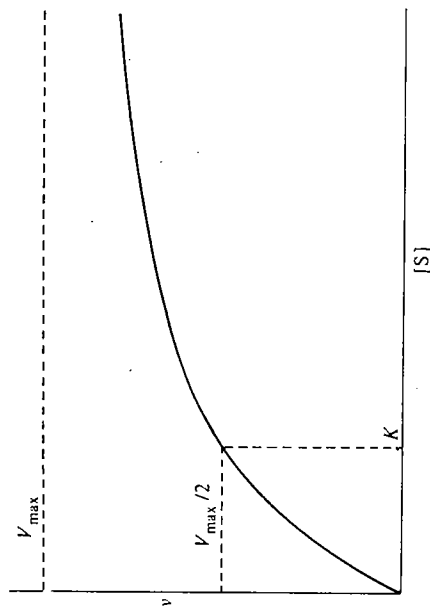


FIG. 4.1. Dependence of velocity (v) on substrate concentration ($[S]$) according to eqn (4.3).

We should note from eqn (4.4) and Fig. 4.1 that when $[S]$ is small compared with K , the reaction is first order in $[S]$, since $v = V_{\text{max}}[S]/K$. When $[S]$ is large compared with K , the reaction is zero order in $[S]$, since $v = V_{\text{max}}$. At intermediate values of $[S]$, the reaction is of a fractional order in $[S]$.

When $v = V_{\text{max}}/2$, $[S] = K$. Thus K corresponds to the concentration of substrate when the velocity is half-maximal. It is known as the Michaelis* constant, and is normally written as K_m . If we have used the equilibrium assumption, we can equate K_m with the dissociation constant of the ES complex (K in eqn (4.1)).

(ii) The steady-state assumption

In this approach, we abandon the assumption that the $E + S \rightleftharpoons ES$ equilibrium is not perturbed by the breakdown of ES. Instead, it is assumed that ES is in a 'steady state', i.e. that the concentration of ES remains constant because the rate of its formation equals the rate of its breakdown. If we were to examine the variation of $[ES]$ with time in a typical experiment we would obtain a graph of the type shown in Fig. 4.2.

After an initial phase (the pre-steady-state period) the concentration of ES remains fairly constant and it is thus in order to apply the steady-state assumption to Scheme 1 to evaluate the fraction of enzyme in the form of the ES complex. (For the steady-state assumption to be valid the rate of change of $[ES]$ must be small compared with the rate of change of $[S]$ or $[P]$. In experiments in which the concentration of substrate is much greater than that of enzyme, the maximum value of $[ES]$, and hence the rate of change of $[ES]$, will be small.)

* Michaelis was one of the first workers to develop the mathematical analysis of enzyme kinetics along the lines indicated here. An equation of the type (4.4) is often referred to as the Michaelis-Menten equation.

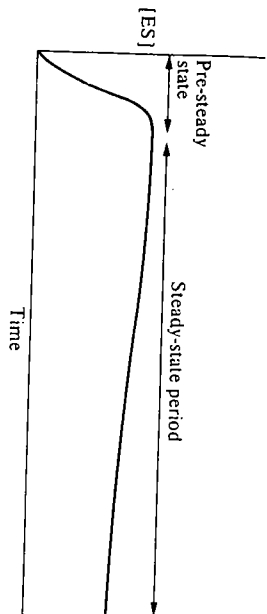


FIG. 4.2. The concentration of the ES complex as a function of time of reaction.

Now, according to the steady-state assumption, the rate of production of $ES (= k_1[E][S])$ must equal the rate of its breakdown $(= k_{-1}[ES] + k_2[ES])$

$$\therefore k_1[E][S] = k_{-1}[ES] + k_2[ES]$$

$$\therefore [ES] = \frac{k_1[E][S]}{k_{-1} + k_2}$$

Proceeding as before,

$$F = \frac{[ES]}{[E] + [ES]}$$

$$= \frac{[S]}{\left[\frac{k_{-1} + k_2}{k_1} \right] + [S]}$$

If the total concentration of enzyme is $[E]_0$, then

$$[ES] = F[E]_0$$

$$= \frac{[E]_0[S]}{\left[\frac{k_{-1} + k_2}{k_1} \right] + [S]}$$

and the rate of product formation, v , is given by:

$$v = k_2[ES]$$

$$= \frac{k_2[E]_0[S]}{\left[\frac{k_{-1} + k_2}{k_1} \right] + [S]}$$

Putting $k_2[E]_0 = V_{\max}$

$$v = \frac{V_{\max}[S]}{\left[\frac{k_{-1} + k_2}{k_1} \right] + [S]}$$

(4.5)*

* This equation is sometimes referred to as the Briggs-Haldane equation because these workers (in 1925) first applied the steady-state assumption to enzyme kinetics.

Thus using the steady-state approximation, the term K in the denominator of eqn (4.4) has been replaced by $(k_{-1} + k_2)/k_1$. Only if $k_2 \ll k_{-1}$, when the *equilibrium assumption* becomes a *limiting case of the steady-state assumption*, does this latter term become equal to the dissociation constant of the ES complex ($K = k_{-1}/k_1$). We should therefore find that the Michaelis constant, the concentration of substrate when the velocity is half-maximal, is not generally equal to this dissociation constant.

Using either assumption, we derive the fundamental equation (4.6) to describe the kinetics of the reaction shown in Scheme 1:

$$v = \frac{V_{\max}[S]}{K_m + [S]} \quad (4.6)$$

4.3.1.2 Treatment of data

It is very difficult to determine the limiting value of v (i.e. $V_{\max}$) directly from a plot of v against $[S]$ (Fig. 4.1) and therefore K_m cannot readily be determined in this way either. To overcome these difficulties, eqn (4.6) can be rearranged in a number of ways to give convenient graphical representations. Four of the best known rearranged forms are given below:

(i) The Lineweaver-Burk equation¹³

This equation is obtained by taking reciprocals of the two sides of eqn (4.6):

$$\frac{1}{v} = \frac{K_m}{V_{\max}} \cdot \frac{1}{[S]} + \frac{1}{V_{\max}}$$

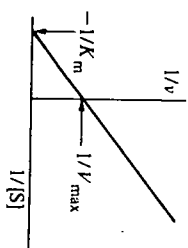
Thus a plot of $1/v$ against $1/[S]$ gives a straight line of slope $K_m/V_{\max}$ and with intercepts on the x and y axes of $-1/K_m$ and $1/V_{\max}$, respectively (Fig. 4.3(a)).

(ii) The Eadie-Hofstee equation^{14, 15}

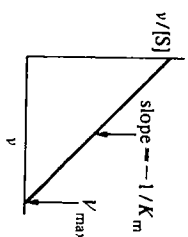
Equation (4.6) is rearranged to give:

$$\frac{v}{[S]} = \frac{V_{\max}}{K_m} - \frac{v}{K_m}$$

(a)



(b)



(c)

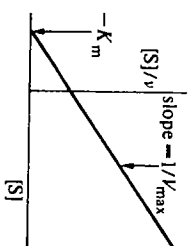


FIG. 4.3. Graphical representations of enzyme kinetic data according to the equations of (a) Lineweaver and Burk, (b) Eadie and Hofstee, and (c) Hanes. It should be remembered that the equation of a straight line is $y = mx + c$, where m equals the slope and c the intercept on the y -axis. The intercept on the x -axis equals $-c/m$.

Thus a plot of $v/[S]$ against v gives a straight line of slope $-1/K_m$ and with an x-axis intercept of $V_{\max}$ (Fig. 4.3(b)).

(iii) *The Hanes equation*¹⁶

Equation (4.6) is rearranged to give:

$$\frac{[S]}{v} = \frac{[S]}{V_{\max}} + \frac{K_m}{V_{\max}}.$$

Thus a plot of $[S]/v$ against $[S]$ is linear with a slope of $1/V_{\max}$ and an x-axis intercept of $-K_m$ (Fig. 4.3(c)).

(iv) *The direct linear plot*^{17, 18}

This method uses a rather different approach from those described in (i)–(iii). Equation (4.6) is arranged to give:

$$V_{\max} = v + \frac{v}{[S]} \cdot K_m \quad (4.7)$$

$V_{\max}$ and K_m are now treated as variables and v and $[S]$ as constants. If the first pair of observed values of v and $[S]$ (i.e. v_1 and $[S]_1$) are plotted as shown in Fig. 4.4(a) then reference to eqn (4.7) shows that the line connecting them describes pairs of values of $V_{\max}$ and K_m that are consistent with the observed values of v and $[S]$.

If we have a second set of values of v and $[S]$ (i.e. v_2 and $[S]_2$), a new line can be drawn (Fig. 4.4(b)). The point of intersection of the lines defines uniquely the values of $V_{\max}$ and K_m that satisfy the two sets of data. This procedure can be repeated for further data points (Fig. 4.4(b)). If there were no experimental error the various lines would all intersect at a common point. In practice a number of points of intersection are obtained (the maximum number of such points is $n(n-1)/2$, where n is the number of observations of v and $[S]$). The correct procedure is then to use the medians* of the values of $V_{\max}$ and K_m as the best-fit values.^{17, 18, 19}

It is not possible to give simple answer to the question of which of the various methods of plotting data should be used. In any case it should be emphasized that the data should be as good as possible, since no graphical transformation allows sound conclusions to be drawn from poor data. The Lineweaver–Burk plot (Fig. 4.3(a)) is still the most commonly used and it has the advantage that the variables v and $[S]$ are plotted on separate axes. However, an analysis of the errors involved in the collection of the data (and hence in the determination of the parameters K_m and $V_{\max}$) shows that there is a highly non-uniform distribution of errors over the range of values of $1/v$ and $1/[S]$ in the Lineweaver–Burk plot.²⁰ For this reason, the use of the Eadie–Hofstee and Hanes plots has been recommended, since in these plots the distribution of errors is more uniform.²⁰

* The median is the middle value of a set of values arranged in order of magnitude, or the mean of the middle pair if the number of values is even.

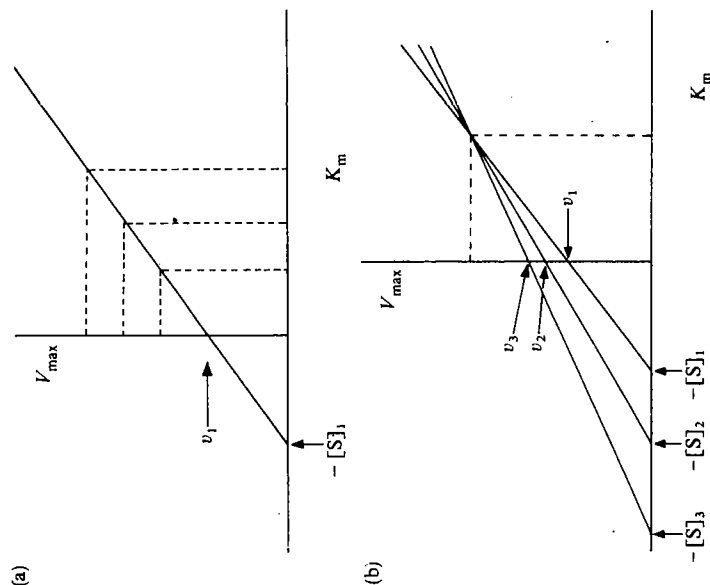


FIG. 4.4. The direct linear plot in which experimental values of v and $[S]$ are plotted directly. (a) A straight line drawn through one set of experimental values; the dashed lines indicate pairs of values of $V_{\max}$ and K_m that are consistent with the experimental values. In (b) a number of sets of experimental values are plotted. The point of intersection of the various lines gives the values of $V_{\max}$ and K_m that uniquely satisfy the experimental data.

The direct linear plot has a number of advantages: (i) Values of v and $[S]$ are plotted directly so that $V_{\max}$ and K_m can be determined without the need for calculations. (ii) It is statistically sound: the use of median values of $V_{\max}$ and K_m minimizes the influence of extreme values of v and $[S]$ on these parameters.¹⁹ However, the direct linear plot has some disadvantages. It is not very suitable for the display of data in multisubstrate reactions (Section 4.3.5) because of the resulting large number of lines. In addition, it is not easy to detect departures from the basic equation (4.6).

In recent years (especially with the greater availability of microcomputers) it has become more common to obtain the 'best fit' values of $V_{\max}$ and K_m by use of computer programs in which the data are fitted directly to eqn (4.6).^{21, 22, 23} A number of such programs are available commercially.

Finally, it should be mentioned that determinations of $V_{\max}$ and K_m can also be made from the integrated form of eqn (4.6). Substrate (or product) concentrations are monitored over a large fraction of the overall reaction, i.e. well beyond

the 'initial rate' period. The values of $V_{\max}$ and K_m can be determined from a single progress curve that essentially represents a series of rate measurements at different values of substrate concentration.^{12, 23, 24, 25} One way in which this is done is described in Appendix 4.1. A major disadvantage of the method is that the accuracy of the estimates of $V_{\max}$ and K_m can be very sensitive to errors in the estimates of the end-point of the reaction.¹² In addition, the progress curve can be used to calculate the true initial rates of reaction in those cases where there has been a significant decline in the rate during the time taken for mixing the assay components and obtaining readings.^{12, 23, 25}

The units of K_m and $V_{\max}$ will be those in which $[S]$ and v are measured. Thus K_m is expressed in units of concentration, i.e. mol dm^{-3} . The velocity, v , can be expressed in a number of ways, depending on the information available. For a purified enzyme, v is expressed as moles substrate consumed per unit time, per weight of enzyme. In the SI system this would be in terms of katal kg^{-1} , whereas in the older system it would be in terms of units mg^{-1} (see Chapter 2, Section 2.3.1). If the M_r of the enzyme is also known, we can calculate the molar concentration of enzyme active sites in the solution and hence evaluate k_{cat} ($= V_{\max}/[E]_0$), which is also known as the *turnover number* of the enzyme (see Section 4.3.1.1).

4.3.1.3 Significance of results

The parameters K_m and $V_{\max}$ (k_{cat}) are of value (i) in characterizing the specificity of an enzyme for a particular substrate, (ii) in deciding between steady-state and equilibrium mechanisms, and (iii) in indicating the role of an enzyme in metabolism.

(i) The specificity of an enzyme for a substrate can be described as follows: From eqn (4.3) we have

$$v = k_{\text{cat}} \frac{[E]_0 [S]}{K_m + [S]}$$

Substituting $[E]_0 = [E] + [ES]$ and noting that $[ES] = [E][S]/K_m$ we have

$$v = \frac{k_{\text{cat}}}{K_m} [E] [S] \quad (4.8)$$

Thus k_{cat}/K_m is an apparent second-order rate constant that describes the rate in terms of the concentrations of the free enzyme and free substrate.

If there are two competing substrates S_1 and S_2 for the enzyme, it follows from eqn (4.8) that the rates of reaction are

$$v_{S_1} = \left(\frac{k_{\text{cat}}}{K_m} \right)_{S_1} [E] [S_1]$$

$$v_{S_2} = \left(\frac{k_{\text{cat}}}{K_m} \right)_{S_2} [E] [S_2]$$

The ratio of these rates of reaction is given by

$$\frac{v_{S_1}}{v_{S_2}} = \frac{(k_{\text{cat}}/K_m)_{S_1} [S_1]}{(k_{\text{cat}}/K_m)_{S_2} [S_2]}$$

Thus, at equal concentrations of S_1 and S_2 , the relative rates of reaction of the two substrates are determined by the relative values of k_{cat}/K_m . The ratio k_{cat}/K_m can thus be used as a measure of the specificity of an enzyme for a substrate. For example, in the reaction catalysed by fumarate hydratase, the ratios k_{cat}/K_m for fumarate, fluorofumarate, chlorofumarate and bromofumarate are 1.6×10^8 , 9.8×10^7 , 2.0×10^5 and $2.5 \times 10^4 \text{ s}^{-1} (\text{mol dm}^{-3})^{-1}$, respectively.²⁶ There is thus high specificity shown towards fumarate and fluorofumarate; substitution by the larger halogens leads to a marked decline in the reactivity of the substrate.

(ii) The ratio of k_{cat}/K_m can be used to test for the applicability of the steady-state or equilibrium mechanisms (Section 4.3.1.1). Using the steady-state assumption, we note from eqns (4.5) and (4.6) that

$$K_m = \frac{k_{-1} + k_2}{k_1}$$

Now, if $k_2 \gg k_{-1}$ (which is the extreme form of the steady-state assumption), then $K_m = k_2/k_1$. Replacing k_2 by k_{cat} , it is clear that k_{cat}/K_m is equal to k_1 , the rate constant for the association of enzyme with substrate. From fast-reaction studies (Section 4.4.3) it is known that the diffusion-controlled rate constant for association of enzyme with substrate is of the order of $10^9 (\text{mol dm}^{-3})^{-1} \text{ s}^{-1}$. Thus, if k_{cat}/K_m is of this order of magnitude, it can be concluded that the steady-state mechanism operates. This is the case for fumarate hydratase, catalase, and triosephosphate isomerase, for which k_{cat}/K_m values are 1.6×10^8 , 4×10^7 and $2.4 \times 10^8 \text{ s}^{-1} (\text{mol dm}^{-3})^{-1}$, respectively.²⁷ However, if the value of k_{cat}/K_m is much lower than these values, it is reasonable to conclude that the equilibrium mechanism is more appropriate.

(iii) The role of an enzyme in metabolism may be judged by relating the K_m value to the prevailing concentration of substrate. This will be discussed further in Chapter 8 (Section 8.3.2) but two examples will serve to illustrate the ideas.

The isoenzyme IV of hexokinase (also known as glucokinase) is confined to the liver and has a high K_m for glucose (10 mmol dm^{-3}), whereas isoenzymes I–III have a much wider tissue distribution and a low K_m ($40 \text{ } \mu\text{mol dm}^{-3}$). Thus, at the prevailing levels of blood glucose in a fasting subject (ca. 3 mmol dm^{-3}), isoenzymes I–III are working essentially at their maximum velocity, whereas isoenzyme IV is only working at about 25 per cent of its maximum velocity. After an intake of carbohydrate, the level of blood glucose rises to about 9.5 mmol dm^{-3} and the isoenzyme IV is now working at about half its maximum velocity (Fig. 4.5). Hence the liver isoenzyme (IV) can 'deal with' this extra glucose, converting it to D-glucose 6-phosphate, which is the first step in the process of storage as glycogen.

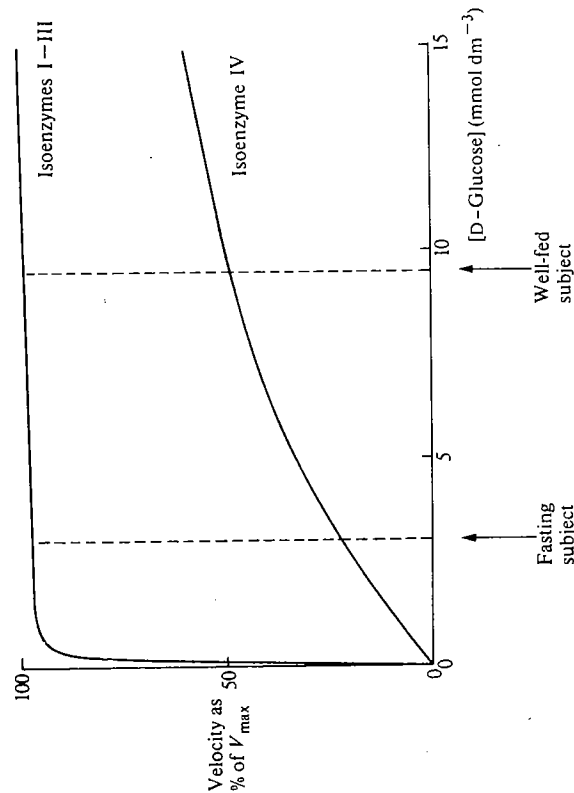


FIG. 4.5. The variation in velocity of the reactions catalysed by hexokinase isoenzymes with concentration of D-glucose.

The studies on ribulose biphosphate carboxylase (see Chapter 2, Section 2.8.2), which catalyses a reaction involving assimilation of CO_2 in photosynthetic organisms, illustrate the need to exercise care in interpreting data on K_m and $V_{\max}$. This enzyme is present in very large amounts in such organisms and is relatively easy to purify. However, early measurements showed that the activity of the purified enzyme was far too low to account for the observed rates of photosynthetic CO_2 fixation, and that, in particular, the K_m for CO_2 was about 50-fold higher than the apparent K_m measured in intact leaves or isolated chloroplasts. The solution to this problem is now thought²⁸ to lie in the fact that the enzyme is slowly activated by CO_2 and Mg^{2+} , and so considerable care has to be taken in performing and interpreting the assays of the purified enzyme. Activated enzyme has a K_m for CO_2 that is comparable with the apparent K_m in the intact system.

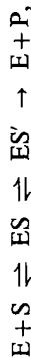
From this second example it is clear that it can be misleading to interpret the parameter K_m in anything other than purely operational terms (i.e. the concentration of substrate at which the velocity is half maximal). It is *sometimes* helpful to think of K_m as a crude measure of the affinity of an enzyme for its substrate (a high K_m implying weak affinity), but the comment made at the end of Section 4.3.1.1 must be borne in mind.

4.3.1.4 Some extensions to the simple model (Scheme 1)

It is worthwhile at this stage to mention some extensions to the relatively simple model (Scheme 1) we have considered up to now.

(i) More than one intermediate

In most cases it is probably unrealistic to suggest that only one intermediate (enzyme-containing complex) is involved in the reaction pathway (Scheme 1). There is likely to be a 'Michaelis complex' representing the initial (non-covalent) association of enzyme and substrate and at least one additional complex in which some bond rearrangement has taken place. In the case of the hydrolysis of ester and amide substrates catalysed by chymotrypsin, there is clear evidence for the involvement of an acyl-enzyme intermediate (Chapter 5, Section 5.4.1.5). If we extend Scheme 1 as follows:



then it can be shown by applying the steady-state assumption to the various intermediates in the pathway (ES and ES') that the rate equation is of the same form as eqn (4.6) but that the interpretation of K_m in terms of the rate constants of the individual steps will be different from that in the simple case (eqn (4.5)).²⁹

(ii) Substrate inhibition

Occasionally it is found that there is a decrease in v at high substrate concentrations. In the Lineweaver-Burk plot (Fig. 4.3(a)), for example, this would be manifested by an upward curvature at low values of $1/[S]$ as in the example of the hydrolysis of D-fructose 1,6-bisphosphate (FBP) catalysed by fructose-bisphosphatase, where the upward curvature is noted at FBP concentrations above 0.1 mmol dm^{-3} (Fig. 4.6).

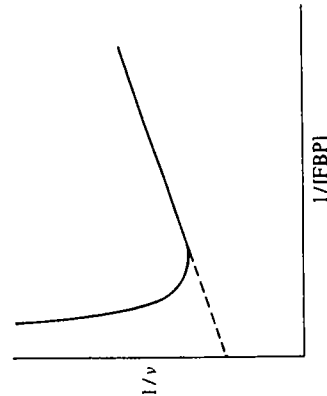


FIG. 4.6. Lineweaver-Burk plot for the reaction catalysed by fructose-bisphosphatase. The solid line represents the experimental data and shows that substrate inhibition occurs at high concentrations of FBP (above about 0.1 mmol dm^{-3}).

This phenomenon is known as substrate inhibition and is usually interpreted in terms of the existence of two types of substrate-binding site in the enzyme. Occupation of the first, high-affinity, type at low $[FBP]$ leads to 'normal' kinetic behaviour (the linear part of Fig. 4.6). At high $[FBP]$, the second, low-affinity, type of site becomes occupied and this is presumed to inhibit the catalytic

reaction taking place at the first type of site. For a fuller discussion of this phenomenon, see reference 30.

(iii) Multiple active sites

Complex kinetics are often observed when an enzyme is composed of a number of subunits and possesses more than one active site. If interactions occur between these various sites non-linearly in the kinetic plots (Fig. 4.3) will be observed, characteristic of positive or negative cooperativity. These points are discussed in more detail in Chapter 6 (Section 6.2.2.2).

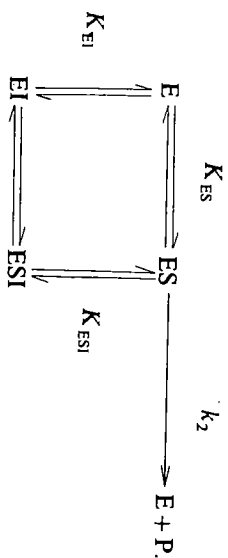
Other more detailed texts should be consulted for details of additional complications to the simple model (Scheme 1) such as inhibition by products^{31, 32} and the occurrence of reverse reactions.^{33, 34, 35}

4.3.2 Inhibition of one-substrate reactions

The study of the effects of inhibitors on enzyme-catalysed reactions is important not only to introduce various terms such as competitive inhibition, but also to give information on the active site of an enzyme (see Chapter 5, Section 5.4.1.3) and on inhibition which may be of possible physiological significance (see Chapter 6, Section 6.2.2.1). We shall confine our attention to *reversible inhibitors*, i.e. inhibitors which combine reversibly with an enzyme* rather than those which cause irreversible covalent modification. The use of the latter in the determination of enzyme mechanisms is discussed in Chapter 5, Section 5.4.4.

4.3.2.1 Theoretical background

One of the ways in which inhibition of enzyme-catalysed reactions can be discussed is in terms of a general scheme shown below (Scheme 2).



Scheme 2. A general scheme for inhibition of enzyme-catalysed reactions. K_{ES} , K_{ESI} , and K_{EI} represent dissociation constants. ESI is assumed to be inactive.

We shall assume that the enzyme-containing complexes are in equilibrium with each other, i.e. that the breakdown of ES to generate product does not significantly disturb the equilibrium. A general kinetic equation for this scheme

* The effect is reversible if it is decreased by lowering the concentration of the inhibitor (e.g. by dilution or dialysis). The distinction between reversible and irreversible inhibition is not absolute and may be difficult to make if the inhibitor binds very tightly to the enzyme and is released very slowly. In these circumstances the inhibitors are termed tight-binding inhibitors.³⁶

can then be derived as described in Appendix 4.2 to this chapter:

$$v = V_{\max} \frac{[S]}{1 + \frac{[S]}{K_{ES}} + \frac{[I]}{K_{EI}} + \frac{[S][I]}{K_{ESI}K_{ES}}}$$

or in reciprocal form:

$$\frac{1}{v} = \frac{1}{V_{\max}} \left[1 + \frac{[I]}{K_{ESI}} \right] + \frac{K_{ES}}{V_{\max}} \left[1 + \frac{[I]}{K_{EI}} \right] \frac{1}{[S]} \quad (4.9)$$

If we apply the steady-state approximation to Scheme 2, a complex kinetic expression is derived that is difficult to test experimentally,³⁷ and will not be discussed further here.

The general equation (4.9) is simplified if certain assumptions are made about the magnitudes of the various dissociation constants. These *limiting cases* will be discussed in terms of the Lineweaver-Burk plots (Fig. 4.3(a)) of kinetic data, but we could, of course, use the other plots shown in Figs. 4.3 and 4.4. A more detailed classification of the various types of inhibition³⁶ and a description of the Dixon plot for the determination of inhibitor constants are given in Appendix 4.2.

4.3.2.2 Treatment of data

Case (i). Competitive inhibition

If we assume that $K_{ESI} = \infty$ (i.e. that the ES complex cannot combine with I nor the EI complex with S), then eqn (4.9) reduces to

$$\frac{1}{v} = \frac{1}{V_{\max}} + \frac{K_{ES}}{V_{\max}} \left[1 + \frac{[I]}{K_{EI}} \right] \frac{1}{[S]}$$

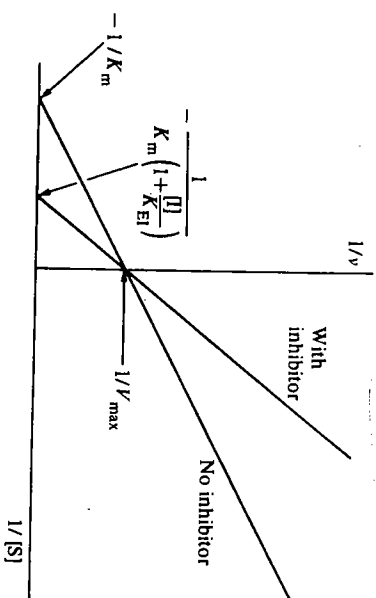
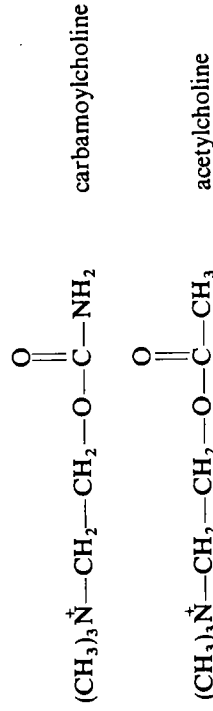


FIG. 4.7. The effect of a competitive inhibitor on a Lineweaver-Burk plot of enzyme kinetic data.

and the effect of the inhibitor on the Lineweaver-Burk plot is as shown in Fig. 4.7. $V_{\max}$ is unaffected but the apparent K_m is increased by a factor $(1 + [I]/K_{EI})$. Effectively, the inhibitor pulls some of the enzyme over into the form of the EI complex. When the concentration of S is increased sufficiently, the effect on the velocity can be overcome (thus giving rise to the term 'competitive'). Many examples of competitive inhibition are found in one-substrate reactions; thus, carbamoylcholine is a competitive inhibitor with respect to acetylcholine in the reaction catalysed by acetylcholinesterase from bovine erythrocytes:



Although in this case the structural similarity between inhibitor and substrate makes it very likely that the two molecules bind to the same site on the enzyme, it is not always a justified conclusion that a competitive inhibitor binds to the active site.³⁸

Case (ii). Non-competitive inhibition

If we assume that $K_{ESI} = K_{EI}$ (i.e. that the binding of S to the enzyme does not affect the binding of I), then eqn (4.9) reduces to

$$\frac{1}{v} = \frac{1}{V_{\max}} \left[1 + \frac{[I]}{K_{EI}} \right] + \frac{K_{ES}}{V_{\max}} \left[1 + \frac{[I]}{K_{EI}} \right] \frac{1}{[S]}$$

and the effect of the inhibitor on the Lineweaver-Burk plot is as shown in Fig. 4.8.

K_m remains unaffected, but $V_{\max}$ is decreased by a factor

$$1 / \left(1 + \frac{[I]}{K_{EI}} \right)^*$$

The inhibitor pulls both E and ES over into inactive forms (EI and ESI, respectively) but does not affect the distribution between them. Hence, increasing the concentration of S does not serve to overcome the effect of the inhibitor on the velocity. Examples of non-competitive inhibition in one substrate reactions are less common than those of competitive inhibition. In the case of fructose-bisphosphatase, AMP acts as a non-competitive inhibitor with respect to the substrate fructose 1,6-bisphosphate. There are many examples, however, of non-competitive inhibition in multisubstrate reactions (see Section 4.3.5.5).

* Reference to Appendix 4.2 shows that this situation is more correctly termed pure non-competitive inhibition.

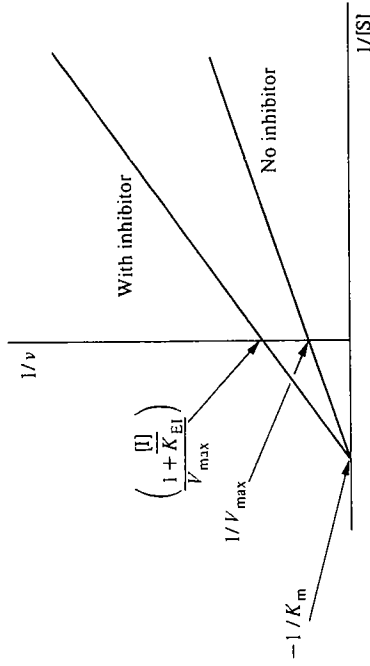


FIG. 4.8. The effect of a non-competitive inhibitor on a Lineweaver-Burk plot of enzyme kinetic data.

Case (iii). Uncompetitive inhibition

If we assume that $K_{EI} = \infty$ (i.e. that I cannot combine with E, but only with the ES complex) then eqn (4.9) reduces to

$$\frac{1}{v} = \frac{1}{V_{\max}} \left[1 + \frac{[I]}{K_{ESI}} \right] + \frac{K_{ES}}{V_{\max}} \frac{1}{[S]}$$

and the effect of the inhibitor on the Lineweaver-Burk plot is as shown in Fig. 4.9. Both K_m and $V_{\max}$ are affected by the inhibitor giving rise to parallel lines. There are very few cases indeed of uncompetitive inhibition in one-substrate reactions (an example is the inhibition of alkaline phosphatase from rat intestine by L-phenylalanine³⁹) but more examples are found in multisubstrate reactions (e.g. S-adenosylmethionine behaves as an uncompetitive inhibitor towards ATP in the reaction catalysed by methionine adenosyltransferase from yeast.⁴⁰

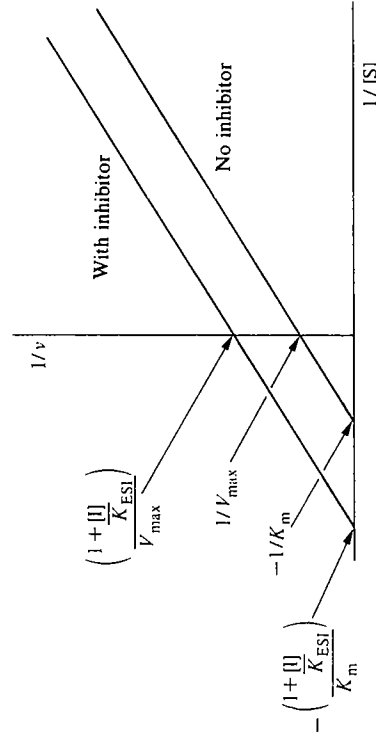


FIG. 4.9. The effect of an uncompetitive inhibitor on a Lineweaver-Burk plot of enzyme kinetic data.

4.3.2.3 Significance of results

We have introduced the terms competitive, non-competitive, and uncompetitive inhibitors by referring to the model depicted in Scheme 2 for one-substrate reactions. In our view it is better to define these types of inhibitors in terms of their effects on the parameters K_m and V_{max} (Figs. 4.7, 4.8, and 4.9) rather than by referring to relationships between the binding sites for S and I on the enzyme, because the latter are much more difficult to determine.³⁸ As we shall see later (Section 4.3.5.4), the terms competitive, non-competitive, and uncompetitive are extremely useful for classifying the effects of inhibitors in multisubstrate reactions, which can be of value in distinguishing between various types of reaction mechanism. In a number of cases (see Chapter 6, Section 6.2.2.1) the effects of an inhibitor do not fall into any of the three limiting cases depicted in Figs. 4.7, 4.8, and 4.9. In such cases the inhibition is usually referred to as being of a 'mixed' type and will require different equations and assumptions from those described in this section.⁴¹ (See the discussion in Appendix 4.2)

4.3.3 The effect of changes in pH on enzyme-catalysed reactions

There are a number of distinct effects that a change in pH can have on enzyme-catalysed reactions, e.g. inactivation of the enzyme outside a certain pH range or a change in the ionization state of the substrate(s). A third possibility is that there could be a change in the equilibrium position if H^+ is involved in the reaction, e.g. in the reaction catalysed by creatine kinase:



In this case, increasing the pH will displace the equilibrium in favour of phosphocreatine synthesis.

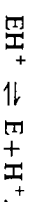
However, the possibility of most interest to us is that there are changes in the ionization state of amino-acid side-chains that are essential for the catalytic activity of the enzyme. In this case we might hope to obtain information about the nature of these side-chains.

4.3.3.1 Theoretical background

The effects of pH on the velocity of an enzyme-catalysed reaction can be complex, since both K_m and V_{max} can be affected, and, in order to undertake a detailed analysis of the ionizations involved, values of both parameters should be obtained over a range of pH values. A number of excellent detailed treatments of the effects of pH on K_m and V_{max} have been given.^{42, 43, 44, 45} We should note that it is usually easier to analyse changes in V_{max} , since this parameter generally reflects a single rate constant, whereas K_m is a function of several rate constants.*

* Analysis of a scheme in which the free enzyme and enzyme-substrate complex each possess two ionizing side-chains shows that changes in V_{max} depend on ionizations of the enzyme-substrate complex; changes in V_{max}/K_m depend on ionizations of the free enzyme, and changes in K_m depend on ionizations of both the free enzyme and the enzyme-substrate complex. For further details of these effects and the analysis of more complex schemes, references 42, 43, 44 and 45 should be consulted.

In order to see how pK_a values can be obtained from the experimental data, we can consider the simple case of a single ionizing side-chain in the enzyme. (Similar equations would result if we considered a single ionizing side-chain in the enzyme-substrate complex.)



We shall assume that EH^{+} is inactive and that E is the active form.

Now the acid dissociation constant, K_a , is given by

$$K_a = \frac{[\text{E}][\text{H}^{+}]}{[\text{EH}^{+}]},$$

$$\therefore [\text{EH}^{+}] = \frac{[\text{E}][\text{H}^{+}]}{K_a}.$$

The fraction, F , of enzyme in the unprotonated (active) form is given by

$$F = \frac{[\text{E}]}{[\text{E}] + [\text{EH}^{+}]} = \frac{K_a}{K_a + [\text{H}^{+}]}.$$

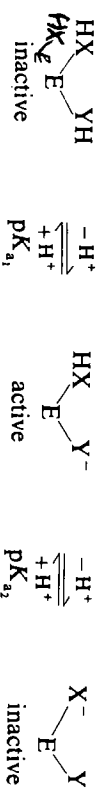
Let $(V_{max})_m$ equal the maximum rate when all the enzyme is in the unprotonated form. Then at any pH, the observed V_{max} is given by

$$V_{max} = (V_{max})_m \cdot F.$$

Thus

$$V_{max} = (V_{max})_m \cdot \frac{K_a}{K_a + [\text{H}^{+}]} \quad (4.10)$$

If we have a second ionizing group in the enzyme, such as shown in the following scheme:



the expression for V_{max} at any pH can be shown to be

$$V_{max} = \frac{[V_{max}]_m}{1 + \frac{[\text{H}^{+}]}{K_{a_1}} + \frac{K_{a_2}}{[\text{H}^{+}]}} \quad (4.11)$$

4.3.3.2 Treatment of data

Consider eqn (4.10). At pH values well below the pK_a (i.e. when $[\text{H}^{+}] \gg K_a$), then

$$V_{max} = (V_{max})_m \cdot \frac{K_a}{[\text{H}^{+}]}.$$

Taking logarithms:

$$\log_{10} V_{\max} = \log_{10} (V_{\max})_m - pK_a + pH,$$

so that a plot of $\log_{10} V_{\max}$ against pH will be linear with a slope of 1. (In practice this will work well up to about 1.5 pH units below the pK_a .)

At pH values well above the pK_a (i.e. when $[H^+] \ll K_a$) then

$$V_{\max} = (V_{\max})_m,$$

i.e. there is no variation of $V_{\max}$ with pH.

The plot of $\log_{10} V_{\max}$ against pH will therefore be of the form shown in Fig. 4.10. Extrapolation of the linear portions of the plot will give the value of pK_a at the point of intersection.

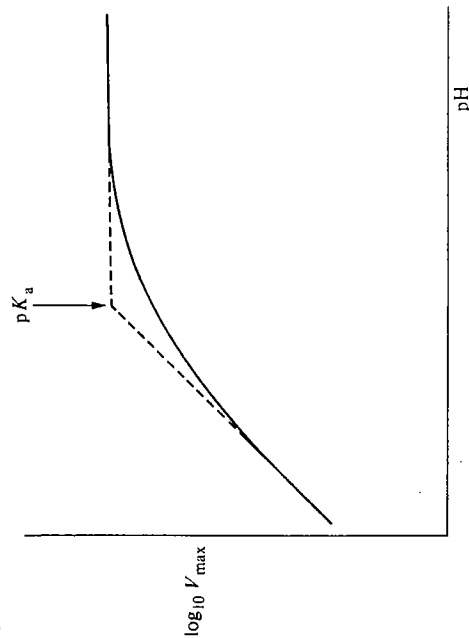


FIG. 4.10. The effect of pH on the $V_{\max}$ of an enzyme-catalysed reaction if only the unprotonated form of the enzyme is catalytically active. The solid curve represents the experimental data.

A similar analysis of eqn (4.11) shows that if two ionizing groups were involved, a plot of $\log_{10} V_{\max}$ against pH would have three distinct regions of slope 1, 0, and -1 as shown in Fig. 4.11. These regions correspond to the cases where $[H^+] \gg K_{a1}$, $K_{a1} \ll [H^+] \gg K_{a2}$, and $K_{a2} \gg [H^+]$ respectively.

As shown in Fig. 4.11, the values of pK_{a1} and pK_{a2} can be obtained from the points of intersection of the extrapolated linear portions of the plot. However, if the two pK_a values are closer than about 1.5 pH units, the ionizations will not be independent and the pK_a values derived may need correction to obtain the true values.⁴⁶

4.3.3.3 Significance of results

A study of the effect of changes in pH on the velocity of an enzyme-catalysed reaction generally leads to the conclusion that there is an 'optimum pH' for that

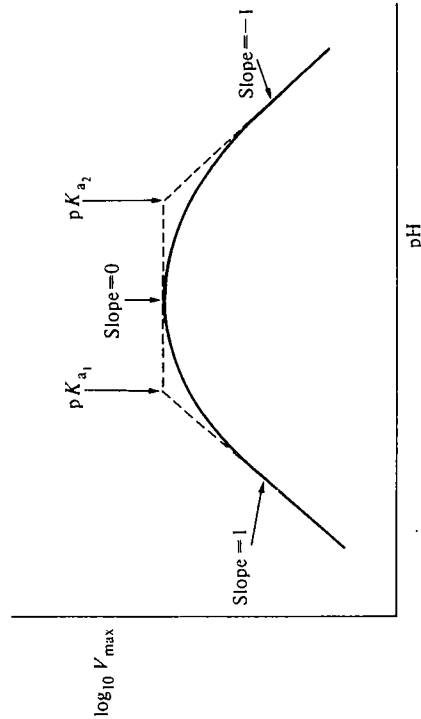


FIG. 4.11. The effect of pH on the $V_{\max}$ of an enzyme-catalysed reaction when two ionizing groups are involved. The solid curve represents the experimental data.

reaction. In view of the various types of effect of pH mentioned in Section 4.3.3, it is clear that such a term should be used with a certain degree of caution. There are cases known in which studies of enzyme kinetics at the optimum pH have given misleading results when considering the *in vivo* situation (see Chapter 8, Section 8.3.2).

It is often tempting to try to assign the measured pK_a values (Figs. 4.10 and 4.11) to particular types of amino-acid side-chain, using the values of pK_a for free amino acids (see Chapter 3, Section 3.3.1). However, there are considerable pitfalls in this procedure, since the environment of a side-chain in an enzyme can be very different from that of the free amino acid and this can cause a large shift in pK_a . Thus, in pepsin, one aspartic acid side-chain has a pK_a of about 1.0, some 3 pH units lower than that of free aspartic acid.⁴⁷ Nevertheless, in some cases it has been possible from pH-rate studies to implicate particular side-chains in the mechanisms of certain enzymes, e.g. histidine side-chains in pancreatic ribonuclease⁴⁸ and chymotrypsin⁴⁹ and carboxylic-acid side-chains in lysozyme.⁵⁰ As described in Chapter 5 (Section 5.4), these conclusions can be supported by the results of other experiments.

4.3.4 The effect of changes in temperature on enzyme-catalysed reactions

In general, the effects of changes in temperature on the rates of enzyme-catalysed reactions do not provide much useful information as far as the mechanism of catalysis is concerned.* However, these effects can be important in indicating

* A notable exception is the study of the chymotrypsin-catalysed hydrolysis of *N*-acetyl-L-tryptophanamide, in which, from the temperature dependence of the reaction rate and other data, an energy profile (see Chapter 5, Fig. 5.1) for the reaction could be constructed.⁵¹



structural changes in enzymes and in considering enzyme activity in poikilothermic organisms, in which intracellular temperatures vary considerably, so a short discussion of the effects will be given in this section. More detailed discussions can be found in references 52 and 53. The effects of temperature on the stability of enzymes have been mentioned in Chapter 3, Sections 3.6.3 and 3.7.1.

4.3.4.1 Theoretical background

From the transition-state theory of chemical reactions, we can derive an expression for the variation of the rate constant, k , with temperature. This is of the form of eqn (4.12), which is sometimes referred to as the Arrhenius expression (see also Chapter 5, Section 5.3).

$$k = Ae^{-E_a/RT} \quad (4.12)$$

where A is known as the pre-exponential factor,

R is the gas constant,

T is the (absolute) temperature, and

E_a is the activation energy for the reaction.

From eqn (4.12) it is clear that there is an exponential increase in reaction rate with temperature. We can introduce a quantity, known as the Q_{10} , which is the ratio (or quotient) of the reaction rate at $(T + 10)^\circ\text{K}$ compared with that at $T^\circ\text{K}$. Using eqn (4.12) it can be shown that at temperatures around 300°K (27°C), Q_{10} is approximately given by $e^{E_a/7500}$. For many chemical reactions, the values of Q_{10} are in the range 2–4, corresponding to activation energies of about 50–100 kJ mol⁻¹.

Similar considerations apply in the case of enzyme-catalysed reactions, but the values of E_a (and hence of Q_{10}) are generally lower than the corresponding values for non-enzyme-catalysed reactions when a comparison can be made. For instance, in the hydrolysis of urea catalysed by acid, the value of E_a is 100 kJ mol⁻¹, whereas the same reaction catalysed by urease has a much lower E_a (42 kJ mol⁻¹).

Above a certain temperature an enzyme will tend to lose the compact three-dimensional structure that is required for catalytic activity. Incubation of many enzymes at temperatures above about 323 K (50°C) leads to a fairly rapid loss of catalytic activity.

4.3.4.2 Treatment of data

According to the Arrhenius expression (eqn (4.10)), a plot of $\ln(\text{velocity})$ against $1/T$ gives a straight line of slope $-E_a/R$. Taking into account the loss of catalytic activity at high temperatures, we would expect the Arrhenius plot for an enzyme-catalysed reaction to resemble that shown in Fig. 4.12(a).*

* Cornish-Bowden⁵⁴ has pointed out that the analysis of Arrhenius plots is more satisfactory if the variations of K_m and V_{max} are analysed separately over a range of temperatures. At any temperature, the observed velocity will depend on both parameters.

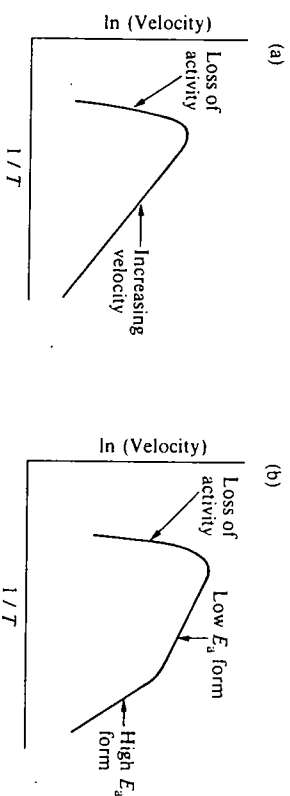


FIG. 4.12. Arrhenius plots for enzyme-catalysed reactions illustrating the loss of catalytic activity at high temperature. (a) Only one form of an enzyme involved; (b) two interconvertible forms with different activation energies involved.

Complications arise when an enzyme can arise in two (or more) interconvertible forms with different activation energies. There will then be a discontinuity in the Arrhenius plot around the temperature where the change over between the two forms becomes significant (Fig. 4.12(b)). An example of this type of behaviour is provided by the enzyme adenosinetriphosphatase (Na^+ , K^+ -activated) for which the transition probably arises from structural changes in the tightly bound phospholipid molecules associated with the enzyme.⁵⁵ A detailed discussion of the causes of discontinuous Arrhenius plots has been given.⁵⁶

4.3.4.3 Significance of results

As mentioned in Section 4.3.4.1 the values of Q_{10} (and hence of E_a) for enzyme-catalysed reactions are generally lower than for reactions not subject to enzyme catalysis. For most enzymes in homoiothermic species (e.g. mammals) the value of Q_{10} is approximately 2. However, in species that have to adapt to cold conditions it is important that the value of Q_{10} is lower than this so that essential metabolic reactions are not slowed down too much. It is found, for instance, that many enzymes from intertidal species such as anemones and winkles have values of Q_{10} of about 1 (i.e. the rates do not change significantly with temperature).⁵⁷ It would be of considerable interest to know how these enzymes have evolved to this state. In a number of species of terrestrial insects there are major alterations in metabolism to coordinate the production of cryoprotectants such as glycerol and sorbitol. In such cases there appear to be differential effects on the kinetics and regulatory properties of the glycolytic enzymes.⁵⁸

Although, as shown in Fig. 4.12, enzymes generally lose activity at temperatures above about 323 K (50°C), it should be noted that the enzymes from thermophilic bacteria (e.g. *Bacillus stearothermophilus*, which grows at temperatures up to 363 K (90°C)) are stable at these high temperatures. The cause of the exceptional stability of these enzymes is currently under investigation; in the case of glyceraldehyde-3-phosphate dehydrogenase (which contains four subunits) it

has been suggested that the stabilization arises from additional inter-subunit ionic bonds and hydrophobic interactions.⁵⁹

4.3.5 Two-substrate reactions

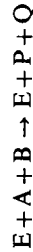
Up to now our discussion has been concerned with enzyme-catalysed reactions involving a single substrate and this has allowed us to introduce terms such as K_m , V_{max} , competitive inhibition, etc. This treatment would appear to be of limited value, since oxidoreductases, transferases, and ligases (Chapter 1, Section 1.5.1) catalyse reactions which by definition involve more than one substrate. Nevertheless, many of the concepts involved in one substrate kinetics can be useful in the analysis of the kinetics of these more complex reactions. In this section we shall explain the broad divisions of types of mechanism of two-substrate reactions and indicate how the necessary equations are derived. We shall then show how to use these equations in the analysis of experimental data, and conclude by describing how it is possible to distinguish between the various types of possible mechanism. More detailed treatments of these topics can be found in a number of textbooks.⁶⁰⁻⁶⁴

4.3.5.1 Theoretical background

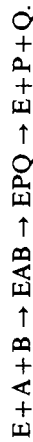
We can divide two-substrate reactions into two main categories.

(i) *Those involving a ternary complex (i.e. a complex containing enzyme and both substrates)*

In these cases the reaction

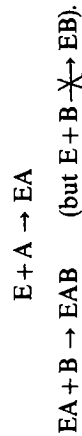


proceeds via ternary complexes of the type EAB and EPQ:

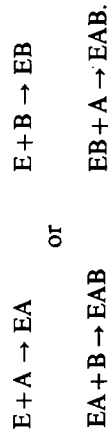


This category can be further subdivided:

(i)(a) Those reactions in which the ternary complex is formed in an *ordered* manner, i.e. the second substrate (say, B) can bind to the enzyme only after A has already bound:

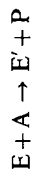


(i)(b) Those reactions in which the ternary complex is formed in a *random* manner (i.e. either substrate can bind first):



(ii) *Those not involving a ternary complex*

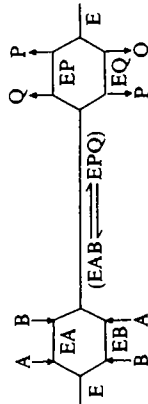
The most important class of reactions in this category proceed by *enzyme substitution* or *ping-pong* mechanisms, i.e. a modified form of the enzyme (E') is formed together with the first product, before the second substrate is bound:



A second class of enzymes in this category operates via a Theorell-Chance mechanism (named after the original investigators) in which a ternary complex is presumably formed but its breakdown to yield the first product is very fast so that the ternary complex is kinetically insignificant. This type of mechanism has been shown to apply in the oxidation of ethanol and other primary alcohols by NAD^+ catalysed by alcohol dehydrogenase from horse liver.^{65, 66}

An alternative way of representing enzyme-catalysed reactions has been proposed by Cleland.⁶⁷ The progress of the reaction is shown by a horizontal line (branched if necessary) with enzyme forms depicted below the line. Successive additions of substrates and release of products are depicted by vertical arrows. The rate constants can be indicated, if necessary adjacent to these arrows.

Thus, the random ternary complex mechanism (type (i)(b) above) would be



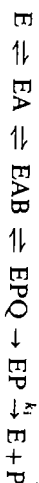
and the enzyme-substitution mechanism (type (ii) above) would be



4.3.5.2 Derivation of equations for two-substrate reactions

The derivation of equations to describe the kinetics of the various types of mechanism does not involve any fundamental principles in addition to those used for one-substrate reactions (Section 4.3.1.1), although the algebra is obviously more complex. (A good account of the derivations is given by Engel; in this book there is also an explanation of the King-Altman procedure, which simplifies the derivations considerably.^{68, 69} Computer-assisted methods for deriving the rate equations have been described.⁷⁰) The basic idea is to evaluate the concentrations of the various enzyme-containing complexes in terms of the total concentration of enzyme, under the stated conditions of substrate concen-

trations. In order to do this we use the steady-state assumption* in the same way as for one substrate reactions (Section 4.3.1.1). The velocity of the overall reaction is then equal to the concentration of the complex that precedes regeneration of free enzyme multiplied by the rate constant for the regeneration step, e.g. in the reaction scheme



the rate of the overall reaction, v , equals $k_1[EP]$. This is because we need to regenerate free enzyme (by the k_1 step above) in order to allow the reaction to continue.

The equations for v , the initial rate of the reaction, which result from these treatments are of the following forms.[†]

For the ternary complex mechanisms ((i)(a) and (i)(b) in Section 4.3.5.1):

$$v = \frac{V_{\max}[A][B]}{K'_A K_B + K_B[A] + K'_A[B] + [A][B]} \quad (4.13)$$

(an equation of this type is also derived for the Theorell-Chance mechanism).

For the enzyme substitution mechanisms ((ii) in Section 4.3.5.1):

$$v = \frac{V_{\max}[A][B]}{K_B[A] + K'_A[B] + [A][B]} \quad (4.14)$$

4.3.5.3 Significance of the parameters in the equations

$V_{\max}$ in eqns (4.13) and (4.14) represents the maximum velocity at saturating levels of substrates A and B.

In a purely practical sense, the constants K'_A and K_B in the eqns (4.13) and (4.14) represent the Michaelis constants (for substrates A and B, respectively) in the presence of saturating concentrations of the other substrate. This is readily shown; e.g. in eqn (4.13) if we divide the numerator and denominator by $[B]$ we obtain

$$v = \frac{V_{\max}[A]}{\frac{K'_A K_B}{[B]} + \frac{K_B[A]}{[B]} + K'_A + [A]}$$

* If we apply the steady-state assumption to the case of the random order ternary complex mechanism (i)(b) in Section 4.3.5.1, the equation obtained is complex, with terms containing the square of the concentrations of substrate. In this case we can apply the equilibrium assumption (Section 4.3.1.1) and assume that E, EA, EB, and EAB are all in equilibrium with one another. The resulting equation (4.13) describes many reactions of this type and to that extent the equilibrium assumption is justified.

† There are a number of different formulations of these equations. Details of some of these are given in Appendix 4.3.

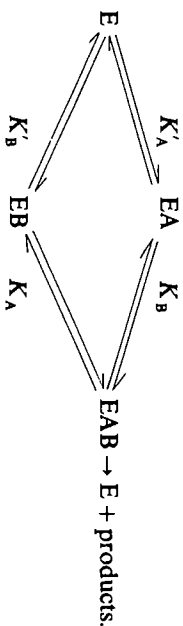
and when $[B] \rightarrow \infty$, $\frac{1}{[B]} \rightarrow 0$, so that

$$v = \frac{V_{\max}[A]}{K'_A + [A]}$$

which is of the same form as eqn (4.6).

K'_A in eqn (4.13) does not have a simple practical meaning.

In terms of the mechanisms of the reactions, the constants K'_A , K_A , and K_B represent combinations of the rate constants of individual steps in the reaction (compare eqn (4.5) for one-substrate reactions). Their precise meanings vary according to the type of mechanism under discussion. In the case of the random-order ternary complex mechanism ((i)(b) in Section 4.3.5.1), K'_A , K_A , and K_B have simple meanings in terms of dissociation constants, because in this case we used the equilibrium assumption to derive eqn (4.13).



K'_A , K_A , and K_B represent the dissociation constants for EA, EAB to yield EB, and EAB to yield EA respectively. (Note that $K'_B = K'_A K_B / K_A$)

4.3.5.4 Treatment of data

The values of the parameters $V_{\max}$, K'_A , K_A , and K_B in eqn (4.13) can be derived from the experimental data by a computer fitting procedure²² analogous to that mentioned for one-substrate reactions (Section 4.3.1.2). However, the values are more commonly obtained by graphical methods involving primary and secondary plots.

The velocity, v , is measured at various values of $[A]$, keeping the concentration of B constant; this procedure is then repeated at other fixed values of $[B]$. By taking the inverse of eqn (4.13), i.e.

$$\frac{1}{v} = \left[1 + \frac{K'_A}{[A]} + \frac{K_B}{[B]} + \frac{K'_A K_B}{[A][B]} \right] \frac{1}{V_{\max}}, \quad (4.15)$$

we see that a primary plot of $1/v$ against $1/[A]$ at a fixed value of $[B]$ will be linear (Fig. 4.13) with

$$\text{slope} = \frac{1}{V_{\max}} \left[\frac{K'_A}{[B]} + \frac{K'_A K_B}{[B]} \right] \quad (4.16)$$

and

$$\text{intercept (on y-axis)} = \frac{1}{V_{\max}} \left[1 + \frac{K_B}{[B]} \right]. \quad (4.17)$$

As $[B]$ increases, both the slope and intercept will decrease.

(The lines in the primary plot (Fig. 4.13) intersect at a point that can be above, on, or below the x-axis, depending on the relative values of K'_A , K_A , and K_B . At any value of $[B]$ an 'apparent' K_m for A can be derived from the intercept on the x-axis. From eqn (4.15) it can be shown that this K_m is $([B]K_A + K'_AK_B)/([B] + K_B)$; K_m will vary with $[B]$, unless $K'_A = K_A$ when the point of intersection is on the x-axis.)

Secondary plots of the slopes and intercepts of the primary plot against $1/[B]$ can then be constructed (Fig. 4.14(a) and (b)).

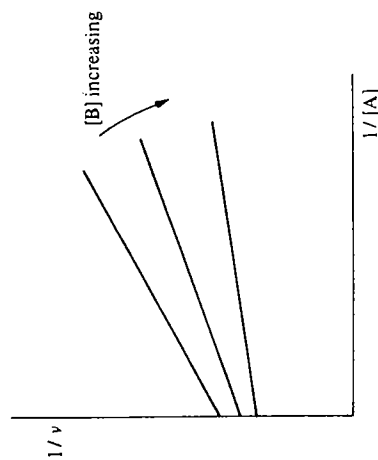


FIG. 4.13. Primary plot of enzyme kinetic data according to eqn (4.15).

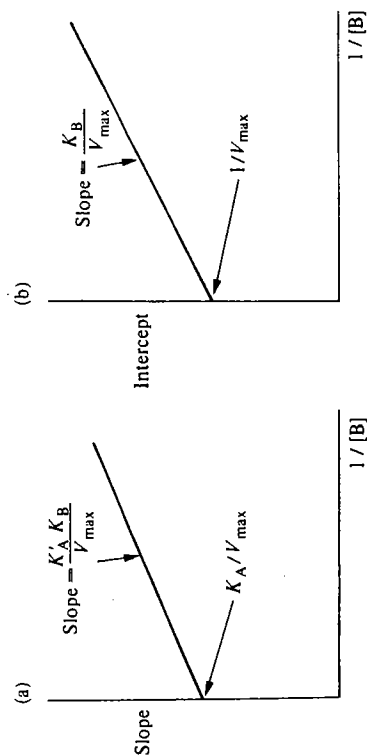


FIG. 4.14. Secondary plots of (a) slopes and (b) intercepts of primary plot (Fig. 4.13) against $1/[B]$.

By inspection of eqn (4.16) we see that a plot of the slope against $1/[B]$ (Fig. 4.14(a)) is linear with a slope of $K'_AK_B/V_{\max}$ and an intercept on the y-axis of $K_A/V_{\max}$.

From eqn (4.17) we see that a plot of the intercepts against $1/[B]$ (Fig. 4.14(b)) is linear with a slope of $K_B/V_{\max}$ and an intercept on the y-axis of $1/V_{\max}$. Thus it is

possible to determine in turn $V_{\max}$, K_B , K_A , and K'_A from the slopes and intercepts of these secondary plots.

As mentioned in Section 4.3.5.3, the parameters K'_A , K_A , and K_B have a simple meaning in the random-order ternary complex mechanism (i)(b) in Section 4.3.5.1. If it is known from other studies (described in Section 4.3.5.5) that this mechanism applies to a particular enzyme then we can draw some conclusions from the relative values of these parameters. Thus, for example, in the case of creatine kinase from rabbit muscle assayed in the direction of phosphocreatine synthesis, it is found that $K'_{\text{MgATP}} > K_{\text{MgATP}}$, which means that the binding of one substrate is enhanced by the binding of the other. (This is known as *substrate synergism*.) By contrast, in the oxidation of ethanol by NAD^+ catalysed by alcohol dehydrogenase from yeast, $K'_{\text{NAD}^+} = K_{\text{NAD}^+}$ and hence the binding of one substrate is independent of the binding of the other.

Turning now to eqn (4.14) for the enzyme substitution mechanism (ii) in Section 4.3.5.1 we find that a primary plot of $1/v$ against $1/[A]$ at fixed values of $[B]$ consists of a set of parallel lines (Fig. 4.15). This can be shown by taking the inverse of eqn (4.14):

$$\frac{1}{v} = \left[1 + \frac{K_A}{[A]} + \frac{K_B}{[B]} \right] \frac{1}{V_{\max}} \quad (4.18)$$

Hence, a plot of $1/v$ against $1/[A]$ has a slope of $K_A/V_{\max}$; this slope is independent of $[B]$, resulting in a set of parallel lines. We can derive the parameters K_B and $V_{\max}$ (and hence K_A) by performing a secondary plot of the y-axis intercepts of the primary plot against $1/[B]$. This secondary plot will have a slope of $K_B/V_{\max}$ and a y-axis intercept of $1/V_{\max}$.

4.3.5.5 Significance of results: distinction between the various mechanisms for two-substrate reactions (Section 4.3.5.1)

From the discussion in Section 4.3.5.4 it can be seen how an enzyme substitution mechanism can be distinguished from a ternary complex mechanism, since the former gives rise to a series of parallel lines in the primary plot (Fig. 4.15). It

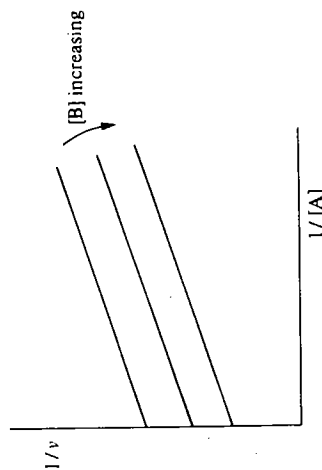
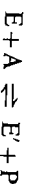


FIG. 4.15. Primary plot of enzyme kinetic data according to eqn (4.18) (the enzyme substitution mechanism).

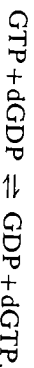
should be emphasized that considerable care is needed to ensure that lines in a plot such as Fig. 4.15 are truly parallel, and it is probably best to make an estimate of the magnitude of the K_A term in the general equation (4.13) and of the error of this estimate (K_A should equal zero for the enzyme substitution mechanism). If the primary plot suggests that an enzyme substitution mechanism is operative in a particular case, confirmatory evidence can be obtained from:

- (1) demonstration of partial reactions, i.e. conversion of A to P in the absence of B:



- (2) isolation of a modified form of the enzyme (E' in the above scheme), often using a suitable radioactive label.

Thus, in a reaction catalysed by nucleoside diphosphate kinase from erythrocytes, an enzyme substitution mechanism is indicated by the kinetic data:



In this case it is possible to isolate a modified radioactive enzyme, in which a histidine side-chain is phosphorylated,⁷¹ after incubation of the enzyme with [γ -³²P]GTP (i.e. GTP labelled at the γ position of the triphosphate group).

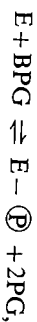


which would be followed by

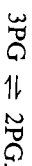


($\textcircled{P}$ represents a phosphoryl group)

Other enzymes that operate via an enzyme substitution mechanism include the amino-transferases, in which the pyridoxal phosphate cofactor (see Chapter 1, Section 1.4) attached to the enzyme undergoes a covalent change to form E' , and phosphoglycerate mutase from yeast or mammalian sources*. The reaction catalysed by the latter enzyme would appear to be a one-substrate reaction in which 2-phospho-D-glycerate (2PG) is converted into 3-phospho-D-glycerate (3PG). However, in this case a primer molecule of 2,3-bisphospho-D-glycerate (BPG) is required. We thus have two partial reactions:



giving a net reaction



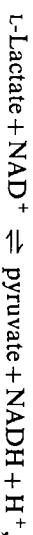
From the steady-state kinetic analysis given in Section 4.3.5.4 we are unable to distinguish between an ordered and a random ternary complex mechanism

* There are in fact two types of phosphoglycerate mutases. The enzyme from plant sources is independent of the 2,3-bisphospho-D-glycerate primer, whereas the enzyme from yeast or mammalian sources requires this primer. However, both types are now classified as isomerases (EC 5.4.2.1).

because both give rise to an expression of the form of eqn (4.13)*. However, a distinction between the mechanisms can be made if other data are available, e.g. from the following types of experiment.

(i) Substrate-binding experiments

In the random-order ternary complex mechanism, each substrate, A and B, should bind separately to the enzyme, whereas in an ordered mechanism the second substrate (e.g. B) cannot bind in the absence of the first substrate, A. For instance, in the reaction catalysed by lactate dehydrogenase,



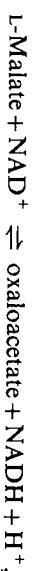
it is found that NAD^+ binds tightly to the enzyme but there is no detectable binding of lactate.⁷³ This suggests that the reaction proceeds via an ordered mechanism with NAD^+ binding preceding that of lactate, a conclusion that is supported by other evidence (see Chapter 5, Section 5.5.4).

(ii) Product-inhibition patterns

The type of inhibition shown by the products P and Q towards the substrates A and B can be used to indicate a probable mechanism for the reaction (or exclude an alternative mechanism). The equations underlying these experiments are complex and full details can be found in more comprehensive textbooks.^{74, 75} An example of the method is to distinguish between the ordered and random ternary complex mechanisms by studying the inhibition caused by product Q. In the random mechanism, Q behaves as a competitive inhibitor with respect to substrate B, whereas in the ordered mechanism, Q behaves as a mixed-type inhibitor with respect to B (i.e. the Lineweaver-Burk plots in the presence and absence of inhibitor intersect at a point to the left of the y-axis).

(iii) Isotope exchange at equilibrium⁷⁶

In this type of experiment we study the enzyme-catalysed rate of exchange of isotope between substrate(s) and product(s) when the components of the reaction mixture are present at their equilibrium concentrations. For instance, in the reaction catalysed by malate dehydrogenase,

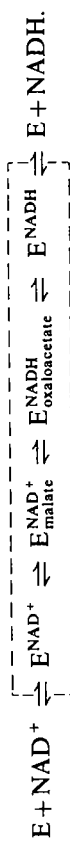


we set up a mixture containing substrates and products so that, at the pH in question, the ratio $[\text{oxaloacetate}][\text{NADH}]/[\text{malate}][\text{NAD}^+]$ equals the previously measured equilibrium constant for the reaction. One of the substrates, say NAD^+ , is isotopically labelled (e.g. with ¹⁴C) and after addition of a catalytic amount of enzyme we study the rate at which this isotope is incorporated into NADH. (Note that there is no net conversion of NAD^+ to NADH, because the

* Although the Theorell-Chance mechanism (Section 4.3.5.1) also gives an expression of the form of eqn. (4.13), it can be distinguished from the ternary complex mechanisms by comparison of the magnitudes of the various parameters in the rate equation with the rates of the forward and reverse reactions (see reference 72).



system is at equilibrium; there will, however, be forward and backward reactions occurring so that *exchange* between substrate and product can occur.) It was found that as the concentrations of malate and oxaloacetate were raised (in a constant ratio so that the equilibrium state was maintained), the rate of isotope exchange between NAD^+ and NADH fell to zero, whereas the rate of exchange between malate and oxaloacetate increased to a plateau value.⁷⁷ This result indicates that the enzyme obeys an ordered mechanism of the type



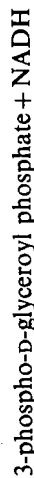
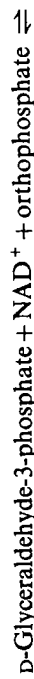
As we raise the concentrations of malate and oxaloacetate, we increase the rate at which the binary complexes (E^{NAD^+} and E^{NADH}) are converted to the ternary complexes. The enzyme thus becomes confined to the central 'box' shown in the above scheme and NAD^+ (or NADH) never dissociates to join the pool of free NAD^+ (or NADH). Since the enzyme is present only in very small amounts compared with the concentrations of substrates, this means that we will observe no exchange of isotope between free NAD^+ and free NADH. The results are not consistent with a random ternary complex mechanism.

As a result of experiments of the type described in (i), (ii), and (iii) it has been possible to assign a number of enzyme-catalysed reactions to the types of mechanism discussed in Section 4.3.5.1. The reactions catalysed by alcohol dehydrogenase (from yeast), creatine kinase, and phosphorylase appear to proceed via random ternary complex mechanisms, whereas the reactions catalysed by lactate dehydrogenase and malate dehydrogenase operate via ordered ternary complex mechanisms. A compilation of mechanisms has been made by Fromm.⁷⁸

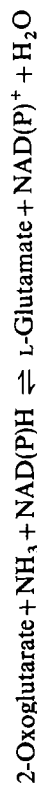
4.3.6 Reactions involving more than two substrates

A number of enzyme-catalysed reactions involve three substrates; some examples are shown below.

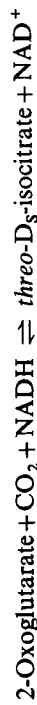
Glyceraldehyde 3-phosphate dehydrogenase



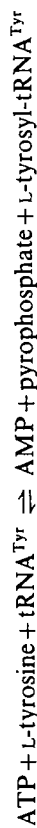
Glutamate dehydrogenase



Isoctrate dehydrogenase



Tyrosyl-tRNA synthetase



In these reactions there are a number of possible mechanisms, e.g. enzyme substitution or quaternary complex formation by ordered, random, or partly ordered mechanisms. The principles involved in the derivation of equations and the treatment of data are similar to those used in the analysis of two-substrate reactions (Section 4.3.5), although the algebra is considerably more complex⁷⁹ and the requirement for accurate data is more stringent. An example of the analysis of a three-substrate reaction is provided by the studies on glutamate dehydrogenase, in which it was shown that the reaction is most likely of the random type with 2-oxoglutarate, NH_4^+ , and NADH binding in any order to form the quaternary complex.⁸⁰

4.4 Pre-steady-state kinetics⁸¹⁻⁸³

4.4.1 Background: the need for special techniques

From kinetic studies performed under steady-state conditions we can obtain only very limited information about the rates of individual steps in an enzyme-catalysed reaction. The maximum velocity (V_{max}) can be used to calculate the value of k_{cat} , or turnover number (see Section 4.3.1.1), according to the equation:

$$k_{\text{cat}} = \frac{V_{\text{max}}}{[\text{E}]_0}$$

where $[\text{E}]_0$ is the concentration of enzyme active sites.

Values of k_{cat} are known for many enzymes and range from about 10 to about 10^7 s^{-1} .⁸⁴ The value of k_{cat} gives an idea of the rate constant of the slowest step in an enzyme-catalysed reaction. In steady-state experiments we generally work with very low concentrations of enzyme (1 nmol dm^{-3} or less is not unusual) and under these conditions the reaction is slow enough to allow the use of 'conventional' methods (e.g. manual addition of enzyme to start the reaction and observation by a spectrophotometer). For instance if $[\text{E}]_0 = 1 \text{ nmol dm}^{-3}$ and $k_{\text{cat}} = 10^3 \text{ s}^{-1}$, V_{max} would equal $60 \mu\text{mol dm}^{-3} \text{ min}^{-1}$, which would be quite convenient to monitor spectrophotometrically.

However, if we wish to examine processes that occur on a time scale of less than about a few seconds (and these will include the steps other than the slowest step of the overall reaction), we need to employ faster methods of mixing and observation. The problem has been neatly solved by the introduction of 'stopped-flow' methods, which can allow observation of reactions that occur on a time scale of a few milliseconds. The basic elements of a 'stopped-flow' apparatus are shown in Fig. 4.16.

The drive barrier is pushed in, usually mechanically, and the contents of the two reactant syringes (e.g. enzyme and substrate(s)) are mixed. The flow of liquid

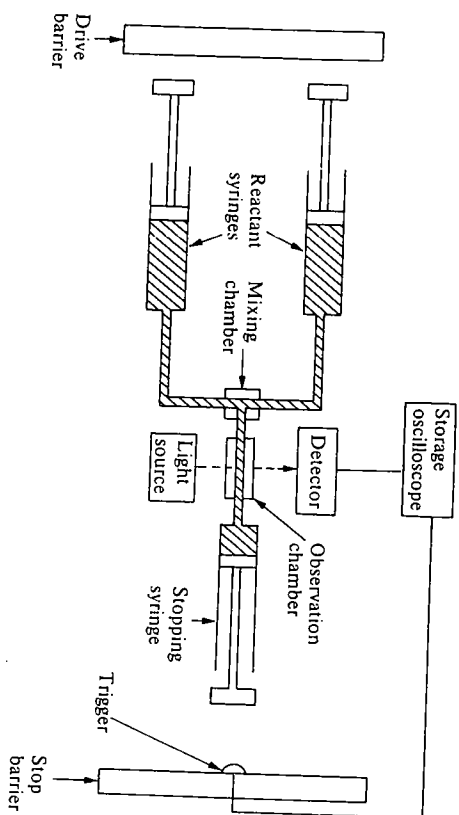


FIG. 4.16. A typical stopped-flow apparatus.

forces the piston of the stopping syringe out until it hits the stop barrier, when the flow stops. At this point the oscilloscope is triggered and the changes in absorbance or fluorescence of the liquid in the observation chamber can be recorded. The trace on the storage oscilloscope is usually photographed; the photograph can then be analysed at leisure. For this type of apparatus there is a 'dead' time, corresponding to the time interval between the start of the reaction by mixing and the stop of the flow of liquid, of about 1 ms.

4.4.2 Applications of the stopped-flow technique

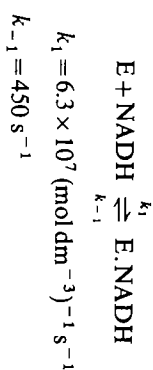
The stopped-flow technique has been valuable in elucidating the details of enzyme-catalysed reactions. We shall describe two of the most important types of study: the determination of rate constants of individual steps in the reaction and the identification of transient species.

4.4.2.1 Determination of rate constants

An example of the determination of the rate constant of an individual step in a reaction is provided by the binding of NADH to lactate dehydrogenase (isoenzyme LDH-5) which was monitored by the increase in the fluorescence of NADH that occurs on binding.⁸⁵ Equal concentrations of enzyme active sites and substrate ($8 \mu\text{mol dm}^{-3}$ of each) were mixed and the fluorescence increase was monitored over a period of about 16 ms. * By analysing the data in terms of equations developed for reversible reactions, the rate constants could be deduced

* For a second-order reaction in which the concentrations of reactants are equal, the half time is given by $1/k[A]$, where $[A]$ is the initial concentration of the reactants and k is the rate constant. In this case (ignoring the reverse reaction) we can see that if $k = 6.3 \times 10^7 \text{ (mol dm}^{-3}\text{)}^{-1} \text{ s}^{-1}$ and $[A] = 8 \mu\text{mol dm}^{-3}$, then $t_{1/2} \approx 2 \text{ ms}$, which is within the time scale of the stopped-flow apparatus.

as:



As will be seen in Chapter 5 (Section 5.5.4.2), the determination of the rate constant for the dissociation step (k_{-1}) is of value in deciding the nature of the slow step of the overall reaction. Further work, using stopped-flow measurements of the fluorescence changes in a mutant form of lactate dehydrogenase, showed that the slow step of the overall reaction corresponds to the movement of a flexible loop region in the enzyme. (This movement causes the loop to close over the active site, with consequent changes in the fluorescence of the tryptophan side-chain in the loop (see Chapter 5, Section 5.5.4.5).)

4.4.2.2 Identification of transient species

In the stopped-flow technique, the concentration of enzyme is high enough to permit the detection and identification of enzyme-containing complexes in the reaction pathway, and to determine the rate constants of the steps that involve formation and decay of these species. Using absorption spectrophotometry, concentrations of the order of $10 \mu\text{mol dm}^{-3}$ can be readily detected; using fluorescence the limit is probably tenfold lower. An early experiment by Chance showed that the oxidation of malachite green by H_2O_2 catalysed by peroxidase involved an enzyme- H_2O_2 complex; this experiment provided the first direct evidence for the participation of an enzyme-substrate complex in an enzyme-catalysed reaction.⁸⁶ In Chapter 5 (Section 5.5.4.2) we shall see that the application of the stopped-flow technique to the reaction catalysed by lactate dehydrogenase (i.e. L-lactate + $\text{NADH}^+ \rightleftharpoons$ pyruvate + $\text{NADH} + \text{H}^+$) has demonstrated the transient production of enzyme-bound NADH. A detailed analysis of this reaction has allowed the rate constants of all the elementary steps in the reaction to be determined.^{87, 88}

An extension of the stopped-flow technique has been described; this involves the use of a rapid scanning detecting device that can scan a spectrum over a 200 nm range in 1 ms .⁸⁹ By this means it is possible to record rapid changes in the absorption spectrum and hence identify transient species with more certainty. The technique has been used to show, for instance, that the NADH produced in the rapid phase of the reaction catalysed by alcohol dehydrogenase from horse liver (i.e. ethanol + $\text{NAD}^+ \rightleftharpoons$ acetaldehyde + $\text{NADH} + \text{H}^+$) is bound to the enzyme, because its absorption maximum (320 nm) is different from that of free NADH (340 nm).

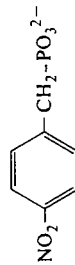
4.4.3 Relaxation methods

Stopped-flow methods (Section 4.4.1) are extremely useful for the study of processes that occur over a period of several milliseconds. The lower limit is set

by the time required to achieve uniform mixing of liquids and this appears to be of the order of 1 ms, or slightly less. We have seen in Section 4.4.2.1 that it is possible to use stopped-flow methods to study the reaction between NADH and lactate dehydrogenase provided that relatively low concentrations of substrate and enzyme are employed. However, if it is not possible to adopt this 'low concentration' procedure to bring the rate of a rapid reaction into a convenient time scale,* we can use the so-called 'relaxation' methods, which do not involve the mixing of reactants. In these methods, a system at equilibrium is perturbed by a sudden change in temperature or other parameter (pressure, pH, etc.) and then allowed to 'relax' to the new position of equilibrium. (By discharging a condenser between electrodes in the solution the temperature can be raised by 5–10 K in 1 μ s. The position of equilibrium will change provided that the enthalpy change (ΔH°) for the reaction is not equal to zero). The rate of the 'relaxation' is monitored by appropriate spectroscopic (or other) techniques, and can be related theoretically to the rate constants for the forward and reverse reactions occurring at equilibrium. From measurements at a variety of concentrations of reactants, the rate constants can be evaluated. More details of these procedures are given in review articles.^{81, 83, 90}

The temperature-jump method has been used, for instance, to show that the rate constant for the association of NADH with malate dehydrogenase ($k = 5 \times 10^8 \text{ (mol dm}^{-3})^{-1} \text{ s}^{-1}$) is close to the limit calculated for diffusion control of association of enzyme with substrate ($k \approx 10^9 \text{ (mol dm}^{-3})^{-1} \text{ s}^{-1}$), i.e. that almost every collision or encounter between enzyme and substrate leads to formation of a complex.

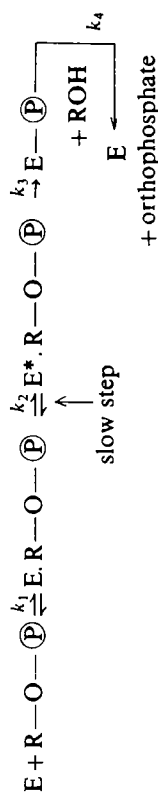
The method was of value in a study of the reaction catalysed by alkaline phosphatase from *E. coli*; this enzyme catalyses the hydrolysis of a variety of phosphate esters, R-O-P(=O)(OH)_2 , via formation and breakdown of a phosphoryl-enzyme intermediate. Because there is little dependence of the rate of reaction on the nature of the R group, it had been postulated that the hydrolysis of the phosphoryl enzyme was the common rate-limiting step for hydrolysis of the different phosphate esters. However, the rate of this step (k_4 in the scheme below) was measured by stopped-flow techniques and found to be faster than the rate of the overall reaction.⁹¹ The nature of the slow step was deduced from experiments in which the rate of binding of a non-hydrolysable substrate analogue, 4-nitrobenzylphosphonate,



to the enzyme was studied by the temperature-jump method.⁹² The results indicated that a structural change in the enzyme occurred after binding the inhibitor and that the rate of this change corresponded to that of the slow step of

* This could be, for instance, because of the small size of an absorbance change at low concentrations or because of the order of the process. The half time of a first-order process does not depend on the concentration of the reactant.

the overall reaction (k_2 in the scheme below):



4.5 Concluding remarks

In this chapter we have introduced the principles and equations that are required to give an understanding of the important features of enzyme kinetics. The application of both steady-state and pre-steady-state measurements is of considerable value in helping to deduce the sequence of enzyme-containing complexes in a reaction. We shall see in Chapter 5 how these data can be used in conjunction with other information to formulate a detailed picture of the mechanism of an enzyme-catalysed reaction in both kinetic and structural terms. As will be shown in Chapters 6 and 8, kinetic information helps in gaining an understanding of the control of enzyme activity and of the role of an enzyme in the metabolic processes in the cell.

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Appendix 4.1

The integrated form of the Michaelis–Menten equation¹

Equation (4.6) states that

$$v = \frac{V_{\max} [S]}{K_m + [S]}$$

The velocity, v , can be expressed as $-d[S]/dt$, therefore

$$\frac{d[S]_p}{dt} = \frac{V_{\max} [S]}{K_m + [S]}$$

Separating variables and integrating between limits at time zero when the substrate concentration = $[S]_0$ and time t where the substrate concentration = $[S]$:

$$\int_{[S]_0}^{[S]} \frac{d[S]}{d[S] + K_m} = -V_{\max} \int_0^t dt,$$

$$\text{therefore } [S] - [S]_0 + K_m \cdot \ln \frac{[S]}{[S]_0} = -V_{\max} \cdot t.$$

Rearranging

$$\frac{[S]_0 - [S]}{t} = -K_m \cdot \frac{1}{t} \cdot \ln \frac{[S]_0}{[S]} + V_{\max}.$$

Thus a plot of $([S]_0 - [S])/t$ against $1/t \cdot \ln([S]_0/[S])$ (Fig. A4.1) gives a straight line of slope $-K_m$ and intercept $V_{\max}$.

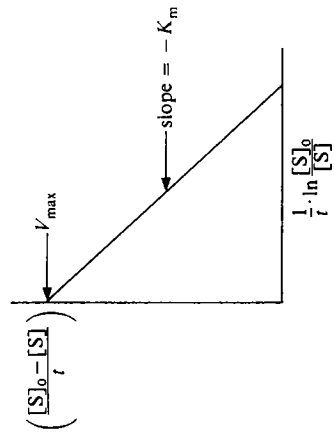


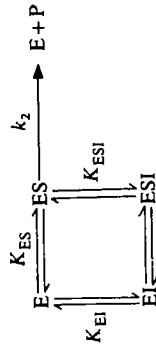
FIG. A4.1. Analysis of the progress curve of an enzyme-catalysed reaction using the integrated form of the Michaelis-Menten equation.

Appendix 4.2

The interaction of an enzyme with substrate (S) and inhibitor (I)

(a) *General considerations*

Consider the scheme below, where E, S, I represent enzyme, substrate, and inhibitor, respectively:



PCI is assumed to be inactive and K_{rel} , K_{rel} and K_{rel} represent dissociation constants.

Now

$$\frac{[E][S]}{[ES]} = \frac{[E][I]}{K_{ES}},$$

and

$$[\text{ESI}] = \frac{[\text{ES}][\text{I}]}{K_{\text{ESI}}} = \frac{[\text{E}][\text{S}][\text{I}]}{K_{\text{ESI}} \cdot K_{\text{ES}}}$$

The fraction (F) of enzyme in the form of the ES complex is given by

$$F = \frac{[\text{ES}]}{[\text{E}] + [\text{ES}] + [\text{EI}] + [\text{ESI}]}$$

$$= \frac{\frac{[S]}{K_{ES}}}{1 + \frac{[S]}{K_{ES}} + \frac{[Y]}{K_{EI}} + \frac{[S][I]}{K_{ESI} \cdot K_{ES}}}$$

Now the observed velocity v is related to the maximum velocity, $V_{\max}$, by

$$v = V_{\max} \cdot F.$$

Hence,

$$v = V_{\max} \frac{\frac{[S]}{K_{ES}}}{1 + \frac{[S]}{K_{ES}} + \frac{[S][I]}{K_{EI} + K_{ES} \cdot K_{ES}}}$$

(b) Classification of inhibition

Because of inconsistencies in the usage of various terms relating to enzyme inhibition, particularly relating to the terms *mixed* and *non-competitive inhibition*, the International Union of Biochemistry has produced a number of recommendations.² The classification of types of inhibition should be in terms of the effects on $V_{\max}$ and $V_{\max}/K_m$:

If the apparent value of $V_{\max}/K_m$ is decreased by the inhibitor, the inhibition is said to have a *competitive* component; if the inhibitor has no effect on the apparent value of $V_{\max}$, the inhibition is said to be *competitive*.

If there is an effect on the apparent value of $V_{\max}$, the inhibition has an *uncompetitive* component; if the inhibitor has no effect on the apparent value of $V_{\max}/K_m$, the inhibition is said to be *uncompetitive*.

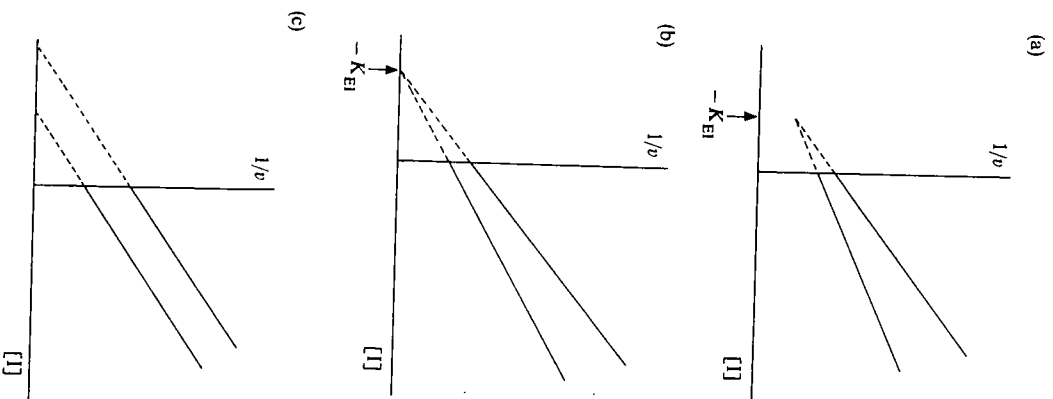


FIG. A.4.2. The Dixon plot for determining inhibitor constants: (a), (b) and (c) show the plots for the cases of competitive, pure non-competitive, and uncompetitive inhibition, respectively. Each line in the plots represents data obtained at a constant $[S]$.

If both competitive and uncompetitive components are present, the inhibition is said to be *mixed*. The case in which the effects on $V_{\max}/K_m$ are greater than on $V_{\max}$ is called *predominantly competitive inhibition*; that in which the effects on $V_{\max}$ are greater than on $V_{\max}/K_m$ is called *predominantly uncompetitive inhibition*. Where the effects on $V_{\max}/K_m$ and $V_{\max}$ are the same (i.e. there is no change in K_m) this may be called *pure non-competitive inhibition* (see Section 4.3.2.2). However, usage of the term *pure non-competitive inhibition* is to be discouraged because of its use for various types of mixed inhibition.

(c) The Dixon plot³

The general equation (4.9) can be rearranged as

$$\frac{1}{v} = \frac{1}{V_{\max}} + \frac{K_{ES}}{V_{\max} [S]} + [I] \left(\frac{1}{V_{\max} K_{ESI}} + \frac{1}{V_{\max} K_{EI}} \cdot \frac{K_{ES}}{[S]} \right).$$

Thus, a plot of $1/v$ against $[I]$ at constant $[S]$ will be a straight line. If two such lines are drawn (from measurements at two different values of $[S]$), the values of K_{EI} can be found from the point of intersection (see Fig. A4.2). (This can be shown by setting the values of $1/v$ equal at two values of $[S]$.) The plot is therefore particularly useful if v is measured at a large number of inhibitor concentrations but only a limited number of substrate concentrations (compare Figs. 4.7, 4.8 and 4.9). In the case of competitive inhibition, the lines intersect above the abscissa (Fig. A4.2(a)), whereas in the case of pure non-competitive inhibition, the point of intersection is on the abscissa (Fig. A4.2(b)). There is no point of intersection in the case of uncompetitive inhibition, since the lines are parallel (Fig. A4.2(c)). The Dixon plot cannot be used to deduce the value of K_{ESI} .

Appendix 4.3

Formulation of equations for two-substrate kinetics

The International Union of Biochemistry² proposes the following formulation for the initial rate, v , of a two-substrate reaction obeying Michaelis-Menten kinetics:

$$v = \frac{[E]_0}{\frac{1}{k_0} + \frac{1}{k_A[A]} + \frac{1}{k_B[B]} + \frac{1}{k_{AB}[A][B]}}.$$

where $[E]_0$ is the total concentration of enzyme and the k s are individual rate constants. In terms of eqn (4.13),

$$k_0 = \frac{V_{\max}}{[E]_0}, \quad k_A = \frac{k_0}{K_A}, \quad k_B = \frac{k_0}{K_B} \quad \text{and} \quad k_{AB} = \frac{k_0}{K_B \cdot K_A}.$$

The formulation of Dalziel⁴ for such a reaction is

$$\frac{[E]_0}{v} = \phi_0 + \frac{\phi_1}{[A]} + \frac{\phi_2}{[B]} + \frac{\phi_{12}}{[A][B]}$$

where ϕ_0 , etc. are known as Dalziel coefficients. These coefficients allow the equation to be written more concisely, since they represent combinations of constants.

In terms of eqn (4.13),

$$\phi_0 = \frac{[E]_0}{V_{\max}}, \quad \phi_1 = \frac{[E]_0 K_A}{V_{\max}},$$

$$\phi_2 = \frac{[E]_0 K_B}{V_{\max}} \quad \text{and} \quad \phi_{12} = \frac{[E]_0 K'_A K_B}{V_{\max}}.$$

For further details of the various formulations of these equations, see references 5, 6, and 7.

References for Appendices

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5

The mechanism of enzyme action

5.1 Introduction

In this chapter, we shall consider the mechanism of action of enzymes and gain some insight into two of the remarkable properties of enzymes referred to in Chapter 1, Section 1.3; namely catalytic power and specificity. In Section 5.3 we introduce some of the basic principles of catalysis and show how the study of the mechanisms of reactions in organic chemistry can provide a framework in which to interpret catalysis by enzymes. The main experimental approaches available to determine the mechanism of enzyme action are described in Section 5.4. Section 5.5 shows how these different approaches have been combined to provide coherent pictures of the mechanism of five particular enzymes, namely chymotrypsin, triosephosphate isomerase, fructose-bisphosphate aldolase, lactate dehydrogenase, and tyrosyl-tRNA synthetase. In the concluding Section (5.6), some more general aspects of enzyme catalysis are discussed.

Enormous progress has been made in the understanding of enzyme mechanism in the last 20 years or so; the major reason for this has been the wealth of structural information that has become available from X-ray crystallographic studies (see Chapter 3, Section 3.5.1). However, although chemically plausible mechanisms have been proposed for a number of enzymes it is doubtful that we are yet in a position to account in quantitative terms, for the catalytic power shown by any one enzyme.

5.2 Definition of the mechanism of an enzyme-catalysed reaction

We can justifiably claim to have deduced the mechanism of an enzyme-catalysed reaction if we have determined (i) the sequence of the enzyme-containing complexes* as substrate(s) is (are) converted to product(s); (ii) the rates at which these complexes are interconverted; and (iii) the structures of these complexes. This definition implies that we require both kinetic and structural information and it is therefore not surprising that kinetic and X-ray crystallographic techniques have made major contributions to our understanding of enzyme mechanisms. Valuable information has also been gained from studies involving detection of intermediates (Section 5.4.2), chemical modification of amino-acid side-chains (Section 5.4.4) and, most recently, by site-directed mutagenesis (Section 5.4.5).

* This sequence constitutes the *elementary steps* of the overall reaction.

5.3 Background to catalysis

Studies of mechanisms in organic chemistry have highlighted various factors that contribute to the enhancement of the rate of reactions. These factors are described (Sections 5.3.1 to 5.3.5) so that their importance in the specific examples of enzyme catalysis discussed in Section 5.5 can be assessed. However, not all the factors are necessarily important in every enzyme-catalysed reaction.

Inspection of the energy profile for a typical reaction (Fig. 5.1(a)) shows that in order to proceed from reactant(s) to product(s) an energy barrier ($\Delta G^\ddagger$) must be surmounted. The highest point of the energy profile is designated the *transition state* of the reaction. From the transition-state theory of reaction rates,¹ an equation can be derived that expresses the relationship between the rate constant, k , and the free energy of activation ($\Delta G^\ddagger$):

$$k = \frac{kT}{h} e^{-\Delta G^\ddagger/RT} = \frac{kT}{h} e^{-\Delta H^\ddagger/RT} \cdot e^{\Delta S^\ddagger/R}$$

where k is the Boltzmann constant,

h is Planck's constant,

T is the temperature,

R is the gas constant,

and $\Delta H^\ddagger$ and $\Delta S^\ddagger$ are the enthalpy and entropy of activation, respectively.

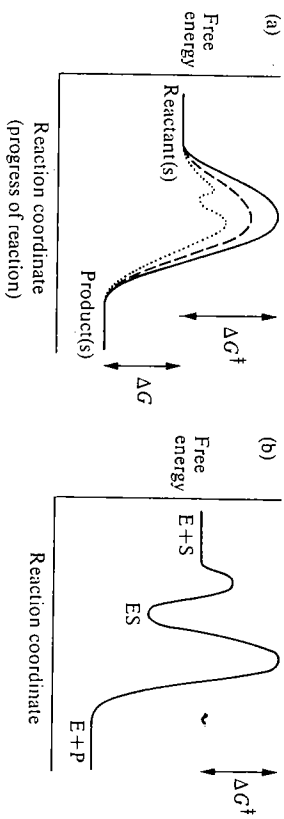


FIG. 5.1. Energy profiles for reactions. (a) Profiles for an uncatalysed reaction (solid line), a catalysed reaction (dashed line), and a catalysed reaction involving intermediate formation (dotted line). (b) Profile for an enzyme-catalysed reaction $E + S \rightleftharpoons ES \rightarrow E + P$. Note that comparisons should not be made between the vertical scales in (a) and (b).

In order to speed up the rate of a reaction at constant temperature, the value of $\Delta G^\ddagger$ must be reduced. Therefore, the function of a catalyst is to provide an alternative reaction pathway so that a lowered energy barrier has to be surmounted (see, for example, the dashed line in Fig. 5.1(a)). One particular way in which a catalyst might function is via the rapid formation and decomposition of an *intermediate*, which is represented by the local minimum in the dotted line in Fig. 5.1(a). This is known as covalent catalysis. In all cases, the catalyst does not alter the magnitude of free-energy change (ΔG in Fig. 5.1(a)) and therefore

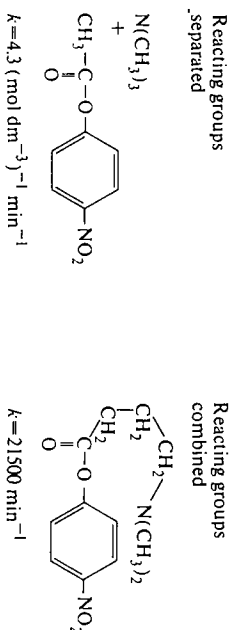
does not cause a shift in the equilibrium between reactant(s) and product(s).^{*} it merely increases the rate at which that equilibrium is attained.

These general considerations also apply to catalysis by enzymes; however, we should note that the energy profile corresponding to even the simplest type of enzyme mechanism is more complex (see Fig. 5.1(b)). The energy profiles of a number of enzyme catalysed reactions have been determined e.g. triosephosphate isomerase (Fig. 5.29), proline racemase² and tyrosyl-tRNA synthetase.³

Various factors leading to rate enhancements that may be relevant to enzyme-catalysed reactions are proximity and orientation effects, acid-base catalysis, covalent catalysis, strain or distortion, and changes in environment. These will now be discussed in turn.

5.3.1 Proximity and orientation effects ★

It seems intuitively obvious that an enzyme could increase the rate of a reaction involving more than one substrate by binding the substrates at adjacent sites and therefore bringing them into close proximity with each other. The reaction would then occur more readily than if it depended on chance encounters between the reacting molecules in solution. Many examples from organic chemistry suggest that reactions can be greatly accelerated when the reacting groups are combined within a single molecule. For example, the following data were obtained for the amine-catalysed hydrolysis of *p*-nitrophenyl esters⁴ (a reaction involving nucleophilic attack by the amine of the carbonyl group of the ester):

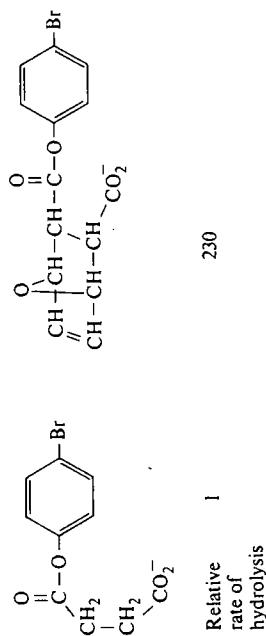


Difficulties arise in such situations when we try to establish the correct basis for comparison; thus, in the above example we are comparing the magnitudes of rate constants that are of different orders and hence of different dimensions. There could also be some differences between the detailed mechanisms of the two reactions. Despite these difficulties, Page and Jencks^{5, 6} have given a detailed thermodynamic treatment of proximity effects in enzyme-catalysed reactions and their conclusions are mentioned at the end of this section after the importance of orientation effects has been discussed.

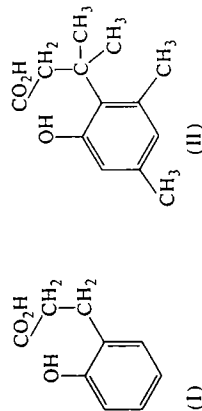
There is no doubt that the orientation of the reacting molecules with respect to each other can greatly influence the rate of reaction. Catalysis could thus be

^{*} This is of course true for catalysis brought about by enzymes. However, if the concentration of enzyme is comparable with that of the substrates and products there can be an *apparent* shift in the equilibrium, see Chapter 8, Section 8.3.4.

achieved by ensuring that the reactants are in the correct orientation as they approach each other. Hydrolysis of the *p*-bromophenyl esters shown below is catalysed in each case by the neighbouring carboxylate group; a process which involves nucleophilic attack by the carboxylate on the carbonyl group of the ester. The presence of a bridge structure that almost totally restricts rotation about the single bond increases the rate of hydrolysis by some 230-fold.⁷

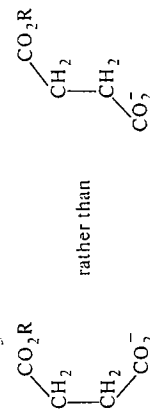


A more spectacular example of rate enhancement is provided by the rates of lactone (internal ester) formation in compounds (I) and (II) below.⁸



The presence of the two methyl groups on the carbon atom adjacent to the aromatic ring in (II) increases the rate of lactone formation by a factor of 4×10^5 .

In these examples the restrictions placed on rotation about single bonds by a bridge structure of bulky substituents ensures that the preferred orientation of the reacting groups closely resembles that of the transition state of the reaction. A simple way of looking at this is to say that the molecule will spend a greater proportion of the time in the conformation that leads to reaction, i.e.



This will mean that less rotational entropy (fewer degrees of freedom) will be lost as the reaction proceeds towards the transition state. The smaller negative entropy of activation will lead to an increase in the rate of reaction.

Estimates of the effects of orientation and proximity on the entropy of activation, and hence on the rate of reaction, have been by Page and Jencks.^{5,6} Their calculations suggest that each factor can contribute a rate enhancement of at least 10^4 -fold for a bimolecular reaction, giving a possible total rate enhancement of at least 10^8 -fold from these two factors combined.

5.3.2 Acid-base catalysis

Many reactions of the type catalysed by enzymes are known to be catalysed by acids and/or bases. An example is provided by the hydrolysis of phenyl- β -D-glucopyranoside⁹ shown in Fig. 5.2. Acid catalysis proceeds (Fig. 5.2(a)) via protonation of the glycosidic oxygen followed by dissociation of phenol to yield a stabilized carbonium ion which is then attacked by water to form glucose. On the other hand, base catalysis proceeds (Fig. 5.2(b)) via abstraction of a proton from the C-2 OH group followed by nucleophilic attack of the O^- group on C-1 to form an epoxide. The epoxide is then attacked by water to form glucose.

Since enzymes contain a number of amino-acid side-chains that are capable of acting as proton donors or acceptors (see Chapter 3, Section 3.3.1), it is reasonable to suppose that acid-base catalysis would be important in enzyme-catalysed reactions. In simple organic reactions, acid catalysis is divided into *specific acid catalysis* (where the rate expression contains contributions from H^+ only) and *general acid catalysis* (where the rate expression includes contributions from H^+ and from other potential proton donors in the solution, e.g.

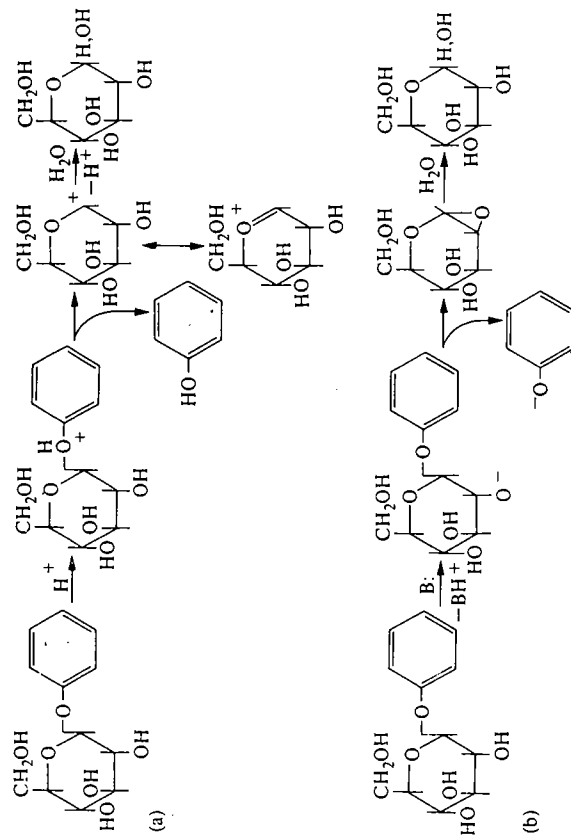


FIG. 5.2. Hydrolysis of phenyl- β -D-glucopyranoside subject to catalysis by acid (a) or base (b).

$\text{CH}_3\text{CO}_2\text{H}$). A similar division is made for base catalysis. As far as enzyme-catalysed reactions are concerned only *general* acid or base catalysis can occur, because enzymes do not possess any mechanism for concentrating H^+ or OH^- . A good example of general acid catalysis occurs in the hydrolysis of oligosaccharides catalysed by lysozyme.¹⁰ In this enzyme the side-chain of Glu 35 is in a highly non-polar environment, and this has the effect of raising its pK_a so that the carboxyl group remains unionized at pH values up to about 6 (the pK_a of the side-chain of free glutamic acid is 4.3). The carboxyl group donates a proton to the glycosidic oxygen of the substrate, facilitating cleavage of the C-O bond, and leading to the formation of a carbonium ion that is stabilized by the neighbouring ionized side-chain of Asp 52 (Fig. 5.3). The analogies with the acid-catalysed hydrolysis of the glucoside in Fig. 5.2(a) should be noted.

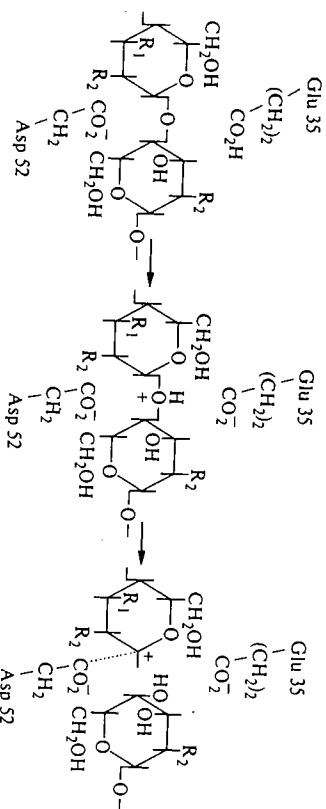


FIG. 5.3. Part of the proposed mechanism of action of lysozyme.¹⁰ The process is completed by reaction of the carbonium ion with water, with overall retention of configuration at C-1.

Histidine side-chains often play important roles as acid-base catalysts in enzyme action (see, for example the cases of chymotrypsin, fructose-bisphosphate aldolase, and lactate dehydrogenase discussed in Section 5.5). This is because the histidine side-chain has a pK_a close to 7, so that near neutral pH there is a reasonable balance between the proton-donating (protonated) form and proton-accepting (deprotonated) forms.

It is conceivable that suitably positioned proton-donating and proton-accepting side-chains in an enzyme could act in a concerted fashion, so that strated in a model system, namely the mutarotation of tetra-*O*-methylglucose (Fig. 5.4) but the importance of the effect in enzyme-catalysed reactions is less clear.¹¹

A fuller discussion of acid-base catalysis and its importance in enzyme-catalysed reactions has been given in the monograph by Fersht.¹²

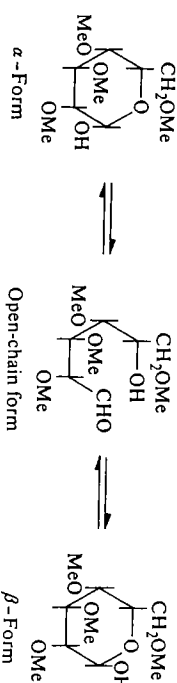


FIG. 5.4. The mutarotation of tetra-*O*-methyl glucose in which α - and β -forms are interconverted via the open-chain form. 2-Hydroxypyridine, in which acid and base groups are combined in one species, acts as a much more effective catalyst than a mixture of pyridine and phenol, indicating that concerted acid-base catalysis occurs.¹¹

5.3.3 Covalent catalysis (intermediate formation)

It has been recognized for many years that reactions can be speeded up by the formation of intermediates, provided that such intermediates are both rapidly formed and rapidly broken down (see the dotted line in Fig. 5.1(a)). An example is provided by the decarboxylation of acetoacetate, which is catalysed by primary amines via the rapid formation and breakdown of a Schiff base or ketimine,¹³ see Fig. 5.5.

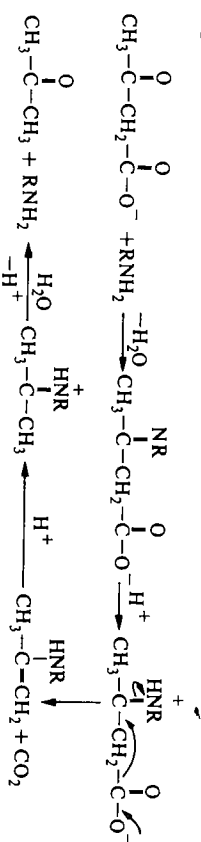


FIG. 5.5. The decarboxylation of acetoacetate catalysed by primary amines.

The advantage conferred by intermediate formation in this case is that the Schiff base is readily protonated, thus providing greater electron-withdrawing power to aid the decarboxylation than would be provided by the original carbonyl group.

Many of the examples of covalent catalysis in enzyme-catalysed reactions involve attack by a nucleophilic side-chain at an electron-deficient centre in the substrate; such attack is termed nucleophilic catalysis. Examples of nucleophilic catalysis are provided by chymotrypsin (Section 5.5.1) and fructose-bisphosphate aldolase (Section 5.5.3); in these cases the nucleophilic side-chains are those of serine and lysine, respectively. Other types of side-chain that have been found to participate in nucleophilic catalysis include cysteine (e.g. papain and glyceraldehyde-3-phosphate dehydrogenase) and histidine (e.g. phosphoglycerate mutase and nucleosidediphosphate kinase).

5.3.4 Strain or distortion

There is the possibility that a substrate may be distorted on binding to the appropriate enzyme. This would speed up the reaction if the distortion (or strain)

lowered the free energy of activation by making the geometry and electronic structure of the substrate more closely resemble that of the postulated transition state.¹⁴ Examples of this effect can be found in model systems, e.g. the very rapid hydrolysis of strained cyclic phosphate esters compared with the open-chain forms.¹⁵ (Fig. 5.6).

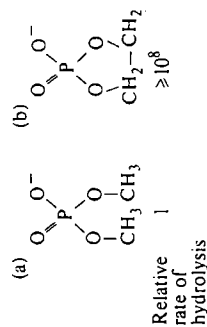


FIG. 5.6. Structures of phosphate esters. The faster rate of hydrolysis of (b) is attributed to the strain in the molecule, which is relieved on hydrolysis.

As is often the case, the evidence is more difficult to interpret in enzyme-catalysed reactions. It has been proposed that the glucose ring at the point of cleavage of an oligosaccharide substrate is distorted on binding to lysozyme, so that the geometry approaches that in the postulated transition state, which involves a carbonium ion¹⁰ (see Figs. 5.3 and 5.7); this should lead to the enhancement of the rate of reaction.* Another example of substrate distortion is provided by studies using i.r. spectroscopy that has shown that dihydroxyacetone phosphate is distorted on binding to triosephosphate isomerase.¹⁷

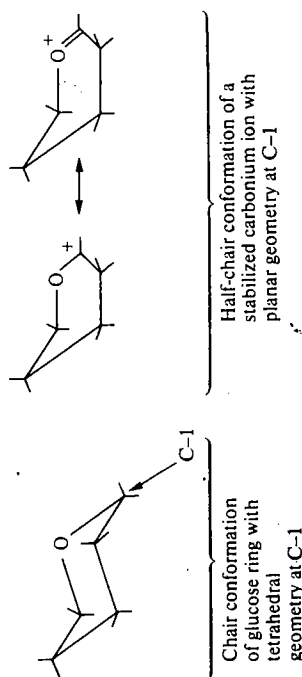


FIG. 5.7. Distortion of glucose ring on formation of a carbonium ion.

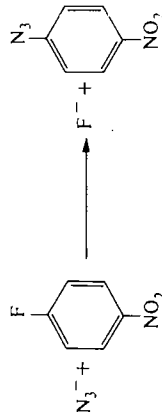
A different way of regarding strain is to think in terms of stabilization of the transition state of the reaction rather than destabilization of the substrate. In other words, it is envisaged that an enzyme is more suited to making favourable contacts with the transition state of the substrate than with the substrate in its

* Detailed calculations of the energy of the lysozyme-substrate complex suggest, however, that this distortion does not in fact contribute greatly to the rate enhancement¹⁶ (see Section 5.3.5).

normal conformation. From this it would follow that a compound that resembles the transition state in geometry and electronic structure will bind more tightly to an enzyme than will the substrate. There has been considerable interest¹⁸ in the search for such transition-state analogues, since these would help to confirm ideas about transition states of reactions. A transition state analogue for triosephosphate isomerase is described in Section 5.5.2.3.

5.3.5 Changes in environment

The rates of many organic reactions are highly sensitive to the nature of the solvents in which they occur. In particular, dipolar aprotic solvents such as dimethylsulphoxide and dimethylformamide, which are not capable of solvating anions, are extremely good solvents for nucleophilic displacement reactions.¹⁹ The reaction below occurs over 12 000 times faster in dimethylsulphoxide than in water.



X-ray crystallographic studies show that enzymes are capable of providing unusual environments for reactions. In the case of lysozyme, the cleft in the molecule that forms the substrate binding site is lined by a number of non-polar amino-acid side-chains that obviously provide an environment markedly different from that of the solvent water. In addition, within the active site there is an appropriately positioned negative charge, the ionized side-chain of Asp 52, which helps to stabilize the positive charge on the carbonium ion in the transition state of the reaction (Fig. 5.3). Calculations¹⁶ suggest that this electrostatic stabilization, which would be strong in a medium of low dielectric constant, may be the most important factor in this case, contributing a rate enhancement of 3×10^6 -fold.

5.3.6 Discussion of the factors likely to be involved in enzyme catalysis

In Sections 5.3.1 to 5.3.5 we have discussed various factors that are generally accepted to be important in bringing about rate enhancement in enzyme-catalysed reactions. The examples of enzyme mechanisms described in Section 5.5 will show how these various factors operate in particular situations. Additional factors may be involved in certain cases; for example, in a metalloenzyme the metal ion can display powerful electron-withdrawing ability thereby polarizing a chemical bond and making it more susceptible to nucleophilic attack. An instance of this type of effect is provided by carboxypeptidase, where a zinc ion serves to polarize the carbonyl group of an amide substrate (see Section 5.5.1.7).

Although the causes of rate enhancement are reasonably well understood in qualitative terms, we are still some way from a *quantitative* understanding of their importance in enzyme-catalysed reactions. However, progress is being made in this direction and recently it has been claimed that the 10^{16} -fold rate enhancement of DNA hydrolysis shown by staphylococcal nuclease can be accounted for by a combination of transition-state stabilization, metal-ion catalysis, general base catalysis, and catalysis by proximity of attacking water.²⁰ It is possible to make practical applications of the knowledge we have concerning rate enhancements both in altering the catalytic properties of enzymes by site-directed mutagenesis (see Section 5.5.5.5) and in designing synthetic 'model' enzymes (or 'synzymes') that are capable of catalysing fairly simple types of reactions. One of the most successful of these attempts has involved the synthesis of an 'artificial' chymotrypsin in which a catalytic site containing imidazole, carboxyl, and hydroxyl groups (see Section 5.5.1.2) has been incorporated into cyclodextrin, which provides the binding site. The artificial enzyme had an M_r only about 5 per cent that of chymotrypsin but showed comparable kinetic features in the esterase reaction and enhanced stability at elevated temperatures and high pH.²¹ Reviews of the design and applications of synthetic enzymes have been given by Wharton²² and Breslow.²³

5.4 Experimental approaches to the determination of enzyme mechanisms

Having discussed the theoretical basis of rate enhancements, we now describe the main experimental approaches that can be used to elucidate the mechanism of action of an enzyme. These approaches will be dealt with under five main headings: kinetic studies (Section 5.4.1), detection of intermediates (Section 5.4.2), X-ray crystallography (Section 5.4.3), chemical modification of amino-acid side-chains (Section 5.4.4), and site-directed mutagenesis (Section 5.4.5). These approaches are to be regarded as complementary; the information gained from one approach should be interpreted in the light of results from the others in order to build up the overall picture of the mechanism.

5.4.1 Kinetic studies

In Chapter 4, we described how the catalytic activity of an enzyme could be assayed and then proceeded to introduce some of the principal concepts and equations of enzyme kinetics. Our object in this section is to show how information about enzyme mechanisms can be gained from these studies. The types of information available from kinetic studies are summarized in Table 5.1 and the various techniques are discussed in turn (Sections 5.4.1.1 to 5.4.1.5).

5.4.1.1 Variation of substrate concentration(s)

Steady-state kinetic studies support the proposal that one-substrate reactions proceed via the formation and decay of one or more enzyme-substrate com-

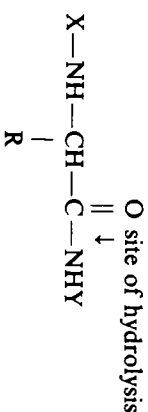
TABLE 5.1
The types of information on enzyme mechanisms available from kinetic studies

Experiment	Information available
Variation of substrate concentration(s)	Sequence of complexes in reaction. Distinction between possible mechanisms
Variation of substrate structure	Structural features responsible for binding and catalytic activity. (Mapping the active site)
Reversible inhibition (see Chapter 4, Section 4.3.2)	Competitive inhibitors can help to define the active site
Variation of pH (see Chapter 4, Section 4.3.3)	The pK_a values of side-chains involved in catalytic activity. Assignment of the pK_a values to particular amino acid side-chains may be possible
Pre-steady-state kinetics (see Chapter 4, Section 4.4)	Detection of enzyme-containing complexes. Rates of elementary steps in the reaction

plexes (see Chapter 4, Section 4.3.1), but cannot by themselves give any indication of the sequence of such complexes. In the case of two-substrate reactions, steady-state kinetic studies can distinguish between ternary complex and enzyme substitution mechanisms (see Chapter 4, Section 4.3.5.1). With additional information from product inhibition, substrate binding, or isotope-exchange experiments, it is possible to distinguish between mechanisms in which a ternary complex is formed in an ordered or a random fashion (see Chapter 4, Section 4.3.5.5).

5.4.1.2 Variation of substrate structure

A good deal has been learnt about the general features of enzyme active sites by correlating the rates of reactions with the structures of the substrates. For instance, by comparing the rates of hydrolysis of a large number of amide derivatives of amino acids it has been demonstrated that chymotrypsin has a strong preference for substrates containing aromatic or bulky hydrophobic R groups, whereas elastase has a strong preference for substrates containing small hydrophobic R groups. Clearly the substrate-binding sites of these enzymes must contain features that account for the observed specificities (see Section 5.5.1.2).



It is possible to develop this approach and map active sites in greater detail. For example exhaustive studies have been carried out on the specificity displayed by papain towards synthetic peptide substrates.²⁴ These studies showed that there are seven 'subsites' on the enzyme (see Fig. 5.8), and that subsite S_2 interacts specifically with an L-phenylalanine side-chain. Later work showed that subsite

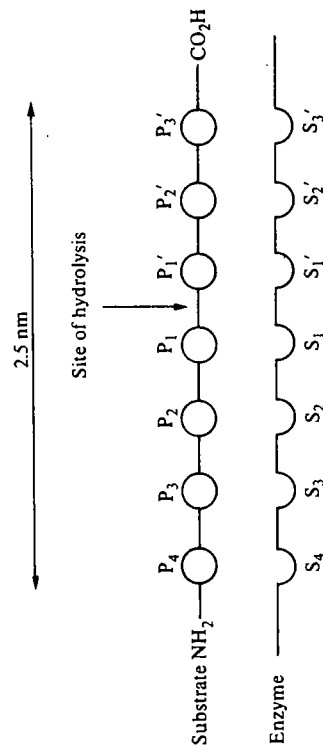


FIG. 5.8. The 'subsites' of papain as revealed by active-site mapping.

S'_1 is stereospecific for L-amino acids with a preference for the hydrophobic side-chains of leucine and tryptophan.²⁵ The explanation in structural terms for the precise interaction between enzyme and substrate in this case presents a formidable challenge to X-ray crystallographers; however, the probable locations of subsites S'_1 , S_1 , and S_2 in the enzyme have been deduced.^{24, 26}

5.4.1.3 Reversible inhibition

The study of inhibition of enzyme-catalysed reactions (see Chapter 4, Section 4.3.2) can give information on the structures of active sites. One of the most likely explanations for competitive inhibition is that substrate and inhibitor bind to the same site on an enzyme. By comparison of the structures of the substrate and the competitive inhibitor it is possible to define the essential structural features of these molecules that are involved in their binding to the active site. A detailed study was carried out in the case of papain;²⁴ thus the tripeptide Ala-Phe-Arg acted as a powerful competitive inhibitor of the enzyme, presumably since it occupied the subsites S_3 , S_2 , and S_1 (Fig. 5.8) and could not therefore undergo hydrolysis. Competitive inhibitors are also of value in X-ray crystallographic work, where it is usually difficult to study the enzyme-substrate complex directly (Section 5.4.3).

5.4.1.4 Variation of pH

The catalytic activity of many enzymes is markedly dependent on pH. As described in Chapter 4, Section 4.3.3, there are a number of explanations for this phenomenon, but the one of most immediate concern here is the ionization of amino-acid side-chains that are involved in the catalytic mechanism. By suitable analysis of plot of reaction rate against pH,²⁷ it is usually possible to deduce the pK_a values of these ionizing side-chains and the identity of the side-chains can be guessed at by comparison with the pK_a values of side-chains of free amino acids (see Chapter 3, Section 3.3.1) or small peptides. However, it should be noted that the environment of a side-chain in an enzyme can shift the pK_a of the side-chain

by up to four units from the value for the free amino acid. More definite deductions about the identities of the side-chains can be made if other information, such as the effect of solvent polarity on the pK_a , is available.* This approach was used, for instance, to implicate two histidine side-chains as being involved in the catalytic mechanism of ribonuclease,²⁸ a conclusion later confirmed by X-ray crystallography and other studies.²⁹

5.4.1.5 Pre-steady-state kinetics

In Chapter 4, Section 4.4 we described how studies of pre-steady-state kinetics could be used to detect enzyme-containing complexes in a reaction and to determine the rates of formation and decay of such complexes. In these experiments, the concentration of enzyme more closely approaches that of the substrates than is the case with steady-state studies, and usually special techniques are required to achieve rapid mixing and rapid detection of the changes occurring.

A simple illustration of the use of pre-steady-state kinetics is afforded by the chymotrypsin-catalysed hydrolysis of *p*-nitrophenyl acetate. The production of *p*-nitrophenol (Fig. 5.9) shows a 'burst' phase³⁰ (the size of which corresponds to approximately 1 mol per mol enzyme) followed by a slower steady-state rate. This observation is consistent with the type of mechanism described in Section 5.5.1.3 for this enzyme, in which a fast step, corresponding to the

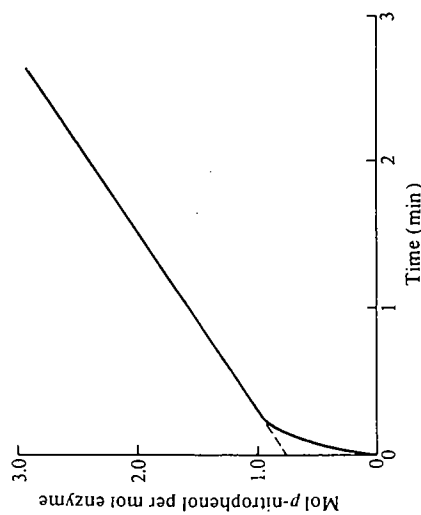
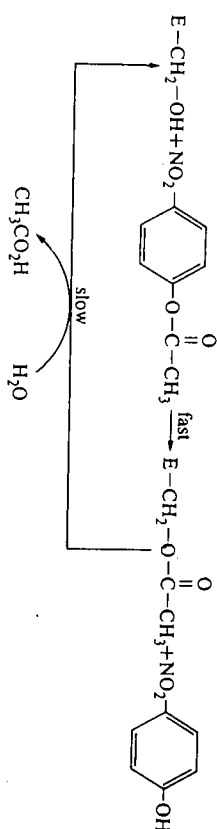


FIG. 5.9. The hydrolysis of *p*-nitrophenylacetate catalysed by chymotrypsin. The solid line represents the experimental data.

* If the dissociating group is neutral, e.g. $-\text{CO}_2\text{H} \rightleftharpoons -\text{CO}_2^- + \text{H}^+$, dissociation is accompanied by charge separation. A decrease in solvent polarity will discourage dissociation, i.e. raise the pK_a . However, if the dissociating group is cationic, e.g. $-\text{ImH}^+ \rightleftharpoons -\text{Im} + \text{H}^+$, there is no separation of charge and the pK_a will be much less sensitive to changes in solvent polarity. (Im = imidazole).



formation of acyl enzyme and release of *p*-nitrophenol, is followed by a slow step, corresponding to the rate of hydrolysis of the acyl enzyme to regenerate enzyme:



This experiment can be performed without the need for specialized apparatus. Clearly, more information on the mechanism would be available from measurements of the early 'burst' phase of the reaction; for example we could measure the rate of formation of the acyl enzyme. The use of fast-reaction techniques is well illustrated by experiments carried out on the reaction catalysed by alcohol dehydrogenase from horse liver.³¹ If the enzyme is mixed rapidly with saturating concentrations of the substrates ethanol and NAD^+ , there is rapid production of 2 mol NADH per mol of the dimeric enzyme, followed by a slower steady-state rate (Fig. 5.10). If the production of NADH is monitored by fluorescence, when free and enzyme-bound NADH can be readily distinguished, it is found that the release of NADH from the enzyme is the rate-determining (i.e. slowest) step in the overall reaction. Values of the rate constants of some of the individual steps in the reaction could be deduced from more detailed measurements.³¹ A further

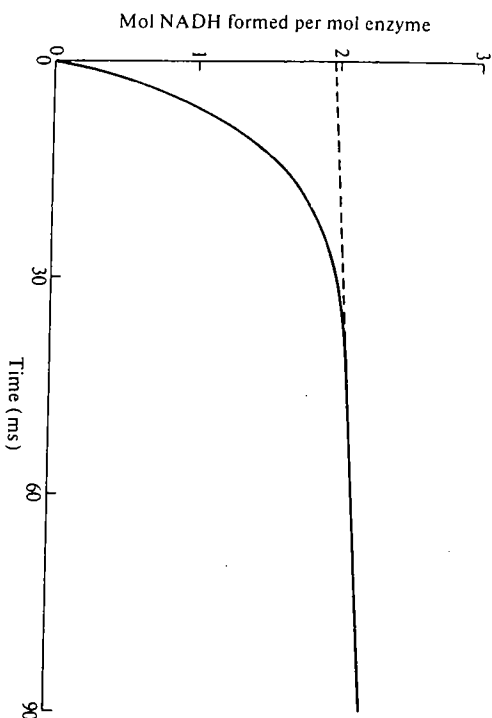
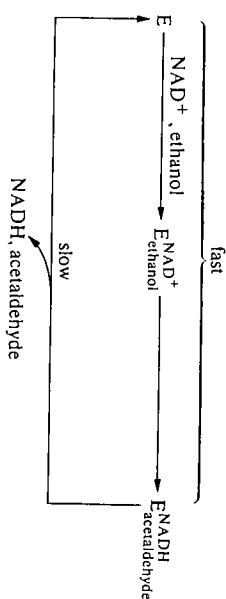


Fig. 5.10. Stopped-flow study of the reaction between ethanol and NAD^+ catalysed by alcohol dehydrogenase from horse liver (a dimeric enzyme).³¹

example of pre-steady-state kinetic studies is provided by the work on the reaction catalysed by lactate dehydrogenase (Section 5.5.4.2).

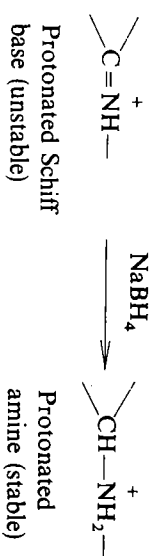


5.4.2 Detection of intermediates

One of the most direct methods for obtaining information about the pathway of a reaction is to detect any intermediates that may be involved in the reaction. In some cases an intermediate is sufficiently stable to be isolated and characterized (the stability will depend on the depth of the minimum in the energy profile shown in Fig. 5.1(a)). In other cases, such as in the reaction catalysed by alcohol dehydrogenase from horse liver, it is possible to infer the existence of an intermediate by spectroscopic means. Whatever the method of detection, it must be shown that the rates of formation and decay of any presumed intermediate are consistent with the overall rate of the reaction. Thus, a postulated intermediate that is so stable that it breaks down more slowly than the overall rate of the reaction cannot be involved in that reaction.

We have already noted (Section 5.4.1.5) that pre-steady-state kinetic studies suggest that an acyl enzyme intermediate is involved in the chymotrypsin-catalysed hydrolysis of esters. The rate of breakdown of the acyl enzyme is very low at acid pH, so that the intermediate can be isolated (and crystallized) if the enzyme and ester substrate are mixed and the pH is then rapidly lowered. X-ray crystallographic work has shown that it is the side-chain of Ser 195 that becomes acylated.³² As we shall see in Section 5.5.1.3, the chymotrypsin-catalysed hydrolysis of amide substrates is also thought to follow a pathway involving formation and breakdown of an acyl enzyme. However, in this case the breakdown step is faster than the formation step, so that the acyl enzyme does not accumulate. It is nevertheless possible to infer its participation in the overall reaction by 'trapping' experiments, as described in Section 5.5.1.3.

Another example of 'trapping' intermediates is provided by the work on fructose-bisphosphate aldolase, where it is possible to reduce the Schiff base intermediate by reduction with sodium borohydride (Section 5.5.3.3). Recently,

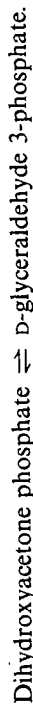


the possibility has been investigated that unstable intermediates can be stabilized at low temperatures, where the rate of breakdown will be decreased.^{33, 34} In such experiments it is usually necessary to work in mixed-solvent systems such as dimethylsulphoxide-water to prevent freezing, although these systems can cause problems by altering reaction mechanisms. In some cases, the use of concentrated solutions of salts such as ammonium acetate may be preferable.³⁵ Considerable success has been achieved with this cryoenzymological approach; thus, in the case of ribonuclease, a series of enzyme-containing complexes on the reaction pathway has been studied by X-ray crystallography.³³ In studies of the elastase-catalysed hydrolysis of a *p*-nitroanilide substrate, the structure of a tetrahedral intermediate (Section 5.5.1.3) has been deduced.³⁴

5.4.3 X-ray crystallographic studies

In Chapter 3 (Section 3.5) we saw that the use of X-ray crystallography has given a wealth of detail about the structures of enzymes. Clearly, we need to be able to locate the active site in an enzyme and then examine the mode of binding of substrate(s) in order to pinpoint the functional amino-acid side-chains that are involved in the catalytic mechanism. Up to fairly recently it has not proved generally possible to study the structure of the catalytically active complex of enzyme and substrate(s) directly, because the collection of diffraction data takes several hours, by which time conversion of substrate(s) to product(s) will have occurred. In the case of an enzyme that catalyses a reaction involving two or more substrates, it is possible to examine the complex formed between the enzyme and one of the substrates; see the example of lactate dehydrogenase in Section 5.5.4.3. A number of alternative approaches have been adopted to gain information on the structure of the catalytically active complex.

Firstly, it may be possible to examine the active complex in the case of a one-substrate reaction when the equilibrium lies very much to one side. Thus, in the case of triosephosphate isomerase, which catalyses the reaction below, it has been possible to examine the crystal structure of the enzyme-dihydroxyacetone phosphate complex (Section 5.5.2.2):



Secondly, an enzyme can be studied in the presence of a very poor substrate or a competitive inhibitor. These molecules are likely to bind to the active site in a manner similar to the substrate but will remain unchanged over the time course of the experiment. By inspection of a three-dimensional model of the enzyme it is then possible to work out how the normal substrate binds to the enzyme. This type of approach is illustrated in the cases of chymotrypsin (Section 5.5.1.2) and lactate dehydrogenase (Section 5.5.4.3).

Thirdly, it may be possible to examine unstable complexes at low temperatures, where the rate of decomposition of such complexes will be decreased. This type of approach has been mentioned in connection with work on ribonuclease and elastase (Section 5.4.2) and will undoubtedly be more widely applied in X-ray crystallographic work in the future.

Recently, there has been considerable interest in the application of synchrotron radiation (the radiation produced during the acceleration of charged particles) to the study of enzyme catalysis. The high intensity of the radiation compared to that of conventional X-ray sources means that the time required for data collection is dramatically reduced and in principle it is possible to obtain information on structural changes or catalytic steps that occur on the millisecond time scale.³⁶ The approach has been used to study the catalysis by phosphorylase *b* in the presence of AMP (see Chapter 6, Section 6.4.2.1) of the reaction between the poor substrate heptenitol and orthophosphate to yield heptulose 2-phosphate. The structure of the ternary complex (which accumulates because the rate of its conversion to product is slow) has been determined.³⁶

The detailed information available from X-ray crystallography is very valuable in providing a framework in which to interpret the results of other experiments. Apart from locating the active site and providing information on the nature of side-chains likely to be involved in the catalytic mechanism, it is also possible to examine the extent of any structural (or conformational*) changes in the enzyme that accompany the binding of substrate(s). These conformational changes may well be of considerable importance in the catalytic process (Section 5.6). However, it should be remembered that even such a powerful technique as X-ray crystallography does have its limitations. Sometimes, attempts to prepare an enzyme-substrate complex by soaking crystals of enzyme in a solution of the substrate leads to cracking of the crystals, thus hampering further crystallographic analysis. The cracking is presumably a result of conformational changes within the enzyme that disrupt the crystal lattice. This type of problem was encountered in the attempt to study the complex between adenylate kinase and AMP.³⁷ In the studies on phosphorylase *a*, the problem posed by the cracking of the crystals on addition of substrates was overcome by prior treatment of the crystals with a cross-linking agent.³⁸ A second type of difficulty is presented by the rather artificial conditions under which crystals are studied. It is possible that in concentrated salt solutions, e.g. 3 mol dm⁻³ ammonium sulphate, the interaction between an enzyme and its substrate may be rather different from that which occurs in solutions of lower ionic strength. Thus, there could be no binding at all or weaker binding to additional sites on the enzyme. Some of the difficulties involved in locating the active site of phosphorylase *a*³⁰ and adenylate kinase^{40, 41} have been discussed.

5.4.4 Chemical modification of amino-acid side-chains

The principle underlying the application of chemical modification techniques to the study of enzyme mechanisms is very simple. If an amino-acid side-chain involved in the catalytic activity is chemically modified, the enzyme will be inactivated. Provided that the identity of the modified side-chain can be

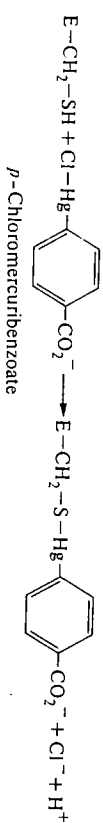
* Conformations of molecules are related to each other by rotations about single bonds. Conformational changes refer to interconversions between different conformations of an enzyme molecule.

established by standard structural techniques, e.g. isolating and sequencing a modified peptide (see Chapter 3, Section 3.4), then it should be possible to determine which side-chain is involved in the catalytic mechanism.

However, chemical modification techniques pose problems both of design and of interpretation (for reviews of the techniques see references 42-44). On the experimental side it is highly desirable that the modification be specific. Thus it should be checked that only one type of side-chain is modified, and further that only one of this type has reacted (e.g. only one of, say, 30 lysines in the enzyme). The requirement for specificity presents considerable problems when it is remembered that many amino-acid side-chains are nucleophilic, so that an electrophile such as 1-fluoro-2,4-dinitrobenzene could be attacked by a variety of side-chains, e.g. those of cysteine, lysine, histidine, tyrosine, etc. In Sections 5.4.4.1 to 5.4.4.3 we shall mention some approaches to improve the specificity of modification reactions; Section 5.4.4.4 will deal with problems encountered in the interpretation of chemical modification experiments.

5.4.4.1 Application of chemical principles

It is well known that mercury has a very strong affinity for sulphur (cf. the very low solubility product of HgS), so it would be expected that mercurial reagents should bring about highly specific modification of cysteine side-chains in enzymes:



We would also expect that the activated aromatic ring in a tyrosine side-chain would be especially susceptible to electrophilic substitution for example by I_2 or by tetranitromethane:



(It has been suggested⁴⁵ that this reaction takes place via initial formation of a charge-transfer complex between tetranitromethane and the phenolate ion, followed by electron transfer.)

It is also useful to take note of the different orders of reactivity observed for the side-chains of free amino acids with various reagents:

acylating reagents (e.g. iodoacetamide, iodoacetate)	Cys > Tyr > His > Lys
aryllating reagents (e.g. 2,4,6-trinitrobenzene sulphonate, 1-fluoro-2,4-dinitrobenzene)	Cys > Lys > Tyr > His

and apply this information in enzyme-modification experiments. Thus, aryllating agents such as 2,4,6-trinitrobenzenesulphonate should be used in preference to acylating agents to modify lysine side-chains.

Using these principles it is possible to draw up a list of reagents that should display reasonable specificity for modifying particular types of amino-acid side-chain (Table 5.2). However, the specificity of any modification reaction should always be checked, e.g. by analysis of the modified enzyme, since the reactivity of a side-chain can be considerably influenced by its local environment within an enzyme.

TABLE 5.2
Some chemical modification procedures that show reasonable specificity for amino-acid side-chains in enzymes

Side-chain	Reagent(s) used
Cysteine	Mercurials, e.g. <i>p</i> -chloromercuribenzoate. Disulphides, e.g. 5,5'-dithiobis-(2-nitrobenzoic acid). Iodoacetamide
Lysine	2,4,6-trinitrobenzenesulphonate. Pyridoxal phosphate ($\pm$ reducing agent such as NaBH_4)
Histidine	Diethylpyrocarbonate Photo-oxidation
Arginine	Phenylglyoxal 2,3-Butanedione
Tyrosine	Tetranitromethane <i>N</i> -acetylhydrazole Iodine
Tryptophan	<i>N</i> -bromosuccinimide
Aspartic acid or Glutamic acid	Water-soluble carbodiimide plus nucleophile, e.g. glycine methyl ester

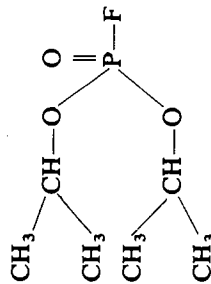
Details of these procedures are given in reference 42.

Changes in the selectivity of certain reagents can be brought about by changing the pH. For instance, at pH values above about 7, the side-chain of cysteine ($\text{pK}_a \approx 8$) is extremely reactive towards iodoacetate because there is a significant proportion of the ionized form, $-\text{CH}_2-\text{S}^-$, which is the reactive nucleophile. At pH values below 6, the fraction of the ionized form of the cysteine side-chain is much less and the side-chain of methionine ($-\text{CH}_2-\text{CH}_2-\text{S}-\text{CH}_3$) is a more reactive nucleophile. By working at pH 5.6 it was possible to react a methionine side-chain with iodoacetate in isocitrate dehydrogenase from pig heart without any modification of cysteine side-chains.⁴⁷

5.4.4.2 Super-reactive side-chains

Sometimes one particular amino-acid side-chain in an enzyme is especially reactive because of its unique environment. In such cases, specific modifications

can be achieved in an 'accidental' manner. A good example of such a super-reactive group is provided by chymotrypsin, where it is found that diisopropylphosphoramide modifies only the side-chain of Ser 195 and does not react with any of the other 27 serine side-chains in the enzyme or with free serine. The reason for the super-reactivity of the side-chain of Ser 195 has been found from



Diisopropylfluorophosphate

X-ray crystallographic studies (Section 5.5.1.2). Another example of a super-reactive group is provided by glutamate dehydrogenase from beef liver, where the side-chain of one lysine (Lys 126) out of a total of over 30 lysine side-chains in each subunit is especially reactive towards 2,4,6-trinitrobenzenesulphonate.⁴⁸ In chymotrypsin and glutamate dehydrogenase these super-reactive groups are involved in the catalytic mechanisms, but this is not always the case. For instance, the highly reactive thiol group in creatine kinase does not appear to be essential for the catalytic activity of the enzyme (Section 5.4.4.4).

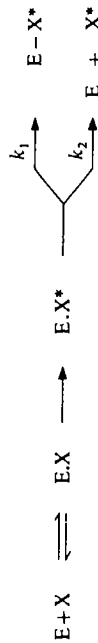
5.4.4.3 Affinity labelling

One way of improving the specificity of a modifying reagent is to incorporate within it some structural feature that will 'direct' it to a certain site on the enzyme, such as the active site. The reactive part of the reagent will then react with an amino-acid side-chain in the vicinity of that site. In the case of triosephosphate isomerase, bromohydroxyacetone phosphate acts as an affinity label,⁴⁹ the resemblance between the affinity label and the substrate for the enzyme is shown below:

Substrate	Affinity label
CH ₂ OH	CH ₂ Br
C=O	C=O
CH ₂ OPO ₃ ²⁻	CH ₂ OPO ₃ ²⁻
Dihydroxyacetone phosphate	Bromohydroxyacetone phosphate

The affinity label binds to the active site of the enzyme and the labile Br atom (activated by the adjacent carbonyl group) can be displaced by a suitably positioned nucleophilic amino-acid side-chain (Section 5.5.2.4). Affinity labelling of chymotrypsin and of trypsin is described in Section 5.5.1.4.

The ideas underlying affinity labelling have been extended to the design of so-called 'suicide substrates' for enzymes.⁵⁰ When acted upon by the appropriate enzyme such a substrate is converted to a product that essentially irreversibly inactivates the enzyme, usually by covalent modification (see the scheme below)



Reaction of an enzyme, E, with a suicide substrate, X. After binding to the enzyme, the substrate is converted by the enzyme to a reactive form (X*). The E.X* complex can either form E-X*, an irreversibly inactivated complex, or can dissociate to form E and X*. The balance between these two processes will depend on the relative values of the rate constants k_1 and k_2 .

A number of suicide substrates (e.g. clavulanic acid and sulbactam) have been designed for penicillinase (the enzyme that confers resistance towards penicillin), and it is clear that this is an area of considerable pharmaceutical importance.⁵⁰

5.4.4.4 Interpretation of chemical modification experiments

Considerable caution is required in interpreting the results of chemical modification experiments. If modification of a particular amino-acid side-chain leads to inactivation of an enzyme, it does not necessarily follow that the side-chain is directly involved in the catalytic mechanism. It could happen, for instance, that the reacting side-chain is some way from the active site, but that modification causes a conformational change in the enzyme leading to loss of activity (Fig. 5.11). There is obviously a whole range of possibilities for the positioning of

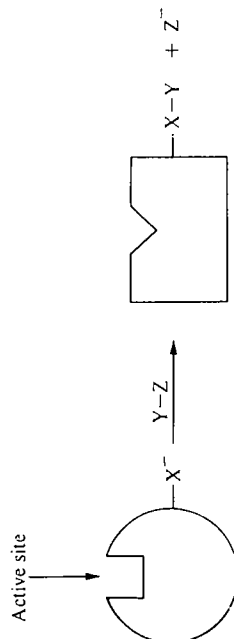
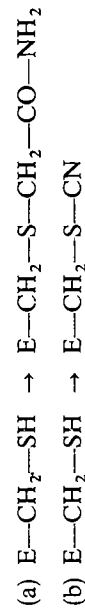


FIG. 5.11. Modification of a group, $-X^-$, outside the active site of an enzyme causes a conformational change leading to loss of activity.

such a side-chain relative to the active site. In many cases it may be possible to make an assessment of the extent of involvement of a side-chain in the catalytic mechanism by comparing the effects of modification by groups of different sizes. Thus, in the case of creatine kinase from rabbit muscle, modification of a single cysteine side-chain by iodoacetamide leads to complete loss of activity (a); however, the smaller perturbation (b) leads to only a 30 per cent loss of activity.⁵¹



The cysteine side-chain cannot therefore be directly involved in the catalytic mechanism. Side-chains that can be ascribed a secondary type of role may play a part in binding of substrates or in conformational changes in the enzyme that are involved in catalysis. Any modification of a side-chain that was directly involved in the catalytic mechanism, such as that of Ser 195 of chymotrypsin which takes part in the formation of the acyl enzyme, would lead to complete inactivation. In a chemical modification experiment there are two criteria that must be fulfilled before it can be stated with any degree of certainty that a modified side-chain is at the active site of an enzyme.⁵²

Firstly, there must be a stoichiometric relationship between the extent of inactivation and the extent of modification, so that complete inactivation corresponds to the modification of one side-chain per active site (Fig. 5.12(a)). Secondly the addition of substrate or competitive inhibitor must protect against inactivation (Fig. 5.12(b)), since the side-chain is no longer accessible to the modifying reagent. The need for proper analysis of such protection experiments has been emphasized.⁵³

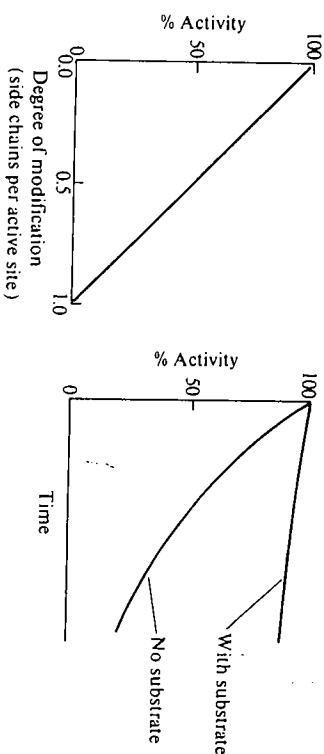


FIG. 5.12. Criteria for modification at the active site of an enzyme. (a) Stoichiometric relationship between extent of inactivation and extent of modification. (b) Protection against inactivation by addition of substrate (or competitive inhibitor).

These criteria should be regarded only as minimal and it is possible that they can be fulfilled even when modification is occurring at a side-chain remote from the active site; the example of the modification of the side-chain of Cys 165 in lactate dehydrogenase is mentioned in Section 5.5.4.4. Despite these problems, chemical modification techniques have been used very extensively to help elucidate details of enzyme mechanisms. We shall refer to chemical modification data in all examples of mechanisms discussed in Section 5.5.

5.4.5 Site-directed mutagenesis

Since the early 1980s the development of recombinant-DNA technology (see Chapter 11, Section 11.5.5) has led to the possibility of introducing amino-acid

replacements at specified positions in a protein, provided that (a) the gene coding for the protein is available, and (b) there is a suitable vector for expression of the gene. The details of this technique, known as site-directed mutagenesis or oligonucleotide-directed mutagenesis are beyond the scope of this book, but can be found in a number of articles and reviews.⁵⁴⁻⁵⁷

A typical procedure is outlined in Fig. 5.13. The gene of interest is cloned into a vector such as the filamentous phage M13. The DNA of this phage, which is in the form of a covalent circle of single-stranded DNA (ssDNA), goes through a covalent double-stranded form (duplex) during replication in the *E. coli* host. One strand of the DNA is isolated. An oligonucleotide is synthesized that is complementary to the region of the gene to be mutated except for a single base (or double base) change that changes the codon from that (those) of the target amino acid(s) to that (those) of the desired replacement amino acid(s). (For example, the change 5'-TTG-3' to 5'-TTC-3' changes the codon from that for Gln to that for Glu.) This oligonucleotide is annealed to the DNA strand and serves as the primer for DNA synthesis by the Klenow fragment of DNA polymerase I (this fragment contains the 5'→3' polymerase and 3'→5' exonuclease activities, but not the 5'→3' exonuclease activity of the intact enzyme). After joining (by DNA ligase) the ends of the new strand of DNA, the heteroduplex is used to transform a host (e.g. the JM strain of *E. coli*). Clones containing the mutant DNA can be selected by their ability to hybridize (form hydrogen bonds) to the

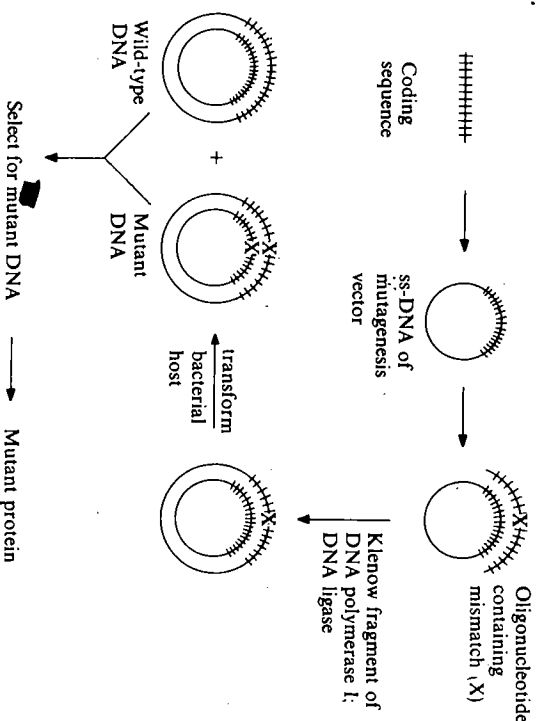


FIG. 5.13. Typical procedure for site-directed mutagenesis. The mismatch is depicted as (X) within a complementary (i.e. base-pairing) region. For further details, see references 54-57.

original oligonucleotide under more forcing conditions of temperature and ionic strength than clones containing the wild-type DNA. The mutant enzyme can then be produced in large quantities in the transformed cells.

A number of techniques are available to increase the frequency of mutants in the transformed cells. One of these involves the incorporation of sulphur-containing nucleotide analogues (phosphothioates) during the *in vitro* synthesis of DNA by the Klenow fragment, which then protects the mutant strand against 'nicking' by certain restriction endonucleases.^{58, 59} This allows the amount of non-mutant strand (which is subject to 'nicking') to be reduced by digestion with exonuclease III.

As an alternative to the use of single-stranded templates such as M13, it is possible to use double-stranded vectors from plasmids in which the region requiring to be mutated is converted to a single-stranded form by 'nicking' with a restriction endonuclease followed by treatment with exonuclease III.^{54, 60} The resuspended portion can then be annealed with the oligonucleotide primer.

The various methods rely on the ready availability of oligonucleotides of defined sequence.⁶¹ Oligonucleotides of 16–21 bases in length are generally used for single base mismatches; longer primers are used for multiple mismatches. The mismatch(es) is (are) best kept towards the middle of the oligonucleotide to protect it (them) from excision by the 3'→5' exonuclease activity of the Klenow fragment.

Site-directed mutagenesis is clearly a very powerful method for assessing the importance of particular amino-acid side-chains in an enzyme. It is necessary to perform some control experiments; thus the mutant gene should be sequenced to confirm that only the desired mutation(s) has (have) been introduced. In addition, it is desirable to check that the overall structural features of the enzyme have not been altered by the mutation, since otherwise it would be difficult to ensure that a change in the catalytic properties is solely a consequence of the replacement amino acid introduced. Spectroscopic techniques such as circular dichroism (see Chapter 3, Section 3.5.2) are useful in this regard, since the overall structural features of an enzyme can be ascertained quickly,⁶² in contrast to the laborious process of structure determination by X-ray crystallography.

Site-directed mutagenesis requires the gene for the enzyme of interest to be available. The results are capable of more detailed interpretation if the X-ray structure of the enzyme is known. Chemical modification of amino-acid side-chains (see Section 5.4.4) is still widely used in cases where these conditions are not fulfilled and to gain an idea of the types of side-chains that may be important for the function of an enzyme. In addition, chemical modification can introduce functionalities into a protein other than those available in the range of amino-acid side-chains. Thus heavy atoms can be introduced for X-ray crystallography (See Chapter 3, Section 3.5.1) and spin labels or fluorescent groups can be introduced for spectroscopic studies.⁶³

The impact of site-directed mutagenesis on studies of enzyme mechanisms is illustrated in most of the examples discussed in Section 5.5.

5.5 Examples of enzyme mechanisms

In this section we shall discuss current ideas about the mechanisms of action of five enzymes; chymotrypsin, triosephosphate isomerase, fructose-bisphosphate aldolase, lactate dehydrogenase and tyrosyl-tRNA synthetase. These enzymes catalyse different types of reactions, i.e. they belong to different classes in the classification scheme described in Chapter 1, Section 1.5.1. In each example it should become apparent how various experimental approaches have been used in a complementary fashion to formulate a reasonably detailed picture of the mechanism. Comments on the mechanisms of related enzymes will be made where appropriate.

5.5.1 Chymotrypsin

Chymotrypsin is a protein-hydrolysing enzyme. It is synthesized in the acinar cells of the pancreas as an inactive precursor (zymogen) known as chymotrypsinogen that is activated in the small intestine by the action of proteases (see Chapter 6, Section 6.2.1.1). The mechanism of chymotrypsin has been intensively studied⁶⁴, and is probably better understood than any other. We shall discuss in turn the information gained from studies of X-ray crystallography, site-directed mutagenesis, detection of intermediates, chemical modification, and kinetics before outlining the proposed mechanism of action.

5.5.1.1 Background information on the structure and action of the enzyme

Chymotrypsin has an M_r of about 25 000 and consists of three polypeptide chains held together by disulphide bonds. This unusual structure arises as a result of the activation of the zymogen by proteolytic cleavage (Fig. 5.14). The activation of chymotrypsinogen occurs in stages.⁶⁴ There is an initial cleavage, catalysed by trypsin, of the Arg 15-Ile 16 bond that yields a fully active, two-chain, enzyme known as π -chymotrypsin. Subsequent cleavage, catalysed by

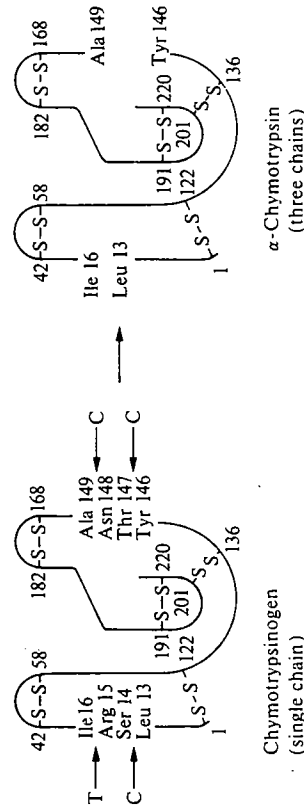
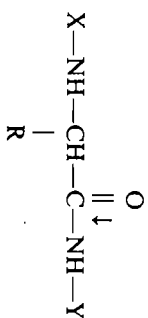


FIG. 5.14. Activation of chymotrypsinogen. T and C represent the actions of trypsin and chymotrypsin, respectively.

chymotrypsin then takes place to yield δ -chymotrypsin (by cleavage of the Leu 13-Ser 14 bond), and finally α -chymotrypsin (by cleavage of the Tyr 146-Thr 147 and Asn 148-Ala 149 bonds). α -Chymotrypsin is the most commonly studied form of the enzyme.

Chymotrypsin catalyses the hydrolysis not only of proteins but also of small substrates and in the latter case will act on both ester and amide bonds. The enzyme shows relatively little specificity when acting on ester substrates: a finding that can be correlated with the inherently higher reactivity of esters. In the case of amide bonds (which are intrinsically less easily hydrolysed) the only substrates that are acted on by chymotrypsin are those with large hydrophobic side-chains, principally those of Trp, Tyr, and Phe, and, to a lesser extent, those of Leu and Met. Specificity is shown for the nature of the R group attached to the carbon atom on the amino side of the amide bond. Despite the differences in specificity as far as amide and ester substrates are concerned, there are good grounds for believing that the mechanism of hydrolysis is the same for both types of substrates (Section 5.5.1.3).



5.5.1.2 X-ray crystallographic studies

The structure of α -chymotrypsin has been discussed briefly in Chapter 3, Section 3.5.6.2, where it was mentioned that the molecule contains a significant amount of antiparallel β -sheet structure. There is a shallow depression at the active site, and also a distinct pocket that plays an important part in the binding of specific substrates.⁶⁵ Of special note is the finding that certain charged groups, e.g. the α -amino group of Ile 16* and the carboxylate side-chains of Asp 102 and Asp 194, are buried in the interior of the molecule.⁶⁵ The significance of these groups is discussed later in this section.

X-ray crystallographic studies have provided answers to a number of questions about chymotrypsin:

- (1) Why is the side-chain of Ser 195 especially reactive?
- (2) How do substrates bind to the enzyme?
- (3) How do the differences in specificity between chymotrypsin and related proteases arise?
- (4) How is the zymogen activated?

* The numbering of amino acids in chymotrypsin refers to the sequence of the intact polypeptide chain of chymotrypsinogen.

(i) Why is the side-chain of Ser 195 especially reactive?

Chymotrypsin is one of a number of enzymes (trypsin, elastase, and thrombin are other examples) that are classified as serine proteases, on the basis that modification of a single serine side-chain in each molecule by a reagent such as diisopropylfluorophosphate leads to inactivation. In chymotrypsin this reactive serine has been identified as Ser 195. The reason for the high reactivity of this side-chain (the other 27 serine side chains in chymotrypsin do not react with diisopropylfluorophosphate, nor does free serine) was unclear until X-ray crystallography showed that there was a precisely arranged group of three side-chains in the enzyme.⁶⁶ This arrangement allows the possibility of partial ionisation of the side-chain of Ser 195 by a 'charge relay' mechanism (Fig. 5.15).

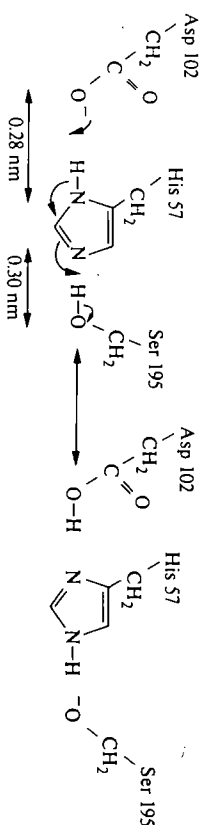


FIG. 5.15. The charge-relay system in chymotrypsin.⁶⁶

Normally ionization of serine side chains is insignificant because of the high pK_a (≈ 14) of the $-\text{CH}_2-\text{OH}$ group. The partial negative charge on the oxygen atom increases the nucleophilicity of the serine side-chain enormously; this is exploited in the catalytic mechanism that proceeds via acyl enzyme formation. All other serine proteases so far examined possess this catalytic triad or charge-relay arrangement of Asp ... His ... Ser side-chains. The extensive similarities of amino-acid sequences around these amino acids in various serine proteases have already been noted (see Chapter 3, Section 3.4.9.5). The details of the charge-relay systems in these enzymes have been extensively investigated. Using n.m.r.^{67, 68} and neutron diffraction⁶⁹ techniques it has been shown that His 57 is the ionizing group of pK_a around 7 and that the buried Asp 102 is of low pK_a . Site-directed mutagenesis has also been used to assess the importance of Asp 102 in the charge-relay system.⁷⁰ The mutant trypsin Asp 102 $\rightarrow$ Asn showed a k_{cat} value some 5000-fold lower than that of the native enzyme, showing that Asp 102 is important but not absolutely required for catalysis. Using affinity labels that modified different side-chains at the active site, it was shown that the Asp 102 $\rightarrow$ Asn mutation had considerably greater effects on the reactivity of Ser 195 than on that of His 57. Other studies using site-directed mutagenesis have shown that the three side-chains in the charge-relay system act synergistically to cause rate enhancement.¹⁵⁶

(ii) How do substrates bind to the enzyme?

A number of crystallographic studies have been made of the binding of poor substrates or competitive inhibitors to the enzyme. The results of such studies⁷¹

on the binding of the inhibitor *N*-formyl tryptophan (amides of which are good substrates for chymotrypsin) are shown in Fig. 5.16. The aromatic side-chain of the inhibitor binds in the pocket, which is lined with non-polar side-chains. The $-\text{NH}-$ group forms a hydrogen bond with the peptide-chain carbonyl group of Ser 214 and the carboxyl group (and hence, by implication, the amide bond of a substrate) is close to the side-chain of Ser 195. The carbonyl group can also form hydrogen bonds with the peptide chain $-\text{NH}-$ group of Ser 195 and/or Gly 193.

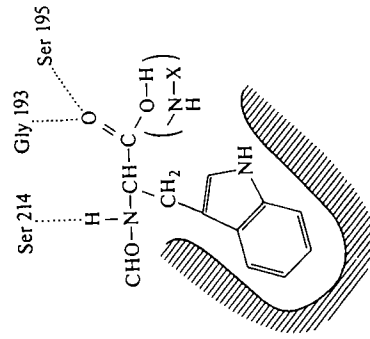


FIG. 5.16. Binding of the inhibitor *N*-formyl tryptophan to chymotrypsin. The structure of the substrate is indicated in brackets.

(iii) How do the differences in specificity between chymotrypsin and related proteases arise?

The serine proteases chymotrypsin, trypsin, and elastase possess very similar three-dimensional structures, but display quite different specificities for substrates. Chymotrypsin is specific for amides with aromatic or other large hydrophobic side-chains, trypsin is specific for amides with positively charged side-chains (Lys or Arg), and elastase has a somewhat broader specificity, showing a preference for amides with small hydrophobic side-chains, e.g. Ala.⁷² The results of X-ray crystallography have shown that these differences in specificity can be correlated with differences in the substrate-binding pockets of the three enzymes (Fig. 5.17). In the case of chymotrypsin, the hydrophobic side-chain of the substrates makes a number of favourable contacts with the non-polar side-chains lining the pocket. At the bottom of the pocket is the uncharged side-chain of Ser 189. In trypsin, this serine is replaced by aspartic acid and the negatively charged side-chain is then exactly placed to bind electrostatically to the positively charged side-chain of a substrate. Elastase has a pocket very similar to that of chymotrypsin; however, access to the pocket is obstructed by the bulky side-chains of Val 216 and Thr 226 and hence bulky side-chains of potential substrates cannot enter the pocket.⁷² Site-directed mutagenesis has been used to investigate the specificity of trypsin.⁷⁰ The side-chains of Gly 216 or

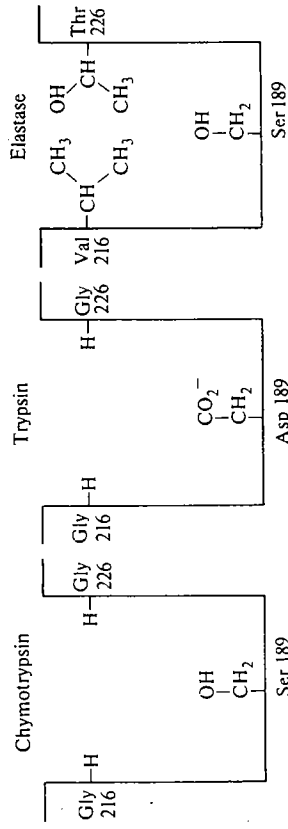


FIG. 5.17. Comparison of the substrate binding pockets of the serine proteases chymotrypsin, trypsin, and elastase.

Gly 226 or both have been replaced by Ala. The mutant enzymes show differing specificities towards lysine and arginine substrates; thus the mutation Gly 226 $\rightarrow$ Ala increases the preference of the enzyme for lysine rather than arginine.

(iv) How is the zymogen activated?

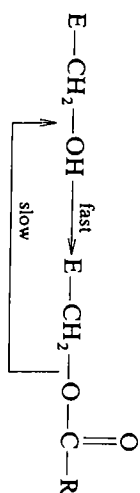
A comparison of the three-dimensional structures of chymotrypsinogen and chymotrypsin allows us to identify the differences between the two molecules and to explain the activation process in structural terms. One perhaps surprising finding is that the Asp ... His ... Ser charge-relay system is already present in the zymogen, with the three side-chains occupying very similar or identical positions to those found in the active enzyme.⁷³ The zymogen, however, is inactive* because the substrate-binding pocket is not properly formed, and thus a substrate cannot be positioned precisely to take advantage of the reactive serine side-chain. Formation of the substrate-binding pocket during the activation process proceeds as follows.⁷³ Cleavage of the Arg 15-Ile 16 bond by trypsin creates a new positive charge at the α -amino group of Ile 16. A strong electrostatic force between this positive charge and that of the side-chain of Asp 194 helps to move other parts of the molecule, such as the side-chains of Arg 145 and Met 192, and to form the substrate-binding pocket. The electrostatic interaction between Ile 16 and Asp 194 is strong because it occurs in a region of low dielectric constant in the interior of the enzyme (see Chapter 3, Section 3.5.5.2).

5.5.1.3 Detection of intermediates

We have already mentioned (Section 5.4.1.5) that the chymotrypsin-catalysed hydrolysis of esters such as *p*-nitrophenyl acetate proceeds via an acyl enzyme intermediate.

* Some careful work has shown that chymotrypsinogen does show very slight activity in catalysing the hydrolysis of esters; the activity is some 10⁶-fold less than that of chymotrypsin. The mechanism of the zymogen-catalysed reaction appears to be similar to that of the active enzyme, proceeding via an acyl enzyme.⁷⁴





In the hydrolysis of amides, it would appear that the deacylation step is more rapid than formation of acyl enzyme; this can be correlated with the lower reactivity of amides compared with esters. There is, therefore, no accumulation of the acyl enzyme and its participation in the reaction can only be inferred indirectly by 'trapping' experiments. A suitable 'trapping' experiment involves conducting the enzyme-catalysed hydrolysis of a substrate in the presence of a high concentration of a nucleophile such as alaninamide or hydroxylamine that can compete with water in the breakdown of the acyl enzyme:

The ratio of products (a)/(b) was determined using an ester substrate for which the involvement of an acyl enzyme intermediate had been established directly. It was then found that, at the same concentration of nucleophile, an identical product ratio was observed using an amide substrate.^{7,5} This provides very good evidence for the participation of an acyl enzyme intermediate in the hydrolysis of amides.

Direct evidence for the existence of tetrahedral intermediates has been gathered from a number of experiments, mainly on enzymes related to chymotrypsin. Firstly, stopped-flow studies (Chapter 4, Section 4.4) on the elastase-catalysed hydrolysis of an amide substrate, a *p*-nitroanilide derivative of a tripeptide, have shown that the decomposition of the first tetrahedral intermediate to the acyl enzyme is the rate-limiting step of the overall hydrolytic reaction.⁷⁶ Secondly, X-ray crystallographic studies have been made of the very strong complexes formed by trypsin with small protein inhibitor molecules such

as bovine pancreatic trypsin inhibitor and soybean trypsin inhibitor.^{77, 78} These studies have shown that the strong interactions arise at least in part from the fact that in these complexes the crucial amide bonds of the inhibitors are distorted so as to resemble a tetrahedral intermediate. Thirdly, as we have already mentioned (Section 5.4.2), it has proved possible by working at low temperatures to stabilize a tetrahedral intermediate in the elastase-catalysed hydrolysis of a *p*-nitroanilide substrate.³⁴

There have been a large number of chemical-modification studies on chymotrypsin, but we shall mention only some of those that have confirmed the roles of side-chains implicated by X-ray crystallographic work.

Some elegant affinity-labelling experiments (Section 5.4.4.3) have shown that the side-chain of His 57 is involved in the catalytic activity of the enzyme. The

close structural relationship between the reagent *N*-*p*-toluenesulphonyl-L-phenylalanine chloromethylketone (TPCK) and the ester substrate is shown in Fig. 5.19. TPCK inactivates chymotrypsin and fulfils the usual criteria for modification at the active site (Section 5.4.4.4). The site of modification is the side-chain of His 57.⁶⁴ That the affinity label does indeed recognize a particular site in chymotrypsin is nicely demonstrated by the finding that TPCK does not inactivate trypsin, which possesses a different substrate-binding pocket (Fig. 5.16). Conversely, the reagent *N*-*p*-toluenesulphonyl-L-lysine chloromethylketone (TLCK) inactivates trypsin but has no effect on chymotrypsin.⁷⁹ The two reagents find considerable application in protein-sequencing studies (Chapter 3, Section 3.4.2), since in the preparation of peptide fragments it is essential to ensure that the chymotrypsin used is not contaminated by trypsin or vice versa. (Thus it is desirable to use TPCK-treated trypsin and TLCK-treated chymotrypsin.)

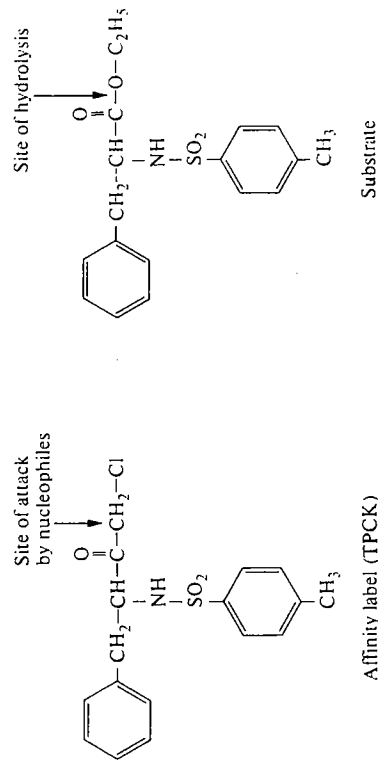


FIG. 5.19. The structure of the affinity label TPCK and its relationship to a substrate of chymotrypsin.

The importance of the α -amino group of Ile 16 in chymotrypsin is suggested by chemical modification studies.⁶⁴ It would normally be extremely difficult to modify this amino group selectively in view of the presence of the other amino groups in the molecule. This problem was overcome by a 'double-labelling' experiment as outlined in Fig. 5.20, in which chymotrypsinogen was first reacted with acetic anhydride (Ac_2O) to acetylate all the amino groups (α - and ϵ -) in the molecule. This acetylated chymotrypsinogen could then be cleaved by trypsin to yield a fully active acetylated δ -chymotrypsin which has a free α -amino group at Ile 16. On treatment with ^{14}C -labelled acetic anhydride it is found that activity is lost in proportion to the amount of radioactivity incorporated into the α -amino group of Ile 16, thus confirming the importance of this group in the activity of the enzyme.⁶⁴

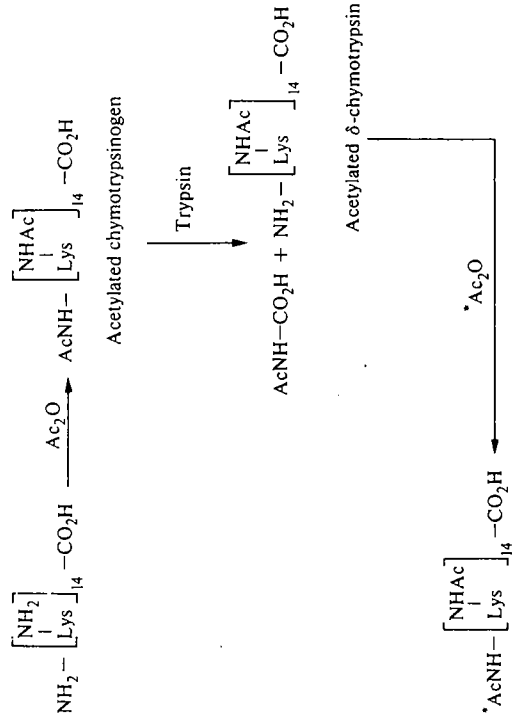


FIG. 5.20. Acetylation of the α -amino group of Ile 16 in chymotrypsin by a double-labelling experiment. Ac_2O is acetic anhydride.

5.5.1.5 Kinetic studies

The kinetics of the reactions catalysed by chymotrypsin have been very intensively investigated. The results of these experiments are more easily interpreted if small synthetic ester or amide substrates are employed; many of these substrates (e.g. *p*-nitrophenyl esters or *p*-nitroanilide derivatives) possess convenient spectroscopic properties that allow the hydrolysis reactions to be monitored easily. The results of steady-state and pre-steady-state kinetic studies have allowed conclusions to be drawn regarding (i) the reaction pathway and the involvement of the acyl enzyme; (ii) the specificity of the enzyme and the nature of the substrate-binding site; and (iii) the magnitudes of the rate constants of a number of the individual steps in Fig. 5.18. For a summary of this work, reference 80 should be consulted.

5.5.1.6 The proposed catalytic mechanism of chymotrypsin

From the various types of experiment described in Section 5.5.1.2 to 5.5.1.5, a mechanism for the chymotrypsin-catalysed hydrolysis of amide (and ester) substrates has been proposed.⁶⁶ The mechanism is outlined in Fig. 5.21, where for the sake of clarity details of the substrate-binding site and the movements of substrate and enzyme have been omitted. A fuller account of these aspects is given in reference 81. In the acyl enzyme, the charge-relay system is disrupted, since no proton is available to link the side-chains of His 57 and Ser 195. However, in the deacylation step, the $\text{R}'\text{NH}_2$ leaving group is replaced by H_2O and the charge-relay system can be re-established.

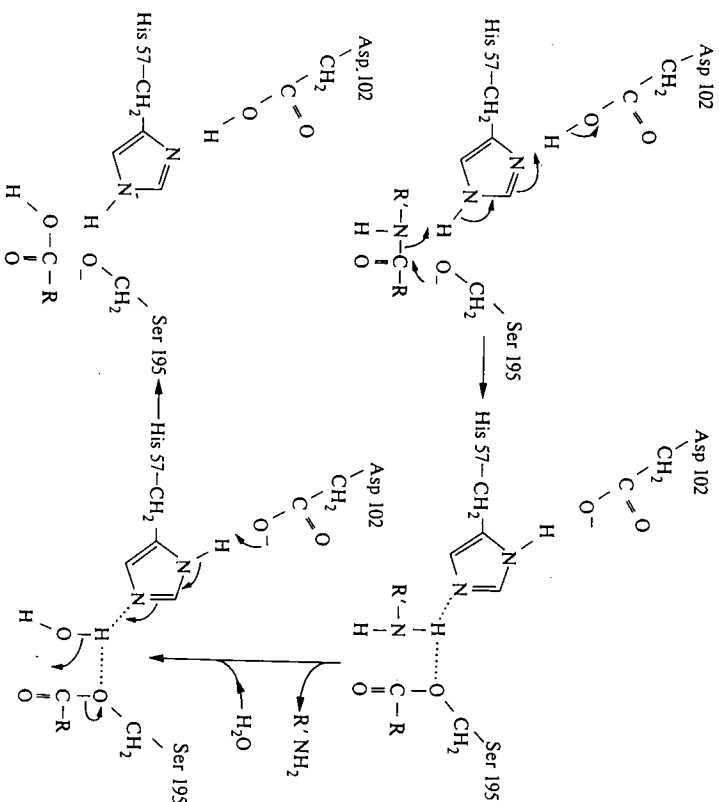
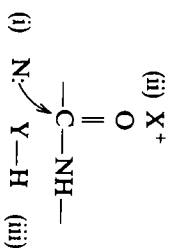


FIG. 5.21. The proposed mechanism of action of chymotrypsin on amide substrates. ⁶⁶

5.5.1.7 The relationship between the mechanism of action of chymotrypsin and those of other proteases

If we consider the hydrolysis of an amide bond, we can formulate the essential chemical requirements of a catalyst as the possession of the following.



- (1) a nucleophilic group, N, to attack the carbonyl group, leading to the formation of a tetrahedral intermediate;
- (2) some positively charged species, X^+ , in the neighbourhood of the oxygen atom of the carbonyl group. This would not only increase the susceptibility of

the carbonyl group to nucleophilic attack but would also stabilize the tetrahedral intermediate;

- (3) a proton donor, $Y \text{---} H$, to make the $NH \text{---}$ moiety a better leaving group.

We have already seen how these requirements are met in the case of chymotrypsin and related serine proteases. The nucleophilic group is the side-chain of Ser 195 activated by the charge-relay mechanism. Polarization of the carbonyl group is enhanced by hydrogen-bonding interactions between the oxygen atom and the peptide chain $\text{---}NH \text{---}$ groups of Gly 193 and Ser 195. The proton-donating species is the side-chain of His 57.

At first sight, other types of proteases possess very different mechanisms, but it is usually possible to interpret these mechanisms in terms of the requirements outlined above. For instance, in carboxypeptidase^{81, 82} (Fig. 5.22), the nucleophilic group is the side-chain of Glu 270, or possibly a water molecule activated by this side-chain and the zinc ion acts as the polarizing influence on the carbonyl group.

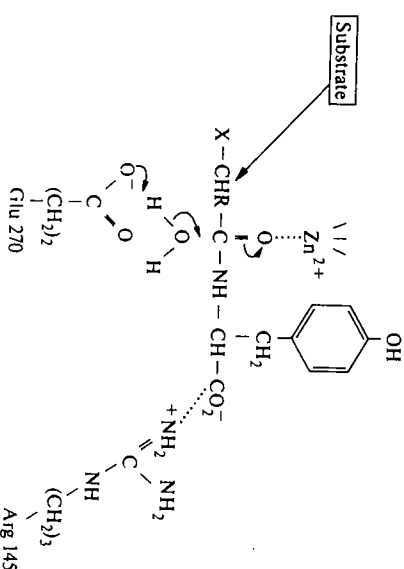


FIG. 5.22. Part of the proposed mechanism of action of carboxypeptidase. ⁸² An alternative mechanism involves direct attack of the side-chain of Glu 270 on the carbonyl group of the substrate.

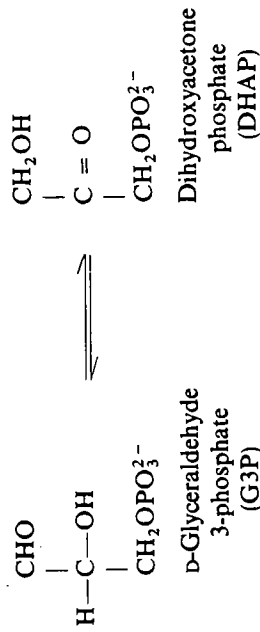
It had been previously thought,⁸² on the basis of chemical-modification and X-ray data, that the phenolic $\text{---}OH$ group of the side-chain of a Tyrosine (probably Tyr 248) acted as a proton donor to the leaving group. However, this possibility has been ruled out by recent site-directed mutagenesis experiments⁸³ in which Tyr 248 has been replaced by Phe. In the mutant the k_{cat} is unchanged although the K_m for peptide and ester substrates is raised. These results suggest that Tyr 248 plays a role in the binding of substrates. Similar results have been found for the mutant in which Tyr 198 (also close to the active site) has been replaced by Phe and in the double mutant (Tyr 248 and Tyr 198 both replaced by Phe).⁸⁴

In thiol proteases such as papain, the nucleophilic group is a cysteine side-chain that is partially ionized as a result of the presence of a neighbouring histidine side-chain. The polarization of the carbonyl group of the amide bond occurs by hydrogen bonding as in the case of chymotrypsin, and the proton donating group is a histidine side-chain.^{81, 85-87} In acid proteases such as pepsin and penicillopepsin, the nucleophilic group appears to be a water molecule activated by an aspartic acid side-chain (Asp 32 in pepsin). The diad of Asp 32 and Asp 215 acts as a proton-donor to the -NH leaving group.⁸⁸

From this discussion it can be appreciated that the various types of proteases have adopted different solutions to the problems associated with the hydrolysis of amide bonds, although all the different solutions can be rationalized in terms of chemical principles. The differences in specificity between the various proteases can be understood in terms of the details of the interactions between enzyme and substrate (see the examples of chymotrypsin, trypsin, and elastase in Section 5.5.1.2). The specificity displayed by carboxypeptidase is for the position of an amide bond in a polypeptide chain, since this enzyme removes the C-terminal amino acid (see Chapter 3, Section 3.4.4.2). By referring to the structure of the active site of the enzyme (Fig. 5.22) we can see that the positively charged side-chain of Arg 145 is in an appropriate position to bind to the C-terminal carboxylate group of the substrate. There are in fact two types of carboxypeptidases, A and B, which are very similar in overall structure. However, carboxypeptidase B, which is specific for the removal of C-terminal Lys or Arg has an appropriately positioned negatively charged side-chain (that of Asp 255) to bind the positively charged side-chains of the substrate.⁸⁹ Carboxypeptidase A will remove all C-terminal amino acids other than Lys or Arg from a polypeptide chain.

5.5.2 Triosephosphate isomerase

Triosephosphate isomerase catalyses the following reaction:



The reaction is of great physiological importance in that it allows inter-conversion of the two three-carbon units produced by the cleavage of D-fructose 1,6-bisphosphate catalysed by fructose-bisphosphate aldolase (Section 5.5.3). D-Glyceraldehyde 3-phosphate is an intermediate in the glycolytic pathway leading from glucose to pyruvate. Dihydroxyacetone phosphate can be reduced

to glycerol 1-phosphate, which acts as a precursor in the synthesis of various lipids.

The equilibrium of the triosephosphate isomerase-catalysed reaction is strongly in favour of dihydroxyacetone phosphate:

$$K = \frac{[\text{DHAP}]}{[\text{G3P}]} = 367^* \text{ at } 298 \text{ K } (25^\circ\text{C}).$$

The enzyme has been intensively studied in recent years and a good understanding of its mechanism has been achieved by the use of X-ray crystallography, affinity labelling, and experiments involving the isotopic labelling of substrates. Attention will be focused on triosephosphate isomerase from chicken breast muscle, although it seems that the enzyme from a wide variety of other sources has very similar properties.

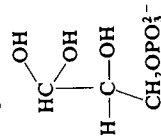
5.5.2.1 Background information on the structure and action of the enzyme

Triosephosphate isomerase has an M_r of 53 000 and is composed of two subunits of identical sequence. It is of special note that the enzyme is an extremely efficient catalyst, with a turnover number, expressed per active site, of about 250 000 min^{-1} in the direction of conversion of D-Glyceraldehyde 3-phosphate to dihydroxyacetone phosphate. As will be mentioned later (Section 5.5.2.5), there are good grounds for believing that triosephosphate isomerase is an almost perfectly evolved enzyme, i.e. that further evolution to produce a more efficient catalyst is not possible.⁹¹

5.5.2.2 X-ray crystallographic studies

Crystallographic studies have shown that each subunit of the enzyme is roughly spherical with a diameter of approximately 3.5 nm. There is a surprisingly regular manner of chain folding so that each subunit consists of an inner cylinder or 'barrel' of eight strands of parallel pleated sheet. The adjacent strands are linked mainly by helical segments, which thus form the outer face of each 'barrel' (Fig. 5.23)⁹². Triosephosphate isomerase is thus an excellent example of the α/β type of structure described by Levitt and Chothia⁹³ (see Chapter 3, Section 3.5.4.6). In the contact area between the subunits there is a significant number of amino acids with polar side-chains.

* Earlier values of K were quoted as being around 20. However, it is now recognized that both G3P and DHAP can exist as hydrated forms in aqueous solution.⁹⁰ These hydrated (diol) forms such as that shown here for G3P do not act as substrates for the enzyme. When the corrections for the amounts of true (unhydrated) substrate and product are made, K becomes much larger.



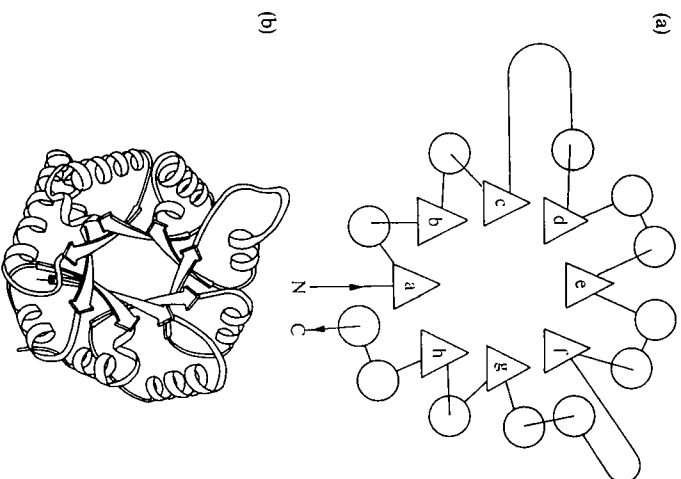


FIG. 5.23. Schematic representations of the chain folding in each subunit of triose-phosphate isomerase.⁹⁴ In (a), circles and triangles represent helices and strands of sheet, respectively. In (b), ribbons and arrows represent helices and strands of sheet, respectively. Part (b) is reproduced with permission from *Advances in Protein Chemistry* 34, 167–339. Copyright J. S. Richardson (1981).

One important aspect of the X-ray crystallographic work is that it is possible to study a true enzyme–substrate complex, because the equilibrium in this one-substrate reaction lies so much over to one side. An analysis of the complex of enzyme with dihydroxyacetone phosphate showed that the substrate is bound to each subunit in a ‘pocket’ that includes a few side-chains from the adjacent subunit, i.e. the active site seems to contain contributions from both subunits. The principal amino-acid side-chains that are within about 0.4 nm of the substrate are shown in Fig. 5.24.^{92,94}

The importance of subunit interactions in the catalytic properties of the enzyme has been emphasized by site-directed mutagenesis experiments in which side-chains near the subunit interface, e.g. Asn 14 and Asn 78, were mutated with consequent effects on k_{cat} .⁹⁵

5.5.2.3 Isotopic labelling of substrates

The effects of isotopic substitution on the rate of a reaction can give valuable information about the elementary steps of that reaction (for reviews, see references 96 and 97. However, we shall confine our attention to experiments in

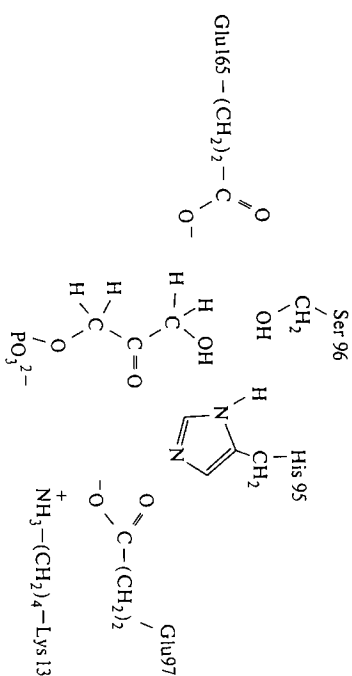
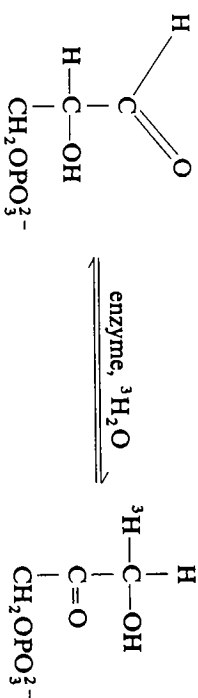


FIG. 5.24. Side-chains within about 0.4 nm of the substrate in the triosephosphate isomerase–dihydroxyacetone phosphate complex.⁹⁴

which isotopes are used as labels to follow the fate of a particular atom or group in a reaction.

Early work established that, in the presence of enzyme, up to one atom of tritium (^3H) can be incorporated into dihydroxyacetone phosphate from the solvent, $^3\text{H}_2\text{O}$. This atom of ^3H is incorporated in a stereospecific manner.⁹⁸



This finding suggests that the reaction proceeds via abstraction of a proton from the substrate by a base (B), followed by an exchange with the solvent. On chemical grounds it seems reasonable that a *cis*-enediol intermediate is involved (Fig. 5.25). The extent of incorporation of ^3H into the product will depend on how the rate of exchange of BH^+ with the solvent compares with the rate at which the proton is returned to the *cis*-enediol to give the product. The *cis*-enediol mechanism implies that the proton (or ^3H) is returned to the same face of the diol from which it is abstracted; this is consistent with the observed stereochemistry of the reaction.

Some information on the nature of the base, B, was gained from an experiment in which specifically labelled dihydroxyacetone phosphate was converted in the presence of enzyme to D-glyceraldehyde 3-phosphate. The product was then ‘trapped’ by an essentially irreversible oxidation with NAD^+ catalysed by glyceraldehyde-3-phosphate dehydrogenase in the presence of arsenate.⁹¹ It was found that there was a small amount (about 5 per cent) of transfer of isotope from C-1 to C-2 (Fig. 5.26). The fact that even a small amount of transfer of label is

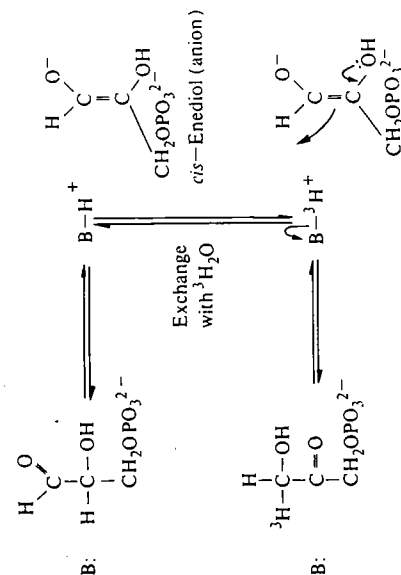


FIG. 5.25. Incorporation of ^3H into dihydroxyacetone phosphate via a *cis*-enediol intermediate in the reaction catalysed by triosephosphate isomerase.

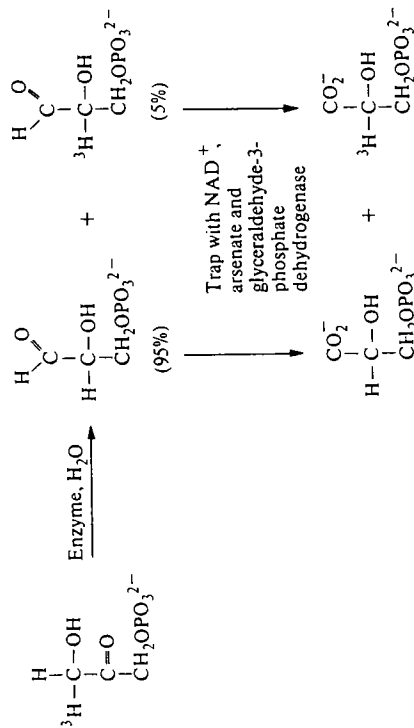
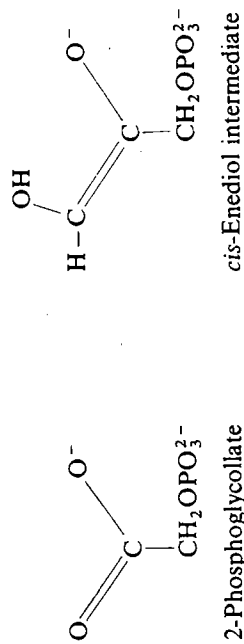


FIG. 5.26. Transfer of ^3H from C-1 to C-2 in the reaction catalysed by triosephosphate isomerase.

observed suggests that only a single base is likely to be involved in the abstraction and return of protons by the *cis*-enediol mechanism shown in Fig. 5.25. If there were more than one base involved, transfer of ^3H could only occur via an elaborate system of proton-exchange reactions within the enzyme. The amount of transfer is small because most of the ^3H on B^3H^+ has exchanged with the solvent (H_2O) to generate BH^+ before the proton is added back to give D-glyceraldehyde 3-phosphate (see Fig. 5.25).

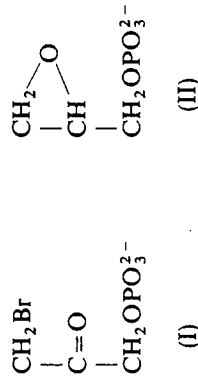
Further support for the idea of a *cis*-enediol intermediate in the enzyme-catalysed reaction was obtained from studies of enzyme inhibition. 2-Phosphoglycolate is a very powerful competitive inhibitor of the enzyme, with a

dissociation constant of $6 \mu\text{mol dm}^{-3}$. The fact that it shows a marked similarity in terms of electron distribution and stereochemistry to the *cis*-enediol intermediate has led to the proposal¹⁸ that 2-phosphoglycolate acts as a transition-state analogue* (Section 5.3.4).



5.5.2.4 Affinity labelling

Some elegant affinity-labelling experiments have helped to identify the base involved in the proton-transfer reactions (Fig. 5.25). Two different affinity labels have been used, bromohydroxyacetone phosphate (I),⁴⁹ and glycidol phosphate (II).⁹⁹



Both inhibitors conformed to the usual criteria for modification at the active site of the enzyme (Section 5.4.4), i.e. stoichiometric inactivation and protection by competitive inhibitors. Inhibitor I clearly resembles dihydroxyacetone phosphate and has an activated Br atom that can be displaced by a nucleophilic group. Inhibitor II resembles the *cis*-enediol intermediate because of the steric requirements of the epoxide ring; this ring can be opened following attack by a suitable nucleophile. (It should be mentioned that the experiments with inhibitor II were conducted with the enzyme isolated from rabbit skeletal muscle.)

In each case the modified, inactivated, enzyme was subjected to proteolytic digestion and the modified peptide isolated. By sequencing the peptides it was

* The value of the dissociation constant ($6 \mu\text{mol dm}^{-3}$) appears very low by comparison with the quoted K_m values for G3P ($320 \mu\text{mol dm}^{-3}$) and DHAP ($620 \mu\text{mol dm}^{-3}$), which give a rough measure of the affinity of the enzyme for these substrates. However, when the corrections are applied for the content of the hydrated forms of these compounds (see footnote on p. 217),⁹⁰ the K_m values for the true substrates become G3P ($11 \mu\text{mol dm}^{-3}$) and DHAP ($340 \mu\text{mol dm}^{-3}$). The value for G3P is not very different from the dissociation constant for the inhibitor, so there is some doubt about the designation of the inhibitor as a transition-state analogue.

The affinity-labelling experiments show that the side-chain of Glu 165 is indeed at the active site of the enzyme and the fact that the side-chain can act as a nucleophile makes it very likely that it acts as the general base involved in proton-transfer reactions (Fig. 5.23). The important role played by Glu 165 has been confirmed by site-directed mutagenesis experiments in which Glu 165 was replaced by Asp.¹⁰⁰ The mutant enzyme had a k_{cat} some 10^3 -fold lower than that of the native enzyme, with an altered K_m . Determination of the rate of the individual steps in the catalytic process showed that the mutation had decreased the rates of the enolization steps ($\text{ES} \rightleftharpoons \text{EZ}$ and $\text{EZ} \rightleftharpoons \text{EP}$ in Fig. 5.29) rather than those of the binding steps ($\text{E} + \text{S} \rightleftharpoons \text{ES}$ and $\text{E} + \text{P} \rightleftharpoons \text{EP}$),

5.5.5.2.5 The mechanism of triosephosphate isomerase

From the evidence discussed above, the outline mechanism of action of the enzyme is clear (Fig. 5.27).

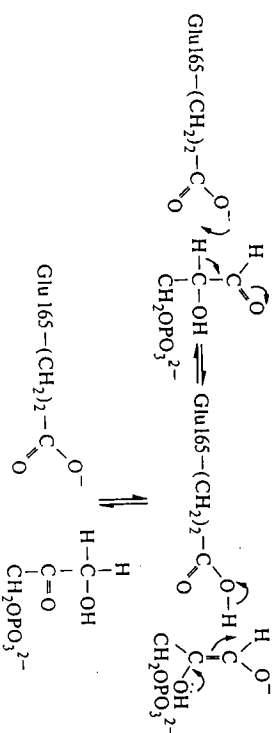


FIG. 5.27. Outline of the mechanism of the reaction catalysed by triosephosphate isomerase. Details of the proton-transfer reactions in the second step have been omitted.

There are, however, a number of points that require clarification. Firstly, what (if any) is the role of the side-chain of His 95 that has been shown by X-ray crystallographic studies^{92, 94} to be close to the substrate-binding site (Fig. 5.24)? This side-chain might well play a part in transferring protons between the

carbonyl and hydroxyl oxygen atoms. Secondly, what positively charged side chain on the enzyme is involved in binding the phosphate group of the substrate? Thirdly, what is the nature of the electrophilic group on the enzyme that serves to polarize the carbonyl group of the substrate? The presence of such a group is not only desirable on chemical grounds, since it helps to make the hydrogen attached to the adjacent carbon atom more acidic, but is also indicated by the fact that dihydroxyacetone phosphate bound to the enzyme is much more readily reduced by hydride ion than is free dihydroxyacetone phosphate, implying polarization of the carbonyl group of the substrate by the enzyme.¹⁰¹ (Fig. 5.28). Additional evidence for the polarization of the carbonyl group of the substrate when it is bound to the enzyme has been obtained from i.r. spectroscopic studies that have provided direct evidence for distortion of the bound substrate.¹⁷ The X-ray crystallographic studies show that the side-chains of Lys 13 and His 95 lie close to the carbonyl oxygen of dihydroxyacetone phosphate; these may well be involved in substrate polarization.⁹⁴

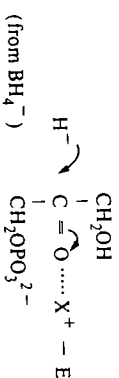


FIG. 5.28. Polarization of the carbonyl group of dihydroxyacetone phosphate bound to triosephosphate isomerase. This would facilitate the reduction of the substrate by borohydride.¹⁰¹

Very detailed measurements have been made of the rates of exchange of isotope between substrate and solvent during the enzyme-catalysed reaction. These results have been used to calculate the rate constants of the individual steps of the reaction and have enabled a free-energy profile of the reaction pathway to be constructed.^{91,102} (Fig. 5.29). The reaction occurs very nearly at a rate that is limited by the rate at which substrate can encounter the enzyme by the normal process of diffusion in solution,^{91,103} i.e. no further 'evolutionary

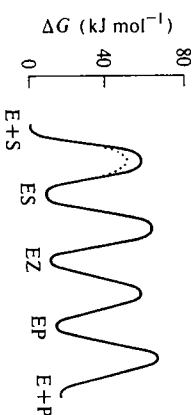


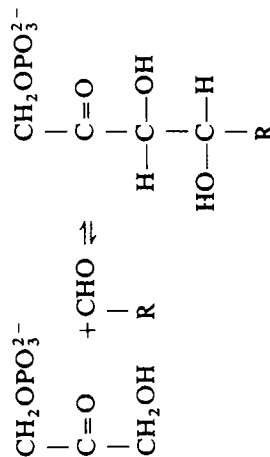
FIG. 5.29. Free-energy profile for the reaction catalysed by triosephosphate isomerase. 102
S is dihydroxyacetone phosphate, P is D-glyceraldehyde 3-phosphate, and Z is the *cis*-
enediol intermediate. The dashed line between E + S and ES represents the diffusion-
controlled encounter rate. Free energies are referred to a standard state of 40 $\mu\text{mol dm}^{-3}$,
which is the concentration of dihydroxyacetone phosphate *in vivo*.

improvement' in catalytic power is possible. (It is, of course, possible that further evolution might lead to 'desirable' changes in the regulatory properties of the enzyme.)

The mechanism of the triosephosphate isomerase-catalysed reaction is probably analogous to those catalysed by a number of other aldose-ketose isomerases. For instance, the mechanism of glucosephosphate isomerase, which catalyses the interconversion of D-glucose 6-phosphate and D-fructose 6-phosphate, is also thought to proceed via a *cis*-enediol intermediate.¹⁰⁴

5.5.3 Fructose-bisphosphate aldolase

Fructose-bisphosphate aldolase catalyses the following aldol condensation reaction:



Of particular interest is the case where R is $-\text{CHOH}-\text{CH}_2\text{OPO}_3^{2-}$ (i.e. where the aldehyde substrate is D-glyceraldehyde 3-phosphate) since this corresponds to the reaction in glycolysis in which the six-carbon unit D-fructose 1,6-bisphosphate (FBP) is cleaved to yield the two three-carbon units D-glyceraldehyde 3-phosphate and dihydroxyacetone phosphate. As befitting an enzyme that is involved in such an important pathway, fructose-bisphosphate aldolase occurs in nearly all plant and animal tissues and in most micro-organisms. The enzyme occurs in a variety of forms^{105,106} (Fig. 5.30).

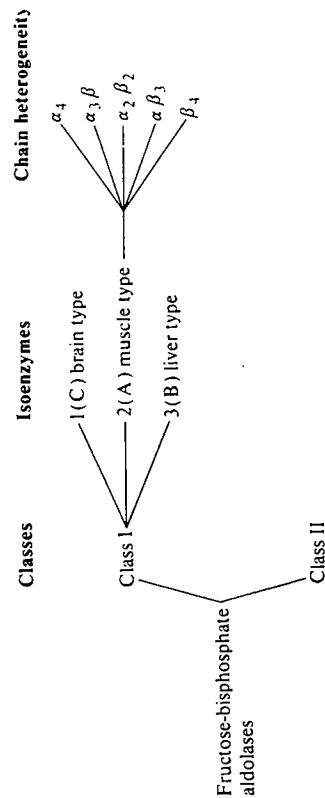


FIG. 5.30. The variety of forms in which fructose-bisphosphate aldolases occur.

There is a clear distinction between the fructose-bisphosphate aldolases from higher plants and animals (Class I), which possess a lysine side-chain at the active site, and those from fungi and bacteria (Class II), which possess a metal ion that is essential for activity. In animal tissues there are three distinct isoenzymes (see Chapter 1, Section 1.5.2). Isoenzyme 2 (also known as A or muscle-type isoenzyme) has a very much higher activity towards FBP than towards D-fructose 1-phosphate (FIP) (the ratio FBP/FIP = 50). By contrast, isoenzyme 3 (also known as B or liver-type isoenzyme) has a ratio of activities FBP/FIP almost equal to 1, and isoenzyme 1 (C or brain-type) has an intermediate ratio (FBP/FIP = 26). The ability of isoenzyme 3 to use FIP as a substrate can be correlated with the role of the liver in the metabolism of fructose. Fructose-bisphosphate aldolase from rabbit muscle consists of five distinct species¹⁰⁵ that can be separated by isoelectric focusing (see Chapter 2, Section 2.6.2.3); this is accounted for by the fact that there are five different ways in which a tetrameric protein can be assembled from two types of subunit (Fig. 5.30). The two types of subunit differ in that the α -type has a C-terminal sequence -Asn-His-Ala-Tyr, and the β -type a corresponding sequence -Asp-His-Ala-Tyr. The conversion $\alpha \rightarrow \beta$ occurs *in vivo*¹⁰⁵ and hence the different forms α_4 , $\alpha_3\beta$, etc.) are not isoenzymes according to the definition given in Chapter 1, Section 1.5.2.

Most of the detailed work on fructose-bisphosphate aldolase has been carried out on the enzyme isolated from rabbit skeletal muscle. We shall discuss the mechanism of the rabbit enzyme, although reference will be made to the enzyme from yeast in Section 5.5.3.6. Despite the fact that the pure enzyme has been available in large quantities for a number of years, it is only recently that the X-ray structure of the rabbit muscle enzyme has been published.¹⁰⁷ Data concerning the mechanism of the enzyme have been obtained from structural studies (Section 5.5.3.1), kinetic studies (Section 5.5.3.2), experiments involving the detection of intermediates (Section 5.5.3.3) and the chemical modification of amino-acid side-chains (Section 5.5.3.4). From these studies a reasonably coherent picture of the mechanism can be gained, although further structural work is required to confirm and extend many of the proposals.

5.5.3.1 Background information on the structure of the enzyme

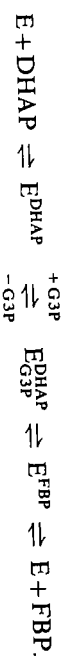
Fructose-bisphosphate aldolase is a tetrameric enzyme of M_r 160 000. The complete amino-acid sequences of a number of Class I aldolases are known.^{108,109,110} These confirm¹⁰⁸ that the only difference between the α -type and β -type subunits is the substitution of Asp for Asn in the C-terminal portion of the molecule (Section 5.5.3).

The X-ray structure¹⁰⁷ shows that each subunit is composed of a single highly compact domain structure with eight strands of parallel β -sheet forming a barrel with flanking helical segments (compare the structure of triosephosphate isomerase, Section 5.5.2.2). The active site appears to be located in the interior of this barrel, which contains hydrophobic amino acids and some charged side-chains with acidic and basic groups arranged in an alternating fashion to allow

charge neutralization. Amongst conserved side-chains that are located in this interior are Lys 229, Lys 107, Lys 146, and Arg 148. The C-terminal portion of the chain (residues 340–363) is not located clearly in the X-ray structure and is thus presumably highly mobile. It is thought that this portion corresponds to an 'arm-like' structure folded over the surface of the subunit to extend to the active site in the interior of the β -barrel.

5.5.3.2 Kinetic studies

By studying the rate of isotope exchange between $^3\text{H}_2\text{O}$ and dihydroxyacetone phosphate (DHAP) at equilibrium, and assessing how the rate was affected by changes in the concentration of D-glyceraldehyde 3-phosphate (G3P), it was shown¹¹¹ that the reaction is of the ordered ternary complex type (see Chapter 4, Section 4.3.5.1) in which DHAP binds to the enzyme first, followed by G3P.



5.5.3.3 Detection of intermediates

The nature of the E^{DHAP} complex was deduced from a 'trapping' experiment. Sodium borohydride was added to a mixture of the enzyme and radioactively labelled (either by ^{14}C or ^{32}P) DHAP; this resulted in incorporation of the radioactivity into the enzyme and loss of catalytic activity. Analysis of the labelled enzyme showed that a Schiff base had been formed between the carbonyl group of DHAP and the ϵ -amino group of a lysine side-chain; this could be reduced by borohydride¹⁰⁵ (Fig. 5.31).

Additional evidence in support of the involvement of a Schiff base is provided by the observation that there is an enzyme-catalysed exchange of ^{18}O between

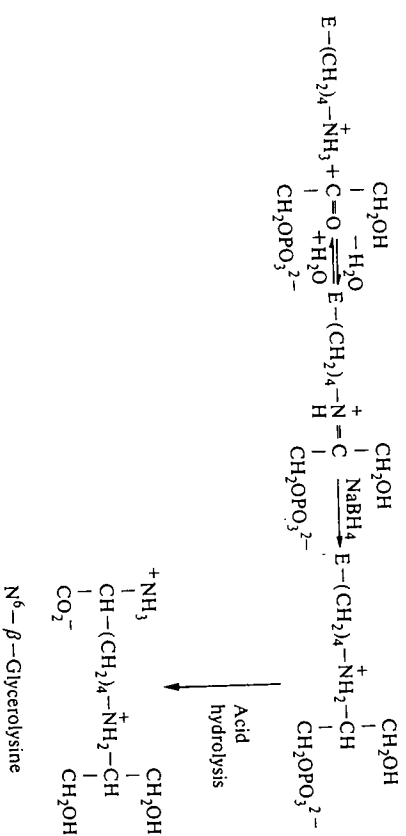
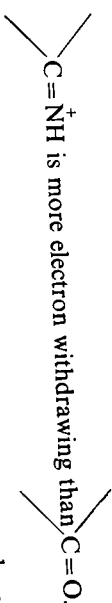


FIG. 5.31. Trapping of the Schiff base formed between fructose-bisphosphate aldolase and dihydroxyacetone phosphate.

the carbonyl oxygen of DHAP and water.¹¹² The side-chain involved in Schiff base formation has been identified as that of Lys 229.^{107,108} There is considerable homology between the structures of active site peptides from a variety of Class I fructose-bisphosphate aldolases.¹⁰⁵

Formation of a Schiff base that is readily protonated is a useful way of generating an electron-withdrawing centre to aid the catalytic process (Section 5.3.3).



In the E^{DHAP} complex, this electrophilic centre serves to make a proton on an adjacent carbon atom more acidic. The carbanion can then attack the carbonyl group of the aldehyde substrate to yield the product (Fig. 5.32). This mechanism is analogous to that observed in a model system, namely the deacidolization of diacetone alcohol that is catalysed by primary and secondary amines via the formation and breakdown of a Schiff base intermediate.¹¹³

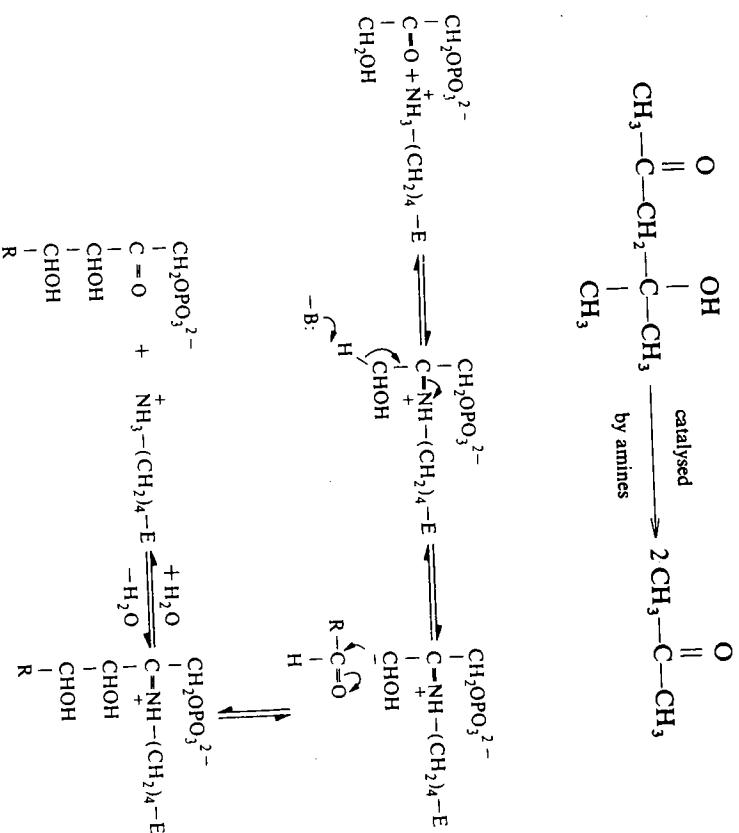


FIG. 5.32. The mechanism of fructose-bisphosphate aldolase proceeding via a protonated Schiff base.

5.5.3.4 Chemical modification of amino-acid side-chains

Numerous chemical-modification experiments have been performed in order to clarify the roles played by various amino-acid side-chains in the mechanism of the enzyme. We shall concentrate on the modifications of lysine and histidine side-chains, as these appear to have provided the most clear cut results.

(i) Lysine

We have already mentioned (Section 5.5.3.3) that the side-chain of Lys 229 is involved in Schiff base formation in the E^{DHAP} complex. Other experiments have suggested that there is a second lysine side-chain at the active site of the enzyme.¹¹⁴ Pyridoxal phosphate inactivates the enzyme by formation of a Schiff base (Fig. 5.33).

That the side-chain of Lys 229 is not involved in this reaction is shown by the fact that the reduced enzyme-pyridoxal phosphate derivative (II) in Fig. 5.33) can still form a Schiff base with DHAP.¹¹⁴ The side-chain modified by pyridoxal phosphate has been identified¹⁰⁸ as that of Lys 107. It has been proposed that this lysine side-chain is involved in electrostatic interactions with the 6-phosphate group of D-fructose 1,6-bisphosphate, and that this binding helps to position the carbonyl group at C-2 so that it can form a Schiff base with the side-chain of Lys 229. This suggestion is in accordance with the results of X-ray crystallography¹⁰⁷ (Section 5.5.3.1).

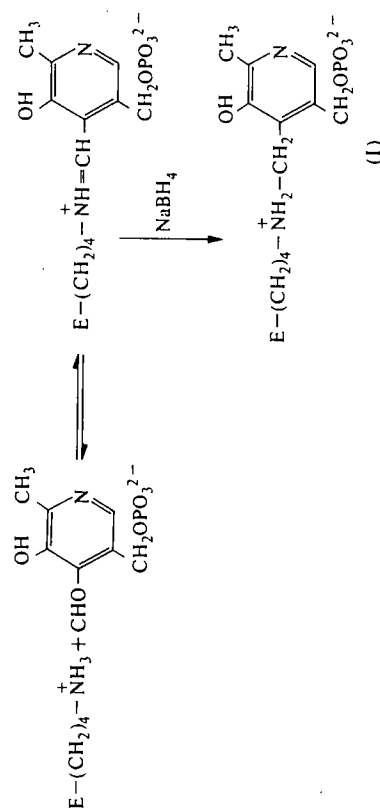


FIG. 5.33. Formation of a Schiff base by reaction of fructose-bisphosphate aldolase with pyridoxal phosphate.

(ii) Histidine

The involvement of histidine side-chains in the catalytic mechanism of fructose-bisphosphate aldolase was suggested by some early experiments in which the enzyme was photo-oxidized in the presence of a dye, rose Bengal.¹¹⁵ This treatment led to inactivation of the enzyme, but the results were rather difficult to

interpret because a number of histidine side-chains were modified. A more clear-cut chemical modification experiment involved the reagent *N*-bromoacetyl-ethanolamine phosphate, which inactivates the enzyme by alkylation of one histidine side-chain per subunit.¹¹⁶ This has been identified as the side-chain of His 361.¹⁰⁸ It was thought that this histidine could well play a role in the proton-transfer reactions in the E^{DHAP} complex (see Fig. 5.32), but this suggestion is now doubtful since it appears that His 361 is not conserved in all Class I aldolases.^{109,110}



5.5.3.5 An outline mechanism for fructose-bisphosphate aldolase

Taking the various pieces of evidence in Sections 5.5.3.2 to 5.5.3.4 we can formulate an outline mechanism for the catalytic process (Fig. 5.34). DHAP binds to the enzyme to form a Schiff base, the carbanion then attacks G3P to form enzyme-bound FBP.

This mechanism should be regarded only as a working hypothesis for the following reasons:

1. The identity of the positively charged group that binds to the 1-phosphate group of the substrate (Fig. 5.34) is not known, although Arg 148 may fulfil this role.^{107,117}
2. Chemical-modification experiments have suggested that cysteine and tyrosine side-chains may be involved in the catalytic mechanism,¹⁰⁵ but the roles of these side chains are not clear.
3. Detailed structures of enzyme-substrate or enzyme-inhibitor complexes are required in order to clarify the identities and roles of some of the side-chains involved.
4. The role played by amino acids in the mobile C-terminal portion of the polypeptide chain needs to be defined more clearly. Of the C-terminal 22 amino acids, only four are conserved in the Class I aldolases sequenced to date.^{107,110} It is possible that this region is important in determining the activity and/or specificity of the various aldolases and their isoenzymes towards the different substrates acted on (Section 5.5.3). The identity of a base thought to be involved in proton transfer reactions remains unresolved (Section 5.5.3.4).

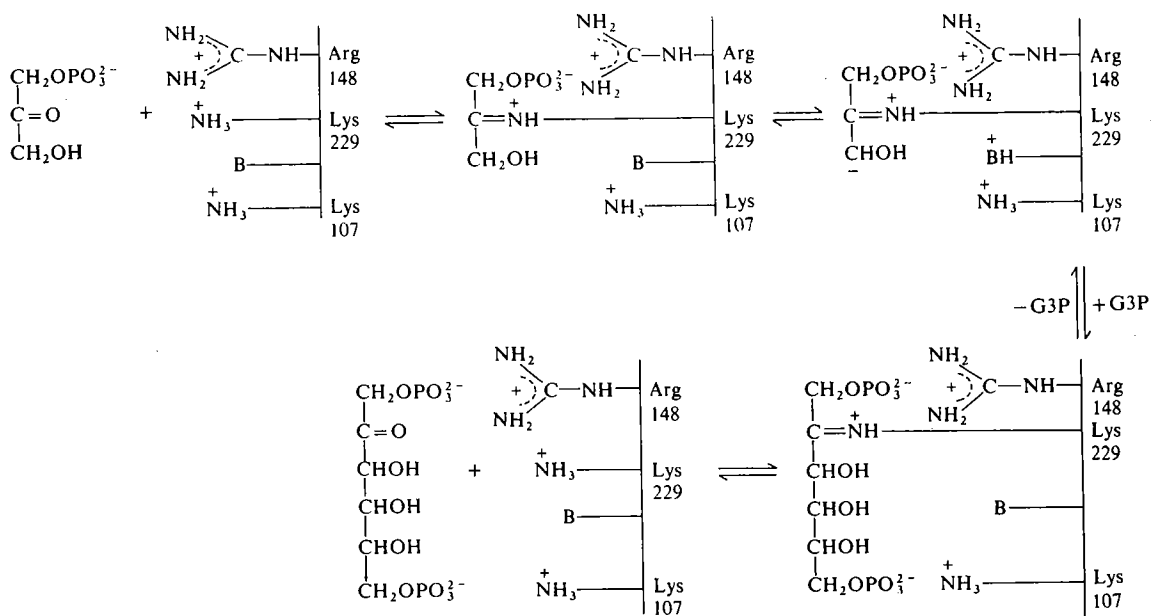
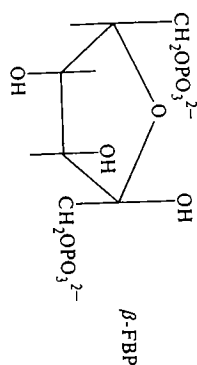


FIG. 5.34. The outline mechanism of fructose-bisphosphate aldolase. For the sake of clarity, details of the side-chains have been omitted. G3P is D-glyceraldehyde 3-phosphate. B represents a base though to be involved in proton-transfer reactions (Section 5.5.3.3).

5. The keto form of D-fructose 1,6-bisphosphate, which forms the Schiff base with the side-chain of Lys 229 (Fig. 5.34), constitutes only a small fraction (<2 per cent) of the total FBP in solution.¹¹⁸ Most of the FBP is present in the furanose forms, e.g. β -FBP, which accounts for 90 per cent of the total FBP.



Does FBP bind as a furanose form and then undergo ring opening to yield the keto form, or does only the keto form bind?

5.5.3.6 The mechanism of Class II fructose-bisphosphate aldolases

We have so far confined the discussion to fructose-bisphosphate aldolase isolated from rabbit skeletal muscle, but it can be assumed that the mechanisms of other Class I enzymes are similar, especially in view of the homology of sequence around the active-site lysine (Section 5.5.3.3). In Class II fructose-bisphosphate aldolases there is no such lysine side-chain but instead a metal ion that is required for activity. Thus, for instance, the enzyme from yeast is a dimer containing one zinc ion per subunit.¹⁰⁵ The metal ion could serve to polarize the carbonyl group of DHAP (Fig. 5.35) in much the same way as the protonated Schiff base in Class I enzymes. This polarization of the carbonyl group could be similar to that already described in the case of carboxypeptidase (Fig. 5.22). However, studies using i.r. spectroscopy¹¹⁹ failed to reveal evidence for polarization of the carbonyl group of FBP or DHAP on binding to the yeast enzyme, although the carbonyl group of G3P appeared to be polarized substantially on binding. There is some evidence that carboxylate side-chains (those of Glu or Asp) can play a role in proton transfer reactions in Class II fructose-bisphosphate aldolases.¹⁰⁵

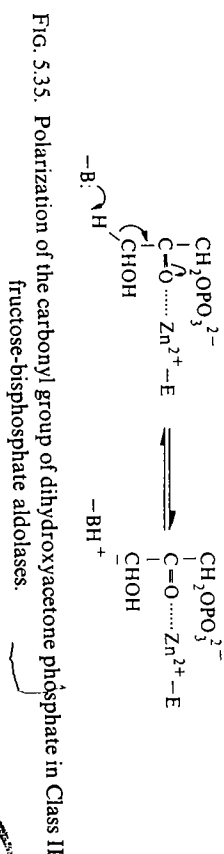
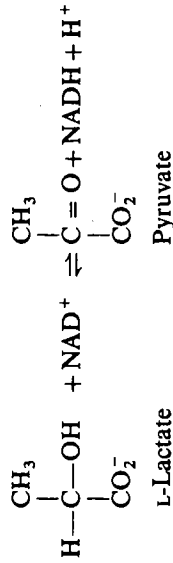


FIG. 5.35. Polarization of the carbonyl group of dihydroxyacetone phosphate in Class II fructose-bisphosphate aldolases.



5.5.4 Lactate dehydrogenase

The enzyme lactate dehydrogenase catalyses the reaction:



In the reverse direction, this reaction represents the last step in the process of anaerobic glycolysis and provides a means of regeneration of NAD^+ required for the reaction catalysed by glyceraldehyde-3-phosphate dehydrogenase. The enzyme is present in the cytosol in sufficiently high concentrations that the reaction is close to equilibrium; this is discussed further in Chapter 8, Sections 8.3.1.4 and 8.5.4. Lactate dehydrogenase has been isolated from many sources, but most of the detailed work relating to the mechanism has been performed on the enzymes isolated from dogfish and pig (for a comprehensive review of this work, see reference 120). We shall discuss the results of kinetic studies (Section 5.5.4.2), X-ray crystallographic work (Section 5.5.4.3), experiments involving chemical modification of amino-acid side-chains (Section 5.5.4.4), and site-directed mutagenesis (Section 5.5.4.5). In Section 5.5.4.6 we shall discuss the mechanism of lactate dehydrogenase in relation to that of other dehydrogenases.

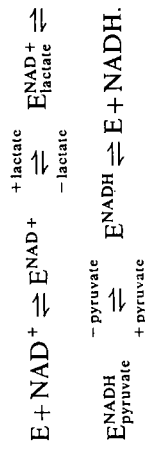
5.5.4.1 Background information on the structure of the enzyme

Lactate dehydrogenase is a tetramer of M_r 140 000. The enzyme provides a good example of the occurrence of isoenzymes (see Chapter 1, Section 1.5.2); in most tissues there are five forms of the enzyme, which can be separated by electrophoresis. The different forms arise from the five possible ways of assembling a tetramer from two types of subunit (α_4 , $\alpha_3\beta$, $\alpha_2\beta_2$, $\alpha\beta_3$, and β_4). It is found that LDH-1 predominates in heart muscle, and is often referred to as the H_4 form, whereas LDH-5 predominates in skeletal muscle, and is often referred to as the M_4 form. Various other isoenzymes (e.g. the C, E, and F forms) are also known,¹²⁰ but they are much less widely distributed than the heart and muscle forms and will not be discussed further here.

5.5.4.2 Kinetic studies

Detailed steady-state kinetic studies on lactate dehydrogenase have shown that the enzyme follows an ordered ternary complex mechanism (see Chapter 4, Section 4.3.5.1) with NAD^+ (or NADH) binding preceding that of lactate (or pyruvate). This is shown in the following scheme, where the proton involved in

the reaction has been omitted:



The ordered ternary complex mechanism was deduced from studies of product inhibition and of substrate binding. For instance, the enzyme will not bind lactate or pyruvate in the absence of the dinucleotide substrate.¹²¹

Steady-state kinetic studies have also shown that oxamate (Fig. 5.36(a)) acts as a competitive inhibitor with respect to pyruvate (Fig. 5.36(b)) (note that the two molecules are isoelectronic) and that oxalate (Fig. 5.36(c)) is competitive with respect to lactate.¹²² These inhibitors are very likely to bind to the active site of the enzyme and have found considerable value as substrate analogues in X-ray crystallographic studies on lactate dehydrogenase, allowing conclusions to be drawn regarding the structure of the catalytically active complex.

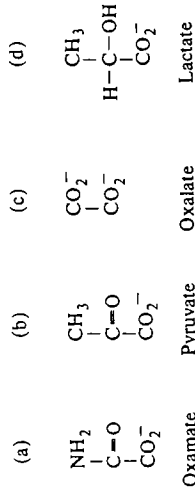


FIG. 5.36. Structures of some substrates and inhibitors of lactate dehydrogenase.

A more detailed analysis of the rates of the individual steps of the reaction has been undertaken using stopped-flow techniques (see Chapter 4, Section 4.4). The enzyme (LDH-1 from pig was used in these experiments) was mixed rapidly with saturating concentrations of the substrates NAD^+ and lactate, and production of NADH was monitored spectrophotometrically¹²³ (Fig. 5.37).

There was an initial burst of NADH production within the 'dead time' of the instrument, followed by a slow steady-state production of NADH. The size of the burst was equal to 1 mol NADH per mol of active sites. From these data, it can be concluded that all the steps in the reaction up to formation of enzyme-bound NADH are very rapid. Similar results were obtained with LDH-5 from pig,¹²⁴ but the size of the burst was only equal to 1 mol NADH per mol active sites at pH values of 8.0 or above.* Since NADH binds very tightly to the enzyme it might be thought that the dissociation of the E^{NADH} complex would be the slowest (rate-determining) step of the overall reaction. However, this is not the case, since the rate constant for the dissociation (measured in a separate experiment as 450 s^{-1})

* At lower pH values, the size of the burst was less; e.g. at pH 6.8 it was $0.5 \text{ mol per mol active sites}$.

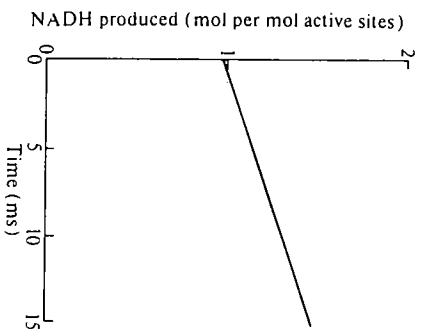


FIG. 5.37. Production of NADH in the reaction catalysed by lactate dehydrogenase.¹²³

is considerably greater than the rate constant for oxidation of lactate in the steady state (80 s^{-1}).¹²⁴ It was therefore concluded that there is an extra step in the overall reaction that occurs after the interconversion of the ternary complexes that produces enzyme-bound NADH, but which precedes the dissociation of NADH. This extra step could be a slow conformational change in the E_{NADH} or E_{pyruvate} complex.¹²⁴ (see Section 5.5.4.5). The use of detailed pre-steady-state kinetic measurements has allowed the rate constants of all the individual steps in the reaction catalysed by LDH-1 from pig to be determined.^{125,126}

5.5.4.3 X-ray crystallographic studies

Crystallographic and amino-acid sequence studies have been performed on LDH-5 from dogfish muscle.¹²⁰ There is a considerable amount of secondary structure (see Chapter 3, Section 3.5.4) in the molecule; approximately 40 per cent of the amino acids occur in helices and a further 23 per cent in various forms of β -structure.

Of particular interest is the occurrence of supersecondary structures in the molecule (see Chapter 3, Section 3.5.4.6). In the N-terminal half of the molecule there is a six-stranded parallel sheet, and in the C-terminal half there are two three-stranded antiparallel sheet structures.¹²⁰ The six-stranded sheet is involved in the binding of NAD^+ and seems to be a general feature of all the dehydrogenases whose structures have been examined so far, and has led to the development of ideas¹²⁷ about the evolutionary relationships between these and other enzymes such as kinases that act on mononucleotide substrates (e.g. ATP).^{*} The six-stranded sheet¹²⁸ is shown schematically in Fig. 5.38, where the

* The tertiary structures of lactate dehydrogenase and malate dehydrogenase are very similar; indeed, by changing three amino acids in the active-site region of the enzyme it is possible to change the specificity of lactate dehydrogenase to that of malate dehydrogenase (see Section 5.5.4.5).

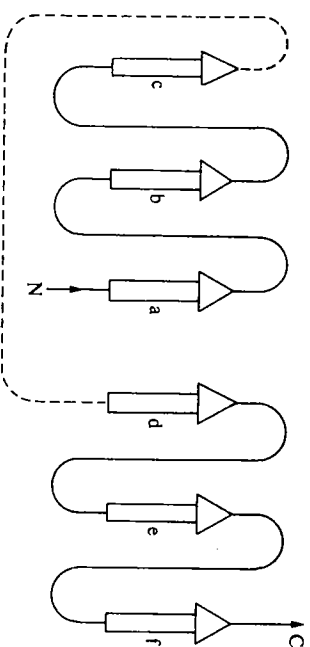


FIG. 5.38. The six-stranded sheet in lactate dehydrogenase. The arrows represent the strands of sheet and the connecting lines represent helices. (Between strands c and d there is a non-helical stretch of peptide chain.) The nicotinamide ring of NAD^+ binds between strands d and e and the adenine ring between strands a and b.¹²⁸

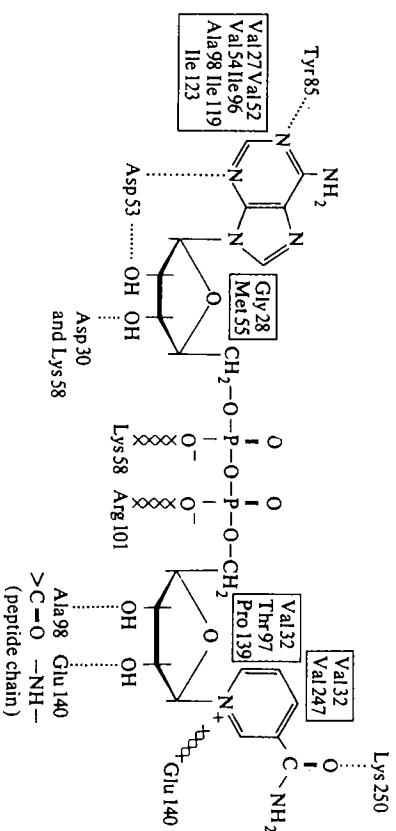


FIG. 5.39. Some of the interactions involved in the binding of NAD^+ to lactate dehydrogenase.¹²⁰ The dotted lines represent hydrogen bonds, the crosses electrostatic interactions, and the boxes hydrophobic interactions.

The adenine ring binds in a hydrophobic pocket provided by side-chains of Val, Ile, and Ala. Hydrogen bonds to the side-chains of Tyr 85 and Asp 53 (Fig. 5.39) impart specificity to the orientation of the bound adenine ring. Both ribose rings are involved in hydrogen bonds, as shown in Fig. 5.39; the specific

strands of β -sheet are depicted by arrows (in the direction $\text{N} \rightarrow \text{C}$) and the thin connecting lines represent helices. The binding of NAD^+ to the enzyme has been examined in great detail.¹²⁰ When we consider the large number of amino-acid side-chains involved in the binding, it is not difficult to appreciate how the remarkable specificity displayed by this and other enzymes arises. Some of the important features of the binding of NAD^+ to lactate dehydrogenase are shown in Fig. 5.39.

interaction with the 2'-OH group of the adenine ribose helps to explain why NADP^+ , which has a phosphate group at this position, is not a substrate for the enzyme.*

As far as the nicotinamide ring is concerned, we should note that there is a very important hydrogen-bonding interaction between the carbonyl oxygen of the side-chain and the side-chain of Lys 250. This interaction serves to orientate the nicotinamide ring in such a way as to expose the 'A'-side of the ring to the other substrate, lactate, thereby accounting for the 'A'-side specificity of the enzyme (Fig. 5.40). In the complex formed between NAD^+ and glyceraldehyde-3-phosphate dehydrogenase, there is a specific hydrogen bond between the carbonyl oxygen of the side-chain of the nicotinamide ring and the side-chain of Asn 313 that serves to expose the 'B'-side of the ring to the other substrate, accounting for the 'B'-side specificity of this enzyme (Fig. 5.40).^{129, 130}

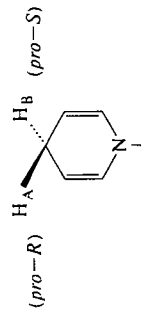


FIG. 5.40. The specificity of NADH-linked dehydrogenases. 'A'-side dehydrogenases (e.g. malate, lactate and alcohol dehydrogenases) select the H_A hydrogen from NADH. 'B'-side dehydrogenases (e.g. glutamate and glyceraldehyde-3-phosphate dehydrogenases) select the H_B hydrogen from NADH. (See also Fig. 1.2.)

The structure of the catalytically active complex of lactate dehydrogenase has been deduced from an examination of the structures of complexes containing inhibitors: E^{NAD^+} , E^{NADH} and $\text{E}^{\text{oxalate}}$ and of the structure of a complex containing an NAD-pyruvate adduct.¹³¹ These structures indicate that the site for lactate is most probably located between the nicotinamide ring and the side-chain of His 195, as shown in Fig. 5.41.

The side-chains of Arg 109 and Arg 171 can bind electrostatically to the carboxylate group of lactate. The side-chain of His 195 can form a hydrogen bond to the hydroxyl group of lactate, permitting a movement of electrons in the catalytic process as shown in Fig. 5.41.

Conformational changes in the enzyme appear to be of crucial importance in the formation of the catalytically active complex.¹²⁰ These changes are particularly marked on binding the second substrate to the binary E^{NAD^+} complex. There is a flexible loop (between amino acids 98 to 120), which in the absence of substrates extends into the solvent, but in the ternary complex moves down to enclose the substrates and thereby exclude water from the active site of the enzyme (see Section 5.5.4.5). For instance, the side-chain of Arg 109 moves 1.4 nm to be close to the substrate (Fig. 5.41). A smaller (0.1–0.2 nm¹³¹) but

* In general, a dehydrogenase acts on either NAD^+ or NADP^+ . Glutamate dehydrogenase is almost unique in that it can act on both (see Chapter 8, Section 8.3.1.4).

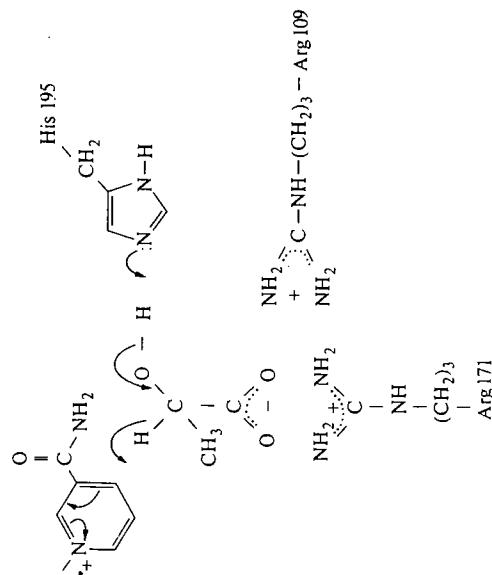


FIG. 5.41. Movement of electrons in the reaction between NAD^+ and lactate catalysed by lactate dehydrogenase.

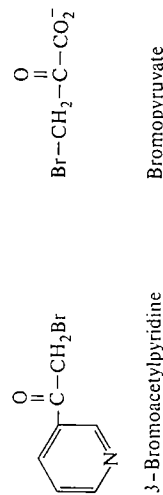
highly important change occurs so as to bring the side-chain of His 195 into contact with the substrate and permit the movement of electrons as shown (Fig. 5.41).

5.5.4.4 Chemical modification of amino-acid side-chains

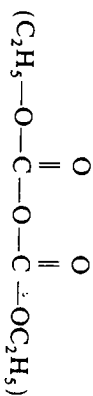
In this section we shall discuss experiments involving the modification of histidine, arginine, and cysteine side-chains. The results of these studies have complemented the information provided by X-ray crystallography.

(i) Histidine

The idea that there is a histidine side-chain at the active site of lactate dehydrogenase was suggested by the results of experiments using the affinity labels 3-bromoacetylpyridine and bromopyruvate.¹²⁰ (The latter acts as an affinity label in the presence of NAD^+ .) In each case it was shown that the side-chain of His 195 was modified.



It is also possible to modify the side-chain of His 195 selectively using diethylpyrocarbonate



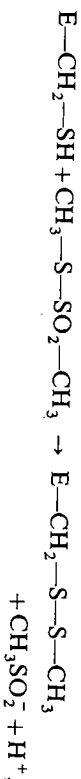
because of the much higher reactivity of this side-chain compared with the other histidine side-chains in the enzyme.¹³² In all three cases modification led to inactivation of the enzyme, in keeping with the proposed role of the side-chain of His 195 (Fig. 5.41).

(ii) Arginine

Phenylglyoxal, which is fairly specific for modification of arginine side-chains (Table 5.2), inactivates lactate dehydrogenase. The findings that only one arginine side-chain per subunit was modified and that formation of the ternary complex provided protection against inactivation suggested that there is an arginine side-chain at the active site of the enzyme.¹²⁰ The X-ray crystallographic data (Fig. 5.41) shows that two arginine side-chains may be involved in the binding of substrate, so it would be of interest to know which one is modified by phenylglyoxal.

(iii) Cysteine

The experiments designed to show whether a cysteine side-chain is involved in the catalytic activity of lactate dehydrogenase provide good examples of the care needed in interpreting chemical-modification data. Reaction of the enzyme with reagents such as substituted maleimides led to inactivation¹²⁰ as a result of modification of a single cysteine side-chain per subunit (later identified as that of Cys 165¹³³). The observations that various combinations of substrates afforded protection against inactivation, and that the amino-acid sequence around this cysteine was somewhat conserved in a variety of dehydrogenases, strengthened the conviction that this side-chain was indeed involved in the catalytic activity. However, X-ray crystallographic studies subsequently showed that the side-chain of Cys 165 was, in fact, some distance (1 nm) from the substrate-binding site.¹²⁰ The inactivation caused by this chemical modification is now interpreted in terms of a secondary, blocking, effect whereby a bulky reagent attached to the side-chain of Cys 165 would prevent the movement of the side-chain of His 195 towards the substrate and hence prevent formation of the catalytically active complex (Fig. 5.41). The X-ray crystallographic work has shown that in a ternary complex (e.g. $E_{pyruvate}^{NAD^+}$) the side-chain of Cys 165 would be inaccessible to modifying reagents; this observation would explain the protection against inactivation noted above. In the light of these comments, it would be expected that, if the side-chain of Cys 165 were modified with a small reagent, at least partial activity would be retained. This expectation has been confirmed in an experiment in which LDH-1 from pig was reacted with methanethiol-sulphonate.¹³⁴ The modified enzyme,



in which the side-chain of Cys 165 was reacted, retained full activity but showed some differences from unmodified enzyme in terms of its affinity for substrates.

5.5.4.5 Site-directed mutagenesis experiments

Two recent sets of experiments on lactate dehydrogenase have contributed to our understanding of the roles played by amino acids in the flexible loop region of the enzyme (Section 5.5.4.3). These experiments have been undertaken on the enzyme from the thermophilic bacterium *Bacillus stearothermophilus* for which the appropriate gene had been isolated. The amino-acid sequence of this enzyme and its catalytic properties (when measured in the presence of the activator FBP) are similar to those of eukaryotic lactate dehydrogenase, indicating that the overall structures of the enzymes are likely to be similar.

In the first experiment,¹³⁵ Arg 109 in the flexible loop was replaced by Gln, thereby changing a positively charged side-chain into a neutral one. The mutation had no effect on the binding of NADH to the enzyme but decreased k_{cat} by some 400-fold. The results suggested that the side-chain of Arg 109 enhances the polarization of the carbonyl group of pyruvate (Fig. 5.41) and stabilizes the transition state of the reaction.

In a second experiment,¹³⁶ the rate of the movement of the loop region that is associated with formation of the catalytically active complex was measured. For this experiment Gly 106 in the loop was replaced by Trp and the other Trp side-chains (Trp 80, Trp 150 and Trp 203) were replaced by Tyr. (It should be noted that in proteins, Trp side-chains generally show much more intense fluorescence than Tyr side-chains because of the quenching of the excited state of the latter.⁶³) The quadruple mutant enzyme thus has effectively a single fluorescent side-chain (Trp 106) within the loop region. The k_{cat} value for the mutant enzyme was approximately 60 per cent of that of the native enzyme and the K_m for pyruvate was unchanged. On mixing the mutant E_{NADH} complex with oxamate to form the ternary complex (Section 5.5.4.3), it was found that the fluorescence of Trp 106 decreased, reflecting movement of the loop. By stopped-flow measurements (see Chapter 4, Section 4.4.1) the rate of this movement was found to be 125 s^{-1} , i.e. very similar to the overall turnover rate of the enzyme as measured by k_{cat} . It is thus clear that the rate-determining step in the overall reaction corresponds to the movement of the loop to close over the active site (Section 5.5.4.3).

In a different type of experiment, the specificity of lactate dehydrogenase has been changed to that of malate dehydrogenase.¹³⁸ To take account of the increased size and additional negative charge of the new substrate, the following changes were introduced into the lactate dehydrogenase from *Bacillus stearothermophilus*: (i) the volume of the active site was increased (Thr 246 → Gly); (ii) an acidic side-chain was replaced by a neutral one (Asp 197 → Asn); (iii) a basic side-chain was introduced (Gln 102 → Arg). These changes led to a new 'malate' dehydrogenase that catalyses the reduction of oxalacetate by NADH 500 times faster than that of pyruvate. (The corresponding ratio for the original lactate dehydrogenase is 0.001.) This experiment confirms that the overall structures

the two enzymes are sufficiently similar that relatively small changes at the active site can cause marked changes in specificity.¹⁵⁸

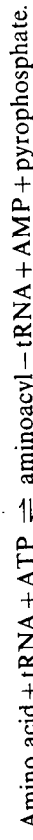
5.5.4.6 The mechanism of lactate dehydrogenase and some other dehydrogenases

The various approaches outlined in Sections 5.5.4.2 to 5.5.4.5 have given a fairly detailed insight into the mechanism of the reaction catalysed by lactate dehydrogenase. The structure of the active complex and the movement of electrons within it have been described (Fig. 5.41) and to this must be added details of the binding sites (e.g. Fig. 5.39) and of the rates of the various elementary steps in the reaction.

The mechanisms of a number of other dehydrogenases (e.g. alcohol dehydrogenase, malate dehydrogenase, and glyceraldehyde-3-phosphate dehydrogenase) are also known, at least in outline, and certain generalizations can be made.⁸¹ All dehydrogenases contain a recognizable nucleotide binding domain in each subunit (see Section 5.5.4.3), with a second (catalytic) domain that can be variable in structure. We have already noted that the mode of binding of the nicotinamide ring of NAD^+ to the various enzymes can explain the specificity of hydrogen transfer (Fig. 5.40). In all the dehydrogenases mentioned, except alcohol dehydrogenase, it appears that a histidine side-chain acts as a general base catalyst by abstracting a proton from the $-\text{O}-\text{H}$ group of the substrate,* thereby facilitating transfer of the hydride ion to NAD^+ (see Fig. 5.41). In the case of alcohol dehydrogenase, a zinc ion is involved in the catalytic activity and it is suggested that this zinc ion, or an ionized water molecule (OH^-) attached to the zinc ion, could play the part of the basic catalyst. It will be of great interest to see how the chemical problems involved in oxidation-reduction reactions have been solved in the cases of other dehydrogenases.

5.5.5 Tyrosyl-tRNA synthetase

The aminoacyl-tRNA synthetases catalyse formation of aminoacyl-tRNA derivatives according to the equation ($\text{tRNA} \equiv \text{transfer RNA}$):



The amino acid is linked to the adenosine at the 3' terminus of its appropriate tRNA. In some cases the attachment is to the 2' hydroxyl, in others to the 3' hydroxyl, and in others to either.¹³⁷ Aminoacyl derivatives are delivered to the ribosomes, the sites of protein synthesis, in the order specified by the sequence of bases on messenger RNA (mRNA) (see Chapter 9, Section 9.2). A sequence of three bases on the mRNA (*a codon*) forms specific hydrogen bonds with a sequence of three bases on the tRNA (*an anti-codon*). The fidelity of protein synthesis depends on the linking of the correct tRNA to the correct amino acid,

* In the case of glyceraldehyde-3-phosphate dehydrogenase, the $-\text{O}-\text{H}$ group is that of a thiohemiacetal attached to the side-chain of Cys 149. The thiohemiacetal is formed by attack of this side-chain on the carbonyl group of the aldehyde substrate.

and this in turn depends on the specificity of the interactions between a given aminoacyl-tRNA synthetase and the appropriate tRNA and amino acid. The aminoacyl-tRNA synthetases (one for each of the twenty amino acids found in proteins) have EC numbers 6.1.1.1–7, 9–12, 14–22. Tyrosyl-tRNA synthetase has the EC number 6.1.1.1.

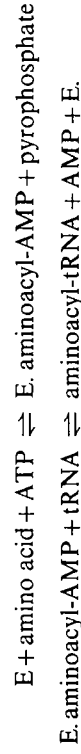
In this section we shall discuss the general aspects of the structures of aminoacyl-tRNA synthetases (Section 5.5.5.1) and the details of the reaction mechanism revealed by kinetic studies (Section 5.5.5.2). The problems posed by the need for recognition of the correct substrates are discussed in Section 5.5.5.3. In Section 5.5.5.4, the X-ray structure of tyrosyl-tRNA synthetase from *Bacillus stearothermophilus* is described. Finally, in Section 5.5.5.5, we shall examine some of the elegant experiments involving site-directed mutagenesis (Section 5.4.5) that have greatly extended our knowledge of the catalytic mechanism of the enzyme and the roles played by various amino acids.

5.5.5.1 Structures of aminoacyl-tRNA synthetases

The aminoacyl-tRNA synthetases for each amino acid have been purified from a variety of prokaryotic and eukaryotic sources.^{137–139} There is a wide variety of structural types including α , α_2 , α_4 and $\alpha_2\beta_2$ with total M_r values ranging from about 50 000 to 300 000. The variety of subunit structures of enzymes catalysing related reactions seems surprising; however, this diversity is also found in the sequences of the enzymes. In general, the subunit M_r of a given enzyme is larger in eukaryotes (yeast) than in prokaryotes (*E. coli*) and this is usually associated with an N-terminal extension (50–300 amino acids) in the former.¹³⁹ In the portions of sequence that can be aligned there is only moderate homology (30–50 per cent) between the corresponding enzymes from the two sources.¹³⁹ There is some evidence for internal repeats in the sequences of the larger monomeric aminoacyl-tRNA synthetases from prokaryotes.¹³⁷ In eukaryotes, nine of the enzymes form a complex of high M_r ($\approx 10^6$), although the significance of this complex is not established.¹³⁸ Detailed X-ray structural information is so far available on only one enzyme, namely tyrosyl-tRNA synthetase from *Bacillus stearothermophilus* (see Section 5.5.5.4).

5.5.5.2 The reaction mechanism—kinetic studies

A number of experiments have shown that the overall reaction catalysed by aminoacyl-tRNA synthetase is the sum of two reactions with an enzyme-bound aminoacyladenylate intermediate:¹³⁷



The enzyme-bound intermediate can readily be isolated; for instance, by mixing isoleucyl-tRNA synthetase with [^3H]isoleucine and ATP in the presence of necessary cofactors, such as Mg^{2+} , followed by gel filtration on Sephadex

G-50.¹⁴⁰ In a separate experiment it was shown that isoleucine was rapidly transferred from this intermediate to give an enzyme-bound product (isoleucyl-tRNA), that then slowly dissociated from the enzyme.

Since ATP, amino acid, and pyrophosphate can each bind to the enzyme, it has been concluded that the first reaction is of a random-order ternary-complex type¹⁴¹ (see Chapter 4, Section 4.3.5.1). The occurrence of the first reaction has also been demonstrated by the observation of an enzyme-catalysed $\text{ATP} \rightleftharpoons \text{pyrophosphate}$ exchange reaction that requires amino acid, but not tRNA.^{137,142} In these experiments [^{32}P]pyrophosphate is added, the ATP is isolated at various intervals, and the extent of ^{32}P incorporation is determined. The rate of the exchange reaction can be measured and compared with the rate of the overall reaction (i.e. synthesis of aminoacyl-tRNA). In the majority of cases the rate of the first reaction is 10–100 times the rate of the overall reaction,¹³⁷ thus implying that the second reaction (aminoacyl transfer or product release) is rate determining. However, in other cases, e.g. tyrosyl-tRNA synthetase from *E. coli*, the rate constants for the two reactions are very similar and thus both contribute to the rate-determining step.¹⁴² In some cases, additional complexities arise: thus, the dimeric tyrosyl-tRNA synthetase from *Bacillus stearothermophilus* shows 'half-of-the-sites' reactivity (see Chapter 6, Section 6.2.2.2), with only one molecule of tyrosyl-AMP formed and one molecule tRNA bound per molecule of enzyme. In this case, the enzyme-bound tyrosyl-AMP intermediate can bind additional tyrosine and ATP and undergo a conformational change.^{143,144}

5.5.5.3 Recognition of substrates

As mentioned in Section 5.5.5, it is crucial for the aminoacyl-tRNA synthetases to recognize their substrates correctly, and much experimental work has been devoted to an analysis of the mechanisms by which this recognition is achieved.

Because of their small size, the recognition of amino acids poses particular problems.¹⁴⁵ For instance, the factor by which isoleucine could be favoured over valine by isoleucyl-tRNA synthetase can be estimated as 20–40 fold (taking into account the binding energy available due to the extra methylene group in isoleucine and the relative intracellular concentrations of the two amino acids). However, the actual rate of misincorporation of valine for isoleucine is very low (1 part in 3000).¹⁴⁶ Experimentally, it has been found that isoleucyl-tRNA synthetase can activate valine, i.e. catalyse the formation of an enzyme-bound valyl-AMP complex in the absence of tRNA.¹⁴⁵ However, the valine cannot be subsequently transferred to tRNA; indeed, addition of tRNA^{Val} to the complex leads to hydrolysis of the valyl-AMP to give valine and AMP. This type of 'editing' or 'proof-reading' mechanism has been found in a number of aminoacyl-tRNA synthetases and involves the operation of a second (editing) active site in the enzyme.¹⁴⁵ In some of these cases there is evidence for the transient formation of the 'wrong' aminoacyl-tRNA (e.g. threonyl-tRNA^{Val}) formed in a reaction catalysed by valyl-tRNA synthetase), but such species are rapidly destroyed before dissociation from the enzyme.¹⁴⁵

Some aminoacyl-tRNA synthetases, do not have an editing function; thus cysteinyl-tRNA and tyrosyl-tRNA synthetases bind the correct amino acids much more tightly than possible competitors.¹⁴⁵ The side-chains responsible for the binding of the correct amino acid by tyrosyl-tRNA synthetase from *Bacillus stearothermophilus* have been identified by X-ray crystallography and site-directed mutagenesis (see Section 5.5.5.5).

The recognition of the correct tRNA would appear to present fewer problems, because of the larger size of this substrate. At present, there is no X-ray structure of a synthetase-tRNA complex available, so most of the conclusions regarding the interaction have been reached by indirect means, including the effects of modification of tRNA bases on binding and the use of photochemical cross-linking to establish which parts of tRNA are in contact with the enzyme.^{137,139} The overall conclusions appear to be that the main points of contact involve the 'inside' of the L-shaped tRNA molecule (Fig. 5.42) including the acceptor stem,¹³⁷ and that in some cases (though not all) the anti-codon seems to be involved in the interaction.^{137,139} The specificity must arise within these regions and the complementary parts of the synthetase. There is evidence that at least part of the recognition process involves a conformational change in the enzyme after the correct tRNA is bound.¹³⁷ The location of amino acids in tyrosyl-tRNA synthetase responsible for the binding of tRNA^{Val} has been examined by site-directed mutagenesis (Section 5.5.5.5).

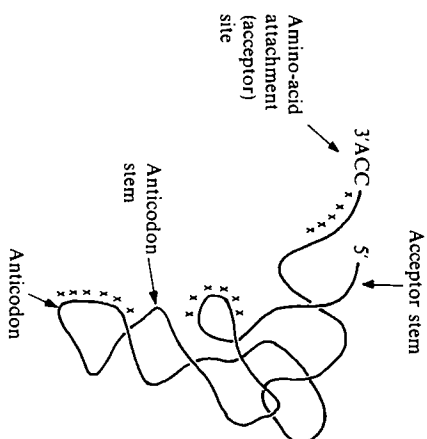


FIG. 5.42. General structure of a tRNA molecule. Regions marked $\times \times \times$ indicate portions of the molecule identified as forming contacts with aminoacyl-tRNA synthetase.¹³⁷

5.5.5.4 The X-ray structure of tyrosyl-tRNA synthetase from *Bacillus stearothermophilus*

The X-ray structure of one subunit of tyrosyl-tRNA synthetase from *Bacillus stearothermophilus* is shown in Fig. 5.43.¹⁴⁷ The enzyme contains 419 amino

acids in each of the two subunits; however, only the N-terminal 319 amino acids can be distinguished in the electron-density map; the C-terminal 100 amino acids are presumably too disordered to be discerned (see Section 5.5.5.5). In the centre of each subunit is a 6-stranded β -sheet structure and there are five long (> 1.2 nm) segments of α -helix with several shorter stretches. The location of the substrate binding site is shown in Fig. 5.43. A number of complexes have been examined by X-ray crystallography, namely with AMP, ATP, tyrosine, and tyrosyl-AMP (the last-named complex could be studied because of its high stability).¹⁴⁴ The amino-acid side-chains that interact with tyrosyl-AMP are described in Section 5.5.5.5.

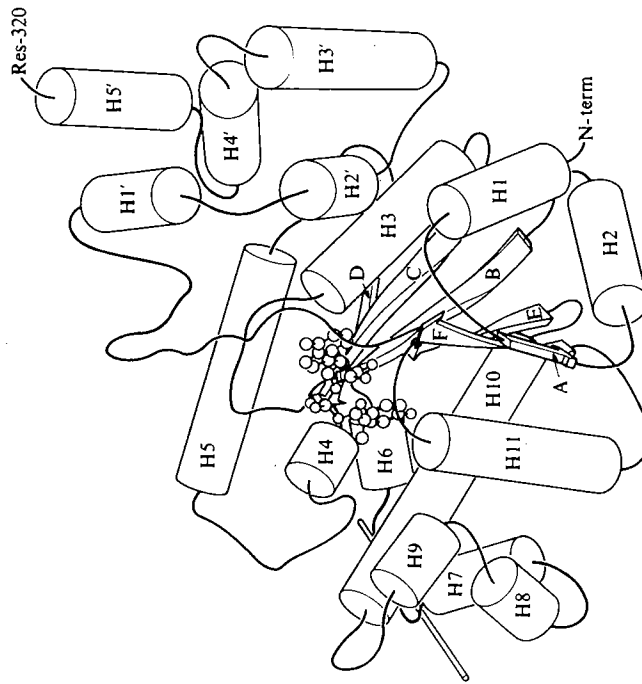


FIG. 5.43. Polypeptide chain conformation of residues 1–320 of the tyrosyl-tRNA synthetase monomer. The binding site for tyrosyl-AMP and the molecular 2-fold axis are shown. The strands of β -sheet are labelled A to F. The α -helices run between the following amino acids:

H1, 2–10; H2, 19–27; H3, 49–60; H4, 71–74; H5, 91–105; H6, 124–128; H7, 132–139; H8, 145–149; H9, 152–156; H10, 164–183; H11, 195–208; H1', 248–256; H2', 263–269; H3', 275–287; H4', 293–306; H5', 309–318.

(The figure was kindly supplied by Dr P. Brick and Prof. D.M. Blow, Imperial College, London, UK.)

5.5.5.5 Studies by site-directed mutagenesis

The powerful technique of site-directed mutagenesis (Section 5.4.5) has been employed by Fersht and his colleagues to undertake a series of very elegant experiments in which the roles of different parts of the tyrosyl-tRNA synthetase from *Bacillus stearothermophilus* have been explored. The design and interpretation of these experiments have drawn heavily on the detailed structural information available from X-ray crystallography. Much of the site-directed mutagenesis work has been summarized in a recent review.¹⁴⁴

(i) Hydrogen bonds and specificity

The X-ray crystallographic work indicated that a number of hydrogen bonds were formed between the enzyme and tyrosyl-AMP (Fig. 5.44).

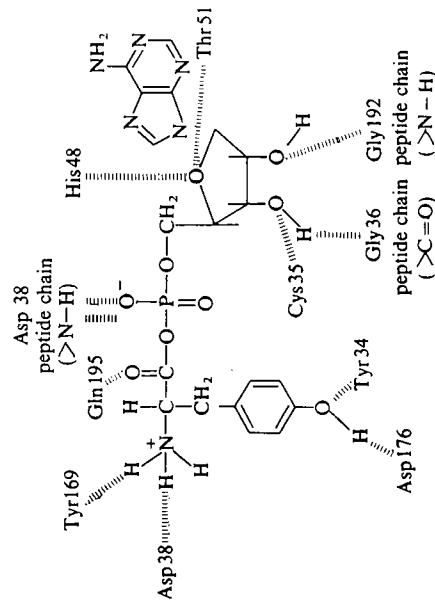


FIG. 5.44. Hydrogen bonds (indicated by dashed lines) involved in the interaction between tyrosyl-AMP and tyrosyl-tRNA synthetase.¹⁴⁸

The side-chains implicated have been systematically replaced by side-chains incapable of forming hydrogen bonds (e.g. Cys 35 $\rightarrow$ Gly, Tyr 34 $\rightarrow$ Phe) and the effects on the kinetics of the activation of tyrosine by ATP have been studied.¹⁴⁸ The results show that each of the contacts identified by X-ray crystallography is important and that other side-chains play a role (e.g. Asp 78 is involved in an interaction with the α -amino group of the tyrosyl-AMP). The side-chains responsible for the specificity of the enzyme for tyrosine as opposed to phenylalanine are those of Tyr 34 and Asp 176¹⁴⁴ (see Fig. 5.44). In the ribose-binding site the side-chains of Cys 35, Thr 51 and His 48 are involved in specific hydrogen-bonding interactions. Cysteine 35 had been implicated by chemical modification experiments as being involved in the catalytic activity and this residue is conserved in other prokaryotic tyrosyl-tRNA synthetases,¹³⁹ however,

replacement by Gly or Ser did not lead to complete loss of activity (k_{cat} was reduced to 30 per cent of the value for wild-type enzyme) and thus Cys 35 is not an essential side-chain.¹⁴⁸

The results of site-directed mutagenesis experiments showed that different types of hydrogen bonds made different contributions to the binding energy.¹⁴⁸ Thus, mutation of an uncharged side-chain (Tyr 169) that forms a hydrogen bond to a charged group on the substrate (the α -amino group) weakens the binding by 15.5 kJ mol⁻¹, whereas mutation of a side-chain (Tyr 34) that forms a hydrogen bond to an uncharged group (the phenolic OH group of tyrosyl-AMP) weakens the binding by only 2.2 kJ mol⁻¹.

In Fig. 5.44, the side-chain of Thr 51 is seen to form a hydrogen bond with the ribose of tyrosyl-AMP. However, this bond is thought to be weak, because of unfavourable geometry. The side-chain of Thr 51 could form a stronger hydrogen bond with water, thereby promoting dissociation of the enzyme-tyrosyl-AMP complex. It is interesting to note that Thr 51 is the only amino acid in the active site that is not conserved in the tyrosyl-tRNA synthetase from other prokaryotic sources, e.g. *E. coli* (Pro) and *Bacillus caldotenax* (Ala). The Thr 51 of the *Bacillus stearothermophilus* enzyme has been replaced by other side-chains, e.g. Pro, Ala, that cannot make the hydrogen bond to ribose.¹⁴⁹ As measured by the ratio k_{cat}/K_m (see Chapter 4, Section 4.3.1.3) the activities of the Thr 51 $\rightarrow$ Pro and Thr 51 $\rightarrow$ Ala mutants are improved over that of the native enzyme by factors of 50 and 2, respectively. This type of experiment indicates that it is possible to change, and even improve, the catalytic properties of enzymes in a rational manner.

(i) Roles of the N- and C-terminal regions of the enzyme

A deletion mutant of tyrosyl-tRNA synthetase has been prepared consisting of the N-terminal 319 amino acids.¹⁵⁰ This mutant was found to be indistinguishable from the native enzyme as far as the activation of tyrosine (to form tyrosyl-AMP) was concerned, but was unable to bind to or aminoacylate tRNA. The conclusion that the C-terminal region of the polypeptide chain is involved in binding tRNA seems to be general for synthetases.¹⁵¹ Some of the amino-acid side-chains of tyrosyl-tRNA synthetase involved in binding tRNA have been identified by site-directed mutagenesis.¹⁵¹ In this work, heterodimers were constructed containing one full-length polypeptide chain (419 amino acids) and one truncated chain (319 amino acids). Positively charged side-chains were replaced by neutral ones (Arg, His $\rightarrow$ Gln; Lys $\rightarrow$ Asn). The interactions with tRNA were found to extend across the two subunits with sets of positively charged side-chains interacting with the acceptor and anti-codon stems (Fig. 5.42), clamping the tRNA in a fixed orientation. The flexible-CCA (3' terminus) could then be aligned at the active site by additional contacts.¹⁵¹

In a further experiment, the nature of the interactions between the subunits of the enzyme was examined. Phe 164 lies on the symmetry axis at the subunit interface and the side-chains from the two subunits form hydrophobic contacts. On replacement of this Phe by Asp, the enzyme was found to dissociate at pH

7.78 because of the introduction of negative charges in close proximity. At lower pH, the Asp side-chains are protonated and the subunits can reassociate.¹⁵²

5.6 Concluding comments on enzyme mechanisms

In this chapter we have shown that it is now possible to propose fairly detailed mechanisms for a number of enzyme-catalysed reactions. Clearly, the detailed structural information available from X-ray crystallography has had a profound influence on the progress made in this area in recent years, but it can be seen that the other techniques including site-directed mutagenesis have had a useful role to play. Many of the factors suggested by model studies (Section 5.3) as leading to rate enhancement, such as acid-base catalysis and intermediate-formation, are seen to occur in the mechanisms of the enzymes discussed in Section 5.5. It is therefore reasonable to state that enzyme catalysis does not involve any new principles in addition to those involved in non-enzyme-catalysed reactions; enzymes merely achieve highly effective combinations of the various established factors leading to rate enhancement. However, such a statement cannot be made with certainty until the observed rate enhancements in a number of enzyme-catalysed reactions can be accounted for quantitatively which, as yet, is not the case.

In the case of lactate dehydrogenase (Section 5.5.4) it is clear that binding of substrates is accompanied by large conformational changes in the enzyme. Such structural changes have been shown to occur in a number of enzymes (e.g. carboxypeptidase, lysozyme, and hexokinase) by X-ray crystallographic or other less-direct spectroscopic methods.¹⁵³ These observations provide support for Koshland's 'induced-fit' hypothesis,¹⁵⁴ which proposed that binding of substrate to enzyme induces a structural change in the enzyme so that the important amino-acid side-chains are brought into the correct spatial relationship for catalysis to occur. This 'induced fit' was viewed as playing a crucial role in enzyme catalysis and specificity (i.e. different substrates would be distinguished from one another by the extents to which they would induce the active conformation of the enzyme). More recently, however, the quantitative significance of 'induced fit' has been called into question,¹⁵⁵ and it has been proposed that specificity arises because of the differences in binding of the transition states of different substrates to the enzyme, i.e. the transition states of better substrates bind more effectively, see Section 5.3.4. It is difficult to believe that there will not be considerable debate on this topic in the future.

References

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